

Sample Problem 13.12

A satellite is launched from an altitude of 500 km in a direction parallel to the surface of the earth with a velocity of 36 900 km/h. Determine (a) the maximum altitude reached by the satellite, (b) the maximum allowable error in the direction of launching if the satellite is to go into orbit and come no closer than 200 km to the surface of the earth.

STRATEGY: Since the only force acting on the satellite is the force of gravity, which is a central force, and you are interested in two positions (the position of the satellite at launch and at its maximum altitude), you can use conservation of angular momentum and conservation of energy.

MODELING: Choose the satellite as your system and model it as a particle.

ANALYSIS:

a. Maximum Altitude. Denote the point of the orbit farthest from the earth by A' and the corresponding distance from the center of the earth by r_1 (Fig. 1). Since the satellite is in free flight between A and A' , you can apply the principle of conservation of energy as

$$T_A + V_A = T_{A'} + V_{A'} \\ \frac{1}{2}mv_0^2 - \frac{GMm}{r_0} = \frac{1}{2}mv_1^2 - \frac{GMm}{r_1} \quad (1)$$

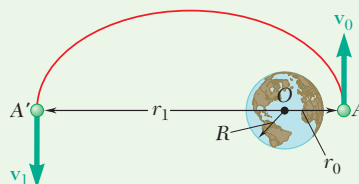


Fig. 1 The system in the two positions of interest.

Now apply the principle of conservation of angular momentum of the satellite about O . Considering points A and A' , you have

$$r_0mv_0 = r_1mv_1 \quad v_1 = v_0 \frac{r_0}{r_1} \quad (2)$$

Substitute this expression for v_1 into Eq. (1), divide each term by the mass m , and rearrange the terms. The result is

$$\frac{1}{2}v_0^2 \left(1 - \frac{r_0^2}{r_1^2} \right) = \frac{GM}{r_0} \left(1 - \frac{r_0}{r_1} \right) \quad 1 + \frac{r_0}{r_1} = \frac{2GM}{r_0v_0^2} \quad (3)$$

Recall that the radius of the earth is $R = 6370$ km. This gives you

$$\begin{aligned} r_0 &= 6370 \text{ km} + 500 \text{ km} = 6870 \text{ km} = 6.87 \times 10^6 \text{ m} \\ v_0 &= 36\,900 \text{ km/h} = (36.9 \times 10^6 \text{ m}) / (3.6 \times 10^3 \text{ s}) = 10.25 \times 10^3 \text{ m/s} \\ GM &= gR^2 = (9.81 \text{ m/s}^2)(6.37 \times 10^6 \text{ m})^2 = 398 \times 10^{12} \text{ m}^3/\text{s}^2 \end{aligned}$$

Substituting these values into Eq. (3), you obtain $r_1 = 66.8 \times 10^6 \text{ m}$.

Maximum altitude $= 66.8 \times 10^6 \text{ m} - 6.37 \times 10^6 \text{ m} = 60.4 \times 10^6 \text{ m} =$
60 400 km ◀

b. Allowable Error in Direction of Launch. The satellite is launched from P_0 in a direction forming an angle ϕ_0 with the vertical OP_0 (Fig. 2). You obtain the value of ϕ_0 corresponding to $r_{\min} = 6370 \text{ km} + 200 \text{ km} = 6570 \text{ km}$ by applying the principles of conservation of energy and of conservation of angular momentum between P_0 and A :

$$\frac{1}{2}mv_0^2 - \frac{GMm}{r_0} = \frac{1}{2}mv_{\max}^2 - \frac{GMm}{r_{\min}} \quad (4)$$

$$r_0mv_0 \sin \phi_0 = r_{\min}mv_{\max} \quad (5)$$

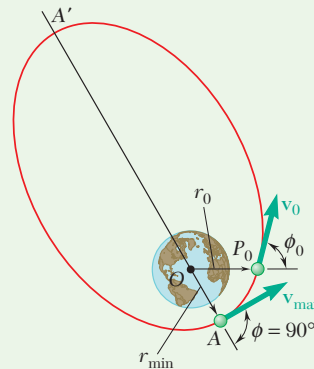


Fig. 2 Two locations used to determine maximum allowable error in direction.

Solving (5) for $v_{\max}$ and then substituting for $v_{\max}$ into (4), you can solve (4) for $\sin \phi_0$. Finally, using the values of v_0 and GM computed in part a and noting that $r_0/r_{\min} = 6870/6570 = 1.0457$, you find

$$\sin \phi_0 = 0.9801 \quad \phi_0 = 90^\circ \pm 11.5^\circ \quad \text{Allowable error} = \pm 11.5^\circ \quad \blacktriangleleft$$

REFLECT and THINK: Space probes and other long-distance vehicles are designed with small rockets to allow for mid-course corrections. Satellites launched from the Space Station usually do not need this kind of fine-tuning.

SOLVING PROBLEMS ON YOUR OWN

In this section you learned that when the work done by a force $\mathbf{F}$ acting on a particle A is independent of the path followed by the particle as it moves from a given position A_1 to a given position A_2 (Fig. 13.11a), then we can define a function V , called **potential energy**, for the force $\mathbf{F}$. Such forces are said to be **conservative forces**, and you can write

$$U_{1 \rightarrow 2} = -(V(x_2, y_2, z_2) - V(x_1, y_1, z_1)) = V(x_1, y_1, z_1) - V(x_2, y_2, z_2) \quad (13.19)$$

or for short,

$$U_{1 \rightarrow 2} = V_1 - V_2 \quad (13.19')$$

The work is negative when the change in potential energy is positive, i.e., when $V_2 > V_1$.

Substituting this expression into the equation for work and energy, you can write

$$T_1 + V_1 = T_2 + V_2 \quad (13.24)$$

or

$$T_1 + V_{g_1} + V_{e_1} = T_2 + V_{g_2} + V_{e_2} \quad (13.24')$$

This equation states that when a particle moves under the action of a conservative force, **the sum of the kinetic and potential energies of the particle remains constant**. We expanded this equation for cases when there are non-conservative forces present:

$$T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{\text{NC}} = T_2 + V_{g_2} + V_{e_2} \quad (13.24'')$$

Your solutions of problems using the above formulas will consist of the following steps.

1. Determine whether all the forces involved are conservative. If some of the forces are not conservative—for example, if friction is involved—you must use the second equation (13.24''), since the work done by such forces depends upon the path followed by the particle and a potential function does not exist for these non-conservative forces. You can then determine the work done by non-conservative forces as:

$$U_{1 \rightarrow 2}^{\text{NC}} = \int_1^2 \mathbf{F}^{\text{NC}} \cdot d\mathbf{s}$$

2. Determine the kinetic energy $T = \frac{1}{2}mv^2$ at each end of the path.

3. Compute the potential energy for all the forces involved at each end of the path. Recall the following expressions for potential energy derived in this section.

a. The potential energy of a weight W close to the surface of Earth and at a height y above a given datum:

$$V_g = Wy \quad (13.16)$$

b. The potential energy of a mass m located at a distance r from the center of the earth, large enough so that the variation of the force of gravity must be taken into account:

$$V_g = -\frac{GMm}{r} \quad (13.17)$$

where the distance r is measured from the center of the earth and V_g is equal to zero at $r = \infty$.

c. The potential energy of a body with respect to an elastic force $F = kx$:

$$V_e = \frac{1}{2}kx^2 \quad (13.18)$$

where the distance x is the deflection of the elastic spring measured from its *undeformed* position and k is the spring constant. Note that V_e depends only upon the deflection x and not upon the path of the body attached to the spring. Also, V_e is always positive, whether the spring is compressed or elongated.

4. Substitute your expressions for the non-conservative work and the kinetic and potential energies into Eq. (13.24"). You will be able to solve this equation for one unknown—for example, for a velocity [Sample Prob. 13.8]. If more than one unknown is involved, you will have to search for another condition or equation, such as Newton's second law [Sample Prob. 13.10], the maximum speed [Sample Prob. 13.9], minimum speed [Sample Prob. 13.10], or the minimum potential energy of the particle. For problems involving a central force, you can obtain a second equation by using conservation of angular momentum [Sample Prob. 13.11]. This is especially useful in space mechanics applications [Sec. 13.2D].

Problems

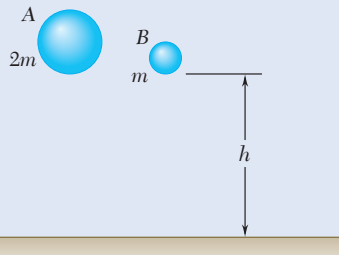


Fig. P13.CQ2

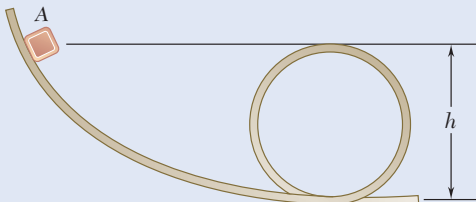


Fig. P13.CQ3

CONCEPT QUESTIONS

13.CQ2 Two small balls A and B with masses $2m$ and m , respectively, are released from rest at a height h above the ground. Neglecting air resistance, which of the following statements is true when the two balls hit the ground?

- The kinetic energy of A is the same as the kinetic energy of B.
- The kinetic energy of A is half the kinetic energy of B.
- The kinetic energy of A is twice the kinetic energy of B.
- The kinetic energy of A is four times the kinetic energy of B.

13.CQ3 A small block A is released from rest and slides down the frictionless ramp to the loop. The maximum height h of the loop is the same as the initial height of the block. Will A make it completely around the loop without losing contact with the track?

- Yes
- No
- Need more information

END-OF-SECTION PROBLEMS

13.55 A force $\mathbf{P}$ is slowly applied to a plate that is attached to two springs and causes a deflection x_0 . In each of the two cases shown, derive an expression for the constant k_e , in terms of k_1 and k_2 , of the single spring equivalent to the given system, that is, of the single spring which will undergo the same deflection x_0 when subjected to the same force $\mathbf{P}$.

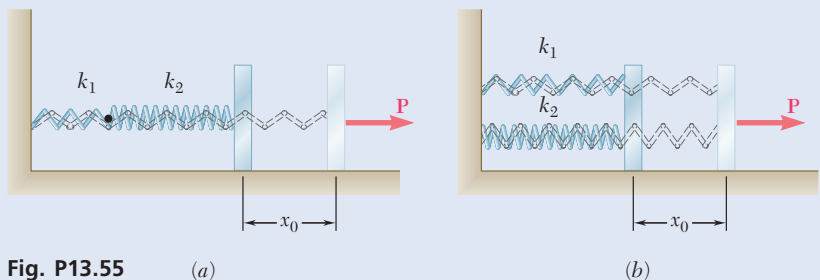


Fig. P13.55

13.56 A loaded railroad car of mass m is rolling at a constant velocity $\mathbf{v}_0$ when it couples with a massless bumper system. Determine the maximum deflection of the bumper assuming the two springs are (a) in series (as shown), (b) in parallel.

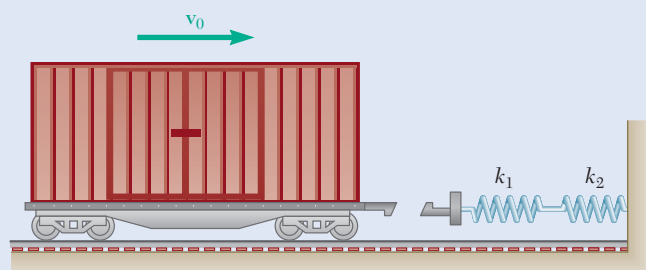


Fig. P13.56

13.57 A 750-g collar can slide along the horizontal rod shown. It is attached to an elastic cord with an undeformed length of 300 mm and a spring constant of 150 N/m. Knowing that the collar is released from rest at A and neglecting friction, determine the speed of the collar (*a*) at B, (*b*) at E.

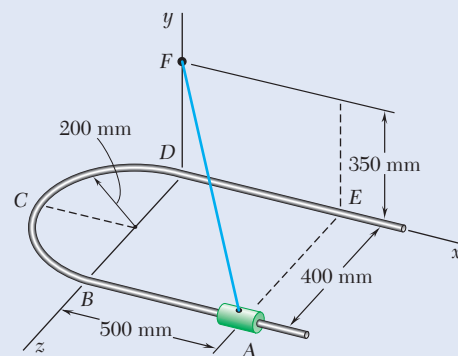


Fig. P13.57

13.58 A 4-lb collar can slide without friction along a horizontal rod and is in equilibrium at A when it is pushed 1 in. to the right and released from rest. The springs are undeformed when the collar is at A and the constant of each spring is 2800 lb/in. Determine the maximum velocity of the collar.

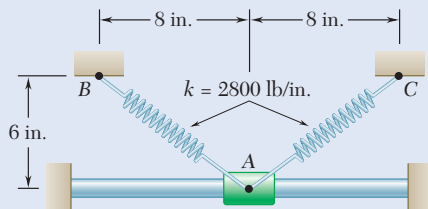


Fig. P13.58 and P13.59

13.59 A 4-lb collar can slide without friction along a horizontal rod and is released from rest at A. The undeformed lengths of springs BA and CA are 10 in. and 9 in., respectively, and the constant of each spring is 2800 lb/in. Determine the velocity of the collar when it has moved 1 in. to the right.

13.60 A 500-g collar can slide without friction on the curved rod BC in a horizontal plane. Knowing that the undeformed length of the spring is 80 mm and that $k = 400$ kN/m, determine (*a*) the velocity that the collar should be given at A to reach B with zero velocity, (*b*) the velocity of the collar when it eventually reaches C.

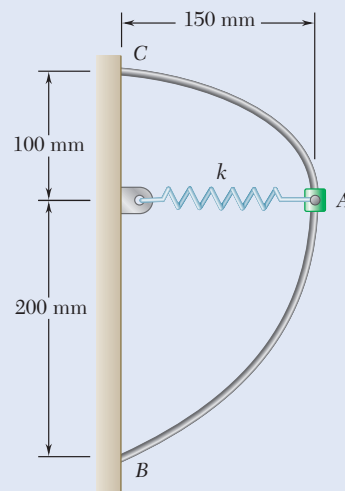


Fig. P13.60

13.61 For the adapted shuffleboard device in Prob 13.28, you decide to utilize an elastic cord instead of a compression spring to propel the puck forward. When the cord is stretched directly between points A and B, the tension is 20 N. The 425-gram puck is placed in the center and pulled back through a distance of 400 mm; a force of 100 N is required to hold it at this location. Knowing that the coefficient of friction is 0.3, determine how far the puck will travel.

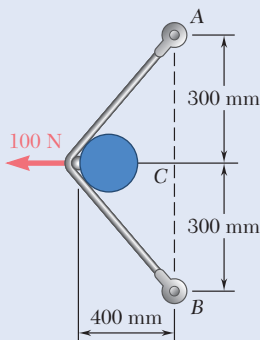


Fig. P13.61

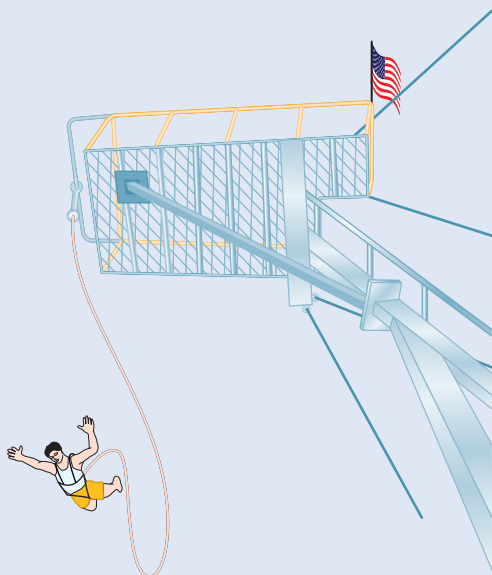


Fig. P13.62

13.62 An elastic cable is to be designed for bungee jumping from a tower 130 ft high. The specifications call for the cable to be 85 ft long when unstretched, and to stretch to a total length of 100 ft when a 600-lb weight is attached to it and dropped from the tower. Determine (a) the required spring constant k of the cable, (b) how close to the ground a 186-lb man will come if he uses this cable to jump from the tower.

13.63 It is shown in mechanics of materials that the stiffness of an elastic cable is $k = AE/L$, where A is the cross-sectional area of the cable, E is the modulus of elasticity, and L is the length of the cable. A winch is lowering a 4000-lb piece of machinery using a constant speed of 3 ft/s when the winch suddenly stops. Knowing that the steel cable has a diameter of 0.4 in., $E = 29 \times 10^6$ lb/in², and when the winch stops $L = 30$ ft, determine the maximum downward displacement of the piece of machinery from the point it was when the winch stopped.

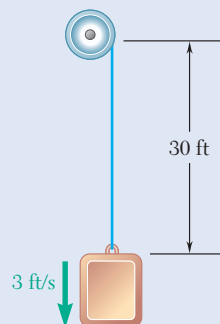


Fig. P13.63

13.64 A 2-kg collar is attached to a spring and slides without friction in a vertical plane along the curved rod ABC . The spring is undeformed when the collar is at C and its constant is 600 N/m. If the collar is released at A with no initial velocity, determine its velocity (a) as it passes through B , (b) as it reaches C .

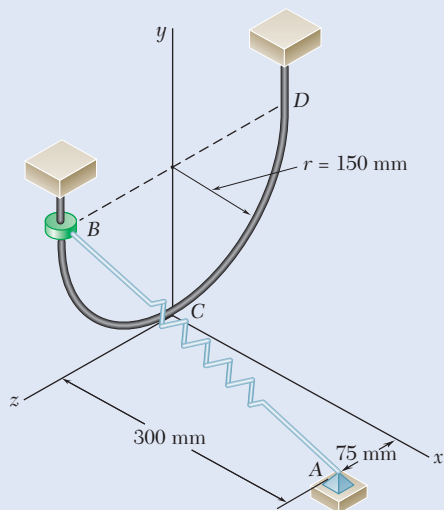


Fig. P13.65

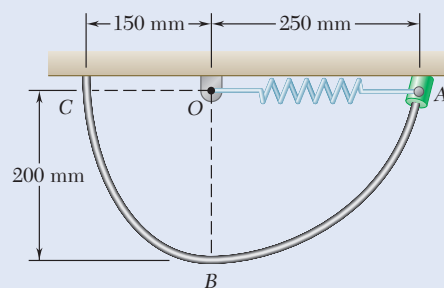


Fig. P13.64

13.65 A 500-g collar can slide without friction along the semicircular rod BCD . The spring is of constant 320 N/m and its undeformed length is 200 mm. Knowing that the collar is released from rest at B , determine (a) the speed of the collar as it passes through C , (b) the force exerted by the rod on the collar at C .

- 13.66** A thin circular rod is supported in a *vertical plane* by a bracket at *A*. Attached to the bracket and loosely wound around the rod is a spring of constant $k = 3 \text{ lb/ft}$ and undeformed length equal to the arc of circle *AB*. An 8-oz collar *C*, not attached to the spring, can slide without friction along the rod. Knowing that the collar is released from rest at an angle θ with the vertical, determine (a) the smallest value of θ for which the collar will pass through *D* and reach point *A*, (b) the velocity of the collar as it reaches point *A*.

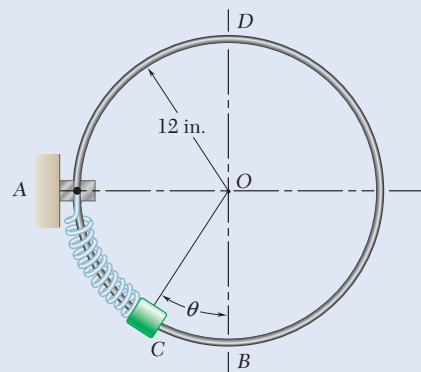


Fig. P13.66

- 13.67** Cornhole is a game that requires you to toss beanbags through a hole in a wooden board. People with limited arm mobility often have difficulty enjoying this favorite tailgating activity. An adapted launching device attaches to a wheelchair so that points *O* and *A* are fixed. The device mimics an underhand throw by utilizing an elastic band to power the arm *OC*, which rotates about pin *O*. The elastic cord has an unstretched length of 1 ft and is attached to the fixed point *A* and to point *B* on the arm. The combined weight of the beanbag and holder at *C* is 4 lbs, and you can neglect the weight of the rod *OB*. Knowing that the starting position is 30° from the horizontal, as shown in the figure, determine the spring constant if the velocity of the bean bag is 31 ft/s when the bag is released at an angle of $\theta = 45^\circ$.

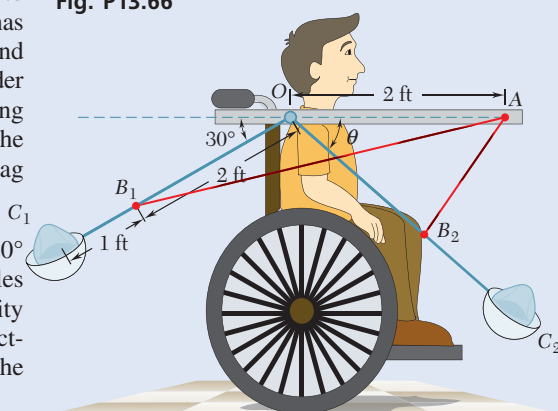


Fig. P13.67

- 13.68** A spring is used to stop a 50-kg package that is moving down a 20° incline. The spring has a constant $k = 30 \text{ kN/m}$ and is held by cables so that it is initially compressed 50 mm. Knowing that the velocity of the package is 2 m/s when it is 8 m from the spring and neglecting friction, determine the maximum additional deformation of the spring in bringing the package to rest.

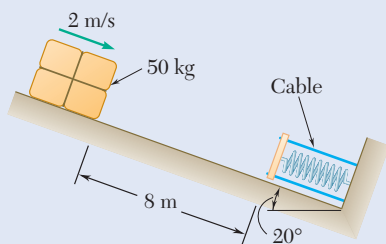


Fig. P13.68

- 13.69** Solve Prob. 13.68 assuming the kinetic coefficient of friction between the package and the incline is 0.2.
- 13.70** A section of track for a roller coaster consists of two circular arcs *AB* and *CD* joined by a straight portion *BC*. The radius of *AB* is 27 m and the radius of *CD* is 72 m. The car and its occupants, of total mass 250 kg, reach point *A* with practically no velocity and then drop freely along the track. Determine the normal force exerted by the track on the car as the car reaches point *B*. Ignore air resistance and rolling resistance.
- 13.71** A section of track for a roller coaster consists of two circular arcs *AB* and *CD* joined by a straight portion *BC*. The radius of *AB* is 27 m and the radius of *CD* is 72 m. The car and its occupants, of total mass 250 kg, reach point *A* with practically no velocity and then drop freely along the track. Determine the maximum and minimum values of the normal force exerted by the track on the car as the car travels from *A* to *D*. Ignore air resistance and rolling resistance.

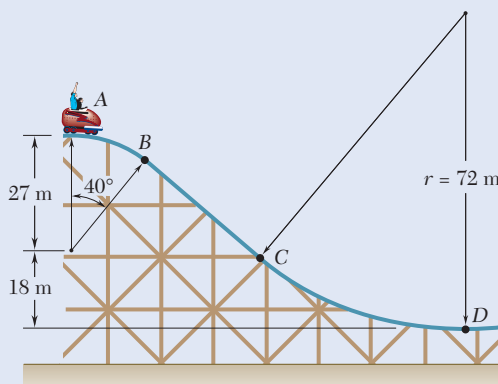


Fig. P13.70 and P13.71

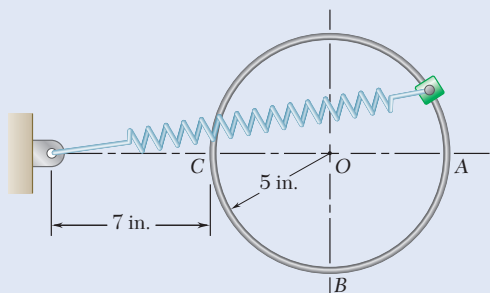


Fig. P13.72

13.72 A 1-lb collar is attached to a spring and slides without friction along a circular rod in a *vertical* plane. The spring has an undeformed length of 5 in. and a constant $k = 10$ lb/ft. Knowing that the collar is released from being held at A, determine the speed of the collar and the normal force between the collar and the rod as the collar passes through B.

13.73 A 10-lb collar is attached to a spring and slides without friction along a fixed rod in a vertical plane. The spring has an undeformed length of 14 in. and a constant $k = 4$ lb/in. Knowing that the collar is released from rest in the position shown, determine the force exerted by the rod on the collar at (a) point A, (b) point B. Both these points are on the curved portion of the rod.

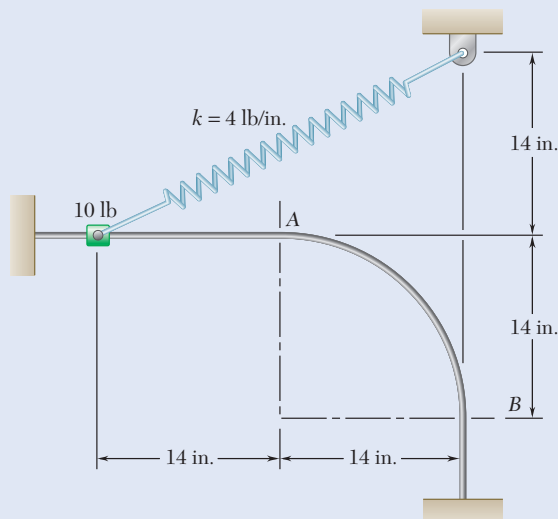


Fig. P13.73

13.74 An 8-oz package is projected upward with a velocity v_0 by a spring at A; it moves around a frictionless loop and is deposited at C. For each of the two loops shown, determine (a) the smallest velocity v_0 for which the package will reach C, (b) the corresponding force exerted by the package on the loop just before the package leaves the loop at C.

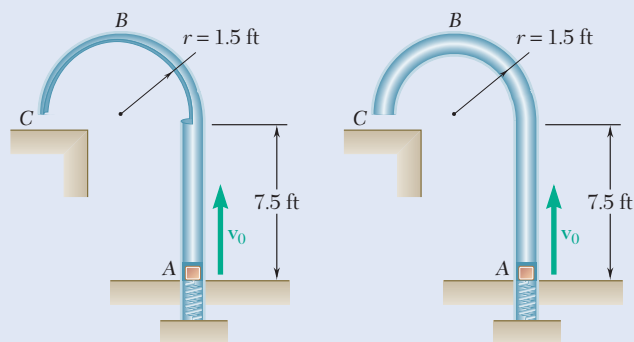


Fig. P13.74 and P13.75

13.75 If the package of Prob. 13.74 is not to hit the horizontal surface at C with a speed greater than 10 ft/s, (a) show that this requirement can be satisfied only by the second loop, (b) determine the largest allowable initial velocity v_0 when the second loop is used.

- 13.76** A small package of weight W is projected into a vertical return loop at A with a velocity v_0 . The package travels without friction along a circle of radius r and is deposited on a horizontal surface at C . For each of the two loops shown, determine (a) the smallest velocity v_0 for which the package will reach the horizontal surface at C , (b) the corresponding force exerted by the loop on the package as it passes point B .

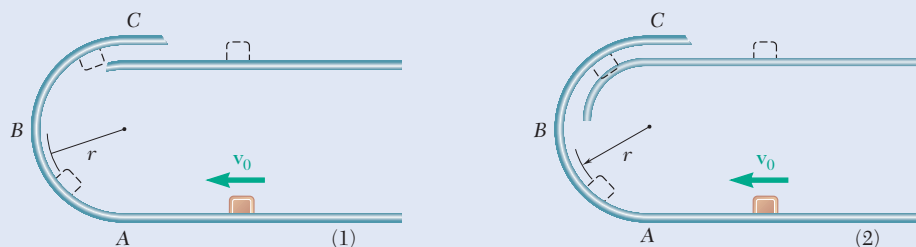


Fig. P13.76

- 13.77** The 1-kg ball at A is suspended by an inextensible cord and given an initial horizontal velocity of 5 m/s. If $l = 0.6$ m and $x_B = 0$, determine y_B so that the ball will enter the basket.

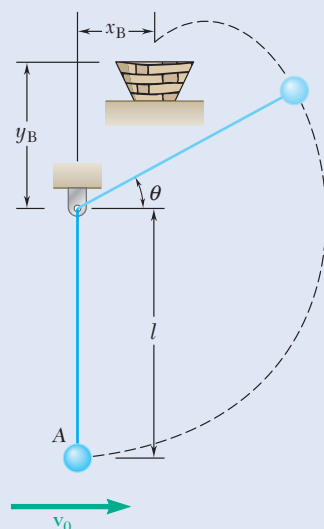


Fig. P13.77

- 13.78** The pendulum shown is released from rest at A and swings through 90° before the cord touches the fixed peg B . Determine the smallest value of a for which the pendulum bob will describe a circle about the peg.

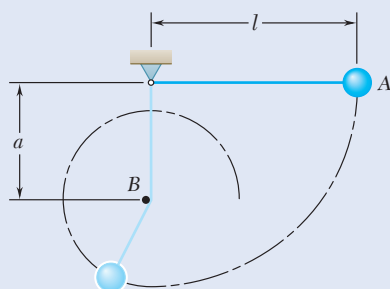


Fig. P13.78

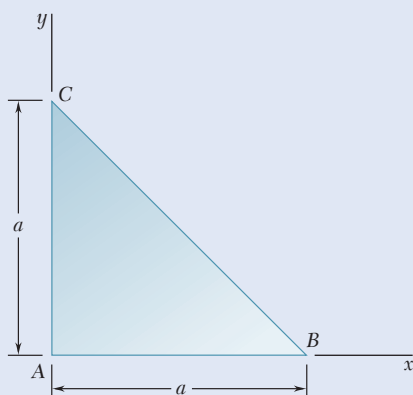


Fig. P13.81

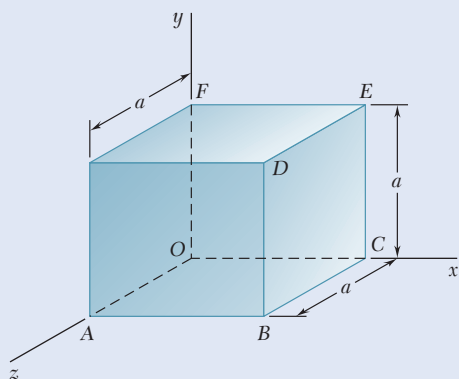


Fig. P13.82

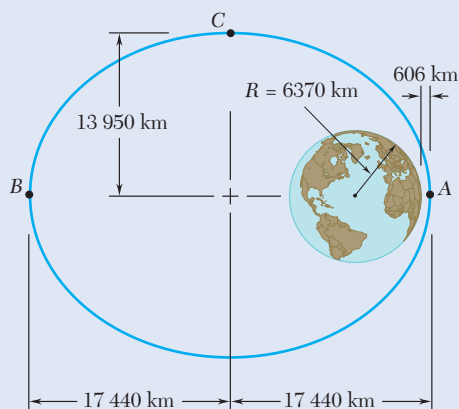


Fig. P13.86

***13.79** Prove that a force $F(x, y, z)$ is conservative if, and only if, the following relations are satisfied:

$$\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x} \quad \frac{\partial F_y}{\partial z} = \frac{\partial F_z}{\partial y} \quad \frac{\partial F_z}{\partial x} = \frac{\partial F_x}{\partial z}$$

13.80 The force $\mathbf{F} = (yz\mathbf{i} + zx\mathbf{j} + xy\mathbf{k})/xyz$ acts on the particle $P(x, y, z)$ which moves in space. (a) Using the relation derived in Prob. 13.79, show that this force is a conservative force. (b) Determine the potential function associated with $\mathbf{F}$.

***13.81** A force $\mathbf{F}$ acts on a particle $P(x, y)$ which moves in the xy plane. Determine whether $\mathbf{F}$ is a conservative force and compute the work of $\mathbf{F}$ when P describes the path $ABCA$ knowing that (a) $\mathbf{F} = (kx + y)\mathbf{i} + (kx + y)\mathbf{j}$, (b) $\mathbf{F} = (kx + y)\mathbf{i} + (x + ky)\mathbf{j}$.

***13.82** The potential function associated with a force $\mathbf{P}$ in space is known to be $V(x, y, z) = -(x^2 + y^2 + z^2)^{1/2}$. (a) Determine the x , y , and z components of $\mathbf{P}$. (b) Calculate the work done by $\mathbf{P}$ from O to D by integrating along the path $OABD$, and show that it is equal to the negative of the change in potential from O to D .

***13.83** (a) Calculate the work done from D to O by the force $\mathbf{P}$ of Prob. 13.82 by integrating along the diagonal of the cube. (b) Using the result obtained and the answer to part b of Prob. 13.82, verify that the work done by a conservative force around the closed path $OABDO$ is zero.

***13.84** The force $\mathbf{F} = (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})/(x^2 + y^2 + z^2)^{3/2}$ acts on the particle $P(x, y, z)$ which moves in space. (a) Using the relations derived in Prob. 13.79, prove that $\mathbf{F}$ is a conservative force. (b) Determine the potential function $V(x, y, z)$ associated with $\mathbf{F}$.

13.85 (a) Determine the kinetic energy per unit mass that a missile must have after being fired from the surface of the earth if it is to reach an infinite distance from the earth. (b) What is the initial velocity of the missile (called the *escape velocity*)? Give your answers in SI units and show that the answer to part b is independent of the firing angle.

13.86 A satellite describes an elliptic orbit of minimum altitude 606 km above the surface of the earth. The semimajor and semiminor axes are 17 440 km and 13 950 km, respectively. Knowing that the speed of the satellite at point C is 4.78 km/s, determine (a) the speed at point A , the perigee, (b) the speed at point B , the apogee.

13.87 While describing a circular orbit 200 mi above the earth, a space vehicle launches a 6000-lb communications satellite. Determine (a) the additional energy required to place the satellite in a geosynchronous orbit at an altitude of 22,000 mi above the surface of the earth, (b) the energy required to place the satellite in the same orbit by launching it from the surface of the earth, excluding the energy needed to overcome air resistance. (A *geosynchronous orbit* is a circular orbit in which the satellite appears stationary with respect to the ground.)

13.88 How much energy per pound should be imparted to a satellite in order to place it in a circular orbit at an altitude of (a) 400 mi, (b) 4000 mi?

- 13.89** Knowing that the velocity of an experimental space probe fired from the earth has a magnitude $v_A = 32.5$ Mm/h at point A, determine the speed of the probe as it passes through point B.
- 13.90** A spacecraft is describing a circular orbit at an altitude of 1500 km above the surface of the earth. As it passes through point A, its speed is reduced by 40 percent and it enters an elliptic crash trajectory with the apogee at point A. Neglecting air resistance, determine the speed of the spacecraft when it reaches the earth's surface at point B.
- 13.91** Observations show that a celestial body traveling at 1.2×10^6 mi/h appears to be describing about point B a circle of radius equal to 60 light years. Point B is suspected of being a very dense concentration of mass called a black hole. Determine the ratio M_B/M_S of the mass at B to the mass of the sun. (The mass of the sun is 330,000 times the mass of the earth, and a light year is the distance traveled by light in 1 year at 186,300 mi/s.)
- 13.92** (a) Show that, by setting $r = R + y$ in the right-hand member of Eq. (13.17') and expanding that member in a power series in y/R , the expression in Eq. (13.16) for the potential energy V_g due to gravity is a first-order approximation for the expression given in Eq. (13.17'). (b) Using the same expansion, derive a second-order approximation for V_g .
- 13.93** Collar A has a mass of 3 kg and is attached to a spring of constant 1200 N/m and of undeformed length equal to 0.5 m. The system is set in motion with $r = 0.3$ m, $v_\theta = 2$ m/s, and $v_r = 0$. Neglecting the mass of the rod and the effect of friction, determine the radial and transverse components of the velocity of the collar when $r = 0.6$ m.

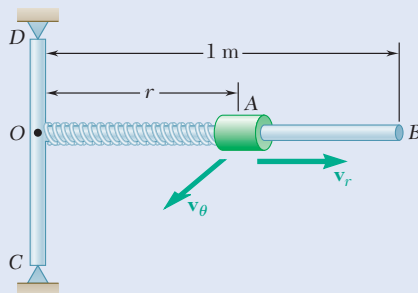


Fig. P13.93 and P13.94

- 13.94** Collar A has a mass of 3 kg and is attached to a spring of constant 1200 N/m and of undeformed length equal to 0.5 m. The system is set in motion with $r = 0.3$ m, $v_\theta = 2$ m/s, and $v_r = 0$. Neglecting the mass of the rod and the effect of friction, determine (a) the maximum distance between the origin and the collar, (b) the corresponding speed. (Hint: Solve the equation obtained for r by trial and error.)
- 13.95** A governor is designed so that the valve of negligible mass at D will open once a vertical force greater than 20 lbs is exerted on it. In initial testing of the device, the two 1-lb masses are at $x = 1$ in. and are prevented from sliding along the rod by stops. Each mass is connected to the valve by 10 lb/in. springs that are both unstretched at $x = 1$ in. The governor rotates so that $v_1 = 30$ ft/s when the stops are removed. When the valve opens, determine the position and velocity of the masses.

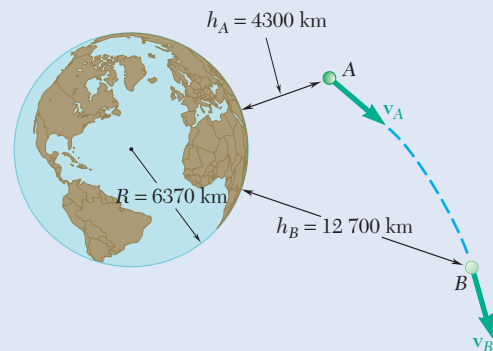


Fig. P13.89

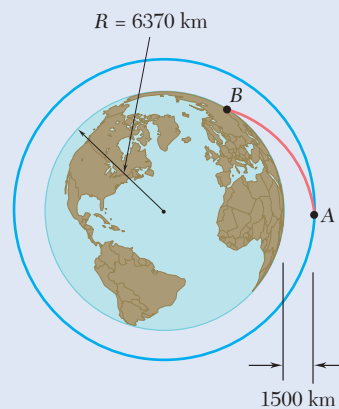


Fig. P13.90

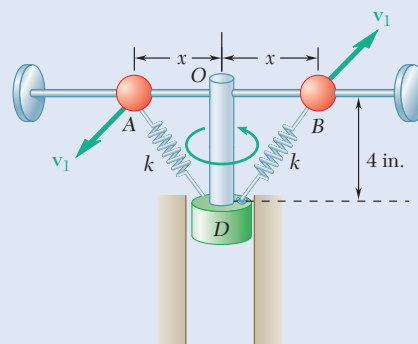


Fig. P13.95

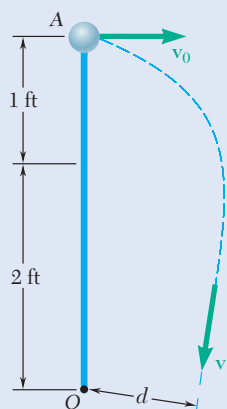


Fig. P13.96 and P13.97

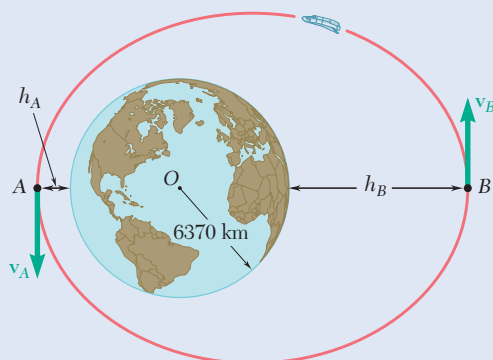


Fig. P13.100

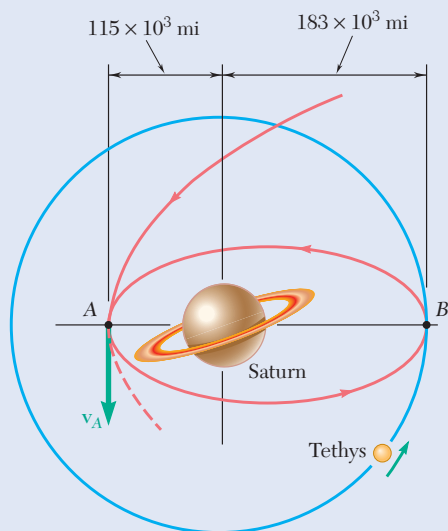


Fig. P13.102

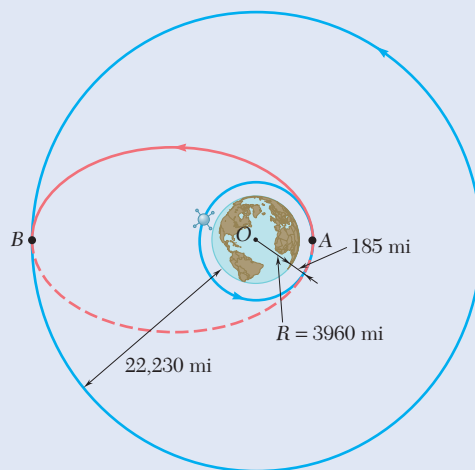


Fig. P13.101

13.96 A 1.5-lb ball that can slide on a *horizontal* frictionless surface is attached to a fixed point O by means of an elastic cord of constant $k = 1$ lb/in. and undeformed length 2 ft. The ball is placed at point A , 3 ft from O , and given an initial velocity v_0 perpendicular to OA . Determine (a) the smallest allowable value of the initial speed v_0 if the cord is not to become slack, (b) the closest distance d that the ball will come to point O if it is given half the initial speed found in part a .

13.97 A 1.5-lb ball that can slide on a *horizontal* frictionless surface is attached to a fixed point O by means of an elastic cord of constant $k = 1$ lb/in. and undeformed length 2 ft. The ball is placed at point A , 3 ft from O , and given an initial velocity v_0 perpendicular to OA , allowing the ball to come within a distance $d = 9$ in. of point O after the cord has become slack. Determine (a) the initial speed v_0 of the ball, (b) its maximum speed.

13.98 Using the principles of conservation of energy and conservation of angular momentum, solve part a of Sample Prob. 12.14.

13.99 Solve Sample Prob. 13.11, assuming that the elastic cord is replaced by a central force $\mathbf{F}$ with a magnitude of $(80/r^2)$ N directed toward O .

13.100 A spacecraft is describing an elliptic orbit of minimum altitude $h_A = 2400$ km and maximum altitude $h_B = 9600$ km above the surface of the earth. Determine the speed of the spacecraft at A .

13.101 While describing a circular orbit, 185 mi above the surface of the earth, a space shuttle ejects at point A an inertial upper stage (IUS) carrying a communications satellite to be placed in a geosynchronous orbit (see Prob. 13.87) at an altitude of 22,230 mi above the surface of the earth. Determine (a) the velocity of the IUS relative to the shuttle after its engine has been fired at A , (b) the increase in velocity required at B to place the satellite in its final orbit.

13.102 A spacecraft approaching the planet Saturn reaches point A with a velocity $\mathbf{v}_A$ of magnitude 68.8×10^3 ft/s. It is to be placed in an elliptic orbit about Saturn so that it will be able to periodically examine Tethys, one of Saturn's moons. Tethys is in a circular orbit of radius 183×10^3 mi about the center of Saturn, traveling at a speed of 37.2×10^3 ft/s. Determine (a) the decrease in speed required by the spacecraft at A to achieve the desired orbit, (b) the speed of the spacecraft when it reaches the orbit of Tethys at B .

13.103 A spacecraft traveling along a parabolic path toward the planet Jupiter is expected to reach point A with a velocity $\mathbf{v}_A$ of magnitude 26.9 km/s. Its engines will then be fired to slow it down, placing it into an elliptic orbit which will bring it to within 100×10^3 km of Jupiter. Determine the decrease in speed Δv at point A which will place the spacecraft into the required orbit. The mass of Jupiter is 319 times the mass of the earth.

13.104 As a first approximation to the analysis of a space flight from the earth to Mars, it is assumed that the orbits of the earth and Mars are circular and coplanar. The mean distances from the sun to the earth and to Mars are 149.6×10^6 km and 227.8×10^6 km, respectively. To place the spacecraft into an elliptical transfer orbit at point A , its speed is increased over a short interval of time to v_A , which is faster than the earth's orbital speed. When the spacecraft reaches point B on the elliptical transfer orbit, its speed v_B is increased to the orbital speed of Mars. Knowing that the mass of the sun is 332.8×10^3 times the mass of the earth, determine the increase in velocity required (a) at A , (b) at B .

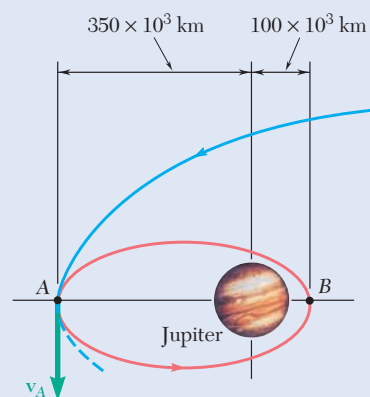


Fig. P13.103

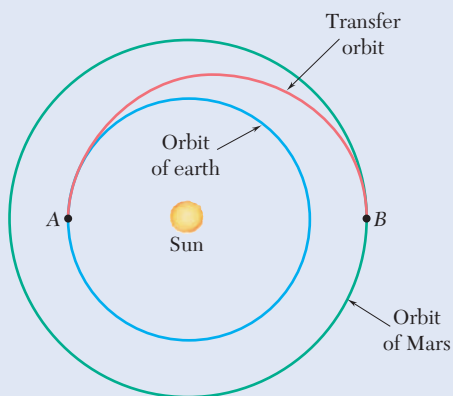


Fig. P13.104

13.105 The optimal way of transferring a space vehicle from an inner circular orbit to an outer coplanar circular orbit is to fire its engines as it passes through A to increase its speed and place it in an elliptic transfer orbit. Another increase in speed as it passes through B will place it in the desired circular orbit. For a vehicle in a circular orbit about the earth at an altitude $h_1 = 200$ mi, which is to be transferred to a circular orbit at an altitude $h_2 = 500$ mi, determine (a) the required increases in speed at A and at B , (b) the total energy per unit mass required to execute the transfer.

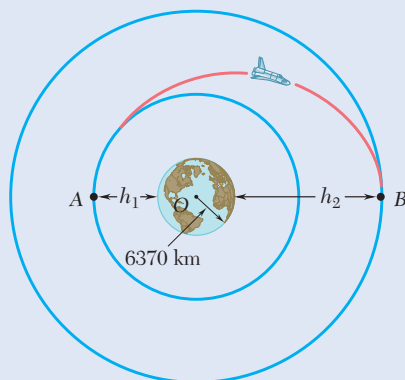


Fig. P13.105

- 13.106** During a flyby of the earth, the velocity of a spacecraft is 10.4 km/s as it reaches its minimum altitude of 990 km above the surface at point A. At point B the spacecraft is observed to have an altitude of 8350 km. Determine (a) the magnitude of the velocity at point B, (b) the angle ϕ_B .

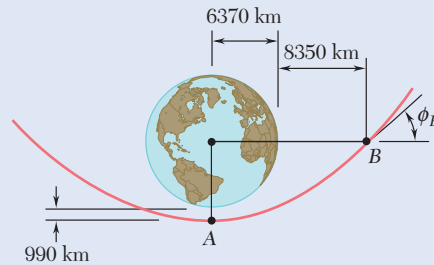


Fig. P13.106

- 13.107** A space platform is in a circular orbit about the earth at an altitude of 300 km. As the platform passes through A, a rocket carrying a communications satellite is launched from the platform with a relative velocity of magnitude 3.44 km/s in a direction tangent to the orbit of the platform. This was intended to place the rocket in an elliptic transfer orbit bringing it to point B, where the rocket would again be fired to place the satellite in a geosynchronous orbit of radius 42 140 km. After launching, it was discovered that the relative velocity imparted to the rocket was too large. Determine the angle γ at which the rocket will cross the intended orbit at point C.

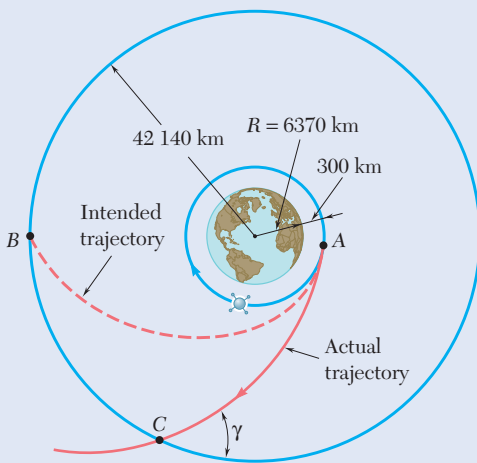


Fig. P13.107

- 13.108** A satellite is projected into space with a velocity $\mathbf{v}_0$ at a distance r_0 from the center of the earth by the last stage of its launching rocket. The velocity $\mathbf{v}_0$ was designed to send the satellite into a circular orbit of radius r_0 . However, owing to a malfunction of control, the satellite is not projected horizontally but at an angle α with the horizontal and, as a result, is propelled into an elliptic orbit. Determine the maximum and minimum values of the distance from the center of the earth to the satellite.

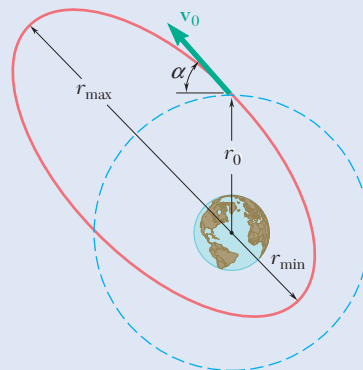


Fig. P13.108

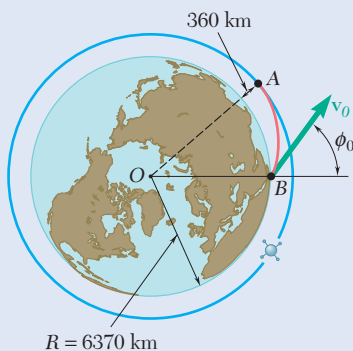


Fig. P13.109

- 13.109** A space vehicle is to rendezvous with an orbiting laboratory that circles the earth at a constant altitude of 360 km. The vehicle has reached an altitude of 60 km when its engine is shut off, and its velocity $\mathbf{v}_0$ forms an angle $\phi_0 = 50^\circ$ with the vertical OB at that time. What magnitude should $\mathbf{v}_0$ have if the vehicle's trajectory is to be tangent at A to the orbit of the laboratory?

13.110 A space vehicle is in a circular orbit at an altitude of 225 mi above the earth. To return to earth, it decreases its speed as it passes through A by firing its engine for a short interval of time in a direction opposite to the direction of its motion. Knowing that the velocity of the space vehicle should form an angle $\phi_B = 60^\circ$ with the vertical as it reaches point B at an altitude of 40 mi, determine (a) the required speed of the vehicle as it leaves its circular orbit at A, (b) its speed at point B.

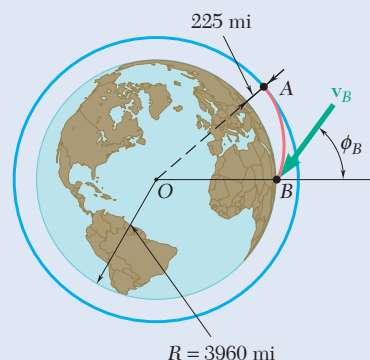


Fig. P13.110

***13.111** In Prob. 13.110, the speed of the space vehicle was decreased as it passed through A by firing its engine in a direction opposite to the direction of motion. An alternative strategy for taking the space vehicle out of its circular orbit would be to turn it around so that its engine would point away from the earth and then give it an incremental velocity $\Delta \mathbf{v}_A$ toward the center O of the earth. This would likely require a smaller expenditure of energy when firing the engine at A, but might result in too fast a descent at B. Assuming this strategy is used with only 50 percent of the energy expenditure used in Prob. 13.110, determine the resulting values of ϕ_B and v_B .

13.112 Show that the values v_A and v_P of the speed of an earth satellite at the apogee A and the perigee P of an elliptic orbit are defined by the relations

$$v_A^2 = \frac{2GM}{r_A + r_P} \frac{r_P}{r_A} \quad v_P^2 = \frac{2GM}{r_A + r_P} \frac{r_A}{r_P}$$

where M is the mass of the earth, and r_A and r_P represent, respectively, the maximum and minimum distances of the orbit to the center of the earth.

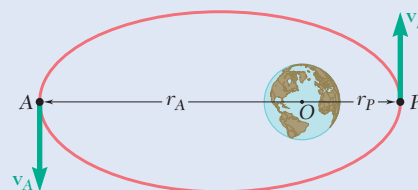


Fig. P13.112 and P13.113

13.113 Show that the total energy E of an earth satellite of mass m describing an elliptic orbit is $E = -GMm/(r_A + r_P)$, where M is the mass of the earth, and r_A and r_P represent, respectively, the maximum and minimum distances of the orbit to the center of the earth. (Recall that the gravitational potential energy of a satellite was defined as being zero at an infinite distance from the earth.)

***13.114** A space probe describes a circular orbit of radius nR with a velocity $\mathbf{v}_0$ about a planet of radius R and center O . Show that (a) in order for the probe to leave its orbit and hit the planet at an angle θ with the vertical, its velocity must be reduced to $\alpha \mathbf{v}_0$, where

$$\alpha = \sin \theta \sqrt{\frac{2(n-1)}{n^2 - \sin^2 \theta}}$$

(b) the probe will not hit the planet if α is larger than $\sqrt{2/(1+n)}$.

13.115 A missile is fired from the ground with an initial velocity $\mathbf{v}_0$ forming an angle ϕ_0 with the vertical. If the missile is to reach a maximum altitude equal to αR , where R is the radius of the earth, (a) show that the required angle ϕ_0 is defined by the relation

$$\sin \phi_0 = (1 + \alpha) \sqrt{1 - \frac{\alpha}{1 + \alpha} \left(\frac{v_{\text{esc}}}{v_0} \right)^2}$$

where v_{esc} is the escape velocity, (b) determine the range of allowable values of v_0 .

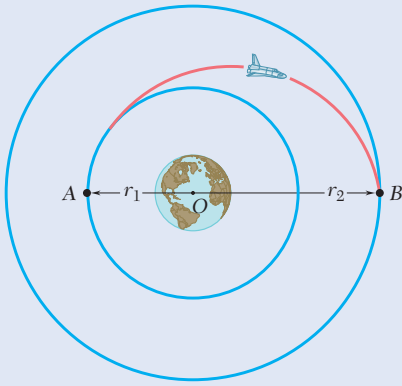


Fig. P13.116

- 13.116** A spacecraft of mass m describes a circular orbit of radius r_1 around the earth. (a) Show that the additional energy ΔE that must be imparted to the spacecraft to transfer it to a circular orbit of larger radius r_2 is

$$\Delta E = \frac{GMm(r_2 - r_1)}{2r_1r_2}$$

where M is the mass of the earth. (b) Further show that if the transfer from one circular orbit to the other is executed by placing the spacecraft on a transitional semielliptic path AB , the amounts of energy ΔE_A and ΔE_B which must be imparted at A and B are, respectively, proportional to r_2 and r_1 :

$$\Delta E_A = \frac{r_2}{r_1 + r_2} \Delta E \quad \Delta E_B = \frac{r_1}{r_1 + r_2} \Delta E$$

- *13.117** Using the answers obtained in Prob. 13.108, show that the intended circular orbit and the resulting elliptic orbit intersect at the ends of the minor axis of the elliptic orbit.

- *13.118** (a) Express in terms of $r_{\min}$ and $v_{\max}$ the angular momentum per unit mass, h , and the total energy per unit mass, E/m , of a space vehicle moving under the gravitational attraction of a planet of mass M (Fig. 13.15). (b) Eliminating $v_{\max}$ between the equations obtained, derive the formula

$$\frac{1}{r_{\min}} = \frac{GM}{h^2} \left[1 + \sqrt{1 + \frac{2E}{m} \left(\frac{h}{GM} \right)^2} \right]$$

(c) Show that the eccentricity ϵ of the trajectory of the vehicle can be expressed as

$$\epsilon = \sqrt{1 + \frac{2E}{m} \left(\frac{h}{GM} \right)^2}$$

(d) Further show that the trajectory of the vehicle is a hyperbola, an ellipse, or a parabola, depending on whether E is positive, negative, or zero.

13.3 IMPULSE AND MOMENTUM

We now consider a third basic method for the solution of problems dealing with the motion of particles. This method is based on the principle of impulse and momentum and can be used to solve problems involving force, mass, velocity, and time. It is of particular interest in the solution of problems involving impulsive motion and problems involving impacts (Secs. 13.3B and 13.4).

13.3A Principle of Impulse and Momentum

Consider a particle of mass m acted upon by a force $\mathbf{F}$. As we saw in Sec. 12.1B, we can express Newton's second law in the form

$$\mathbf{F} = \frac{d}{dt}(m\mathbf{v}) \quad (13.27)$$

where $m\mathbf{v}$ is the linear momentum of the particle. Multiplying both sides of Eq. (13.27) by dt and integrating from a time t_1 to a time t_2 , we have

$$\begin{aligned} \mathbf{F} dt &= d(m\mathbf{v}) \\ \int_{t_1}^{t_2} \mathbf{F} dt &= m\mathbf{v}_2 - m\mathbf{v}_1 \end{aligned}$$

Moving $m\mathbf{v}_1$ to the left side of this equation gives us

$$m\mathbf{v}_1 + \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2 \quad (13.28)$$

The integral in Eq. (13.28) is a vector known as the **linear impulse**, or simply the **impulse**, of the force $\mathbf{F}$ during the interval of time considered. Resolving $\mathbf{F}$ into rectangular components, we have

$$\begin{aligned} \text{Imp}_{1 \rightarrow 2} &= \int_{t_1}^{t_2} \mathbf{F} dt \\ &= \mathbf{i} \int_{t_1}^{t_2} F_x dt + \mathbf{j} \int_{t_1}^{t_2} F_y dt + \mathbf{k} \int_{t_1}^{t_2} F_z dt \end{aligned} \quad (13.29)$$

Note that the components of the impulse of force $\mathbf{F}$ are, respectively, equal to the areas under the curves obtained by plotting the components F_x , F_y , and F_z against t (Fig. 13.16). In the case of a force $\mathbf{F}$ of constant magnitude and direction, the impulse is represented by the vector $\mathbf{F}(t_2 - t_1)$, which has the same direction as $\mathbf{F}$.

If we use SI units, the magnitude of the impulse of a force is expressed in N·s. However, recalling the definition of the newton, we have

$$\text{N} \cdot \text{s} = (\text{kg} \cdot \text{m/s}^2) \cdot \text{s} = \text{kg} \cdot \text{m/s}$$

which is the unit obtained in Sec. 12.1C for the linear momentum of a particle. This verifies that Eq. (13.28) is dimensionally correct. If we use U.S. customary units, the impulse of a force is expressed in lb·s, which is also the unit obtained in Sec. 12.1C for the linear momentum of a particle.



Photo 13.4 This impact test between an F-4 Phantom and a rigid reinforced target was to determine the impact force as a function of time.

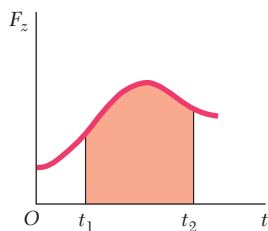
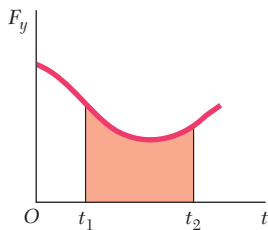
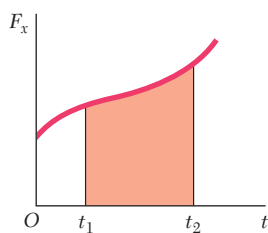


Fig. 13.16 Components of the impulse of a force $\mathbf{F}$ acting from times t_1 to t_2 .

Equation (13.28) states that when a particle is acted upon by a force $\mathbf{F}$ during a given time interval, **we can obtain the final momentum $m\mathbf{v}_2$ of the particle by adding vectorially its initial momentum $m\mathbf{v}_1$ and the impulse of the force $\mathbf{F}$ during the time interval considered.** This can be expressed as:

Principle of impulse and momentum

$$m\mathbf{v}_1 + \mathbf{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2 \quad (13.30)$$

Figure 13.17 is a pictorial representation of this principle and is called an *impulse-momentum diagram*. To obtain an analytic solution, it is thus necessary to replace Eq. (13.30) with the corresponding component equations. Note that whereas kinetic energy and work are scalar quantities, momentum and impulse are vector quantities.

$$\begin{aligned} (mv_x)_1 + \int_{t_1}^{t_2} F_x dt &= (mv_x)_2 \\ (mv_y)_1 + \int_{t_1}^{t_2} F_y dt &= (mv_y)_2 \\ (mv_z)_1 + \int_{t_1}^{t_2} F_z dt &= (mv_z)_2 \end{aligned} \quad (13.31)$$

When several forces act on a particle, we must consider the impulse of each of the forces. We have

$$m\mathbf{v}_1 + \Sigma \mathbf{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2 \quad (13.32)$$

Again, this equation represents a relation between vector quantities; in the actual solution of a problem, it should be replaced by the corresponding component equations.

When a problem involves two or more particles, we can consider each particle separately and write Eq. (13.32) for each particle. We can also add vectorially the momenta of all the particles and the impulses of all the forces involved. We then have

$$\Sigma m\mathbf{v}_1 + \Sigma \mathbf{Imp}_{1 \rightarrow 2} = \Sigma m\mathbf{v}_2 \quad (13.33)$$

Since the forces of action and reaction exerted by the particles on each other form pairs of equal and opposite forces, and since the time interval from t_1 to t_2 is common to all of the forces involved, the impulses of the forces of action and reaction cancel out. Thus, we need consider only the impulses of the external forces.[†]

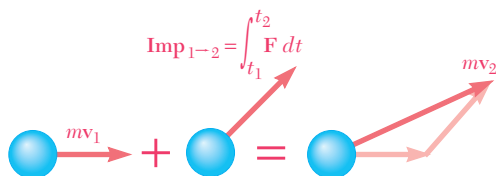


Fig. 13.17 Impulse-momentum diagram. Initial momentum plus impulse of a force $\mathbf{F}$ equals final momentum.

[†]Note the difference between this statement and the corresponding statement in Sect. 13.1C regarding the work of the forces of action and reaction between several particles. Although the sum of the impulses of these forces is always zero, the sum of their work is zero only under special circumstances, e.g., when the particles involved are connected by inextensible cords or links and thus are constrained to move through equal distances.

If no external force is exerted on the particles or, more generally, if the sum of the external forces is zero, the second term in Eq. (13.33) vanishes and the equation reduces to

Conservation of linear momentum

$$\Sigma m\mathbf{v}_1 = \Sigma m\mathbf{v}_2 \quad (13.34)$$

For two particles A and B , this is

$$m_A\mathbf{v}_A + m_B\mathbf{v}_B = m_A\mathbf{v}'_A + m_B\mathbf{v}'_B \quad (13.34')$$

where $\mathbf{v}'_A$ and $\mathbf{v}'_B$ represent the velocities of the bodies at the second time. This equation says that **the total momentum of the particles is conserved**. Consider, for example, two boats with masses of m_A and m_B , initially at rest, that are being pulled together (Fig. 13.18). If we neglect the resistance of the water, the only external forces acting on the boats are their weights and the buoyant forces exerted on them. Since these forces are balanced, we have

$$\begin{aligned} \Sigma m\mathbf{v}_1 &= \Sigma m\mathbf{v}_2 \\ 0 &= m_A\mathbf{v}'_A + m_B\mathbf{v}'_B \end{aligned}$$

where $\mathbf{v}'_A$ and $\mathbf{v}'_B$ represent the velocities of the boats after a finite interval of time. This equation indicates that the boats move in opposite directions (toward each other) with velocities inversely proportional to their masses.[†]

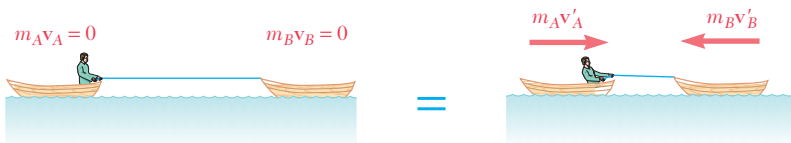


Fig. 13.18 Neglecting the resistance of the water, linear momentum is conserved for two boats being pulled together.

13.3B Impulsive Motion

A force acting on a particle during a very short time interval but large enough to produce a definite change in momentum is called an **impulsive force**. The resulting motion is called an **impulsive motion**. For example, when a baseball is struck, the contact between bat and ball takes place during a very short time interval Δt . But the average value of the force $\mathbf{F}_{\text{avg}}$ exerted by the bat on the ball is very large, and the resulting impulse $\mathbf{F}_{\text{avg}}\Delta t$ is large enough to change the sense of motion of the ball (Fig. 13.19).

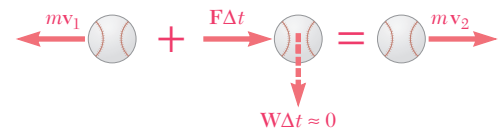


Fig. 13.19 When an impulsive force (i.e., a large force that acts over a short time) acts on a system, we can often neglect non-impulsive forces, such as weight.

[†]We use blue equals signs in Fig. 13.18 and throughout the remainder of this chapter to indicate that two systems of vectors are *equipollent*; i.e., that they have the same resultant and moment resultant (cf. Sec. 3.4B). We continue to use red equals signs to indicate that two systems of vectors are *equivalent*; i.e., they have the same effect. We will discuss this and the concept of the conservation of momentum for a system of particles in greater detail in Chap. 14.

When impulsive forces act on a particle, Eq. (13.32) becomes

**Impulse-momentum principle
for impulsive motion**

$$m\mathbf{v}_1 + \Sigma \mathbf{F}_{\text{avg}} \Delta t = m\mathbf{v}_2 \quad (13.35)$$

We can neglect any force that is not an impulsive force because the corresponding impulse $\mathbf{F}_{\text{avg}}\Delta t$ is very small. Non-impulsive forces include the weight of the body, the force exerted by a spring, or any other force that is known to be small compared with an impulsive force. Unknown reactions may or may not be impulsive; their impulses therefore should be included in Eq. (13.35) as long as they have not been proved negligible. For example, we may neglect the impulse of the weight of the baseball considered previously. If we analyze the motion of the bat, we can neglect the impulse of the weight of the bat. The impulses of the reactions of the player's hands on the bat, however, should be included; these impulses are not negligible if the ball is incorrectly hit.

Note that the method of impulse and momentum is particularly effective in analyzing the impulsive motion of a particle, since it involves only the initial and final velocities of the particle and the impulses of the forces exerted on the particle. The direct application of Newton's second law, on the other hand, would require determining the forces as functions of time and integrating the equations of motion over the time interval Δt .

In the case of the impulsive motion of several particles, we can use Eq. (13.33). It reduces to

$$\Sigma m\mathbf{v}_1 + \Sigma \mathbf{F}_{\text{avg}} \Delta t = \Sigma m\mathbf{v}_2 \quad (13.36)$$

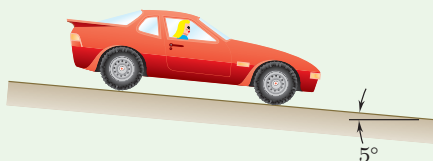
where the second term involves only impulsive, external forces. If all of the external forces acting on the various particles are non-impulsive, the second term in Eq. (13.36) vanishes, and this equation reduces to Eq. (13.34):

$$\Sigma m\mathbf{v}_1 = \Sigma m\mathbf{v}_2 \quad (13.34)$$

As before, for two particles, this reduces to

$$m_A\mathbf{v}_A + m_B\mathbf{v}_B = m_A\mathbf{v}'_A + m_B\mathbf{v}'_B \quad (13.34')$$

In other words, the total momentum of the particles is conserved. This situation occurs, for example, when two freely moving particles collide with one another. We should note, however, that although the total momentum of the particles is conserved, their total energy is generally *not* conserved. Problems involving the collision or *impact* of two particles are discussed in detail in Sec. 13.4.



Sample Problem 13.13

An automobile weighing 4000 lb is moving down a 5° incline at a speed of 60 mi/h when the brakes are applied, causing a constant total braking force (applied by the road on the tires) of 1500 lb. Determine the time required for the automobile to come to a stop.

STRATEGY: Since you are given velocities at two different times, use the principle of impulse and momentum.

MODELING: Choose the automobile to be your system and assume you can model it as a particle. The impulse-momentum diagram for this system is shown in Fig. 1.

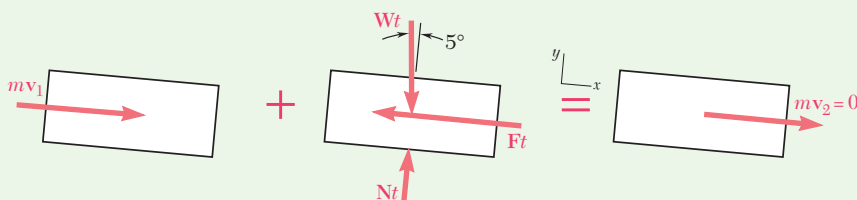


Fig. 1 Impulse-momentum diagram for the car.

ANALYSIS: The general impulse-momentum principle is

$$mv_1 + \Sigma \text{Imp}_{1 \rightarrow 2} = mv_2$$

This is a vector equation, and since the impulsive force is constant, the impulse is simply equal to the force multiplied by its time duration. You can obtain scalar equations by using Fig. 1. In the direction down the incline, you get

$$\begin{aligned} \text{+}\searrow \text{ components: } mv_1 + (W \sin 5^\circ)t - Ft &= 0 \\ (4000/32.2)(88 \text{ ft/s}) + (4000 \sin 5^\circ)t - 1500t &= 0 \quad t = 9.49 \text{ s} \quad \blacktriangleleft \end{aligned}$$

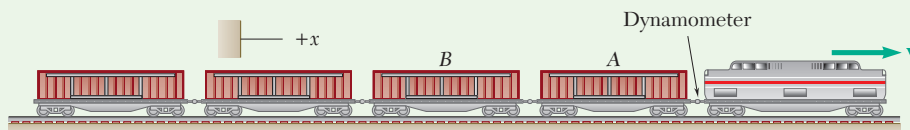
REFLECT and THINK: You could use Newton's second law to solve this problem. First, you would determine the car's deceleration, separate variables, and then integrate $a = dv/dt$ to relate the velocity, deceleration, and time. You could not use conservation of energy to solve this problem, because this principle does not involve time.



Sample Problem 13.14

In order to determine the weight of a freight train of 40 identical boxcars, an engineer attaches a dynamometer between the train and the locomotive. The train starts from rest, travels over a straight, level track, and reaches a speed of 30 mi/h after three minutes. During this time interval, the average reading of the dynamometer is 120 tons. Knowing that the effective coefficient of friction in the system is 0.03 and air resistance is negligible, determine (a) the weight of the train (in tons), (b) the coupling force between boxcars A and B.

(continued)



STRATEGY: This problem could be solved using Newton's second law and kinematic relationships, but since you are given velocities at two times and asked to find the force, you can also use impulse and momentum.

MODELING: Choose the system to be the 40 boxcars behind the engine. An impulse-momentum diagram for this system is shown in Fig. 1, where $\mathbf{F}$ is the dynamometer force.

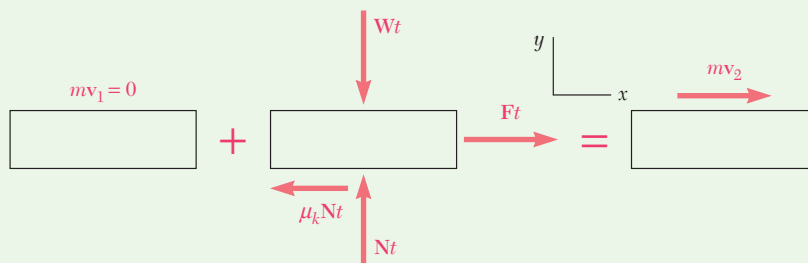


Fig. 1 Impulse-momentum diagram for the 40 boxcars.

ANALYSIS: Apply the principle of impulse and momentum

$$m\mathbf{v}_1 + \Sigma \text{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2$$

You can obtain scalar equations by using Fig. 1 and looking at the x and y directions.

$$+\uparrow y \text{ components:} \quad Nt - Wt = 0 \quad N = W$$

$$\rightarrow x \text{ components:} \quad 0 + Ft - \mu_k Nt = mv_2$$

$$0 + (120 \text{ ton})(2000 \text{ lb/ton})(180 \text{ s}) - 0.03(W)(180 \text{ s}) \\ = \left(\frac{W}{32.2 \text{ ft/s}^2} \right) (30 \text{ mi/h}) \left(\frac{1 \text{ h}}{3600 \text{ s}} \right) \left(\frac{5280 \text{ ft}}{\text{mi}} \right)$$

Solving for W , you obtain

$$W = 6.384 \times 10^6 \text{ lb} = 3190 \text{ tons} \quad \blacktriangleleft$$

Coupling Force Between Cars A and B. You need to define a new system where the force of interest is an external force. Therefore, choose car A to be your system and define F_A as the coupling force between cars A and B. The impulse-momentum diagram for this system is shown in Fig. 2.

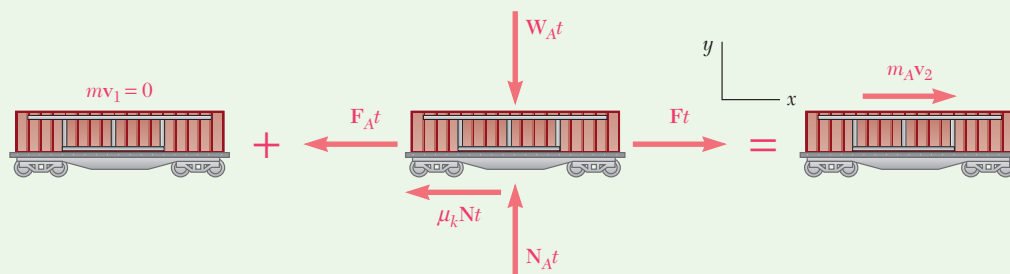


Fig. 2 Impulse-momentum diagram for car A.

Since all the cars weigh the same amount, the weight of A is $W_A = W/40 = 159,600$ lb. Applying impulse-momentum in the y -direction gives you $N_A = W_A$. Considering the x -direction,

$$\rightarrow x \text{ components: } 0 + Ft - \mu_k N_A t - F_A t = m_A v_2$$

Substituting in numbers and solving for F_A gives

$$F_A = 117.0 \text{ tons} \quad \blacktriangleleft$$

REFLECT and THINK: Rather than using A as your system, you could have chosen the remaining 39 cars to be your system. In this case, you would find

$$0 - \mu_k N_{39} t + F_A t = m_{39} v_2$$

where N_{39} and m_{39} are the normal force and the mass, respectively, for the remaining 39 cars. The answer, as you would expect, is the same.

Sample Problem 13.15

A hammer and punch is used by a surgeon when inserting a hip implant. To better understand this process, an instrumented implant is inserted into a fixed replicate femur. The upward resisting force from the replicate femur on the hip implant can be neglected during the impact, and the impact force from the punch can be approximated by a half sine wave. Determine the speed of the 0.3-kg implant immediately after impact.

STRATEGY: Since you are relating force, time, and velocities, you should use the principle of impulse and momentum.

MODELING: Choose the system to be the implant. An impulse-momentum diagram for this system is shown in Fig. 1. The resisting force is left off Fig. 1 since it is assumed to be negligible.

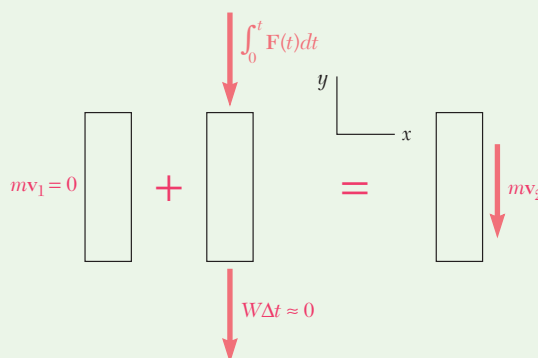
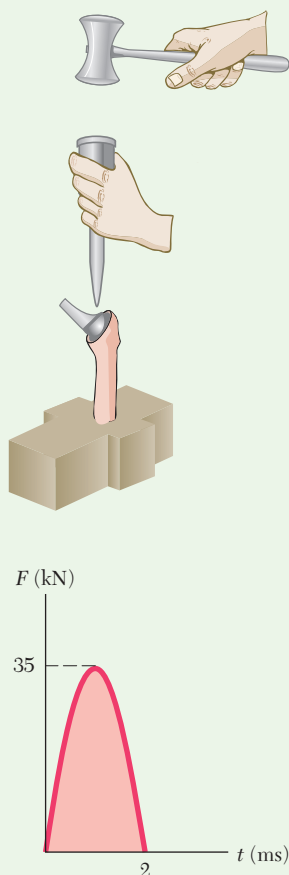


Fig. 1 Impulse-momentum diagram for the implant.

(continued)

ANALYSIS: Apply the principle of impulse and momentum

$$m\mathbf{v}_1 + \Sigma \mathbf{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2$$

You can obtain scalar equations by looking at the vertical components.

$$\downarrow + y \text{ components:} \quad 0 + \int_0^t F(t) dt = mv_2 \quad (1)$$

where

$$\begin{aligned} \int_0^t F(t) dt &= \int_0^{0.002} 35\,000 \sin\left(\frac{2\pi}{0.004}t\right) dt = -35\,000 \frac{0.004}{2\pi} \cos\left(\frac{2\pi}{0.004}t\right) \Big|_0^{0.002} \\ &= 45.56 \text{ N}\cdot\text{s} \end{aligned}$$

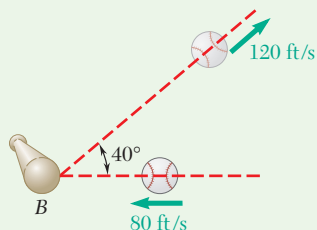
Substituting this into Eq. (1) and solving for v_2 gives

$$v_2 = \frac{\int_0^t F(t) dt}{m} = \frac{45.56 \text{ N}\cdot\text{s}}{0.3 \text{ kg}}$$

$$v_2 = 148.5 \text{ m/s} \quad \blacktriangleleft$$

REFLECT and THINK: This problem is similar to Sample Prob. 13.5, where the drop-hammer pile driver hits the pile, then the hammer and pile move down, and the earth resists the motion. In that problem, you analyzed the motion *after* the impact; in this problem, you are analyzing the motion *during* the impact. In reality, you would need to do some experimental measurements to determine if the resisting force really is negligible during the impact. If you knew the force relationship of the femur on the implant, you could solve this as a two-part problem to first find the velocity of the implant immediately after the impact using impulse and momentum, and then determine how far the implant moves down into the femur using work and energy.

Sample Problem 13.16



A 4-oz baseball is pitched with a velocity of 80 ft/s toward a batter. After the ball is hit by the bat B , it has a velocity of 120 ft/s in the direction shown. If the bat and ball are in contact for 0.015 s, determine the average impulsive force exerted on the ball during the impact.

STRATEGY: This situation features an impact, and therefore impulsive forces, so apply the principle of impulse and momentum to the ball.

MODELING: Choose the ball as your system and treat it as a particle. The impulse-momentum diagram for this system is shown in Fig. 1. Because the weight of the ball is a non-impulsive force that is typically much smaller than the impulsive force, you can neglect it.

ANALYSIS: Apply the principle of impulse and momentum

$$m\mathbf{v}_1 + \Sigma \text{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2$$

Applying this in the x and y directions gives

$\rightarrow x$ components: $-mv_1 + F_x \Delta t = mv_2 \cos 40^\circ$

$$-\frac{\frac{4}{16}}{32.2} (80 \text{ ft/s}) + F_x (0.015 \text{ s}) = \frac{\frac{4}{16}}{32.2} (120 \text{ ft/s}) \cos 40^\circ$$

$$F_x = +89.0 \text{ lb}$$

$\uparrow y$ components: $0 + F_y \Delta t = mv_2 \sin 40^\circ$

$$F_y (0.015 \text{ s}) = \frac{\frac{4}{16}}{32.2} (120 \text{ ft/s}) \sin 40^\circ$$

$$F_y = +39.9 \text{ lb}$$

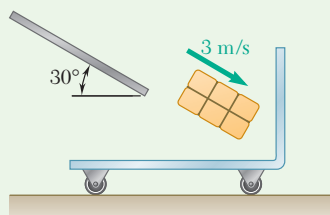
Fig. 1 Impulse-momentum diagram for the ball.

From its components F_x and F_y you can determine the magnitude and direction of the average impulsive force $\mathbf{F}$ as

$$\mathbf{F} = 97.5 \text{ lb} \nearrow 24.2^\circ \quad \blacktriangleleft$$

REFLECT and THINK: In this problem, we neglected the impulse due to the weight. This would have had a magnitude of $(4/16 \text{ lb})(0.015 \text{ s}) = 0.00288 \text{ lb}\cdot\text{s}$. This indeed is much smaller than the impulse exerted on the ball by the bat, which is $(97.4 \text{ lb})(0.015 \text{ s}) = 1.463 \text{ lb}\cdot\text{s}$.

Sample Problem 13.17



A 10-kg package drops from a chute into a 25-kg cart with a velocity of 3 m/s. The cart is initially at rest and can roll freely. Determine (a) the final velocity of the cart, (b) the impulse exerted by the cart on the package, (c) the fraction of the initial energy lost in the impact.

STRATEGY: Since you have an impact, and therefore impulsive forces, use the principle of impulse and momentum.

MODELING: Choose the package and the cart to be your system, and assume that both can be treated as particles. The impulse-momentum diagram for this system is shown in Fig. 1. Note that a vertical impulse occurs between the cart and the ground, because the cart is constrained to move horizontally.

ANALYSIS: Apply the principle of impulse and momentum

$$m\mathbf{v}_1 + \Sigma \text{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2$$

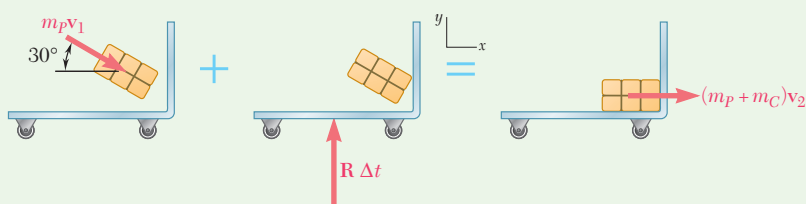


Fig. Impulse-momentum diagram for the system.

(continued)

a. Package and Cart. Applying this principle in the x -direction gives

$$\begin{aligned} \rightarrow x \text{ components: } \quad m_P v_1 \cos 30^\circ + 0 &= (m_P + m_C) v_2 \\ (10 \text{ kg})(3 \text{ m/s}) \cos 30^\circ &= (10 \text{ kg} + 25 \text{ kg}) v_2 \end{aligned}$$

$$v_2 = 0.742 \text{ m/s} \rightarrow \blacktriangleleft$$

In Fig. 1, the force between the package and the cart is not shown because it is internal to the defined system. To determine this force, you need a new system; that is, just the package by itself. The impulse-momentum diagram for the package alone is shown in Fig. 2.

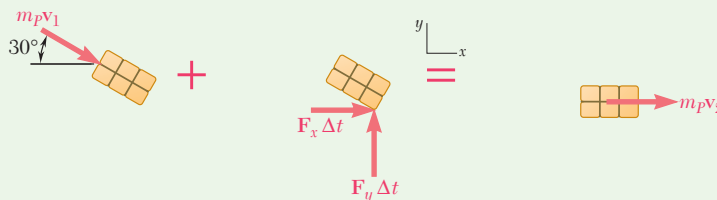


Fig. 2 Impulse-momentum diagram for the package.

b. Impulse-Momentum Principle: Package. The package moves in both x and y directions, so write the conservation of momentum equation for each component of the motion.

$$\begin{aligned} \rightarrow x \text{ components: } \quad -m v_1 + F_x \Delta t &= m v_2 \cos 40^\circ \\ (10 \text{ kg})(3 \text{ m/s}) \cos 30^\circ + F_x \Delta t &= (10 \text{ kg})(0.742 \text{ m/s}) \\ F_x \Delta t &= -18.56 \text{ N}\cdot\text{s} \\ +\uparrow y \text{ components: } \quad -m_P v_1 \sin 30^\circ + F_y \Delta t &= 0 \\ -(10 \text{ kg})(3 \text{ m/s}) \sin 30^\circ + F_y \Delta t &= 0 \\ F_y \Delta t &= +15 \text{ N}\cdot\text{s} \end{aligned}$$

The impulse exerted on the package is

$$\mathbf{F} \Delta t = 23.9 \text{ N}\cdot\text{s} \searrow 38.9^\circ \blacktriangleleft$$

c. Fraction of Energy Lost. The initial and final energies are

$$\begin{aligned} T_1 &= \frac{1}{2} m_P v_1^2 = \frac{1}{2} (10 \text{ kg})(3 \text{ m/s})^2 = 45 \text{ J} \\ T_2 &= \frac{1}{2} (m_P + m_C) v_2^2 = \frac{1}{2} (10 \text{ kg} + 25 \text{ kg})(0.742 \text{ m/s})^2 = 9.63 \text{ J} \end{aligned}$$

The fraction of energy lost is

$$\frac{T_1 - T_2}{T_1} = \frac{45 \text{ J} - 9.63 \text{ J}}{45 \text{ J}} = 0.786 \blacktriangleleft$$

REFLECT and THINK: Except in the purely theoretical case of a “perfectly elastic” collision, mechanical energy is never conserved in a collision between two objects, even though linear momentum may be conserved. Note that, in this problem, momentum was conserved in the x direction but was not conserved in the y direction because of the vertical impulse on the wheels of the cart. Whenever you deal with an impact, you need to use impulse-momentum methods.

SOLVING PROBLEMS ON YOUR OWN

In this section, we integrated Newton's second law to derive the **principle of impulse and momentum** for a particle. Recalling that we defined the *linear momentum* of a particle as the product of its mass m and its velocity $\mathbf{v}$ [Sec. 12.1B], we have

$$m\mathbf{v}_1 + \Sigma \text{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2 \quad (13.32)$$

This equation states that we can obtain the linear momentum $m\mathbf{v}_2$ of a particle at time t_2 by adding its linear momentum $m\mathbf{v}_1$ at time t_1 to the **impulses** of the forces exerted on the particle during the time interval t_1 to t_2 . For computing purposes, we can express the momenta and impulses in terms of their rectangular components and replace Eq. (13.32) by the equivalent scalar equations. The units of momentum and impulse are N·s in the SI system of units and lb·s in U.S. customary units. To solve problems using this equation you can follow these steps.

1. Draw an impulse-momentum diagram showing the particle, its momentum at t_1 and at t_2 , and the impulses of the forces exerted on the particle during the time interval t_1 to t_2 .

2. Calculate the impulse of each force, expressing it in terms of its rectangular components if more than one direction is involved. You may encounter the following cases:

a. The time interval is finite and the force is constant.

$$\text{Imp}_{1 \rightarrow 2} = \mathbf{F}(t_2 - t_1)$$

b. The time interval is finite and the force is a function of t .

$$\text{Imp}_{1 \rightarrow 2} = \int_{t_1}^{t_2} \mathbf{F}(t) dt$$

c. The time interval is very small and the force is very large. The force is called an **impulsive force**, and its impulse over the time interval $t_2 - t_1 = \Delta t$ is

$$\text{Imp}_{1 \rightarrow 2} = \mathbf{F}_{\text{avg}} \Delta t$$

Note that this impulse is assumed to be zero for a non-impulsive force such as the weight of a body, the force exerted by a spring, or any other force that is known to be small by comparison with the impulsive forces. However, we *cannot* assume unknown reactions are non-impulsive, and you should take their impulses into account.

3. Substitute the values obtained for the impulses into Eq. (13.32) or into the equivalent scalar equations. You will find that the forces and velocities in the problems of this section are contained in a plane. Therefore, you can write two scalar equations and solve these equations for two unknowns. These unknowns may be a *time* [Sample Prob. 13.13], a *force* [Sample Prob. 13.14], a *velocity* [Sample Prob. 13.15], an *average impulsive force* [Sample Prob. 13.16], or an *impulse* [Sample Prob. 13.17].

4. When several particles are involved, it is often necessary to draw a separate diagram for each particle showing the initial and final momentum of the particle as well as the impulses of the forces exerted on the particle.

a. It is usually convenient, however, to first consider a system that includes all of the particles. This system leads to

$$\Sigma m\mathbf{v}_1 + \Sigma \mathbf{Imp}_{1 \rightarrow 2} = \Sigma m\mathbf{v}_2 \quad (13.33)$$

where you need to consider the impulses of only the forces external to the system. Therefore, the two equivalent scalar equations will not contain any of the impulses of the unknown internal forces.

b. If the sum of the impulses of the external forces is zero, Eq. (13.33) reduces to

$$\Sigma m\mathbf{v}_1 = \Sigma m\mathbf{v}_2 \quad (13.34)$$

or for two particles as

$$m_A\mathbf{v}_A + m_B\mathbf{v}_B = m_A\mathbf{v}'_A + m_B\mathbf{v}'_B \quad (13.34')$$

which says that *the total linear momentum of the particles is conserved*. This occurs when the time interval is very short and the external forces are negligible compared to the impulsive forces. Keep in mind, however, that the total momentum may be conserved in one direction, but not in another [Sample Prob. 13.17].

Problems

CONCEPT QUESTIONS

13.CQ4 A large insect impacts the front windshield of a sports car traveling down a road. Which of the following statements is true during the collision?

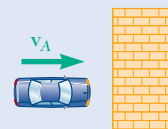
- The car exerts a greater force on the insect than the insect exerts on the car.
- The insect exerts a greater force on the car than the car exerts on the insect.
- The car exerts a force on the insect, but the insect does not exert a force on the car.
- The car exerts the same force on the insect as the insect exerts on the car.
- Neither exerts a force on the other; the insect gets smashed simply because it gets in the way of the car.

13.CQ5 The expected damages associated with two types of perfectly plastic collisions are to be compared. In the first case, two identical cars traveling at the same speed impact each other head-on. In the second case, the car impacts a massive concrete wall. In which case would you expect the car to be more damaged?

- Case 1
- Case 2
- The same damage in each case



Case 1



Case 2

Fig. P13.CQ5

IMPULSE-MOMENTUM DIAGRAM PRACTICE PROBLEMS

13.F1 The initial velocity of the block in position *A* is 30 ft/s. The coefficient of kinetic friction between the block and the plane is $\mu_k = 0.30$. Draw the impulse-momentum diagram that can be used to determine the time it takes for the block to reach *B* with zero velocity, if $\theta = 20^\circ$.

13.F2 A 4-lb collar which can slide on a frictionless vertical rod is acted upon by a force $\mathbf{P}$ which varies in magnitude as shown. Knowing that the collar is initially at rest, draw the impulse-momentum diagram that can be used to determine its velocity at $t = 3$ s.

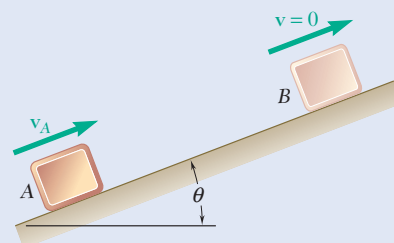


Fig. P13.F1

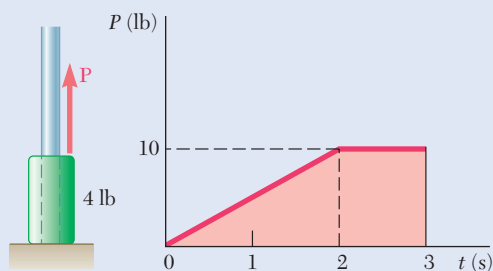


Fig. P13.F2

- 13.F3** The 15-kg suitcase A has been propped up against one end of a 40-kg luggage carrier B and is prevented from sliding down by other luggage. When the luggage is unloaded and the last heavy trunk is removed from the carrier, the suitcase is free to slide down, causing the 40-kg carrier to move to the left with a velocity v_B of magnitude 0.8 m/s. Neglecting friction, draw the impulse-momentum diagrams that can be used to determine (a) the velocity of A as it rolls on the carrier, (b) the velocity of the carrier after the suitcase hits the right side of the carrier without bouncing back.

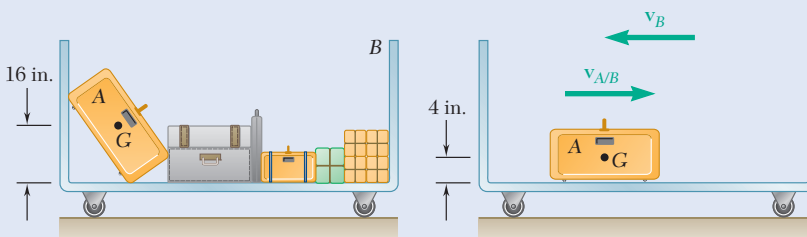


Fig. P13.F3

- 13.F4** Car A was traveling west at a speed of 15 m/s and car B was traveling north at an unknown speed when they slammed into each other at an intersection. Upon investigation it was found that after the crash the two cars got stuck and skidded off at an angle of 50° north of east. Knowing the masses of A and B are m_A and m_B , respectively, draw the impulse-momentum diagram that can be used to determine the velocity of B before impact.

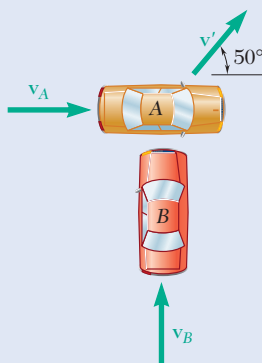


Fig. P13.F4

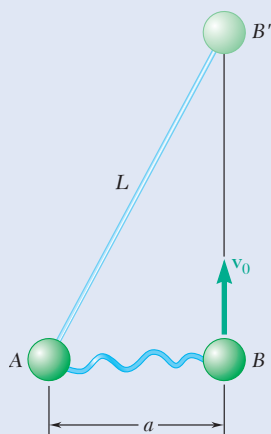


Fig. P13.F5

- 13.F5** Two identical spheres A and B , each of mass m , are attached to an inextensible inelastic cord of length L and are resting at a distance a from each other on a frictionless horizontal surface. Sphere B is given a velocity v_0 in a direction perpendicular to line AB and moves without friction until it reaches B' where the cord becomes taut. Draw the impulse-momentum diagram that can be used to determine the magnitude of the velocity of each sphere immediately after the cord has become taut.

END-OF-SECTION PROBLEMS

- 13.119** A 35 000-Mg ocean liner has an initial velocity of 4 km/h. Neglecting the frictional resistance of the water, determine the time required to bring the liner to rest by using a single tugboat that exerts a constant force of 150 kN.

13.120 A 2500-lb automobile is moving at a speed of 60 mi/h when the brakes are fully applied, causing all four wheels to skid. Determine the time required to stop the automobile (a) on dry pavement ($\mu_k = 0.75$), (b) on an icy road ($\mu_k = 0.10$).

13.121 A sailboat weighing 980 lb with its occupants is running downwind at 8 mi/h when its spinnaker is raised to increase its speed. Determine the net force provided by the spinnaker over the 10-s interval that it takes for the boat to reach a speed of 12 mi/h.

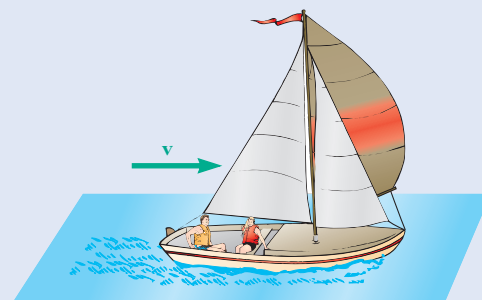


Fig. P13.121

13.122 A truck is hauling a 300-kg log out of a ditch using a winch attached to the back of the truck. Knowing the winch applies a constant force of 2500 N and the coefficient of kinetic friction between the ground and the log is 0.45, determine the time for the log to reach a speed of 0.5 m/s.

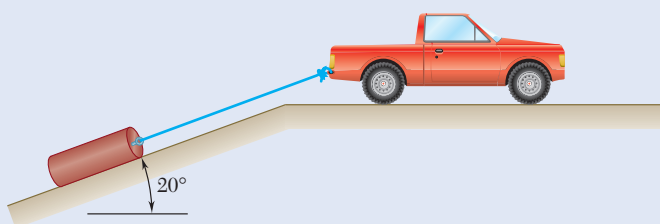


Fig. P13.122

13.123 The coefficients of friction between the load and the flatbed trailer shown are $\mu_s = 0.40$ and $\mu_k = 0.35$. Knowing that the speed of the rig is 55 mi/h, determine the shortest time in which the rig can be brought to a stop if the load is not to shift.

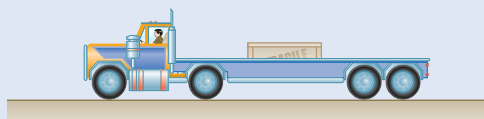


Fig. P13.123

13.124 Steep safety ramps are built beside mountain highways to enable vehicles with defective brakes to stop. A 10-ton truck enters a 15° ramp at a high speed $v_0 = 108$ ft/s and travels for 6 s before its speed is reduced to 36 ft/s. Assuming constant deceleration, determine (a) the magnitude of the braking force, (b) the additional time required for the truck to stop. Neglect air resistance and rolling resistance.

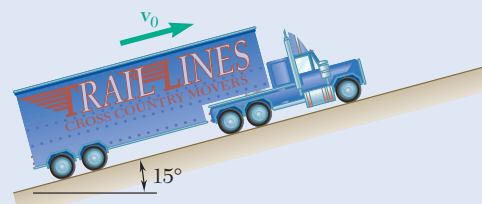


Fig. P13.124

13.125 Baggage on the floor of the baggage car of a high-speed train is not prevented from moving other than by friction. The train is traveling down a 5-percent grade when it decreases its speed at a constant rate from 120 mi/h to 60 mi/h in a time interval of 12 s. Determine the smallest allowable value of the coefficient of static friction between a trunk and the floor of the baggage car if the trunk is not to slide.

- 13.126** The 18 000-kg F-35B uses thrust vectoring to allow it to take off vertically. In one maneuver, the pilot reaches the top of her static hover at 200 m. The combined thrust and lift force on the airplane applied at the end of the static hover can be expressed as $\mathbf{F} = (44t + 2500t^2)\mathbf{i} + (250t^2 + t + 176\,580)\mathbf{j}$, where $\mathbf{F}$ and t are expressed in newtons and seconds, respectively. Determine (a) how long it will take the airplane to reach a cruising speed of 1000 km/hr (cruising speed is defined to be in the x -direction only), (b) the altitude of the plane at this time.



Fig. P13.126

- 13.127** A truck is traveling down a road with a 4-percent grade at a speed of 60 mi/h when its brakes are applied to slow it down to 20 mi/h. An antiskid braking system limits the braking force to a value at which the wheels of the truck are just about to slide. Knowing that the coefficient of static friction between the road and the wheels is 0.60, determine the shortest time needed for the truck to slow down.
- 13.128** In anticipation of a long 6° upgrade, a bus driver accelerates at a constant rate from 80 km/h to 100 km/h in 8 s while still on a level section of the highway. Knowing that the speed of the bus is 100 km/h as it begins to climb the grade at time $t = 0$ and that the driver does not change the setting of the throttle or shift gears, determine (a) the speed of the bus when $t = 10$ s, (b) the time when the speed is 60 km/h.

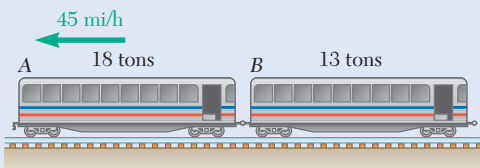


Fig. P13.129

- 13.129** A light train made of two cars travels at 45 mi/h. Car A weighs 18 tons, and car B weighs 13 tons. When the brakes are applied, a constant braking force of 4300 lb is applied to each car. Determine (a) the time required for the train to stop after the brakes are applied, (b) the force in the coupling between the cars while the train is slowing down.
- 13.130** Solve Problem 13.129, assuming that a constant braking force of 4300 lb is applied to car B but that the brakes on car A are not applied.

- 13.131** A tractor-trailer rig with a 2000-kg tractor, a 4500-kg trailer, and a 3600-kg trailer is traveling on a level road at 90 km/h. The brakes on the rear trailer fail, and the antiskid system of the tractor and front trailer provide the largest possible force that will not cause the wheels to slide. Knowing that the coefficient of static friction is 0.75, determine (a) the shortest time for the rig to come to a stop, (b) the force in the coupling between the two trailers during that time. Assume that the force exerted by the coupling on each of the two trailers is horizontal.

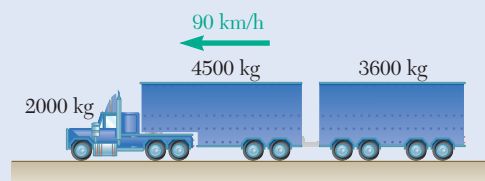


Fig. P13.131

- 13.132** The system shown is at rest when a constant 150-N force is applied to collar *B*. Neglecting the effect of friction, determine (a) the time at which the velocity of collar *B* will be 2.5 m/s to the left, (b) the corresponding tension in the cable.

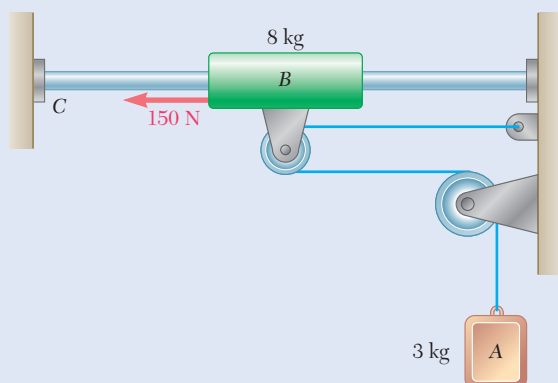


Fig. P13.132

- 13.133** An 8-kg cylinder *C* rests on a 4-kg platform *A* supported by a cord that passes over the pulleys *D* and *E* and is attached to a 4-kg block *B*. Knowing that the system is released from rest, determine (a) the velocity of block *B* after 0.8 s, (b) the force exerted by the cylinder on the platform.
- 13.134** An estimate of the expected load on over-the-shoulder seat belts is to be made before designing prototype belts that will be evaluated in automobile crash tests. Assuming that an automobile traveling at 45 mi/h is brought to a stop in 110 ms, determine (a) the average impulsive force exerted by a 200-lb man on the belt, (b) the maximum force F_m exerted on the belt if the force-time diagram has the shape shown.

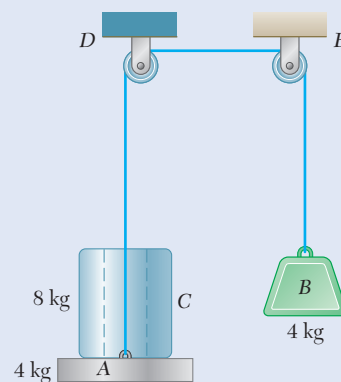


Fig. P13.133

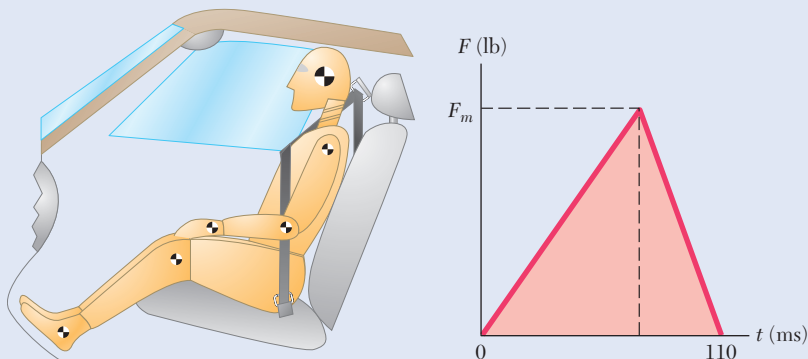


Fig. P13.134

- 13.135** A 60-g model rocket is fired vertically. The engine applies a thrust $\mathbf{P}$ which varies in magnitude as shown. Neglecting air resistance and the change in mass of the rocket, determine (a) the maximum speed of the rocket as it goes up, (b) the time for the rocket to reach its maximum elevation.

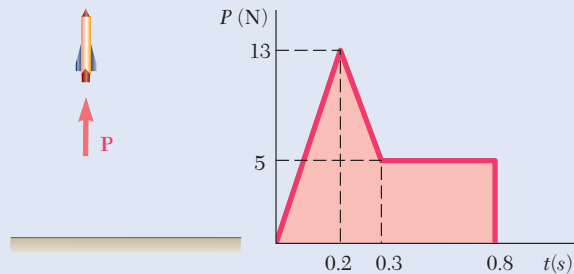


Fig. P13.135

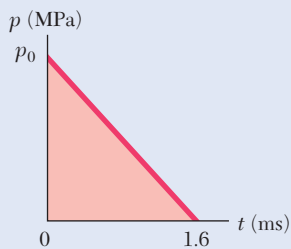


Fig. P13.136

- 13.136** A simplified model consisting of a single straight line is to be obtained for the variation of pressure inside the 10-mm-diameter barrel of a rifle as a 20-g bullet is fired. Knowing that it takes 1.6 ms for the bullet to travel the length of the barrel and that the velocity of the bullet upon exit is 700 m/s, determine the value of p_0 .

- 13.137** A crash test is performed between an SUV A and a 2500-lb compact car B. The compact car is stationary before the impact and has its brakes applied. A transducer measures the force during the impact, and the force $\mathbf{P}$ varies as shown. Knowing that the coefficients of friction between the tires and road are $\mu_s = 0.9$ and $\mu_k = 0.7$, determine (a) the time at which the compact car will start moving, (b) the maximum speed of the car, (c) the time at which the car will come to a stop.

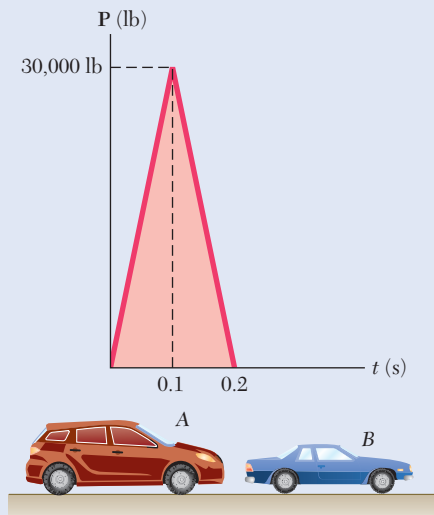


Fig. P13.137 and P13.138

- 13.138** A crash test is performed between a 4500 lb SUV A and a compact car B. A transducer measures the force during the impact, and the force $\mathbf{P}$ varies as shown. Knowing that the SUV is travelling 30 mph when it hits the car, determine the speed of the SUV immediately after the impact.

13.139 A baseball player catching a ball can soften the impact by pulling his hand back. Assuming that a 5-oz ball reaches his glove at 90 mi/h and that the player pulls his hand back during the impact at an average speed of 30 ft/s over a distance of 6 in., bringing the ball to a stop, determine the average impulsive force exerted on the player's hand.

13.140 A 1.62-oz golf ball is hit with a golf club and leaves it with a velocity of 100 mi/h. We assume that for $0 \leq t \leq t_0$, where t_0 is the duration of the impact, the magnitude F of the force exerted on the ball can be expressed as $F = F_m \sin(\pi t/t_0)$. Knowing that $t_0 = 0.5$ ms, determine the maximum value F_m of the force exerted on the ball.

13.141 The triple jump is a track-and-field event in which an athlete gets a running start and tries to leap as far as he can with a hop, step, and jump. Shown in the figure is the initial hop of the athlete. Assuming that he approaches the takeoff line from the left with a horizontal velocity of 10 m/s, remains in contact with the ground for 0.18 s, and takes off at a 50° angle with a velocity of 12 m/s, determine the vertical component of the average impulsive force exerted by the ground on his foot. Give your answer in terms of the weight W of the athlete.

13.142 The last segment of the triple jump track-and-field event is the jump, in which the athlete makes a final leap, landing in a sand-filled pit. Assuming that the velocity of a 80-kg athlete just before landing is 9 m/s at an angle of 35° with the horizontal and that the athlete comes to a complete stop in 0.22 s after landing, determine the horizontal component of the average impulsive force exerted on his feet during landing.

13.143 The design for a new cementless hip implant is to be studied using an instrumented implant and a fixed simulated femur. Assuming the punch applies an average force of 2 kN over a time of 2 ms to the 200-g implant, determine (a) the velocity of the implant immediately after impact, (b) the average resistance of the implant to penetration if the implant moves 1 mm before coming to rest.

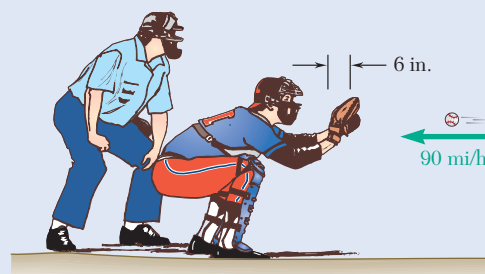


Fig. P13.139

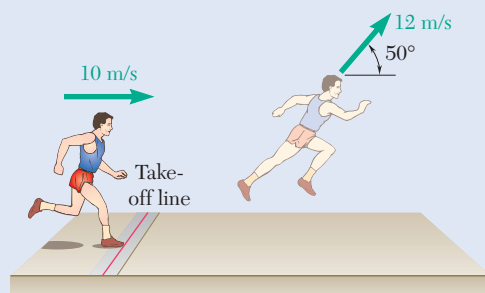


Fig. P13.141

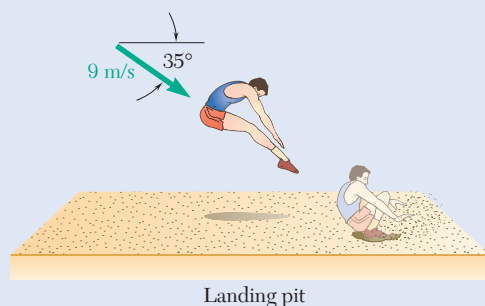


Fig. P13.142

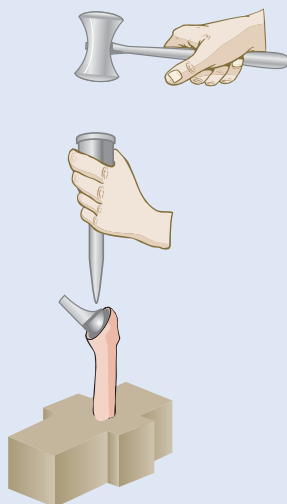


Fig. P13.143

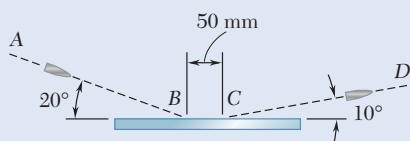


Fig. P13.144

13.144 A 28-g steel-jacketed bullet is fired with a velocity of 650 m/s toward a steel plate and ricochets along path CD with a velocity of 500 m/s. Knowing that the bullet leaves a 50-mm scratch on the surface of the plate and assuming that it has an average speed of 600 m/s while in contact with the plate, determine the magnitude and direction of the impulsive force exerted by the plate on the bullet.

13.145 A 25-ton railroad car moving at 2.5 mi/h is to be coupled to a 50-ton car that is at rest with locked wheels ($\mu_k = 0.30$). Determine (a) the velocity of both cars after the coupling is completed, (b) the time it takes for both cars to come to rest.

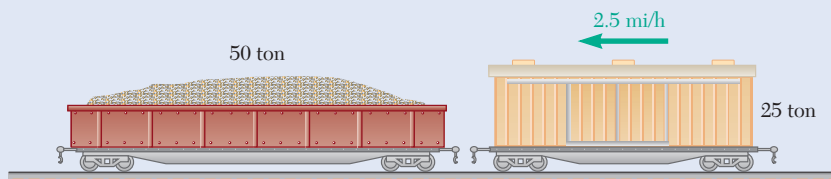


Fig. P13.145

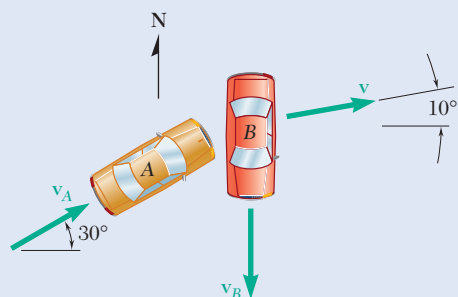


Fig. P13.146

13.146 At an intersection, car B was traveling south and car A was traveling 30° north of east when they slammed into each other. Upon investigation it was found that after the crash the two cars got stuck and skidded off at an angle of 10° north of east. Each driver claimed that he was going at the speed limit of 50 km/h and that he tried to slow down but couldn't avoid the crash because the other driver was going a lot faster. Knowing that the masses of cars A and B were 1500 kg and 1200 kg, respectively, determine (a) which car was going faster, (b) the speed of the faster of the two cars if the slower car was traveling at the speed limit.

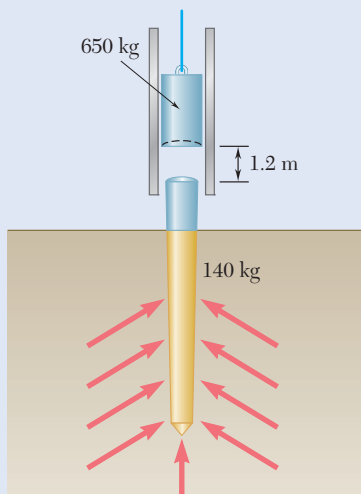


Fig. P13.147

13.147 The 650-kg hammer of a drop-hammer pile driver falls from a height of 1.2 m onto the top of a 140-kg pile, driving it 110 mm into the ground. Assuming perfectly plastic impact ($e = 0$), determine the average resistance of the ground to penetration.

13.148 A small rivet connecting two pieces of sheet metal is being clinched by hammering. Determine the impulse exerted on the rivet and the energy absorbed by the rivet under each blow, knowing that the head of the hammer has a weight of 1.5 lb and that it strikes the rivet with a velocity of 20 ft/s. Assume that the hammer does not rebound and that the anvil is supported by springs and (a) has an infinite mass (rigid support), (b) has a weight of 9 lb.

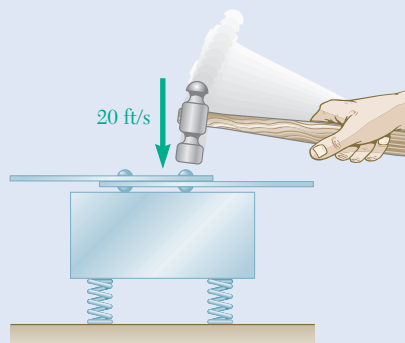


Fig. P13.148

- 13.149** Bullet B weighs 0.5 oz and blocks A and C both weigh 3 lb. The coefficient of friction between the blocks and the plane is $\mu_k = 0.25$. Initially the bullet is moving at v_0 and blocks A and C are at rest (Fig. 1). After the bullet passes through A it becomes embedded in block C and all three objects come to stop in the positions shown (Fig. 2). Determine the initial speed of the bullet v_0 .

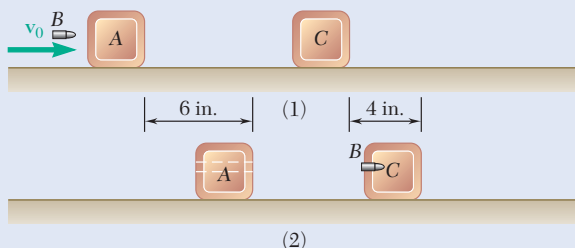


Fig. P13.149

- 13.150** A 180-lb man and a 120-lb woman stand at opposite ends of a 300-lb boat, ready to dive, each with a 16-ft/s velocity relative to the boat. Determine the velocity of the boat after they have both dived, if (a) the woman dives first, (b) the man dives first.



Fig. P13.150

- 13.151** A 75-g ball is projected from a height of 1.6 m with a horizontal velocity of 2 m/s and bounces from a 400-g smooth plate supported by springs. Knowing that the height of the rebound is 0.6 m, determine (a) the velocity of the plate immediately after the impact, (b) the energy lost due to the impact.

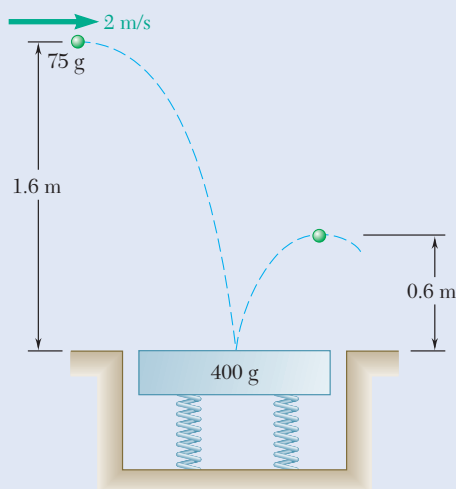


Fig. P13.151

- 13.152** A ballistic pendulum is used to measure the speed of high-speed projectiles. A 6-g bullet *A* is fired into a 1-kg wood block *B* suspended by a cord with a length of $l = 2.2$ m. The block then swings through a maximum angle of $\theta = 60^\circ$. Determine (a) the initial speed of the bullet v_0 , (b) the impulse imparted by the bullet on the block, (c) the force on the cord immediately after the impact.

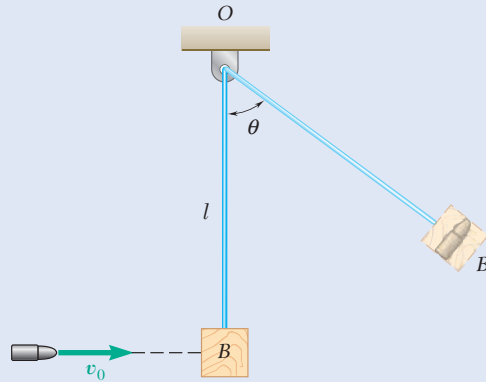


Fig. P13.152

- 13.153** A 1-oz bullet is traveling with a velocity of 1400 ft/s when it impacts and becomes embedded in a 5-lb wooden block. The block can move vertically without friction. Determine (a) the velocity of the bullet and block immediately after the impact, (b) the horizontal and vertical components of the impulse exerted by the block on the bullet.

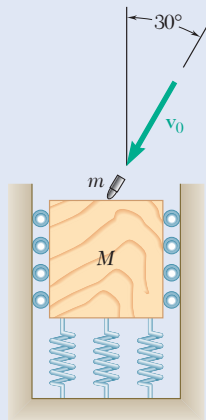


Fig. P13.153

- 13.154** In order to test the resistance of a chain to impact, the chain is suspended from a 240-lb rigid beam supported by two columns. A rod attached to the last link is then hit by a 60-lb block dropped from a 5-ft height. Determine the initial impulse exerted on the chain and the energy absorbed by the chain, assuming that the block does not rebound from the rod and that the columns supporting the beam are (a) perfectly rigid, (b) equivalent to two perfectly elastic springs.

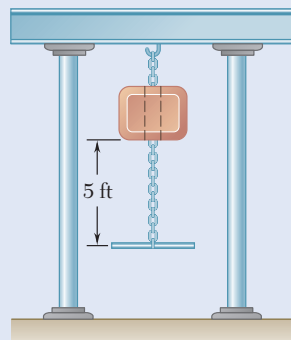


Fig. P13.154

13.4 IMPACTS

A collision between two bodies that occurs in a very small interval of time, and during which the two bodies exert relatively large forces on each other, is called an **impact**. The common normal to the surfaces in contact during the impact is called the **line of impact**. If the mass centers of the two colliding bodies are located on this line, the impact is called a **central impact**. Otherwise, the impact is said to be **eccentric**. Our present study is limited to the central impact of two particles. In Chapter 17, we consider the analysis of the eccentric impact of two rigid bodies.

If the velocities of the two particles are directed along the line of impact, the impact is said to be a **direct impact** (Fig. 13.20a). If either or both particles move along a line other than the line of impact, the impact is said to be an **oblique impact** (Fig. 13.20b).

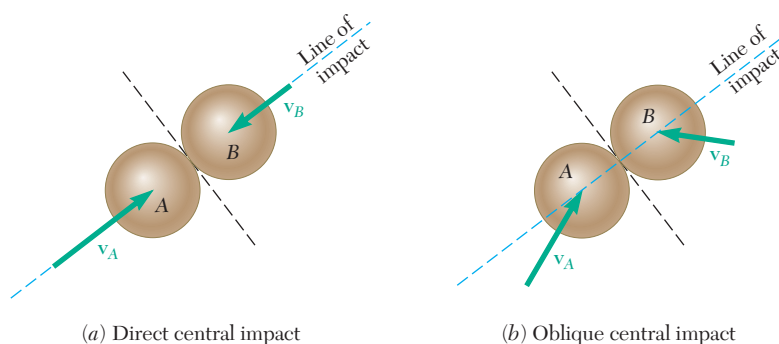


Fig. 13.20 Central impacts can be (a) direct (or “head-on”) or (b) oblique.

13.4A Direct Central Impact

Consider two particles A and B with mass m_A and m_B that are moving in the same straight line and to the right with known velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ (Fig. 13.21a). If $\mathbf{v}_A$ is larger than $\mathbf{v}_B$, particle A eventually strikes particle B . Under the impact, the two particles **deform**, and at the end of the period of deformation, they have the same velocity $\mathbf{u}$ (Fig. 13.21b). A period of **restitution** then takes place. At the end of this period, depending upon the magnitude of the impact forces and upon the materials involved, the two particles either have regained their original shape or will stay permanently deformed. Our purpose here is to determine the velocities $\mathbf{v}'_A$ and $\mathbf{v}'_B$ of the particles at the end of the period of restitution (Fig. 13.21c).

Considering first the two particles as a single system, we note that there is no impulsive, external force. Thus, the total linear momentum of the two particles is conserved, and we have

$$m_A \mathbf{v}_A + m_B \mathbf{v}_B = m_A \mathbf{v}'_A + m_B \mathbf{v}'_B \quad (13.34')$$

Since all of the velocities considered are directed along the same axis, we can replace this equation by the following relation involving only scalar components, as

$$m_A v_A + m_B v_B = m_A v'_A + m_B v'_B \quad (13.37)$$

A positive value for any of the scalar quantities v_A , v_B , v'_A , or v'_B means that the corresponding vector is directed to the right; a negative value indicates that the corresponding vector is directed to the left.

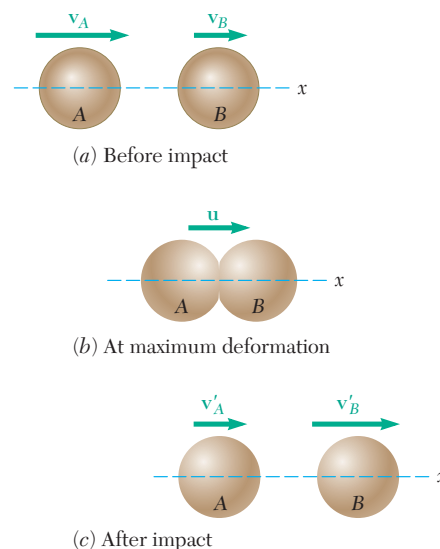


Fig. 13.21 Every impact has three stages: (a) before the impact, (b) a maximum deformation when the particles have the same velocity, and (c) after the impact.

To obtain the velocities $\mathbf{v}'_A$ and $\mathbf{v}'_B$, it is necessary to establish a second relation between the scalars $\mathbf{v}'_A$ and $\mathbf{v}'_B$. For this purpose, let us now consider the motion of particle A during the period of deformation and apply the principle of impulse and momentum. Since the only impulsive force acting on A during this period is the force $\mathbf{P}$ exerted by B (Fig. 13.22a), we have, again using scalar components,

$$m_A v_A - \int P dt = m_A u \quad (13.38)$$

where the integral extends over the period of deformation. Considering now the motion of A during the period of restitution and denoting the force exerted by B on A during this period by $\mathbf{R}$ (Fig. 13.22b), we have

$$m_A u - \int R dt = m_A v'_A \quad (13.39)$$

where the integral extends over the period of restitution.

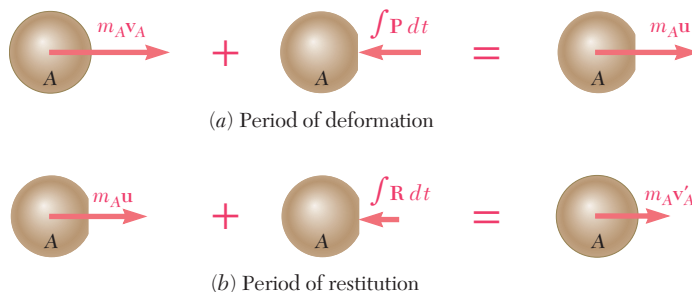


Fig. 13.22 Impulse-momentum diagram for particle A during (a) the period of deformation, and (b) during the period of restoration.

In general, the force $\mathbf{R}$ exerted on A during the period of restitution differs from the force $\mathbf{P}$ exerted during the period of deformation, and the magnitude $\int R dt$ of its impulse is smaller than the magnitude $\int P dt$ of the impulse of $\mathbf{P}$. The ratio of the magnitudes of the impulses, corresponding, respectively, to the period of restitution and to the period of deformation, is called the **coefficient of restitution** and is denoted by e . We have

$$e = \frac{\int R dt}{\int P dt} \quad (13.40)$$

The value of the coefficient e is always between 0 and 1. It depends to a large extent on the two materials involved, but it also varies considerably with the impact velocity and the shape and size of the two colliding bodies.

Solving Eqs. (13.38) and (13.39) for the two impulses and substituting into Eq. (13.40), we obtain

$$e = \frac{u - v'_A}{v_A - u} \quad (13.41)$$

A similar analysis of particle B leads to the relation

$$e = \frac{v'_B - u}{u - v_B} \quad (13.42)$$

Since the quotients in Eqs. (13.41) and (13.42) are equal, they are also equal to the quotient obtained by adding, respectively, their numerators and their denominators. We therefore have

$$e = \frac{(u - v'_A) + (v'_B - u)}{(v_A - u) + (u - v_B)} = \frac{v'_B - v'_A}{v_A - v_B}$$

and

Coefficient of restitution

$$v'_B - v'_A = e(v_A - v_B) \quad (13.43)$$

Since $v'_B - v'_A$ represents the relative velocity of the two particles after impact and $v_A - v_B$ represents their relative velocity before impact, formula (13.43) says:

We can obtain the relative velocity of the two particles after impact by multiplying their relative velocity before impact by the coefficient of restitution.

This property is used to determine experimentally the value of the coefficient of restitution of two given materials.

We can now obtain the velocities of the two particles after impact by solving Eqs. (13.37) and (13.43) simultaneously for v'_A and v'_B . Recall that the derivations of Eqs. (13.37) and (13.43) were based on the assumption that particle B is located to the right of A and that both particles are initially moving to the right. If particle B is initially moving to the left, the scalar v_B should be considered negative. The same sign convention holds for the velocities after impact: A positive sign for v'_A indicates that particle A moves to the right after impact, and a negative sign indicates that it moves to the left.

Two particular cases of impact are of special interest.

1. **$e = 0$, Perfectly Plastic Impact.** When $e = 0$, Eq. (13.43) yields $v'_B = v'_A$. There is no period of restitution, and both particles stay together after impact. Substituting $v'_B = v'_A = v'$ into Eq. (13.37), which expresses that the total momentum of the particles is conserved, we have

$$m_A v_A + m_B v_B = (m_A + m_B) v' \quad (13.44)$$

We can solve this equation for the common velocity v' of the two particles after impact.

2. **$e = 1$, Perfectly Elastic Impact.** When $e = 1$, Eq. (13.43) reduces to

$$v'_B - v'_A = v_A - v_B \quad (13.45)$$

This equation says that the relative velocities before and after impact are equal. This means that the impulses received by each particle during the period of deformation and during the period of restitution are equal. We can obtain the velocities v'_A and v'_B by solving Eqs. (13.37) and (13.45) simultaneously.

It is worth noting that **in the idealized case of a perfectly elastic impact, the total energy of the two particles**, as well as their total momentum, **is conserved**. We can write Eqs. (13.37) and (13.45) as

$$m_A(v_A - v'_A) = m_B(v'_B - v_B) \quad (13.37')$$

$$v_A + v'_A = v_B + v'_B \quad (13.45')$$

Multiplying Eqs. (13.37') and (13.45') member by member, we have

$$\begin{aligned} m_A(v_A - v'_A)(v_A + v'_A) &= m_B(v'_B - v_B)(v'_B + v_B) \\ m_A v_A^2 - m_A (v'_A)^2 &= m_B (v'_B)^2 - m_B v_B^2 \end{aligned}$$

Rearranging the terms in this equation and multiplying by $1/2$, we obtain

$$\frac{1}{2} m_A v_A^2 + \frac{1}{2} m_B v_B^2 = \frac{1}{2} m_A (v'_A)^2 + \frac{1}{2} m_B (v'_B)^2 \quad (13.46)$$

This equation states that the kinetic energy of the particles is conserved.

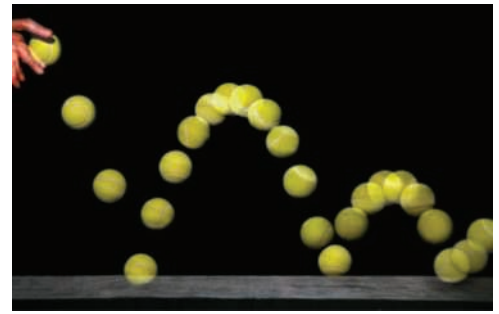


Photo 13.5 The height the tennis ball bounces decreases after each impact because it has a coefficient of restitution less than one and energy is lost with each bounce.

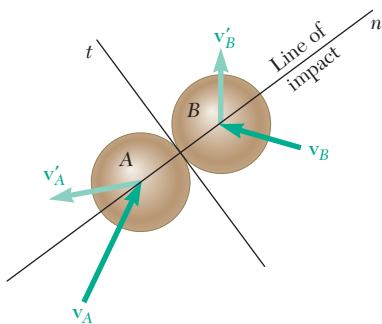


Fig. 13.23 In an oblique central impact, the velocities of the colliding particles are not directed along the line of impact.

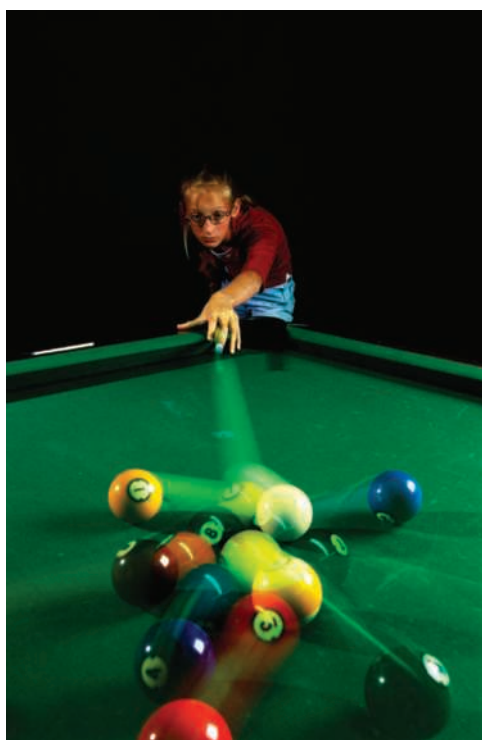


Photo 13.6 When pool balls strike each other there is a transfer of momentum.

Note, however, that **in the general case of impact**, i.e., when e is not equal to 1, **the total energy of the particles is not conserved**. This can be shown in any given case by comparing the kinetic energies before and after impact. The lost kinetic energy may be transformed into other forms of energy, such as heat, sound, generation of elastic waves within the two colliding bodies, or permanent deformation of the bodies.

13.4B Oblique Central Impact

Let us now consider the case when the velocities of the two colliding particles are *not* directed along the line of impact (Fig. 13.23). As mentioned earlier, the impact is said to be **oblique**. Since the velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ of the particles after impact are unknown in direction as well as in magnitude, their determination requires the use of four independent equations.

We choose as coordinate axes the n axis along the line of impact (i.e., along the common normal to the surfaces in contact) and the t axis along their common tangent. In very special cases where we can assume that the particles are perfectly smooth and frictionless, we observe that the only impulses exerted on the particles during the impact are due to internal forces directed along the line of impact, i.e., along the n axis (Fig. 13.24). This leads to the following results.

1. The component along the t axis of the momentum of each particle, considered separately, is conserved because no impulses act in the t direction; hence the t component of the velocity of each particle remains unchanged. We have

$$(v_A)_t = (v'_A)_t \quad (v_B)_t = (v'_B)_t \quad (13.47)$$

2. The component along the n axis of the total momentum of the two particles is conserved because the two impulses are equal and opposite to one another. We have

$$m_A(v_A)_n + m_B(v_B)_n = m_A(v'_A)_n + m_B(v'_B)_n \quad (13.48)$$

3. We obtain the component along the n axis of the relative velocity of the two particles after impact by multiplying the n component of their relative velocity before impact by the coefficient of restitution. Indeed, a derivation similar to that given in Sec. 13.4A for direct central impact yields

$$(v'_B)_n - (v'_A)_n = e[(v_A)_n - (v_B)_n] \quad (13.49)$$

We have thus obtained four independent equations that can be solved for the components of the velocities of A and B after impact. This method of solution is illustrated in Sample Prob. 13.20.

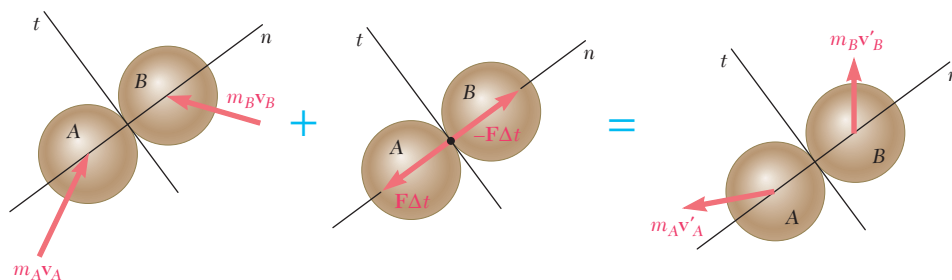


Fig. 13.24 Impulse-momentum diagram for an oblique impact. By including the internal impulses as equal and opposite, you also have the impulse-momentum diagram for each individual object (just ignore the other object).

Our analysis of the oblique central impact of two particles has been based so far on the assumption that both particles move freely before and after the impact. Let us now examine the case when one or both of the colliding particles is constrained in its motion. Consider, for instance, the collision between block A , which is constrained to move on a horizontal surface, and ball B , which is free to move in the plane of the figure (Fig. 13.25). Assuming no friction between the block and the ball or between the block and the horizontal surface, we note that the impulses exerted on the system consist of the impulses of the internal forces $\mathbf{F}$ and $-\mathbf{F}$ directed along the line of impact, i.e., along the n axis, and of the impulse of the external force $\mathbf{F}_{\text{ext}}$ exerted by the horizontal surface on block A and directed along the vertical, as shown in the impulse-momentum diagram (Fig. 13.26).

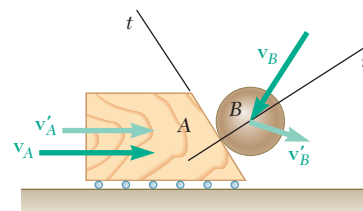


Fig. 13.25 An impact between a block moving on a horizontal surface and a ball moving in the vertical plane is called a “constrained impact.”

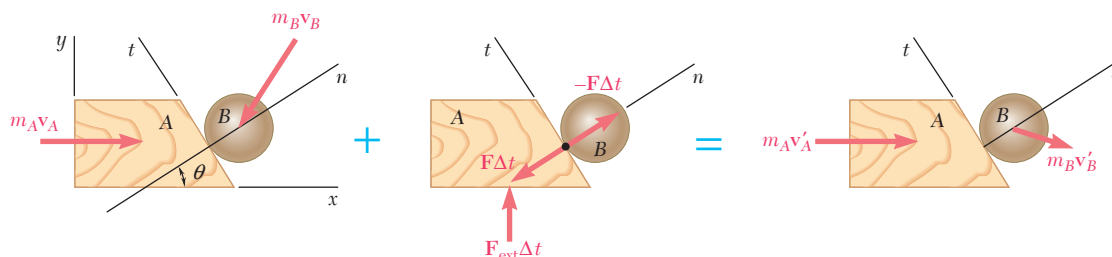


Fig. 13.26 Impulse-momentum diagram for a constrained impact between block A and ball B .

The velocities of block A and ball B immediately after the impact are represented by three unknowns: the magnitude of the velocity $\mathbf{v}'_A$ of block A , which is known to be horizontal, and the magnitude and direction of the velocity $\mathbf{v}'_B$ of ball B . We must therefore write three equations. We do this by using the impulse-momentum diagram and observing the following behavior.

1. The component along the t axis of the momentum of ball B is conserved because no impulses act on the ball in the t direction; hence, the t component of the velocity of ball B remains unchanged. We have

$$(v_B)_t = (v'_B)_t \quad (13.50)$$

2. The component along the horizontal x axis of the total momentum of block A and ball B is conserved because no external impulses act in the x -direction. We write this as

$$m_A v_A + m_B (v_B)_x = m_A v'_A + m_B (v'_B)_x \quad (13.51)$$

3. We obtain the component along the n axis of the relative velocity of block A and ball B after impact by multiplying the n component of their relative velocity before impact by the coefficient of restitution. We again have

$$(v'_B)_n - (v'_A)_n = e[(v_A)_n - (v_B)_n] \quad (13.49)$$

Note, however, that in the case considered here, we cannot establish the validity of Eq. (13.49) through a mere extension of the derivation given in Sec. 13.4A for the direct central impact of two particles moving in a straight line. Indeed, these particles were not subjected to any external impulse, whereas block A in the present analysis is subjected to the impulse exerted by the horizontal surface. To prove that Eq. (13.49) is still valid, we first apply the principle of impulse and momentum to block A

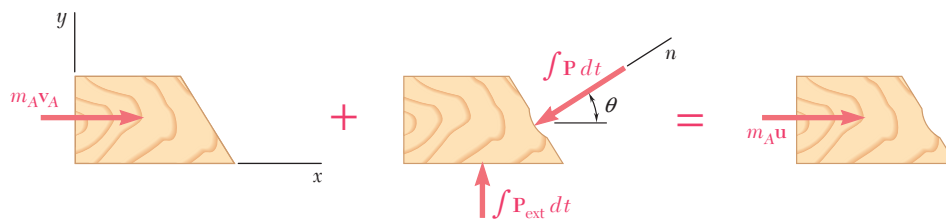


Fig. 13.27 Impulse-momentum diagram for block A.

over the period of deformation (Fig. 13.27). Considering only the horizontal components, we have

$$m_A v_A - (\int P dt) \cos \theta = m_A u \quad (13.52)$$

where the integral extends over the period of deformation and $\mathbf{u}$ represents the velocity of block A at the end of that period. Considering now the period of restitution, we have similarly

$$m_A u - (\int R dt) \cos \theta = m_A v'_A \quad (13.53)$$

where the integral extends over the period of restitution.

Recalling from Sec. 13.4A the definition of the coefficient of restitution, we have

$$e = \frac{\int R dt}{\int P dt} \quad (13.40)$$

Solving Eqs. (13.52) and (13.53) for the integrals $\int P dt$ and $\int R dt$ and substituting into Eq. (13.40), we have, after reductions,

$$e = \frac{u - v'_A}{v_A - u}$$

Then multiplying all velocities by $\cos \theta$ to obtain their projections on the line of impact gives

$$e = \frac{u_n - (v'_A)_n}{(v_A)_n - u_n} \quad (13.54)$$

Note that Eq. (13.54) is identical to Eq. (13.41) except for the subscripts n that we use here to indicate that we are considering velocity components along the line of impact. Since the motion of ball B is unconstrained, we can complete the proof of Eq. (13.49) in the same manner as the derivation of Eq. (13.43). Thus, we conclude that the relation in Eq. (13.49) between the components along the line of impact of the relative velocities of two colliding particles remains valid when one of the particles is constrained in its motion. The validity of this relation is easily extended to the case when both particles are constrained in their motion.

13.4C Problems Involving Multiple Principles

You now have at your disposal three different methods for the solution of kinetics problems.

- The direct application of Newton's second law, $\Sigma \mathbf{F} = m\mathbf{a}$.
- The method of work and energy, $T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{\text{NC}} = T_2 + V_{g_2} + V_{e_2}$, where $U_{1 \rightarrow 2}^{\text{NC}}$ is the work of external non-conservative forces such as friction.
- The method of impulse and momentum, $m\mathbf{v}_1 + \mathbf{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2$.

To derive maximum benefit from these three methods, you should be able to choose the method best suited for the solution of a given problem. You also should be prepared to solve problems that require you to use multiple principles.

You have already seen that the method of work and energy is in many cases more expeditious than the direct application of Newton's second law. As indicated in Sec. 13.1C, however, the method of work and energy has limitations, and it must sometimes be supplemented by the use of $\Sigma \mathbf{F} = m\mathbf{a}$. This is the case, for example, when you wish to determine an acceleration or a normal force.

For the solution of problems involving no impulsive forces, usually the equation $\Sigma \mathbf{F} = m\mathbf{a}$ yields a solution just as fast as the method of impulse and momentum, and the method of work and energy (if it applies) is more rapid and more convenient. However, in problems involving impact, the method of impulse and momentum is the only practicable method. A solution based on the direct application of $\Sigma \mathbf{F} = m\mathbf{a}$ would be unwieldy, and the method of work and energy cannot be used, because impact (unless perfectly elastic) involves a loss of mechanical energy.

Many problems involve only conservative forces except for a short impact phase during which impulsive forces act. The solution of such problems can be divided into several parts. The part corresponding to the impact phase calls for the use of the method of impulse and momentum and of the relation between relative velocities. The other parts usually can be solved by using the method of work and energy. If the problem involves the determination of a normal force, however, the use of $\Sigma \mathbf{F} = m\mathbf{a}$ is necessary.

Consider, for example, a pendulum A , with a mass m_A and a length l , that is released with no velocity from a position A_1 (Fig. 13.28a). The pendulum swings freely in a vertical plane and hits a second pendulum B , with a mass m_B and the same length l , that is initially at rest. After the impact (with coefficient of restitution e), pendulum B swings through an angle θ that we wish to determine.

The solution of the problem can be divided into three parts:

- 1. Pendulum A Swings from A_1 to A_2 .** Use the principle of conservation of energy to determine the velocity $(v_A)_2$ of the pendulum at A_2 (Fig. 13.28b).

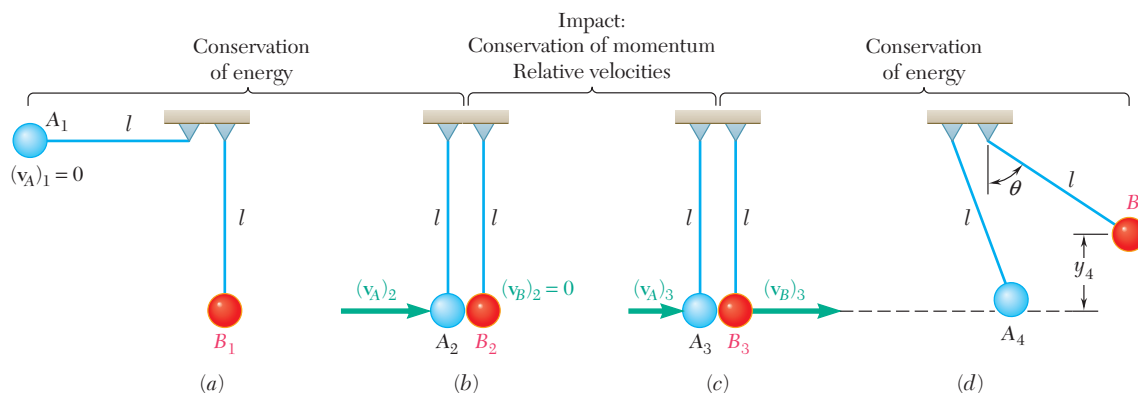


Fig. 13.28 Analyzing an impact between two pendulum bobs by conservation of energy and conservation of momentum.

2. **Pendulum A Hits Pendulum B.** Use the fact that the total momentum of the two pendulums is conserved, and use the relation between their relative velocities—that is, the coefficient of restitution—to determine the velocities $(\mathbf{v}_A)_3$ and $(\mathbf{v}_B)_3$ of the two pendulums after impact (Fig. 13.28c).
3. **Pendulum B Swings from B_3 to B_4 .** Apply the principle of conservation of energy to pendulum B to determine the maximum elevation y_4 reached by that pendulum (Fig. 13.28d). You can then determine the angle θ by trigonometry.

Note that if you need to determine the tensions in the cords holding the pendulums, the method of solution just described should be supplemented by the use of $\Sigma \mathbf{F} = m\mathbf{a}$. A summary of all the kinetics principles we have discussed so far and some clues as to when to apply them are shown in Fig. 13.29.

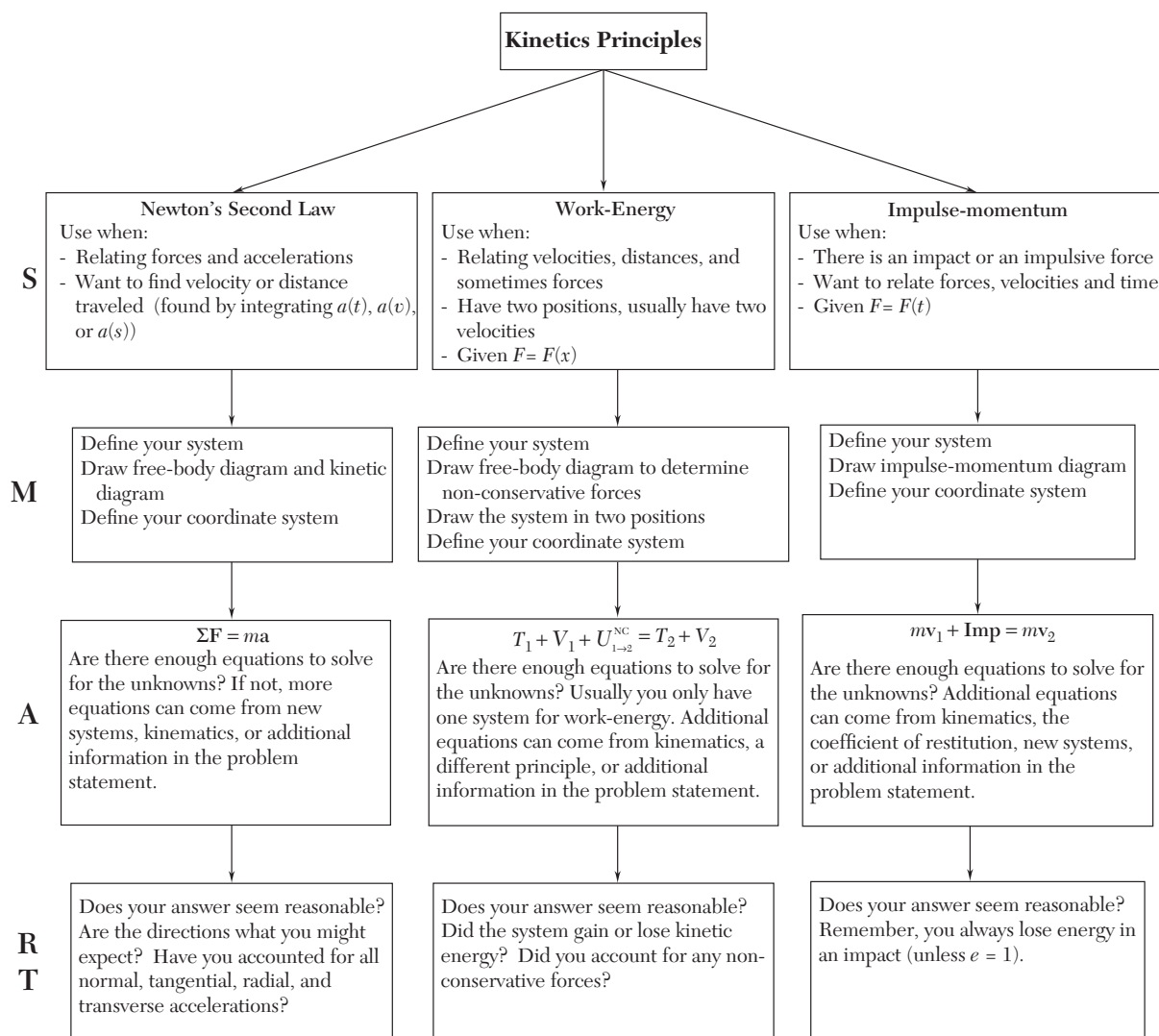


Fig. 13.29 The three kinetics principles using the SMART methodology.

Sample Problem 13.18

A 20-Mg railroad car moving at a speed of 0.5 m/s to the right collides with a 35-Mg car at rest. After the collision, the 35-Mg car moves to the right at a speed of 0.3 m/s. Determine the coefficient of restitution between the two cars.

STRATEGY: Since there is an impact and no external impulses, use the conservation of linear momentum. You will also need to use the equation for the coefficient of restitution.

MODELING: Choose your system to be both railroad cars and model them as particles. The impulse-momentum diagram for this system is shown in Fig. 1. There are no external impulses acting on this system.

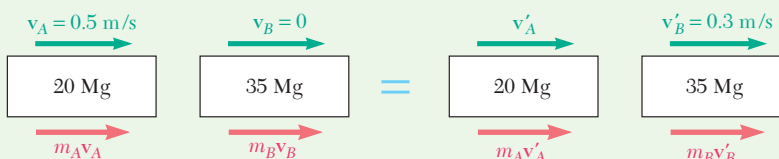


Fig. 1 Velocities and linear momenta of the cars before and after impact.

ANALYSIS: The total momentum of the two cars is conserved, so

$$m_A \mathbf{v}_A + m_B \mathbf{v}_B = m_A \mathbf{v}'_A + m_B \mathbf{v}'_B$$

Substituting in the known values gives

$$(20 \text{ Mg})(+0.5 \text{ m/s}) + (35 \text{ Mg})(0) = (20 \text{ Mg})v'_A + (35 \text{ Mg})(+0.3 \text{ m/s})$$

$$v'_A = -0.025 \text{ m/s} \quad \mathbf{v}'_A = 0.025 \text{ m/s} \leftarrow$$

You can obtain the coefficient of restitution from its definition as

$$e = \frac{v'_B - v'_A}{v_A - v_B} = \frac{+0.3 - (-0.025)}{+0.5 - 0} = \frac{0.325}{0.5}$$

$$e = 0.65 \quad \blacktriangleleft$$

REFLECT and THINK: The railroad cars are constrained to move along the track, so this is a one-dimensional direct central impact. The interaction forces are large, but they last for only a very short time. Mechanical energy is lost during this impact, so you could not have used the conservation of energy.

Sample Problem 13.19

A ball is thrown against a frictionless, vertical wall. Immediately before the ball strikes the wall, its velocity has a magnitude of v and forms an angle of 30° with the horizontal. Knowing that $e = 0.90$, determine the magnitude and direction of the velocity of the ball as it rebounds from the wall.

STRATEGY: An impact occurs, and you are given the coefficient of restitution, so use conservation of momentum and the definition of the coefficient of restitution.

MODELING: Choose your system to be the ball, and model it as a particle. The impulse-momentum diagram for this system is shown in Fig. 1.

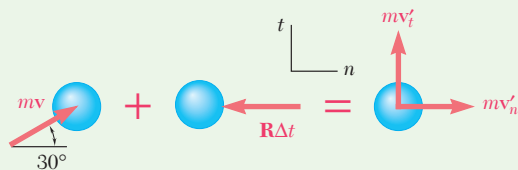


Fig. 1 Impulse-momentum diagram for the ball.

ANALYSIS: Resolve the initial velocity of the ball into components perpendicular and parallel to the wall, as shown in Fig. 2.

$$v_n = v \cos 30^\circ = 0.866v \quad v_t = v \sin 30^\circ = 0.500v$$

Motion Parallel to the Wall. Since the wall is frictionless, the impulse it exerts on the ball is perpendicular to the wall. Thus, the component of the momentum of the ball parallel to the wall is conserved. You have

$$\mathbf{v}'_t = \mathbf{v}_t = 0.500v \uparrow$$

Motion Perpendicular to the Wall. Since the mass of the wall (and of the earth) is essentially infinite, writing an equation for conservation of the total momentum of the ball and wall would yield no useful information. However, using the equation for coefficient of restitution, you have

$$0 - v'_n = e(v_n - 0) \\ v'_n = -0.90(0.866v) = -0.779v \quad \mathbf{v}'_n = 0.779v \leftarrow$$

Resultant Motion. Adding vectorially the components $\mathbf{v}'_n$ and $\mathbf{v}'_t$ (Fig. 3), you find

$$\mathbf{v}' = 0.926v \searrow 32.7^\circ \blacktriangleleft$$

REFLECT and THINK: Tests similar to this are done to make sure that sporting equipment—such as tennis balls, golf balls, and basketballs—are consistent and fall within certain specifications. Testing modern golf balls and clubs shows that the coefficient of restitution actually decreases with increasing club speed (from about 0.84 at a speed of 90 mph to about 0.80 at club speeds of 130 mph).

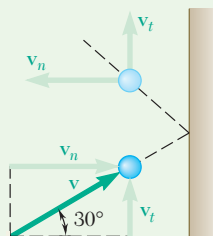


Fig. 2 Components of the initial velocity.

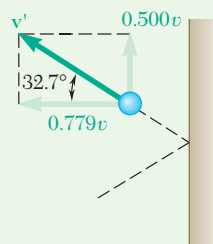


Fig. 3 Finding the magnitude and direction for the final velocity.

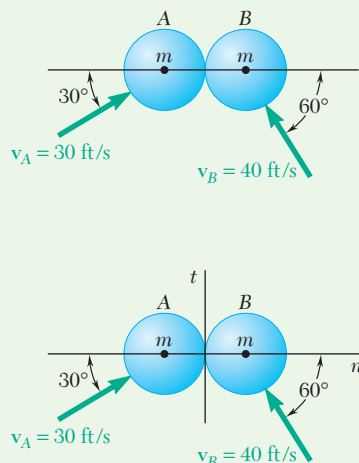


Fig. 1 Initial velocities of *A* and *B* and the coordinate system to be used.

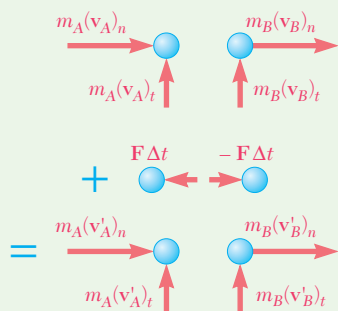


Fig. 2 Impulse-momentum diagram for the system.

Sample Problem 13.20

The magnitudes and directions of the velocities of two identical frictionless balls are shown before they strike each other. Assuming $e = 0.90$, determine the magnitude and direction of the velocity of each ball after the impact.

STRATEGY: Since an impact occurs, use the principle of impulse and momentum. You also need the equation for the coefficient of restitution.

MODELING: Choose your system to be both balls. Assuming they are small and do not rotate, you can model them as particles. Figure 1 shows the normal and tangential directions; Fig. 2 shows the impulse-momentum diagram for this system. The impulsive forces that the balls exert on each other during the impact are directed along a line joining the centers of the balls (the line of impact). Therefore, it is best to resolve the velocities into components directed, respectively, along the line of impact and along the common tangent to the surfaces in contact. Thus,

$$(v_A)_n = v_A \cos 30^\circ = +26.0 \text{ ft/s}$$

$$(v_A)_t = v_A \sin 30^\circ = +15.0 \text{ ft/s}$$

$$(v_B)_n = -v_B \cos 60^\circ = -20.0 \text{ ft/s}$$

$$(v_B)_t = v_B \sin 60^\circ = +34.6 \text{ ft/s}$$

ANALYSIS:

Motion Along the Common Tangent. Considering only the t components, apply the principle of impulse and momentum to each ball *separately*. Since the impulsive forces are directed along the line of impact, the t component of the momentum, and hence the t component of the velocity of each ball, is unchanged. You have

$$(v'_A)_t = 15.0 \text{ ft/s} \uparrow \quad (v'_B)_t = 34.6 \text{ ft/s} \uparrow$$

Motion Along the Line of Impact. In the n direction, consider the two balls as a single system. By Newton's third law, the internal impulses are, respectively, $\mathbf{F} \Delta t$ and $-\mathbf{F} \Delta t$, so they cancel. Thus, the total momentum of the balls is conserved as

$$\begin{aligned} m_A(v_A)_n + m_B(v_B)_n &= m_A(v'_A)_n + m_B(v'_B)_n \\ m(26.0) + m(-20.0) &= m(v'_A)_n + m(v'_B)_n \\ (v'_A)_n + (v'_B)_n &= 6.0 \quad (1) \end{aligned}$$

Using the equation for the coefficient of restitution relating the relative velocities, you have

$$(v'_B)_n - (v'_A)_n = e[(v_A)_n - (v_B)_n]$$

You can now substitute the known quantities into this equation. It is important to use the signs correctly when substituting into this equation, e.g. $(v_B)_n = -20$. This gives

$$\begin{aligned} (v'_B)_n - (v'_A)_n &= (0.90)[26.0 - (-20.0)] \\ (v'_B)_n - (v'_A)_n &= 41.4 \quad (2) \end{aligned}$$

(continued)

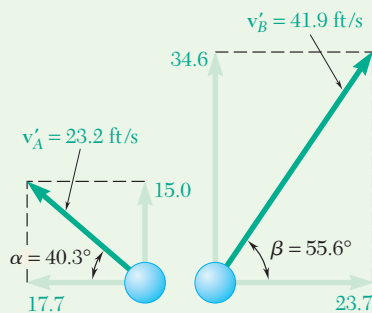


Fig. 3 The velocity components can be resolved into their magnitudes and directions.

Solving Eqs. (1) and (2) simultaneously yields

$$\begin{aligned}(v'_A)_n &= -17.7 & (v'_B)_n &= +23.7 \\ (v'_A)_t &= 17.7 \text{ ft/s} \leftarrow & (v'_B)_t &= 23.7 \text{ ft/s} \rightarrow\end{aligned}$$

Resultant Motion. Adding the velocity components of each ball vectorially (Fig. 3), you obtain

$$\mathbf{v}'_A = 23.2 \text{ ft/s} \searrow 40.3^\circ \quad \mathbf{v}'_B = 41.9 \text{ ft/s} \nearrow 55.6^\circ \quad \blacktriangleleft$$

REFLECT and THINK: Rather than choosing your system to be both balls, you could have applied impulse-momentum along the line of impact for each ball individually. This would have resulted in two equations and one additional unknown, $F\Delta t$. To determine the impulsive force F , you would need to be given the time for the impact, Δt .

Sample Problem 13.21

Ball B is hanging from an inextensible cord BC . An identical ball A is released from rest when it is just touching the cord and acquires a velocity $\mathbf{v}_0$ before striking ball B . Assuming a perfectly elastic impact ($e = 1$) and no friction, determine the velocity of each ball immediately after impact.

STRATEGY: Since an impact occurs, use the impulse-momentum principle. You also need the equation for coefficient of restitution.

MODELING: You have several choices of systems in this problem. If you choose A as your system, you obtain the impulse-momentum diagram shown in Fig. 1. Choosing the system to be both balls results in the impulse-momentum diagram shown in Fig. 2.

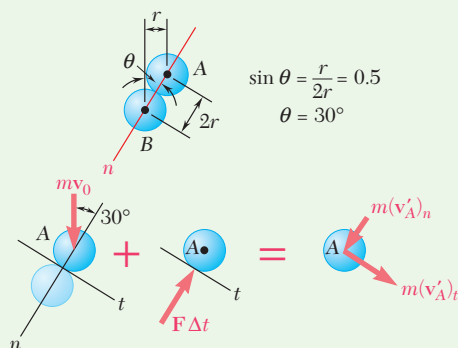


Fig. 1 Impulse-momentum diagram for ball A .

ANALYSIS:

Impulse-Momentum Principle: Ball A. Applying the conservation of momentum of ball A along the common tangent to balls A and B (Fig. 1) gives

$$\begin{aligned}
 m\mathbf{v}_A + \mathbf{F} \Delta t &= m\mathbf{v}'_A \\
 + \nearrow t \text{ components: } m v_0 \sin 30^\circ + 0 &= m(v'_A)_t \\
 (v'_A)_t &= 0.5v_0
 \end{aligned} \tag{1}$$

Impulse-Momentum Principle: Balls A and B. Since ball B is constrained to move in a circle with center C, its velocity $\mathbf{v}_B$ after impact must be horizontal. Applying the conservation of momentum for a system containing both balls (Fig. 2) gives

$$\begin{aligned}
 m\mathbf{v}_A + \mathbf{T} \Delta t &= m\mathbf{v}'_A + m\mathbf{v}'_B \\
 \rightarrow x \text{ components: } 0 &= m(v'_A)_t \cos 30^\circ - m(v'_A)_n \sin 30^\circ - mv'_B
 \end{aligned}$$

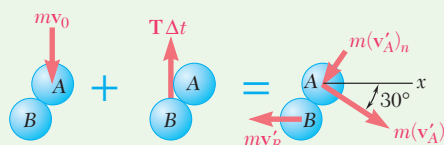


Fig. 2 Impulse-momentum diagram for both balls.

This equation expresses the conservation of total momentum in the x direction. Substituting for $(v'_A)_t$ from Eq. (1) and rearranging terms, you have

$$0.5(v'_A)_n + v'_B = 0.433v_0 \tag{2}$$

Relative Velocities Along the Line of Impact. Since $e = 1$, the equation for the coefficient of restitution gives

$$\begin{aligned}
 (v'_B)_n - (v'_A)_n &= (v_A)_n - (v_B)_n \\
 v'_B \sin 30^\circ - (v'_A)_n &= v_0 \cos 30^\circ - 0 \\
 0.5v'_B - (v'_A)_n &= 0.866v_0
 \end{aligned} \tag{3}$$

It is important to note that the coefficient of restitution always uses the components of the velocities along the line of impact; that is, the n -direction. Solving Eqs. (2) and (3) simultaneously, you obtain

$$\begin{aligned}
 (v'_A)_n &= -0.520v_0 & v'_B &= 0.693v_0 \\
 \mathbf{v}'_B &= 0.693v_0 \leftarrow
 \end{aligned}$$

Recalling Eq. (1), draw a sketch (Fig. 3) and obtain by trigonometry

$$\begin{aligned}
 v'_A &= 0.721v_0 & \beta &= 46.1^\circ & \alpha &= 46.1^\circ - 30^\circ = 16.1^\circ \\
 \mathbf{v}'_B &= 0.721v_0 \nearrow 16.1^\circ
 \end{aligned}$$

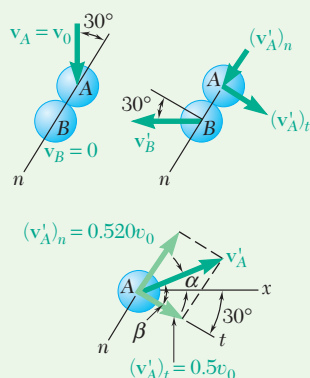
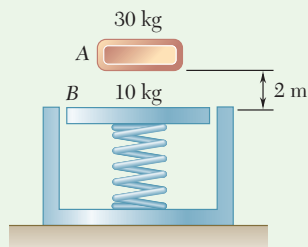


Fig. 3 Diagram to find the magnitude and direction for the final velocity of B.

REFLECT and THINK: Since $e = 1$, the impact between A and B is perfectly elastic. Therefore, rather than using the coefficient of restitution, you could have used the conservation of energy as your final equation.

Sample Problem 13.22



A 30-kg block is dropped from a height of 2 m onto the 10-kg pan of a spring scale. The constant of the spring is $k = 20 \text{ kN/m}$. Assuming the impact to be perfectly plastic, determine the maximum deflection of the pan.

STRATEGY: This problem has three distinct phases, as shown in Fig. 1. In phase 1, A falls (use the conservation of energy); in phase 2, A hits B (use the conservation of momentum); and in phase 3, A and B move down together (use the conservation of energy).

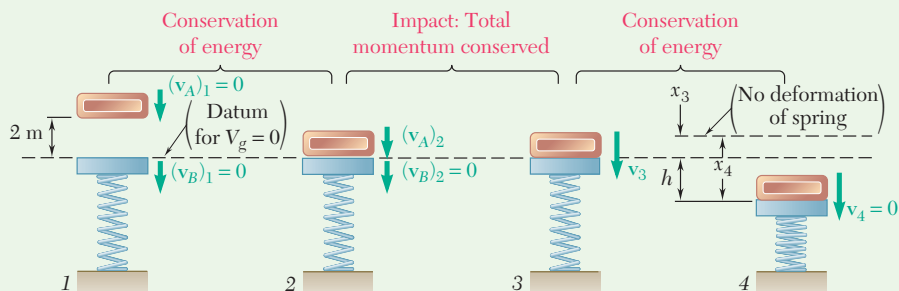


Fig. 1 Three phases of the motion.

MODELING: For each phase of the motion, define a different system. For phase 1, choose A as your system, and for phase 2, define your system as A and B together. For phase 3, your system is A, B, and the spring.

ANALYSIS:

Conservation of Energy A. Block A weighs

$$W_A = (30 \text{ kg})(9.81 \text{ m/s}^2) = 294 \text{ N}.$$

Thus,

$$\begin{aligned} T_1 &= \frac{1}{2}m_A(v_A)_1^2 = 0 & V_1 &= W_A y = (294 \text{ N})(2 \text{ m}) = 588 \text{ J} \\ T_2 &= \frac{1}{2}m_A(v_A)_2^2 = \frac{1}{2}(30 \text{ kg})(v_A)_2^2 & V_2 &= 0 \\ T_1 + V_1 &= T_2 + V_2: & 0 + 588 \text{ J} &= \frac{1}{2}(30 \text{ kg})(v_A)_2^2 + 0 \\ & & (v_A)_2 &= +6.26 \text{ m/s} & (v_A)_2 &= 6.26 \text{ m/s} \downarrow \end{aligned}$$

Impact: Conservation of Momentum for A and B. The impact is perfectly plastic, so $e = 0$; the block and pan move together after the impact.

$$\begin{aligned} m_A(v_A)_2 + m_B(v_B)_2 &= (m_A + m_B)v_3 \\ (30 \text{ kg})(6.26 \text{ m/s}) + 0 &= (30 \text{ kg} + 10 \text{ kg})v_3 \\ v_3 &= +4.70 \text{ m/s} & \mathbf{v}_3 &= 4.70 \text{ m/s} \downarrow \end{aligned}$$

Conservation of Energy for A, B, and the Spring. Initially, the spring supports the weight W_B of the pan; thus the initial deflection of the spring is

$$x_3 = \frac{W_B}{k} = \frac{(10 \text{ kg})(9.81 \text{ m/s}^2)}{20 \times 10^3 \text{ N/m}} = \frac{98.1 \text{ N}}{20 \times 10^3 \text{ N/m}} = 4.91 \times 10^{-3} \text{ m}$$

Denoting the total maximum deflection of the spring by x_4 , you have

$$T_3 = \frac{1}{2}(m_A + m_B)v_3^2 = \frac{1}{2}(30 \text{ kg} + 10 \text{ kg})(4.70 \text{ m/s})^2 = 442 \text{ J}$$

$$V_3 = V_g + V_e = 0 + \frac{1}{2}kx_3^2 = \frac{1}{2}(20 \times 10^3)(4.91 \times 10^{-3})^2 = 0.241 \text{ J}$$

$$T_4 = 0$$

$$V_4 = V_g + V_e = (W_A + W_B)(-h) + \frac{1}{2}kx_4^2 = -(392)h + \frac{1}{2}(20 \times 10^3)x_4^2$$

The displacement of the pan is $h = x_4 - x_3$, so the final result is

$$T_3 + V_3 = T_4 + V_4:$$

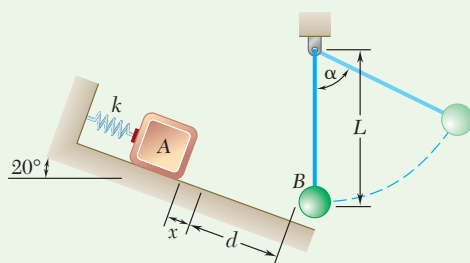
$$442 + 0.241 = 0 - 392(x_4 - 4.91 \times 10^{-3}) + \frac{1}{2}(20 \times 10^3)x_4^2$$

$$x_4 = 0.230 \text{ m} \quad h = x_4 - x_3 = 0.230 \text{ m} - 4.91 \times 10^{-3} \text{ m}$$

$$h = 0.225 \text{ m} \quad \quad \quad h = 225 \text{ mm} \quad \blacktriangleleft$$

REFLECT and THINK: The spring constant for this scale is pretty large, but the block is fairly massive and is dropped from a height of 2 m. From this perspective, the deflection seems reasonable.

Sample Problem 13.23



A 2-kg block A is pushed up against a spring, compressing it a distance $x = 0.1 \text{ m}$. The block is then released from rest and slides down the 20° incline until it strikes a 1-kg sphere B that is suspended by a 1-m inextensible rope. The spring constant $k = 800 \text{ N/m}$, the coefficient of friction between A and the ground is 0.2, block A slides from the unstretched length of the spring a distance $d = 1.5 \text{ m}$, and the coefficient of restitution between A and B is 0.8. When $\alpha = 40^\circ$, determine (a) the speed of B, (b) the tension in the rope.

STRATEGY: A lot of things are going on in this problem, so you need to break the motion into steps.

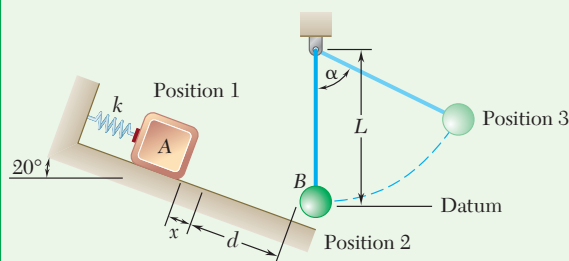


Fig. 1 Three positions of interest for this problem.

Step 1: Block A slides down the incline, so there are two positions. Therefore, use the work–energy principle between position 1 and position 2 to find the velocity of A just before it strikes ball B (Fig. 1).

Step 2: Block A hits B, so an impact occurs. Therefore, use impulse–momentum and the equation for the coefficient of restitution.

Step 3: Ball B is swinging up, so you have two positions (position 2 and position 3 in Fig. 1). You are asked to find the speed at position 3, therefore, use the conservation of energy.

Step 4: To find the tension when $\alpha = 40^\circ$, use Newton’s second law with normal and tangential coordinates.

(continued)

MODELING: Each step requires a different system. For Step 1, your system is A and the spring. For Step 2, it is A and B . Finally, for Steps 3 and 4, it is B . We model A and B as particles and draw the appropriate figures in the analysis section.

ANALYSIS:

Step 1. Block Slides Down the Incline. The principle of work and energy between where the block is released to the point it strikes B is

$$T_1 + V_{g1} + V_{e1} + U_{1 \rightarrow 2}^{\text{NC}} = T_2 + V_{g2} + V_{e2} \quad (1)$$

Work. The only non-conservative force that does work is the friction force. A free-body diagram for A is shown in Fig. 2. Applying Newton's second law gives

$$\begin{aligned} +\nearrow \Sigma F_y &= 0: & N - m_A g \cos \theta &= 0 & \text{or} \\ N &= m_A g \cos \theta = (2 \text{ kg})(9.81 \text{ m/s}^2) \cos 20^\circ = 18.437 \text{ N} \end{aligned}$$

and the friction force is

$$F_f = \mu_k N = (0.2)(18.437 \text{ N}) = 3.687 \text{ N}$$

So the work is

$$U_{1 \rightarrow 2}^{\text{NC}} = -F_f(x + d) = -(3.687 \text{ N})(1.6 \text{ m}) = -5.900 \text{ J}$$

Position 1. Place your datum for V_g at the impact point near B (see Fig. 1). Calculate the initial energy as

$$\begin{aligned} T_1 &= 0, & V_{e1} &= \frac{1}{2} k x_1^2 = \frac{1}{2} (800) (0.1)^2 = 4.00 \text{ J} \\ V_{g1} &= m_A g h_1 = m_A g (x + d) \sin \theta = (2)(9.81)(1.6) \sin 20^\circ = 10.737 \text{ J} \end{aligned}$$

Position 2. The energy at position 2 is

$$T_2 = \frac{1}{2} m_A v_A^2 = \frac{1}{2} (2) v_A^2 = 1.000 v_A^2 \quad V_2 = 0$$

Substituting into Eq. (1) gives $0 + 10.737 \text{ J} + 4.00 \text{ J} - 5.900 \text{ J} = 1.00 v_A^2 + 0$. Solving for v_A gives $v_A = 2.973 \text{ m/s}$.

Step 2. Impact. The impulse-momentum diagram for A and B is shown in Fig. 3.

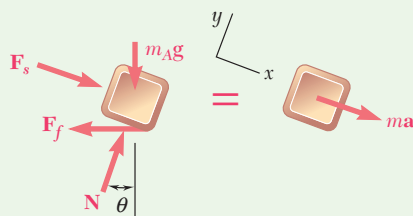


Fig. 2 Free-body diagram and kinetic diagram for block A .

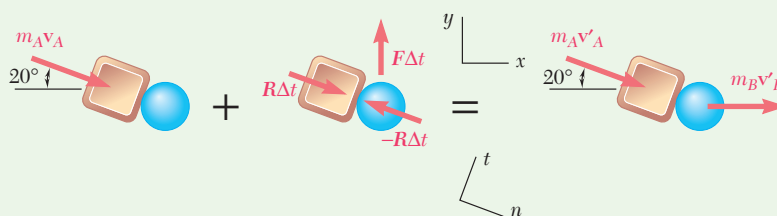


Fig. 3 Impulse-momentum diagram for A and B .

Note that two coordinate systems are defined: n defines the line of impact between the block and ball and y is in the direction of the impulsive force of the rope. Since no impulsive forces act in the horizontal direction, apply impulse-momentum in the x direction. Thus,

$$\rightarrow x \text{ components: } m_A v_A \cos \theta + 0 = m_A v'_A \cos \theta + m_B v'_B \quad (2)$$

Coefficient of Restitution.

$$(v'_B)_n - (v'_A)_n = e[(v_A)_n - (v_B)_n] \quad \text{or} \quad v_B \cos \theta - v'_A = e v_A \quad (3)$$

In Eqs. (2) and (3) you can solve for two unknowns, v'_A and v'_B . This gives

$$\begin{aligned} v'_A &= 1.0382 \text{ m/s} \\ v'_B &= 3.6356 \text{ m/s} \end{aligned}$$

Step 3. Sphere B Rises. The tension does no work, so use the conservation of energy for B between positions 2 and 3. Again, define the datum as shown in Fig 1.

$$T_2 + V_{g_2} + V_{e_2} = T_3 + V_{g_3} + V_{e_3} \quad (4)$$

Position 2.

$$T_2 = \frac{1}{2} m_B (v'_B)^2, \quad V_{g_2} = 0, \quad V_{e_2} = 0$$

Position 3.

$$T_3 = \frac{1}{2} m_B v_3^2, \quad V_{g_3} = m_B g L (1 - \cos \alpha), \quad V_{e_3} = 0$$

Substituting these into Eq. (4) and solving for v_{B_3} gives

$$v_{B_3} = 2.94 \text{ m/s} \quad \blacktriangleleft$$

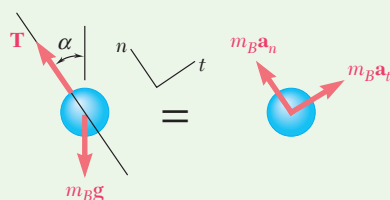


Fig. 4

Step 4. Tension in the Rope. A free-body diagram and kinetic diagram for the sphere at position 3 are shown in Fig. 4. Applying Newton's second law in the normal direction gives

$$+\nearrow \Sigma F_n = m_B a_n: \quad T - m_B g \cos \alpha = m_B a_n = m_B \frac{v_{B_3}^2}{L}$$

Solving for T , you find

$$T = 16.14 \text{ N} \quad \blacktriangleleft$$

REFLECT and THINK: You cannot use work–energy from position 1 to position 3 because a loss of energy occurs when A hits B . If the coefficient of friction had been larger, say $\mu_k = 0.4$, you would find that after the impact, B has a speed of 2.10 m/s. Plugging this into Eq. (4) gives an imaginary number for the speed at $\alpha = 40^\circ$, meaning sphere B does not reach this angle.

SOLVING PROBLEMS ON YOUR OWN

This section deals with the **impact of two smooth bodies**, i.e., with a collision occurring in a very small interval of time. You solved several impact problems by noting that the total momentum of the two bodies is conserved and by expressing the relationship between the relative velocities of the two bodies before and after impact.

1. As a first step in your solution, you should select and draw two coordinate axes: the t axis, which is tangent to the surfaces of contact of the two colliding bodies; and the n axis, which is normal to the surfaces of contact and defines the line of impact. In all of the problems in this section, the line of impact passes through the mass centers of the colliding bodies, and the impact is referred to as a **central impact**.

2. Next draw an impulse-momentum diagram showing the momenta of the bodies before impact, the impulses exerted on the bodies during impact, and the final momenta of the bodies after impact (Fig. 13.24). Then observe whether the impact is a **direct central impact** or an **oblique central impact**.

3. Direct central impact [Sample Prob. 13.18]. This occurs when the velocities of bodies A and B are both directed along the line of impact before impact (Fig. 13.20a).

a. Conservation of momentum. Since the impulsive forces are internal to the system, the total momentum of A and B is conserved as

$$m_A v_A + m_B v_B = m_A v'_A + m_B v'_B \quad (13.37)$$

where v_A and v_B denote the velocities of bodies A and B before impact and v'_A and v'_B denote their velocities after impact.

b. Coefficient of restitution. You can also write the relation between the relative velocities of the two bodies before and after impact as

$$v'_B - v'_A = e(v_A - v_B) \quad (13.43)$$

where e is the coefficient of restitution between the two bodies.

Note that Eqs. (13.37) and (13.43) are scalar equations that you can solve for two unknowns. Also, be careful to adopt a consistent sign convention for all velocities.

4. Oblique central impact [Sample Prob. 13.20]. This occurs when one or both of the initial velocities of the two bodies is not directed along the line of impact (Fig. 13.20b). Again, these solution steps are only applicable to problems where the impulsive forces in the tangential direction are negligible (e.g., you would not use these to solve Prob. 13.146). To solve problems of this type, you should first resolve the momenta and impulses shown in your diagram into components along the t axis and the n axis.

a. Conservation of momentum. Since the impulsive forces act along the line of impact, i.e., along the n axis, the component along the t axis of the momentum of *each body* is conserved. Therefore, for each body, you can write that the t components of its velocity before and after impact are equal. So,

$$(v_A)_t = (v'_A)_t \quad (v_B)_t = (v'_B)_t \quad (13.47)$$

Also, the component along the n axis of the total momentum of the system is conserved as

$$m_A(v_A)_n + m_B(v_B)_n = m_A(v'_A)_n + m_B(v'_B)_n \quad (13.48)$$

b. Coefficient of restitution. The relation between the relative velocities of the two bodies before and after impact can be written in the n direction only. Hence,

$$(v'_B)_n - (v'_A)_n = e[(v_A)_n - (v_B)_n] \quad (13.49)$$

You now have four equations that you can solve for four unknowns. Note that after finding all of the velocities, you can determine the impulse exerted by body A on body B by drawing an impulse-momentum diagram for B alone and equating components in the n direction.

c. When the motion of one of the colliding bodies is constrained, you must include the impulses of the external forces in your diagram [Sample Probs. 13.21 and 13.23]. You will then observe that some of the previous relations do not hold. However, in the example shown in Fig. 13.26, the total momentum of the system is conserved in a direction perpendicular to the external impulse. Also note that, when a body A bounces off a fixed surface B , the only conservation of momentum equation that you can use is the first of Eq. (13.47) [Sample Prob. 13.19].

5. Remember that energy is lost during most impacts. The only exception is for **perfectly elastic** impacts ($e = 1$), where energy is conserved. Thus, in the general case of impact where $e < 1$, mechanical energy is not conserved. Therefore, be careful *not to apply* the principle of conservation of energy through an impact situation. Instead, apply this principle separately to the motions preceding and following the impact [Sample Probs. 13.22 and 13.23].

Problems

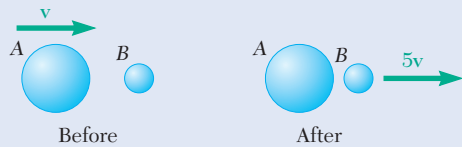


Fig. P13.CQ6

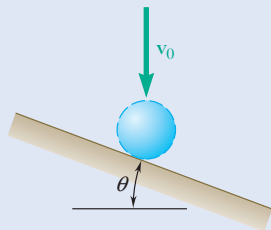


Fig. P13.F6

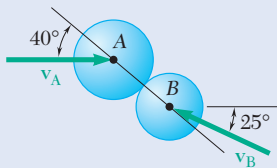


Fig. P13.F8

CONCEPT QUESTIONS

13.CQ6 A 5-kg ball A strikes a 1-kg ball B that is initially at rest. Is it possible that after the impact A is not moving and B has a speed of $5v$?

- Yes
- No

Explain your answer.

IMPULSE-MOMENTUM DIAGRAM PRACTICE PROBLEMS

13.F6 A sphere with a speed v_0 rebounds after striking a frictionless inclined plane as shown. Draw the impulse-momentum diagram that can be used to find the velocity of the sphere after the impact.

13.F7 An 80-Mg railroad engine A coasting at 6.5 km/h strikes a 20-Mg flatcar C carrying a 30-Mg load B which can slide along the floor of the car ($\mu_k = 0.25$). The flatcar was at rest with its brakes released. Instead of A and C coupling as expected, it is observed that A rebounds with a speed of 2 km/h after the impact. Draw impulse-momentum diagrams that can be used to determine (a) the coefficient of restitution and the speed of the flatcar immediately after impact, (b) the time it takes the load to slide to a stop relative to the car.

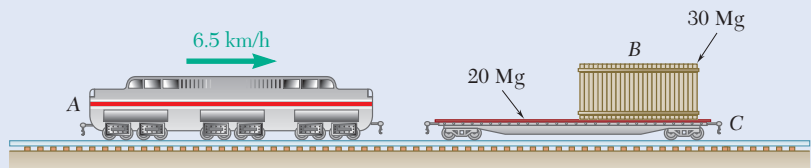


Fig. P13.F7

13.F8 Two frictionless balls strike each other as shown. The coefficient of restitution between the balls is e . Draw the impulse-momentum diagram that could be used to find the velocities of A and B after the impact.

13.F9 A 10-kg ball A moving horizontally at 12 m/s strikes a 10-kg block B . The coefficient of restitution of the impact is 0.4 and the coefficient of kinetic friction between the block and the inclined surface is 0.5. Draw the impulse-momentum diagram that can be used to determine the speeds of A and B after the impact.

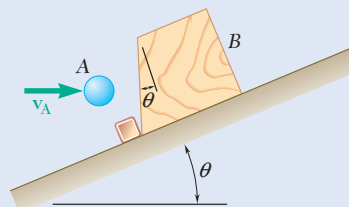


Fig. P13.F9

- 13.F10** Block A of mass m_A strikes ball B of mass m_B with a speed of v_A as shown. Draw the impulse-momentum diagram that can be used to determine the speeds of A and B after the impact and the impulse during the impact.

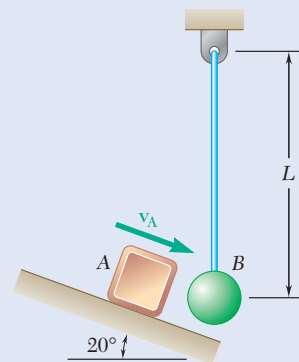


Fig. P13.F10

END-OF-SECTION PROBLEMS

- 13.155** The coefficient of restitution between the two collars is known to be 0.70. Determine (a) their velocities after impact, (b) the energy loss during impact.

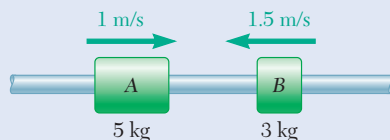


Fig. P13.155

- 13.156** Collars A and B , of the same mass m , are moving toward each other with identical speeds as shown. Knowing that the coefficient of restitution between the collars is e , determine the energy lost in the impact as a function of m , e , and v .

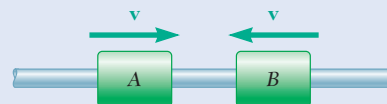


Fig. P13.156

- 13.157** One of the requirements for tennis balls to be used in official competition is that, when dropped onto a rigid surface from a height of 100 in., the height of the first bounce of the ball must be in the range 53 in. $\leq h \leq 58$ in. Determine the range of the coefficients of restitution of the tennis balls satisfying this requirement.

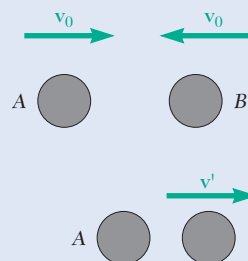


Fig. P13.158

- 13.158** Two disks sliding on a frictionless horizontal plane with opposite velocities of the same magnitude v_0 hit each other squarely. Disk A is known to have a weight of 6 lb and is observed to have zero velocity after impact. Determine (a) the weight of disk B , knowing that the coefficient of restitution between the two disks is 0.5, (b) the range of possible values of the weight of disk B if the coefficient of restitution between the two disks is unknown.

- 13.159** To apply shock loading to an artillery shell, a 20-kg pendulum A is released from a known height and strikes impactor B at a known velocity v_0 . Impactor B then strikes the 1-kg artillery shell C . Knowing the coefficient of restitution between all objects is e , determine the mass of B to maximize the impulse applied to the artillery shell C .

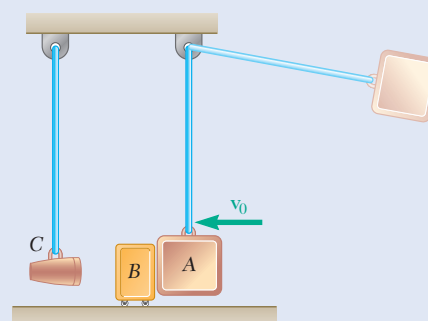


Fig. P13.159

- 13.160** Packages in an automobile parts supply house are transported to the loading dock by pushing them along on a roller track with very little friction. At the instant shown, packages B and C are at rest, and package A has a velocity of 2 m/s. Knowing that the coefficient of restitution between the packages is 0.3, determine (a) the velocity of package C after A hits B and B hits C , (b) the velocity of A after it hits B for the second time.

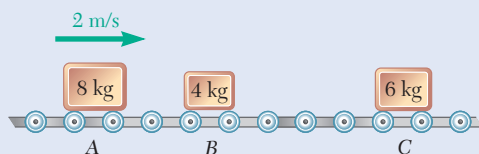


Fig. P13.160

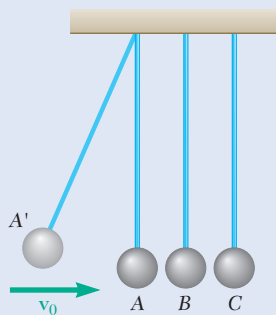


Fig. P13.161

13.161 Three steel spheres of equal mass are suspended from the ceiling by cords of equal length that are spaced at a distance slightly greater than the diameter of the spheres. After being pulled back and released, sphere A hits sphere B , which then hits sphere C . Denoting the coefficient of restitution between the spheres by e and the velocity of A just before it hits B by v_0 , determine (a) the velocities of A and B immediately after the first collision, (b) the velocities of B and C immediately after the second collision. (c) Assuming now that n spheres are suspended from the ceiling and that the first sphere is pulled back and released as described here, determine the velocity of the last sphere after it is hit for the first time. (d) Use the result of part c to obtain the velocity of the last sphere when $n = 8$ and $e = 0.9$.

13.162 At an amusement park there are 200-kg bumper cars A , B , and C that have riders with masses of 40 kg, 60 kg, and 35 kg, respectively. Car A is moving to the right with a velocity $v_A = 2$ m/s and car C has a velocity $v_C = 1.5$ m/s to the left, but car B is initially at rest. The coefficient of restitution between each car is 0.8. Determine the final velocity of each car, after all impacts, assuming (a) cars A and C hit car B at the same time, (b) car A hits car B before car C does.



Fig. P13.162 and P13.163

13.163 At an amusement park there are 200-kg bumper cars A , B , and C that have riders with masses of 40 kg, 60 kg, and 35 kg, respectively. Car A is moving to the right with a velocity $v_A = 2$ m/s when it hits stationary car B . The coefficient of restitution between each car is 0.8. Determine the velocity of car C so that after car B collides with car C the velocity of car B is zero.

13.164 Two identical billiard balls can move freely on a horizontal table. Ball A has a velocity v_0 as shown and hits ball B , which is at rest, at a point C defined by $\theta = 45^\circ$. Knowing that the coefficient of restitution between the two balls is $e = 0.8$ and assuming no friction, determine the velocity of each ball after impact.

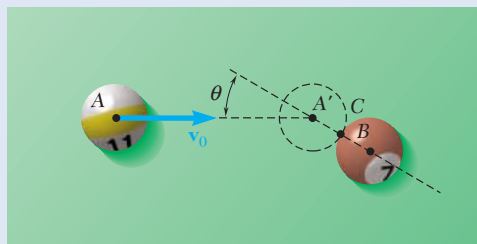


Fig. P13.164

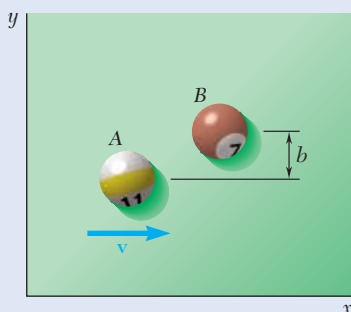


Fig. P13.165

13.165 Two identical pool balls with a 2.37-in. diameter may move freely on a pool table. Ball B is at rest, and ball A has an initial velocity of $\mathbf{v} = v_0 \mathbf{i}$. (a) Knowing that $b = 2$ in. and $e = 0.7$, determine the velocity of each ball after impact. (b) Show that if $e = 1$, the final velocities of the balls form a right angle for all values of b .

13.166 A 600-g ball *A* is moving with a velocity of magnitude 6 m/s when it is hit as shown by a 1-kg ball *B* that has a velocity of magnitude 4 m/s. Knowing that the coefficient of restitution is 0.8 and assuming no friction, determine the velocity of each ball after impact.

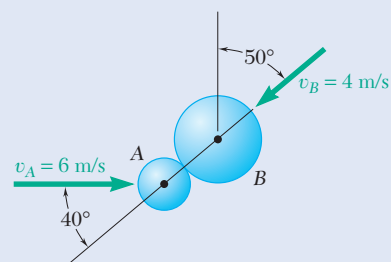


Fig. P13.166

13.167 Two identical hockey pucks are moving on a hockey rink at the same speed of 3 m/s and in perpendicular directions when they strike each other as shown. Assuming a coefficient of restitution $e = 0.9$, determine the magnitude and direction of the velocity of each puck after impact.

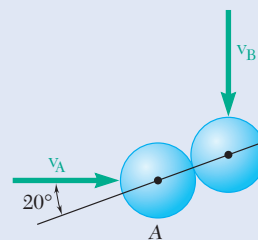


Fig. P13.167

13.168 The coefficient of restitution is 0.9 between the two 60-mm-diameter billiard balls *A* and *B*. Ball *A* is moving in the direction shown with a velocity of 1 m/s when it strikes ball *B*, which is at rest. Knowing that after impact *B* is moving in the x direction, determine (a) the angle θ , (b) the velocity of *B* after impact.

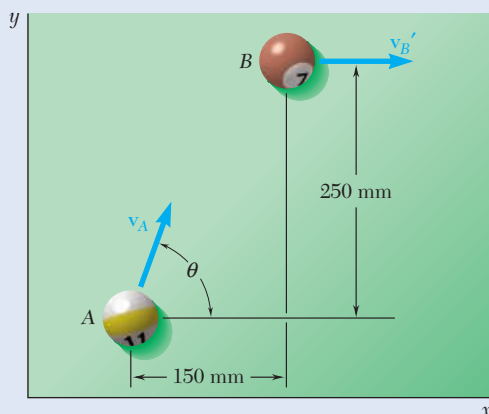


Fig. P13.168

13.169 A boy located at point *A* halfway between the center *O* of a semicircular wall and the wall itself throws a ball at the wall in a direction forming an angle of 45° with *OA*. Knowing that after hitting the wall the ball rebounds in a direction parallel to *OA*, determine the coefficient of restitution between the ball and the wall.

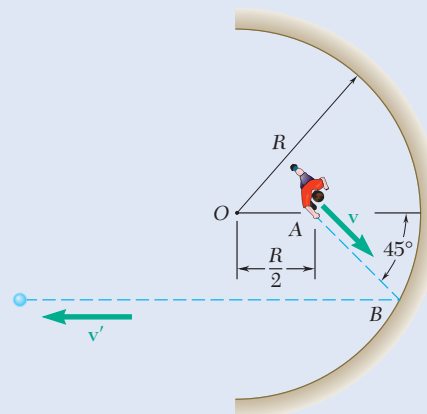


Fig. P13.169

13.170 The Mars Pathfinder spacecraft used large airbags to cushion its impact with the planet's surface when landing. Assuming the spacecraft had an impact velocity of 18.5 m/s at an angle of 45° with respect to the horizontal, the coefficient of restitution is 0.85 and neglecting friction, determine (a) the height of the first bounce, (b) the length of the first bounce. (Acceleration of gravity on Mars = 3.73 m/s^2 .)

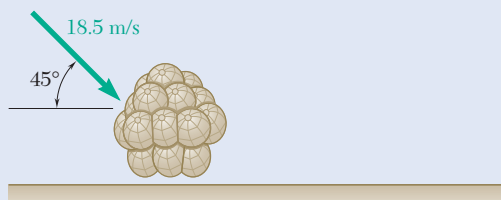


Fig. P13.170

- 13.171** A girl throws a ball at an inclined wall from a height of 3 ft, hitting the wall at A with a horizontal velocity v_0 of magnitude 25 ft/s. Knowing that the coefficient of restitution between the ball and the wall is 0.9 and neglecting friction, determine the distance d from the foot of the wall to the point B where the ball will hit the ground after bouncing off the wall.

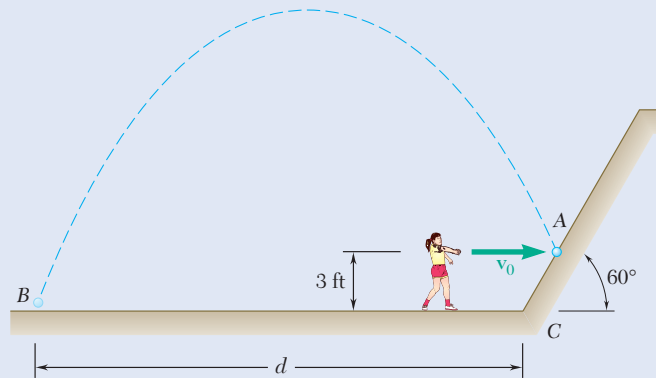


Fig. P13.171

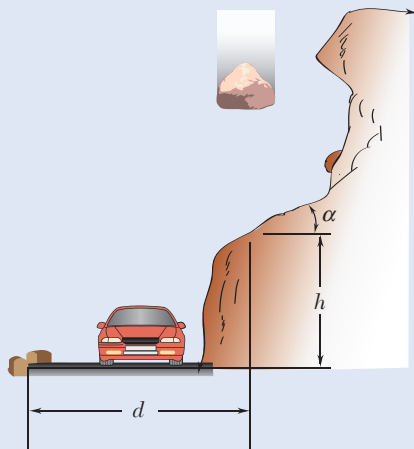


Fig. P13.172 and P13.173

- 13.172** Rockfalls can cause major damage to roads and infrastructure. To design mitigation bridges and barriers, engineers use the coefficient of restitution to model the behavior of the rocks. Rock A falls a distance of 20 m before striking an incline with a slope of $\alpha = 40^\circ$. Knowing that the coefficient of restitution between rock A and the incline is 0.2, determine the velocity of the rock after the impact.

- 13.173** From experimental tests, smaller boulders tend to have a greater coefficient of restitution than larger boulders. Rock A falls a distance of 20 meters before striking an incline with a slope of $\alpha = 45^\circ$. Knowing that $h = 30$ m and $d = 20$ m, determine if a boulder will land on the road or beyond the road for a coefficient of restitution of (a) $e = 0.2$, (b) $e = 0.1$.

- 13.174** Two cars of the same mass run head-on into each other at C . After the collision, the cars skid with their brakes locked and come to a stop in the positions shown in the lower part of the figure. Knowing that the speed of car A just before impact was 5 mi/h and that the coefficient of kinetic friction between the pavement and the tires of both cars is 0.30, determine (a) the speed of car B just before impact, (b) the effective coefficient of restitution between the two cars.

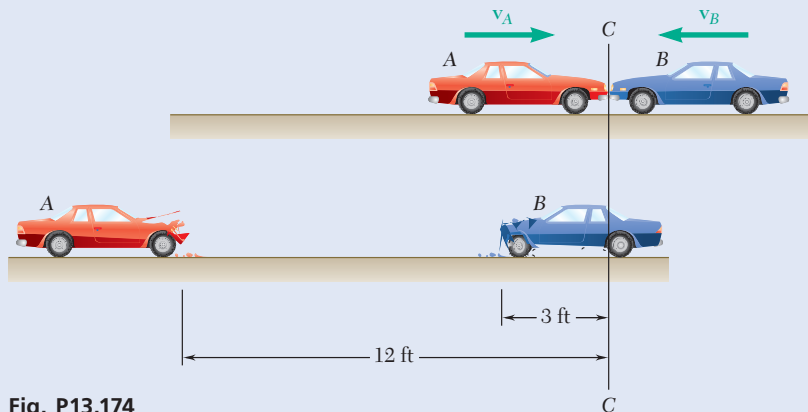


Fig. P13.174

- 13.175** A 1-kg block B is moving with a velocity $\mathbf{v}_0$ of magnitude $v_0 = 2$ m/s as it hits the 0.5-kg sphere A , which is at rest and hanging from a cord attached at O . Knowing that $\mu_k = 0.6$ between the block and the horizontal surface and $e = 0.8$ between the block and the sphere, determine after impact (a) the maximum height h reached by the sphere, (b) the distance x traveled by the block.

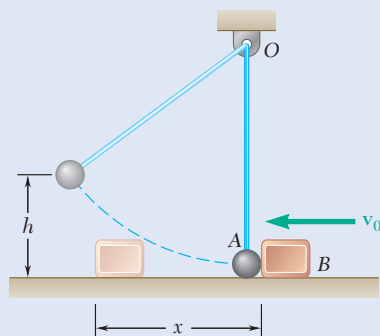


Fig. P13.175

- 13.176** A 0.25-lb ball thrown with a horizontal velocity $\mathbf{v}_0$ strikes a 1.5-lb plate attached to a vertical wall at a height of 36 in. above the ground. It is observed that after rebounding, the ball hits the ground at a distance of 24 in. from the wall when the plate is rigidly attached to the wall (Fig. 1) and at a distance of 10 in. when a foam-rubber mat is placed between the plate and the wall (Fig. 2). Determine (a) the coefficient of restitution e between the ball and the plate, (b) the initial velocity $\mathbf{v}_0$ of the ball.

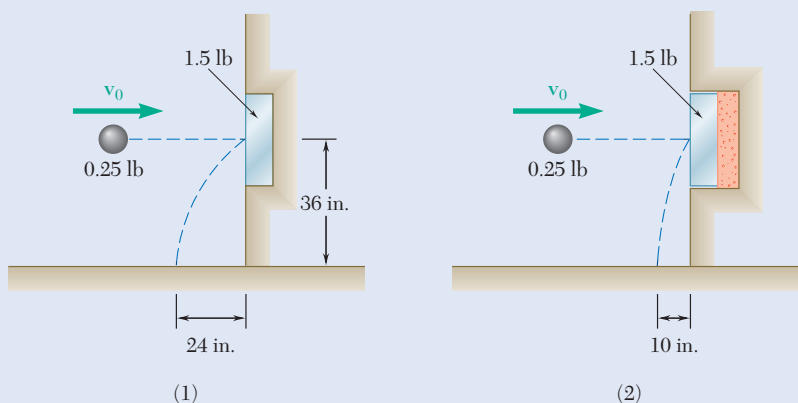


Fig. P13.176

- 13.177** After having been pushed by an airline employee, an empty 40-kg luggage carrier A hits with a velocity of 5 m/s an identical carrier B containing a 15-kg suitcase equipped with rollers. The impact causes the suitcase to roll into the left wall of carrier B . Knowing that the coefficient of restitution between the two carriers is 0.80 and that the coefficient of restitution between the suitcase and the wall of carrier B is 0.30, determine (a) the velocity of carrier B after the suitcase hits its wall for the first time, (b) the total energy lost in that impact.

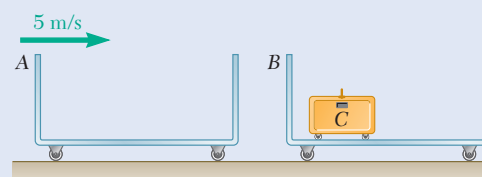


Fig. P13.177

- 13.178** Blocks *A* and *B* each weigh 0.8 lb and block *C* weighs 2.4 lb. The coefficient of friction between the blocks and the plane is $\mu_k = 0.30$. Initially block *A* is moving at a speed $v_0 = 15$ ft/s and blocks *B* and *C* are at rest (Fig. 1). After *A* strikes *B* and *B* strikes *C*, all three blocks come to a stop in the positions shown (Fig. 2). Determine (a) the coefficients of restitution between *A* and *B* and between *B* and *C*, (b) the displacement x of block *C*.

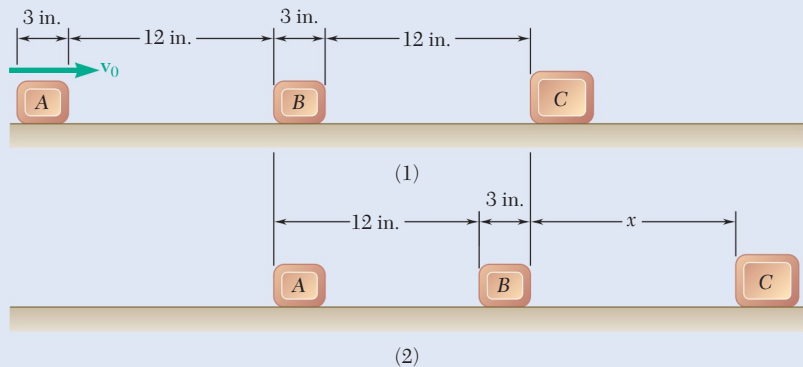


Fig. P13.178

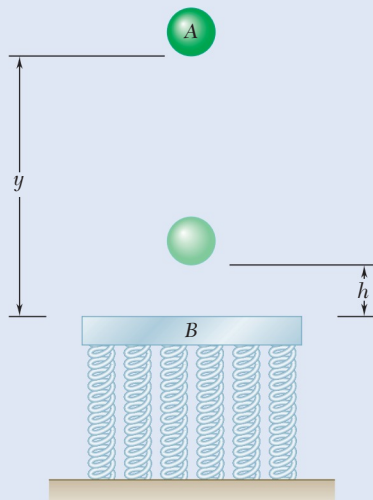


Fig. P13.179 and P13.180

- 13.179** A 5-kg sphere is dropped from a height of $y = 2$ m to test newly designed spring floors used in gymnastics. The mass of the floor section is 10 kg, and the effective stiffness of the floor is $k = 120$ kN/m. Knowing that the coefficient of restitution between the ball and the platform is 0.6, determine (a) the height h reached by the sphere after rebound, (b) the maximum force in the springs.
- 13.180** A 5-kg sphere is dropped from a height of $y = 3$ m to test a new spring floor used in gymnastics. The mass of floor section *B* is 12 kg, and the sphere bounces back upwards a distance of 44 mm. Knowing that the maximum deflection of the floor section is 33 mm from its equilibrium position, determine (a) the coefficient of restitution between the sphere and the floor, (b) the effective spring constant k of the floor section.
- 13.181** The three blocks shown are identical. Blocks *B* and *C* are at rest when block *A* is hit by block *A*, which is moving with a velocity v_A of 3 ft/s. After the impact, which is assumed to be perfectly plastic ($e = 0$), the velocity of blocks *A* and *B* decreases due to friction, while block *C* picks up speed, until all three blocks are moving with the same velocity v . Knowing that the coefficient of kinetic friction between all surfaces is $\mu_k = 0.20$, determine (a) the time required for the three blocks to reach the same velocity, (b) the total distance traveled by each block during that time.

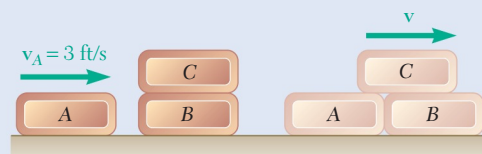


Fig. P13.181

- 13.182** Block *A* is released from rest and slides down the frictionless surface of *B* until it hits a bumper on the right end of *B*. Block *A* has a mass of 10 kg and object *B* has a mass of 30 kg and *B* can roll freely on the ground. Determine the velocities of *A* and *B* immediately after impact when (a) $e = 0$, (b) $e = 0.7$.

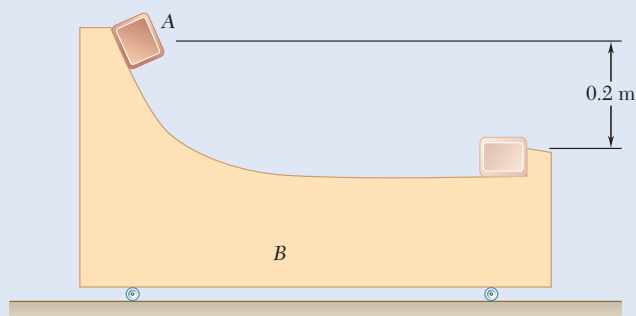


Fig. P13.182

- 13.183** A 340-g ball *B* is hanging from an inextensible cord attached to a support *C*. A 170-g ball *A* strikes *B* with a velocity $\mathbf{v}_0$ with a magnitude of 1.5 m/s at an angle of 60° with the vertical. Assuming perfectly elastic impact ($e = 1$) and no friction, determine the height h reached by ball *B*.

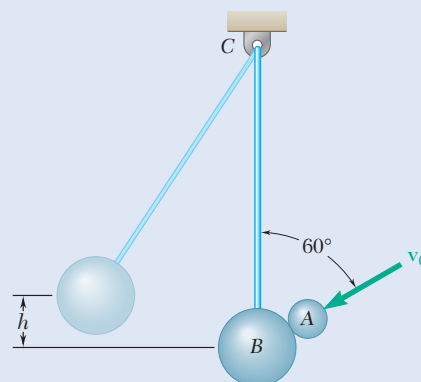


Fig. P13.183

- 13.184** A test machine that kicks soccer balls has a 5-lb simulated foot attached to the end of a 6-ft long pendulum arm of negligible mass. Knowing that the arm is released from the horizontal position and that the coefficient of restitution between the foot and the 1-lb ball is 0.8, determine the exit velocity of the ball (a) if the ball is stationary, (b) if the ball is struck when it is rolling towards the foot with a velocity of 10 ft/s.

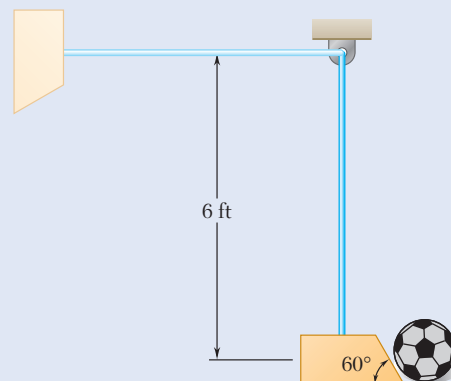


Fig. P13.184

- 13.185** Ball *B* is hanging from an inextensible cord. An identical ball *A* is released from rest when it is just touching the cord and drops through the vertical distance $h_A = 8$ in. before striking ball *B*. Assuming $e = 0.9$ and no friction, determine the resulting maximum vertical displacement h_B of the ball *B*.

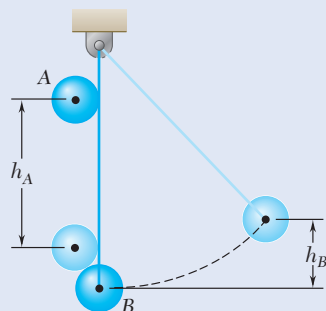


Fig. P13.185

- 13.186** A 70-g ball *B* dropped from a height $h_0 = 1.5$ m reaches a height $h_2 = 0.25$ m after bouncing twice from identical 210-g plates. Plate *A* rests directly on hard ground, while plate *C* rests on a foam-rubber mat. Determine (a) the coefficient of restitution between the ball and the plates, (b) the height h_1 of the ball's first bounce.

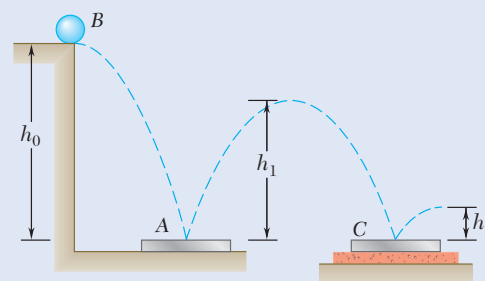


Fig. P13.186

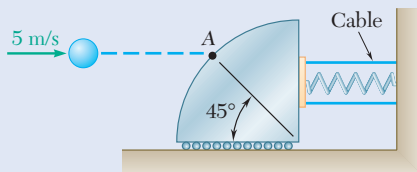


Fig. P13.187

13.187 A 2-kg sphere moving to the right with a velocity of 5 m/s strikes at A, which is on the surface of a 9-kg quarter cylinder that is initially at rest and in contact with a spring with a constant of 20 kN/m. The spring is held by cables, so it is initially compressed 50 mm. Neglecting friction and knowing that the coefficient of restitution is 0.6, determine (a) the velocity of the sphere immediately after impact, (b) the maximum compressive force in the spring.

13.188 When the rope is at an angle of $\alpha = 30^\circ$, the 1-lb sphere A has a speed $v_0 = 4$ ft/s. The coefficient of restitution between A and the 2-lb wedge B is 0.7 and the length of rope $l = 2.6$ ft. The spring constant has a value of 2 lb/in. and $\theta = 20^\circ$. Determine (a) the velocities of A and B immediately after the impact, (b) the maximum deflection of the spring, assuming A does not strike B again before this point.

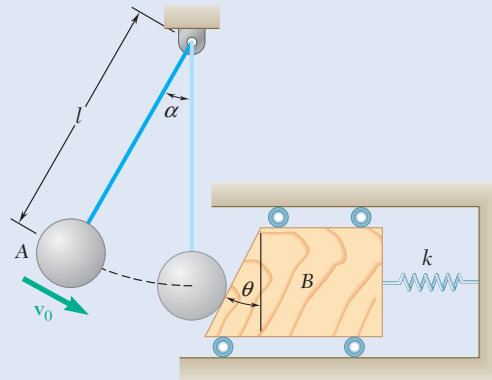


Fig. P13.188 and P13.189

13.189 When the rope is at an angle of $\alpha = 30^\circ$, the 1-kg sphere A has a speed $v_0 = 0.6$ m/s. The coefficient of restitution between A and the 2-kg wedge B is 0.8 and the length of rope $l = 0.9$ m. The spring constant has a value of 1500 N/m and $\theta = 20^\circ$. Determine (a) the velocities of A and B immediately after the impact, (b) the maximum deflection of the spring, assuming A does not strike B again before this point.

Review and Summary

This chapter was devoted to presenting the method of work and energy and the method of impulse and momentum. In the first half of the chapter, we studied the method of work and energy and its application to the analysis of the motion of particles.

Work of a Force

We first considered a force $\mathbf{F}$ acting on a particle A and defined the **work of $\mathbf{F}$ corresponding to the small displacement $d\mathbf{r}$** [Sec. 13.1] as the quantity

$$dU = \mathbf{F} \cdot d\mathbf{r} \quad (13.1)$$

or recalling the definition of the scalar product of two vectors, as

$$dU = F ds \cos \alpha \quad (13.1')$$

where α is the angle between $\mathbf{F}$ and $d\mathbf{r}$ (Fig. 13.30). We obtained the work of $\mathbf{F}$ during a finite displacement from A_1 to A_2 , denoted by $U_{1 \rightarrow 2}$, by integrating Eq. (13.1) along the path described by the particle as

$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} \mathbf{F} \cdot d\mathbf{r} \quad (13.2)$$

For a force defined by its rectangular components, we wrote

$$U_{1 \rightarrow 2} = \int_{A_1}^{A_2} (F_x dx + F_y dy + F_z dz) \quad (13.2'')$$

Work of a Weight

We obtain the work of the weight $\mathbf{W}$ of a body as its center of gravity moves from the elevation y_1 to y_2 (Fig. 13.31) by substituting $F_x = F_z = 0$ and $F_y = -W$ into Eq. (13.2'') and integrating. We found

$$U_{1 \rightarrow 2} = - \int_{y_1}^{y_2} W dy = Wy_1 - Wy_2 \quad (13.4)$$

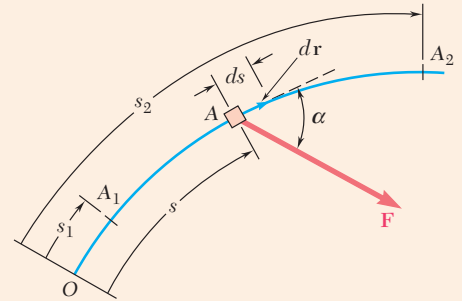


Fig. 13.30

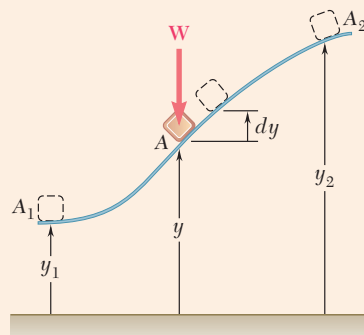


Fig. 13.31

Work of the Force Exerted by a Spring

The work of a force $\mathbf{F}$ exerted by a spring on a body A during a finite displacement of the body (Fig. 13.32) from A_1 ($x = x_1$) to A_2 ($x = x_2$) was obtained by

$$dU = -F dx = -kx dx$$

$$U_{1 \rightarrow 2} = - \int_{x_1}^{x_2} kx dx = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2 \quad (13.6)$$

The work of $\mathbf{F}$ is therefore positive when the spring is returning to its undeformed position.

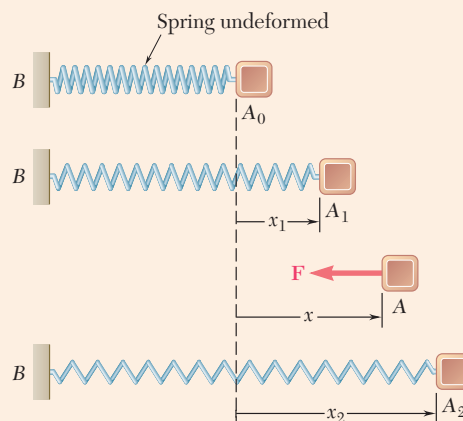


Fig. 13.32

Work of the Gravitational Force

We obtained the **work of the gravitational force $\mathbf{F}$** exerted by a particle of mass M located at O on a particle of mass m as the latter moves from A_1 to A_2 (Fig. 13.33) by recalling from Sec. 12.2C the expression for the magnitude of $\mathbf{F}$ and writing

$$U_{1 \rightarrow 2} = - \int_{r_1}^{r_2} \frac{GMm}{r^2} dr = \frac{GMm}{r_2} - \frac{GMm}{r_1} \quad (13.7)$$

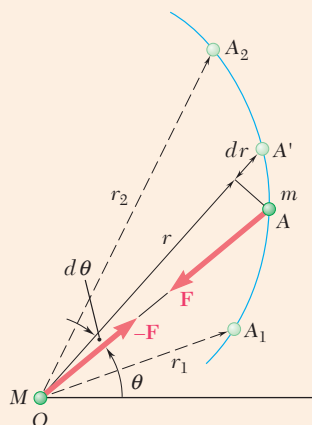


Fig. 13.33

Kinetic Energy of a Particle

We defined the **kinetic energy of a particle** of mass m moving with a velocity $\mathbf{v}$ [Sec. 13.1B] as the scalar quantity

$$T = \frac{1}{2}mv^2 \quad (13.9)$$

Principle of Work and Energy

From Newton's second law, we derived the **principle of work and energy**, which states that we can obtain the kinetic energy of a particle at A_2 by adding its kinetic energy at A_1 to the work done during the displacement from A_1 to A_2 by the force $\mathbf{F}$ exerted on the particle as

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (13.11)$$

Method of Work and Energy

The method of work and energy simplifies the solution of many problems dealing with forces, displacements, and velocities, since it does not require the determination of accelerations [Sec. 13.1C]. We also note that it involves only scalar quantities, and we do not need to consider forces that do no work [Sample Probs. 13.1 and 13.4]. However, this method should be supplemented by the direct application of Newton's second law to determine a force normal to the path of the particle [Sample Prob. 13.6].

Power and Mechanical Efficiency

The power developed by a machine and its mechanical efficiency were discussed in Sec. 13.1D. We defined power as the time rate at which work is done by

$$\text{Power} = \frac{dU}{dt} = \mathbf{F} \cdot \mathbf{v} \quad (13.12, 13.13)$$

where $\mathbf{F}$ is the force exerted on the particle and $\mathbf{v}$ is the velocity of the particle [Sample Prob. 13.7]. The **mechanical efficiency**, denoted by η , was expressed as

$$\eta = \frac{\text{power output}}{\text{power input}} \quad (13.15)$$

Conservative Force and Potential Energy

When the work of a force $\mathbf{F}$ is independent of the path followed [Secs. 13.2A and 13.2B], the force $\mathbf{F}$ is said to be a **conservative force**, and its work is equal to minus the change in the potential energy V associated with $\mathbf{F}$

$$U_{1 \rightarrow 2} = V_1 - V_2 \quad (13.19')$$

We obtained the following expressions for the potential energy associated with each of the forces considered earlier.

Force of gravity (weight):

$$V_g = Wy \quad (13.16)$$

Gravitational force:

$$V_g = -\frac{GMm}{r} \quad (13.17)$$

Elastic force exerted by a spring:

$$V_e = \frac{1}{2}kx^2 \quad (13.18)$$

Principle of Conservation of Energy

Substituting for $U_{1 \rightarrow 2}$ from Eq. (13.19') into Eq. (13.11) and rearranging the terms [Sec. 13.2C], we obtained

$$T_1 + V_1 = T_2 + V_2 \quad (13.24)$$

or

$$T_1 + V_{g1} + V_{e1} = T_2 + V_{g2} + V_{e2} \quad (13.24')$$

This is the **principle of conservation of energy**, which states that, when a particle moves under the action of conservative forces, the sum of its kinetic and potential energies remains constant. The application of this principle facilitates the solution of problems involving only conservative forces [Sample Probs. 13.8 and 13.9].

Alternative Expression for the Principle of Work and Energy

Rather than finding the work due to all external forces, you can write an alternative expression for the work-energy principle such that

$$T_1 + V_{g1} + V_{e1} + U_{1 \rightarrow 2}^{NC} = T_2 + V_{g2} + V_{e2} \quad (13.24'')$$

where $U_{1 \rightarrow 2}^{NC}$ is the work of external non-conservative forces such as friction [Sample Prob. 13.10].

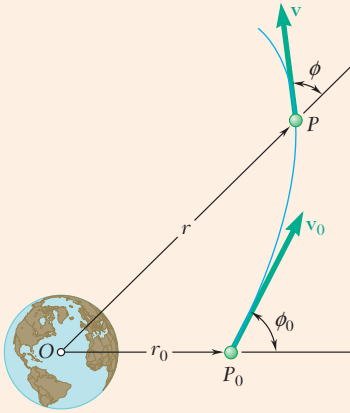


Fig. 13.34

Motion Under a Gravitational Force

Recalling from Sec. 12.2B that when a particle moves under a central force $\mathbf{F}$ its angular momentum about the center of force O remains constant, we observed [Sec. 13.D] that, if the central force $\mathbf{F}$ is also conservative, the principles of conservation of angular momentum and of conservation of energy can be used jointly to analyze the motion of the particle [Sample Prob. 13.11]. Since the gravitational force exerted by the earth on a space vehicle is both central and conservative, this approach was used to study the motion of such vehicles [Sample Prob. 13.12] and was found particularly effective in the case of an **oblique launching**. Considering the initial position P_0 and an arbitrary position P of the vehicle (Fig. 13.34), we have

$$(H_O)_0 = H_O: \quad r_0 m v_0 \sin \phi_0 = r m v \sin \phi \quad (13.25)$$

$$T_0 + V_0 = T + V: \quad \frac{1}{2} m v_0^2 - \frac{G M m}{r_0} = \frac{1}{2} m v^2 - \frac{G M m}{r} \quad (13.26)$$

where m is the mass of the vehicle and M the mass of the earth.

Principle of Impulse and Momentum for a Particle

The second half of this chapter was devoted to the method of impulse and momentum and to its application to the solution of various types of problems involving the motion of particles.

We defined the **linear momentum of a particle** [Sec. 13.3A] as the product $m\mathbf{v}$ of the mass m of the particle and its velocity $\mathbf{v}$. From Newton's second law, $\mathbf{F} = m\mathbf{a}$, we derived the relation

$$m\mathbf{v}_1 + \int_{t_1}^{t_2} \mathbf{F} dt = m\mathbf{v}_2 \quad (13.28)$$

where $m\mathbf{v}_1$ and $m\mathbf{v}_2$ represent the momentum of the particle at a time t_1 and a time t_2 , respectively, and where the integral defines the **linear impulse of the force $\mathbf{F}$** during the corresponding time interval. Therefore, we have

$$m\mathbf{v}_1 + \mathbf{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2 \quad (13.30)$$

which expresses the principle of impulse and momentum for a particle.

When the particle considered is subjected to several forces, we need to use the sum of the impulses of these forces; we have

$$m\mathbf{v}_1 + \Sigma \mathbf{Imp}_{1 \rightarrow 2} = m\mathbf{v}_2 \quad (13.32)$$

Since Eqs. (13.30) and (13.32) involve vector quantities, it is necessary to consider their x and y components separately when applying them to the solution of a given problem [Sample Probs. 13.13 through 13.15].

Impulsive Motion

The method of impulse and momentum is particularly effective in the study of the **impulsive motion** of a particle, when very large forces, called **impulsive forces**, are applied for a very short interval of time Δt , since this method involves the impulses $\mathbf{F}_{\text{avg}}\Delta t$ of the forces, rather than the forces themselves [Sec. 13.3B]. Assuming that all non-impulsive forces (e.g., weight) are negligible, we wrote

$$m\mathbf{v}_1 + \Sigma \mathbf{F}_{\text{avg}}\Delta t = m\mathbf{v}_2 \quad (13.35)$$

In the case of the impulsive motion of several particles, we had

$$\Sigma m\mathbf{v}_1 + \Sigma \mathbf{F}_{\text{avg}}\Delta t = \Sigma m\mathbf{v}_2 \quad (13.36)$$

where the second term involves only impulsive, external forces [Sample Probs. 13.16 and 13.17].

In the particular case when the sum of the impulses of the external forces is zero, Eq. (13.36) reduces to $\Sigma m\mathbf{v}_1 = \Sigma m\mathbf{v}_2$; that is, the total momentum of the particles is conserved. For two particles, this reduces to

$$m_A\mathbf{v}_A + m_B\mathbf{v}_B = m_A\mathbf{v}'_A + m_B\mathbf{v}'_B \quad (13.34')$$

Direct Central Impact

In Sec. 13.4, we considered the **central impact** of two colliding bodies. In the case of a **direct central impact** [Sec. 13.4A], the two colliding bodies A and B were moving along the **line of impact** with velocities $\mathbf{v}_A$ and $\mathbf{v}_B$, respectively (Fig. 13.35). Two equations could be used to determine their velocities $\mathbf{v}'_A$ and $\mathbf{v}'_B$ after the impact. The first expressed conservation of the total momentum of the two bodies as

$$m_A\mathbf{v}_A + m_B\mathbf{v}_B = m_A\mathbf{v}'_A + m_B\mathbf{v}'_B \quad (13.37)$$

where a positive sign indicates that the corresponding velocity is directed to the right. The second equation, called the coefficient of restitution equation, related the *relative velocities* of the two bodies before and after the impact as

$$v'_B - v'_A = e(v_A - v_B) \quad (13.43)$$

The constant e is known as the **coefficient of restitution**; its value lies between 0 and 1 and depends in a large measure on the materials involved. When $e = 0$, the impact is said to be **perfectly plastic**; when $e = 1$, it is said to be **perfectly elastic**. Eq. (13.43) is only valid for direct central impact. [Sample Prob. 13.18].

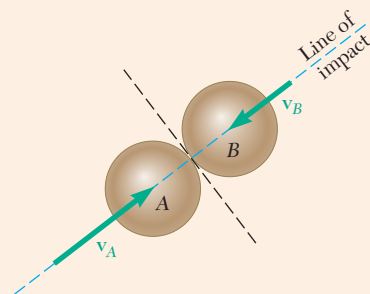


Fig. 13.35

Oblique Central Impact

In the case of an **oblique central impact** [Sec. 13.4B], the velocities of the two colliding smooth bodies before and after the impact were resolved into n components along the line of impact and t components along the common tangent to the surfaces in contact (Fig. 13.36). We observed that the t component of the velocity of each body remained unchanged, while the n components satisfied equations similar to Eqs. (13.37) and (13.43) [Sample Probs. 13.19 and 13.20]. We showed that, although this method was developed for bodies moving freely before and after the impact, it could be extended to the case when one or both of the colliding bodies is constrained in its motion [Sample Prob. 13.21]. When the velocities are not along the line of impact, the coefficient of restitution equation uses the normal component,

$$(v'_B)_n - (v'_A)_n = e[(v_A)_n - (v_B)_n] \quad (13.49)$$

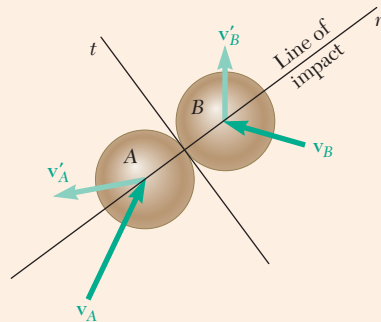


Fig. 13.36

Using the Three Fundamental Methods of Kinetic Analysis

In Sec. 13.4C, we discussed the relative advantages of the three fundamental methods presented in this chapter and the preceding one, namely, Newton's second law, work and energy, and impulse and momentum. We noted that we can combine the method of work and energy and the method of impulse and momentum to solve problems involving a short impact phase during which impulsive forces must be taken into consideration [Sample Probs. 13.22 and 13.23].

Review Problems

- 13.190** A 32,000-lb airplane lands on an aircraft carrier and is caught by an arresting cable. The cable is inextensible and is paid out at A and B from mechanisms located below deck and consisting of pistons moving in long oil-filled cylinders. Knowing that the piston-cylinder system maintains a constant tension of 85 kips in the cable during the entire landing, determine the landing speed of the airplane if it travels a distance $d = 95$ ft after being caught by the cable.

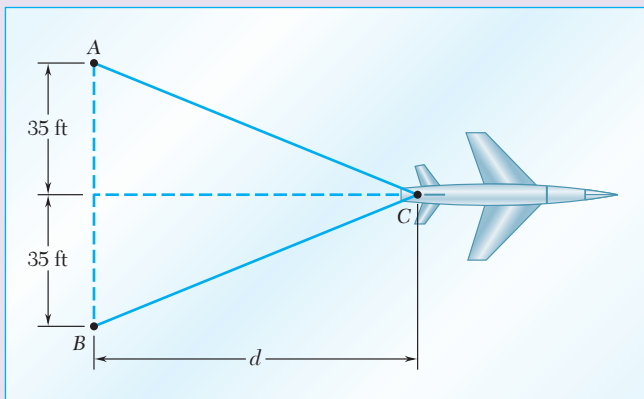


Fig. P13.190

- 13.191** A 2-oz pellet shot vertically from a spring-loaded pistol on the surface of the earth rises to a height of 300 ft. The same pellet shot from the same pistol on the surface of the moon rises to a height of 1900 ft. Determine the energy dissipated by aerodynamic drag when the pellet is shot on the surface of the earth. (The acceleration of gravity on the surface of the moon is 0.165 times that on the surface of the earth.)

- 13.192** A satellite describes an elliptic orbit about a planet of mass M . The minimum and maximum values of the distance r from the satellite to the center of the planet are, respectively, r_0 and r_1 . Use the principles of conservation of energy and conservation of angular momentum to derive the relation

$$\frac{1}{r_0} + \frac{1}{r_1} = \frac{2GM}{h^2}$$

where h is the angular momentum per unit mass of the satellite and G is the constant of gravitation.

- 13.193** A 60-g steel sphere attached to a 200-mm cord can swing about point O in a vertical plane. It is subjected to its own weight and to a force $\mathbf{F}$ exerted by a small magnet embedded in the ground. The magnitude of that force expressed in newtons is $F = 3000/r^2$, where r is the distance from the magnet to the sphere expressed in millimeters. Knowing that the sphere is released from rest at A , determine its speed as it passes through point B .

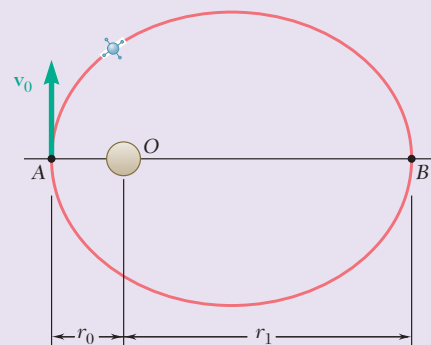


Fig. P13.192

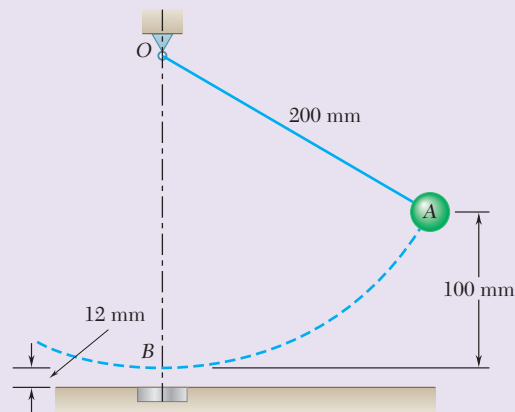


Fig. P13.193

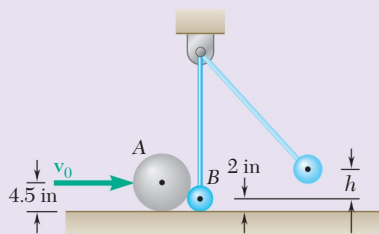


Fig. P13.194

13.194 A 50-lb sphere A with a radius of 4.5 in. is moving with a velocity of magnitude $v_0 = 6$ ft/s. Sphere A strikes a 4.6-lb sphere B that has a radius of 2 in., is hanging from an inextensible cord, and is initially at rest. Knowing that sphere B swings to a maximum height of $h = 0.75$ ft, determine the coefficient of restitution between the two spheres.

13.195 A 300-g block is released from rest after a spring of constant $k = 600$ N/m has been compressed 160 mm. Determine the force exerted by the loop $ABCD$ on the block as the block passes through (a) point A , (b) point B , (c) point C . Assume no friction.

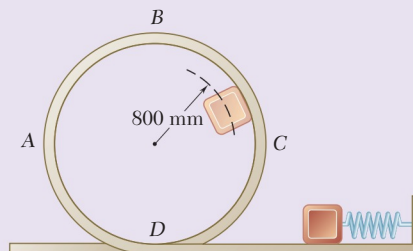


Fig. P13.195

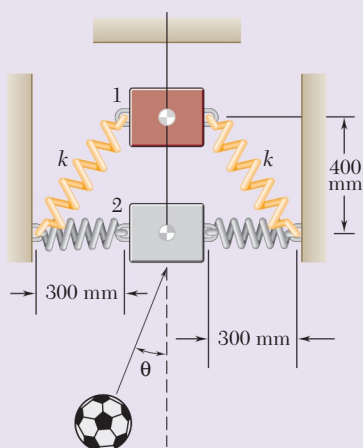


Fig. P13.196

13.196 A kicking-simulation attachment goes on the front of a wheelchair, allowing athletes with mobility impairments to play soccer. The athletes load up the spring shown through a ratchet mechanism that pulls the 2-kg "foot" back to the position 1. They then release the "foot" to impact the 0.45-kg soccer ball that is rolling towards the "foot" with a speed of 2 m/s at an angle $\theta = 30^\circ$, as shown in the figure. The impact occurs with a coefficient of restitution $e = 0.75$ when the foot is at position 2, where the springs are unstretched. Knowing that the effective friction coefficient during rolling is $\mu_k = 0.1$, determine (a) the necessary spring coefficient to make the ball roll 30 m, (b) the direction the ball will travel after it is kicked.

13.197 A 300-g collar A is released from rest, slides down a frictionless rod, and strikes a 900-g collar B that is at rest and supported by a spring of constant 500 N/m. Knowing that the coefficient of restitution between the two collars is 0.9, determine (a) the maximum distance collar A moves up the rod after impact, (b) the maximum distance collar B moves down the rod after impact.

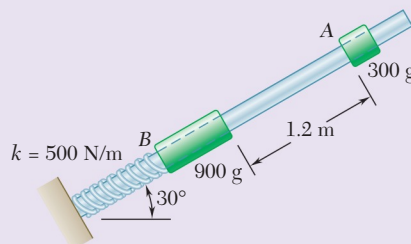


Fig. P13.197

- 13.198** Blocks A and B are connected by a cord which passes over pulleys and through a collar C . The system is released from rest when $x = 1.7$ m. As block A rises, it strikes collar C with perfectly plastic impact ($e = 0$). After impact, the two blocks and the collar keep moving until they come to a stop and reverse their motion. As A and C move down, C hits the ledge and blocks A and B keep moving until they come to another stop. Determine (a) the velocity of the blocks and collar immediately after A hits C , (b) the distance the blocks and collar move after the impact before coming to a stop, (c) the value of x at the end of one complete cycle.

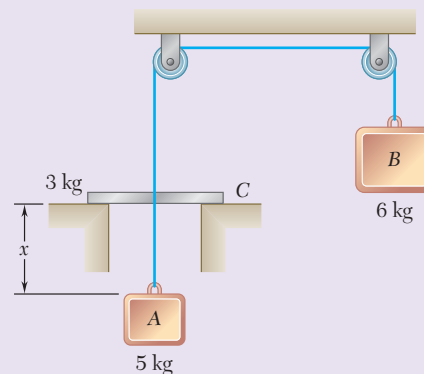


Fig. P13.198

- 13.199** A 2-kg ball B is traveling horizontally at 10 m/s when it strikes 2-kg ball A . Ball A is initially at rest and is attached to a spring with constant 100 N/m and an unstretched length of 1.2 m. Knowing the coefficient of restitution between A and B is 0.8 and friction between all surfaces is negligible, determine the normal force between A and the ground when it is at the bottom of the hill.

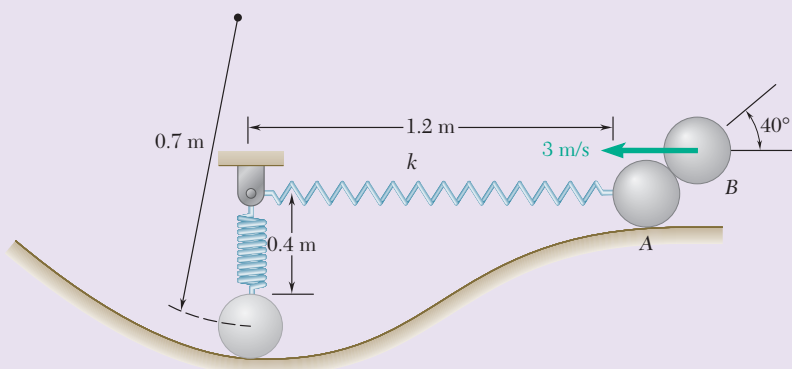


Fig. P13.199

- 13.200** A 2-kg block A is pushed up against a spring compressing it a distance x . The block is then released from rest and slides down the 20° incline until it strikes a 1-kg sphere B which is suspended from a 1-m inextensible rope. The spring constant $k = 800$ N/m, the coefficient of friction between A and the ground is 0.2, the distance A slides from the unstretched length of the spring $d = 1.5$ m, and the coefficient of restitution between A and B is 0.8. Knowing the tension in the rope is 20 N when $\alpha = 30^\circ$, determine the initial compression x of the spring.

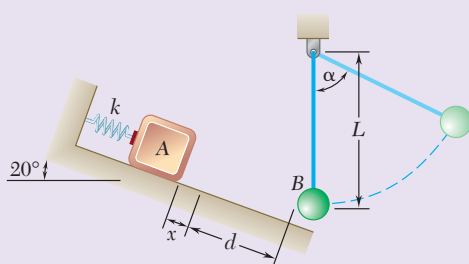


Fig. P13.200

***13.201** The 2-lb ball at A is suspended by an inextensible cord and given an initial horizontal velocity of $\mathbf{v}_0$. If $l = 2$ ft, $x_B = 0.3$ ft, and $y_B = 0.4$ ft, determine the initial velocity $\mathbf{v}_0$ so that the ball will enter the basket. (*Hint:* Use a computer to solve the resulting set of equations.)

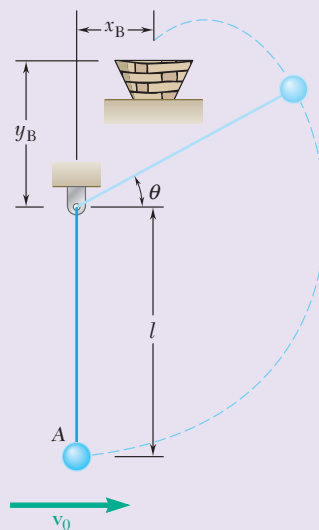


Fig. P13.201



14

Systems of Particles

The thrust for this XR-5M15 prototype engine is produced by gas particles being ejected at a high velocity. The determination of the forces on the test stand is based on the analysis of the motion of a *variable system of particles*, i.e., the motion of a large number of air particles considered together rather than separately.

Introduction

14.1 APPLYING NEWTON'S SECOND LAW AND MOMENTUM PRINCIPLES TO SYSTEMS OF PARTICLES

- 14.1A Newton's Second Law for a System of Particles
- 14.1B Linear and Angular Momentum of a System of Particles
- 14.1C Motion of the Mass Center of a System of Particles
- 14.1D Angular Momentum of a System of Particles About its Mass Center
- 14.1E Conservation of Momentum for a System of Particles

14.2 ENERGY AND MOMENTUM METHODS FOR A SYSTEM OF PARTICLES

- 14.2A Kinetic Energy of a System of Particles
- 14.2B Work-Energy Principle and Conservation of Energy for a System of Particles
- 14.2C Impulse-Momentum Principle and Conservation of Momentum for a System of Particles

14.3 VARIABLE SYSTEMS OF PARTICLES

- 14.3A Steady Stream of Particles
- 14.3B Systems Gaining or Losing Mass

Objectives

- **Apply** Newton's second law to a system of particles.
- **Calculate** the linear momentum and the angular momentum about a point of a system of particles.
- **Describe** the motion of the center of mass of a system of particles.
- **Determine** the kinetic energy of a system of particles.
- **Analyze** the motion of a system of particles by using the principle of work and energy and the principle of impulse and momentum.
- **Analyze** the motion of steady streams of particles
- **Analyze** systems of particles gaining or losing mass.

Introduction

In this chapter, you will study the motion of **systems of particles**; that is, the motion of a large number of particles considered together. In the first part of the chapter, we examine systems consisting of well-defined particles, like a set of billiard balls or a projectile that fragments into pieces. In the second part, we consider the motion of variable systems; these are systems that are continually gaining or losing particles or doing both at the same time. This could describe the motion of a stream of water or of a rocket during launch.

We start by applying Newton's second law to each particle of the system. We show that the *external forces* acting on the various particles form a system equipollent to the system of $m_i \mathbf{a}_i$ for the various particles. In other words, both systems have the same resultant and the same moment resultant about any given point. We further show that the resultant and moment resultant of the external forces are equal, respectively, to the rate of change of the total linear momentum and to the rate of change of the total angular momentum of the particles of the system.

We then define the *mass center* of a system of particles and describe the motion of that point, along with an analysis of the motion of the particles about their mass center. We discuss the conditions under which the linear momentum and the angular momentum of a system of particles are conserved and apply these results to the solution of various problems.

In Sec. 14.2, we apply the work–energy principle to a system of particles, and then we apply the impulse–momentum principle. We use these ideas to solve several problems of practical interest.

Note that although the derivations given in the first part of this chapter are carried out for a system of independent particles, they remain valid when the particles of the system are rigidly connected, i.e., when they form a rigid body. In fact, these results form the foundation of our discussion of the kinetics of rigid bodies in Chaps. 16 through 18.

In Sec. 14.3, we consider steady streams of particles, such as a stream of water diverted by a fixed vane or the flow of air through a

jet engine. We show how to determine the force exerted by the stream on the vane and the thrust developed by the engine. Finally, we analyze systems that gain mass by continually absorbing particles or lose mass by continually expelling particles. Among the various practical applications of this analysis is the determination of the thrust developed by a rocket engine.

14.1 APPLYING NEWTON'S SECOND LAW AND MOMENTUM PRINCIPLES TO SYSTEMS OF PARTICLES

In statics, we studied the effects of forces on particles and on rigid bodies in equilibrium. However, when you consider particles in motion, the situation of particles acting together but not forming a rigid body occurs in several important and practical applications. We analyze this kind of problem by applying Newton's laws to the system. The results are an interesting middle ground between the dynamics of particles and the dynamics of rigid bodies, which we will study next.

14.1A Newton's Second Law for a System of Particles

In order to derive the equations of motion for a system of n particles, let us begin by writing Newton's second law for each individual particle of the system. Consider the particle P_i , where $1 \leq i \leq n$. Let m_i be the mass of P_i and let $\mathbf{a}_i$ be its acceleration with respect to the newtonian frame of reference $Oxyz$. The force exerted on P_i by another particle P_j of the system (Fig. 14.1), called an **internal force**, is denoted by $\mathbf{f}_{ij}$. The resultant of the internal forces exerted on P_i by all the other particles of the system is thus $\sum_{j=1}^n \mathbf{f}_{ij}$ (where $\mathbf{f}_{ii}$ has no meaning and is assumed to be equal to zero). On the other hand, denoting the resultant of all the **external forces** acting on P_i by $\mathbf{F}_i$, we write Newton's second law for the particle P_i as

$$\mathbf{F}_i + \sum_{j=1}^n \mathbf{f}_{ij} = m_i \mathbf{a}_i \quad (14.1)$$

Denoting the position vector of P_i by $\mathbf{r}_i$ and taking the moments about O of the various terms in Eq. (14.1), we also have

$$\mathbf{r}_i \times \mathbf{F}_i + \sum_{j=1}^n (\mathbf{r}_i \times \mathbf{f}_{ij}) = \mathbf{r}_i \times m_i \mathbf{a}_i \quad (14.2)$$

Repeating this procedure for each particle P_i of the system, we obtain n equations of the type in Eq. (14.1) and n equations of the type in Eq. (14.2), where i takes successively the values $1, 2, \dots, n$. Thus, these equations state that the external forces $\mathbf{F}_i$ and the internal forces $\mathbf{f}_{ij}$ acting on the various particles form a system equivalent to the system of

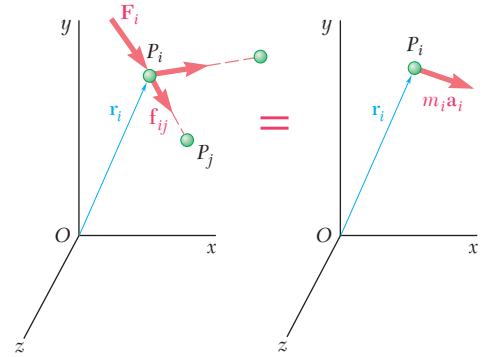


Fig. 14.1 Newton's second law for the i th particle in a system of particles.

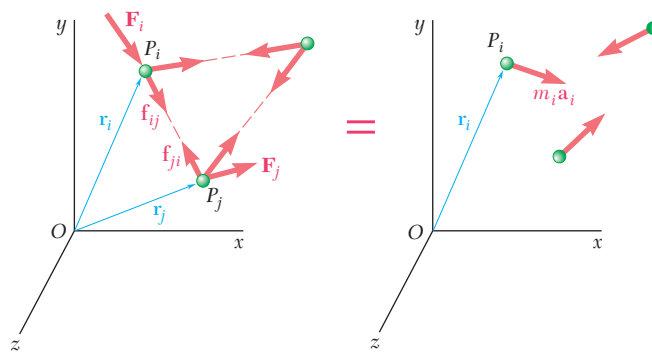


Fig. 14.2 The sum of internal forces equals zero, and the sum of external forces equals the sum of the mass times acceleration for every particle in the system.

the $m_i \mathbf{a}_i$ terms (i.e., one system may be replaced by the other) (Fig. 14.2).

Before proceeding further with our derivation, let us examine the internal forces $\mathbf{f}_{ij}$. These forces occur in pairs as $\mathbf{f}_{ij}$, $\mathbf{f}_{ji}$, where $\mathbf{f}_{ij}$ represents the force exerted by the particle P_j on the particle P_i and $\mathbf{f}_{ji}$ represents the force exerted by P_i on P_j (see Fig. 14.2). Now, according to Newton's third law (Sec. 6.1), as extended by Newton's law of gravitation to particles acting at a distance (Sec. 12.2C), the forces $\mathbf{f}_{ij}$ and $\mathbf{f}_{ji}$ are equal and opposite and have the same line of action. Their sum is therefore $\mathbf{f}_{ij} + \mathbf{f}_{ji} = 0$, and the sum of their moments about O is

$$\mathbf{r}_i \times \mathbf{f}_{ij} + \mathbf{r}_j \times \mathbf{f}_{ji} = \mathbf{r}_i \times (\mathbf{f}_{ij} + \mathbf{f}_{ji}) + (\mathbf{r}_j - \mathbf{r}_i) \times \mathbf{f}_{ji} = 0$$

since the vectors $\mathbf{r}_j - \mathbf{r}_i$ and $\mathbf{f}_{ji}$ in the last term are collinear. Adding all of the internal forces of the system and summing their moments about O , we obtain the equations

$$\sum_{i=1}^n \sum_{j=1}^n \mathbf{f}_{ij} = 0 \quad \sum_{i=1}^n \sum_{j=1}^n (\mathbf{r}_i \times \mathbf{f}_{ij}) = 0 \quad (14.3)$$

These equations state that the resultant and the moment resultant of the internal forces of the system are zero.

Returning now to the n equations (14.1), where $i = 1, 2, \dots, n$, we sum their left-hand sides and sum their right-hand sides. Taking into account the first of Eqs. (14.3), we obtain

$$\sum_{i=1}^n \mathbf{F}_i = \sum_{i=1}^n m_i \mathbf{a}_i \quad (14.4)$$

Proceeding similarly with Eq. (14.2) and taking into account the second of Eqs. (14.3), we have

$$\sum_{i=1}^n (\mathbf{r}_i \times \mathbf{F}_i) = \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{a}_i) \quad (14.5)$$

Equations (14.4) and (14.5) express the fact that the system of the

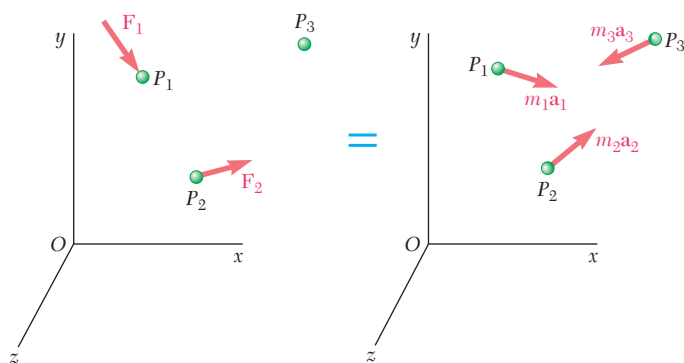


Fig. 14.3 The free-body diagram for a system of particles is equal to the kinetic diagram for a system of particles.

external forces $\mathbf{F}_i$ and the system of $m_i\mathbf{a}_i$ have the same resultant and the same moment resultant. Referring to the definition given in *Statics* Sec. 3.4B for two equipollent systems of vectors, we can therefore state that **the system of the external forces acting on the particles and the system of the $m_i\mathbf{a}_i$ terms of the particles are equipollent** (Fig. 14.3). Figure 14.3 basically shows that a free-body diagram for a system of particles is equal to its kinetic diagram.

Equations (14.3) state that the system of internal forces $\mathbf{f}_{ij}$ is equipollent to zero. Note, however, that it does *not* follow that the internal forces have no effect on the individual particles under consideration. Indeed, the gravitational forces that the sun and the planets exert on one another are internal to the solar system and are equipollent to zero. Yet these forces are responsible for the motion of the planets about the sun.

Similarly, it does not follow from Eqs. (14.4) and (14.5) that two systems of external forces that have the same resultant and that the same moment resultant will have the same effect on a given system of particles. Clearly, the systems shown in Figs. 14.4a and 14.4b have the same resultant and the same moment resultant; yet the first system accelerates particle A and leaves particle B unaffected, whereas the second system accelerates B and does not affect A. It is important to recall that when we stated in Sec. 3.4B that two equipollent systems of forces acting on a rigid body are also equivalent, we specifically noted that this property could *not* be extended to a system of forces acting on a set of independent particles such as those considered in this chapter.

In order to avoid any confusion, we use blue equals signs to connect equipollent systems of vectors, such as those shown in Figs. 14.3 and 14.4. These signs indicate that the two systems of vectors have the same resultant and the same moment resultant. We continue to use red equals signs to indicate that two systems of vectors are equivalent, i.e., that one system can actually be replaced by the other (Fig. 14.2).

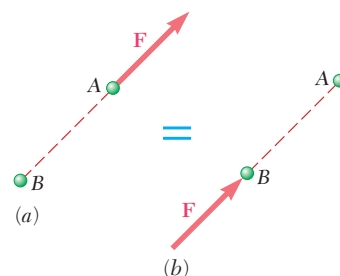


Fig. 14.4 (a) A system of resultant force and moment applied to particle A is not equivalent to (b) the same force and moment applied to particle B.

14.1B Linear and Angular Momentum of a System of Particles

We can express Eqs. (14.4) and (14.5) in a more condensed form by introducing the linear and the angular momentum of the system of

particles. We define the linear momentum $\mathbf{L}$ of the system of particles as the sum of the linear momenta of the various particles of the system (Sec. 12.1B). Then we have

**Linear momentum,
system of particles**

$$\mathbf{L} = \sum_{i=1}^n m_i \mathbf{v}_i \quad (14.6)$$

Defining the angular momentum $\mathbf{H}_O$ about O of the system of particles in a similar way (Sec. 12.2A) gives us

**Angular momentum,
system of particles**

$$\mathbf{H}_O = \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{v}_i) \quad (14.7)$$

Differentiating both sides of Eqs. (14.6) and (14.7) with respect to t , we have

$$\dot{\mathbf{L}} = \sum_{i=1}^n m_i \dot{\mathbf{v}}_i = \sum_{i=1}^n m_i \mathbf{a}_i \quad (14.8)$$

and

$$\begin{aligned} \dot{\mathbf{H}}_O &= \sum_{i=1}^n (\dot{\mathbf{r}}_i \times m_i \mathbf{v}_i) + \sum_{i=1}^n (\mathbf{r}_i \times m_i \dot{\mathbf{v}}_i) \\ &= \sum_{i=1}^n (\mathbf{v}_i \times m_i \mathbf{v}_i) + \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{a}_i) \end{aligned}$$

Because the vectors $\mathbf{v}_i$ and $m_i \mathbf{v}_i$ are collinear, this last equation reduces to

$$\dot{\mathbf{H}}_O = \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{a}_i) \quad (14.9)$$

Note that the right-hand sides of Eqs. (14.8) and (14.9) are identical to the right-hand sides of Eqs. (14.4) and (14.5), respectively. It follows that the left-hand sides of these equations are also equal. Recall that the left-hand side of Eq. (14.5) represents the sum of the moments $\mathbf{M}_O$ about O of the external forces acting on the particles of the system. So, omitting the subscript i from the sums, we have

$$\Sigma \mathbf{F} = \dot{\mathbf{L}} \quad (14.10)$$

$$\Sigma \mathbf{M}_O = \dot{\mathbf{H}}_O \quad (14.11)$$

These equations state:

The resultant and the moment resultant about the fixed point O of the external forces are equal to the rates of change of the linear momentum and of the angular momentum about O , respectively, of the system of particles.

14.1C Motion of the Mass Center of a System of Particles

We can write Eq. (14.10) in an alternative form by considering the **mass center** of the system of particles. The mass center of the system is the point G defined by the position vector $\bar{\mathbf{r}}$, which satisfies the relation

$$m\bar{\mathbf{r}} = \sum_{i=1}^n m_i \mathbf{r}_i \quad (14.12)$$

where m represents the total mass $m = \sum_{i=1}^n m_i$ of the particles. Resolving the position vectors $\bar{\mathbf{r}}$ and $\mathbf{r}_i$ into rectangular components, we obtain the following three scalar equations, which we can use to determine the coordinates $\bar{x}$, $\bar{y}$, $\bar{z}$ of the mass center:

$$m\bar{x} = \sum_{i=1}^n m_i x_i \quad m\bar{y} = \sum_{i=1}^n m_i y_i \quad m\bar{z} = \sum_{i=1}^n m_i z_i \quad (14.12')$$

Since $m_i g$ represents the weight of the particle P_i , and mg is the total weight of the particles, G is also the center of gravity of the system of particles. However, in order to avoid any confusion, we refer to G as the *mass center* of the system of particles when we are discussing properties associated with the *mass* of the particles, and as the *center of gravity* of the system when we consider properties associated with the *weight* of the particles. Particles located outside the gravitational field of the earth, for example, have a mass but no weight. We can then properly refer to their mass center, but obviously not to their center of gravity.[†]

Differentiating both members of Eq. (14.12) with respect to t , we obtain

$$m\dot{\bar{\mathbf{r}}} = \sum_{i=1}^n m_i \dot{\mathbf{r}}_i$$

or

$$m\bar{\mathbf{v}} = \sum_{i=1}^n m_i \mathbf{v}_i \quad (14.13)$$

where $\bar{\mathbf{v}}$ represents the velocity of the mass center G of the system of particles. But the right-hand side of Eq. (14.13) is, by definition, the linear momentum $\mathbf{L}$ of the system [see Eq. (14.6)]. We therefore have

$$\mathbf{L} = m\bar{\mathbf{v}} \quad (14.14)$$

and, differentiating both members with respect to t ,

$$\dot{\mathbf{L}} = m\bar{\mathbf{a}} \quad (14.15)$$

[†]We should also point out that the mass center and the center of gravity of a system of particles do not exactly coincide, since the weights of the particles are directed toward the center of the earth and thus do not truly form a system of parallel forces. For particles on the earth, this difference is extremely small.

where $\bar{\mathbf{a}}$ represents the acceleration of the mass center G . Substituting for $\dot{\mathbf{L}}$ from Eq. (14.15) into Eq. (14.10), we obtain

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}} \quad (14.16)$$

which defines the motion of the mass center G of the system of particles.

Note that Eq. (14.16) is identical to the equation we would obtain for a particle of mass m equal to the total mass of the particles of the system, acted upon by all the external forces. We therefore state:

The mass center of a system of particles moves as if the entire mass of the system and all of the external forces were concentrated at that point.

This principle is best illustrated by the motion of an exploding projectile. We know that if air resistance is neglected, we can assume that a projectile will travel along a parabolic path. After it has exploded, the mass center G of the fragments of the projectile will continue to travel along the same path. Indeed, point G must move as if the mass and the weight of all fragments were concentrated at G ; it must therefore move as if the projectile had not exploded.

Also note that the preceding derivation does not involve the moments of the external forces. Therefore, *it would be wrong to assume* that the external forces are equipollent to a vector $m\bar{\mathbf{a}}$ attached at the mass center G . In general, this is not the case since, as you will see next, the sum of the moments about G of the external forces is not in general equal to zero.

14.1D Angular Momentum of a System of Particles About Its Mass Center

In some applications (for example, in analyzing the motion of a rigid body), it is convenient to consider the motion of the particles of the system with respect to a centroidal frame of reference $Gx'y'z'$ that translates with respect to the newtonian frame of reference $Oxyz$ (Fig. 14.5). Although a centroidal frame is not, in general, a Newtonian frame of reference, we will show that the fundamental relation in Eq. (14.11) holds when the frame $Oxyz$ is replaced by $Gx'y'z'$.

Let's denote the position vector and the velocity of the particle P_i relative to the moving frame of reference $Gx'y'z'$ by $\mathbf{r}'_i$ and $\mathbf{v}'_i$, respectively. We then define the **angular momentum** $\mathbf{H}'_G$ of the system of particles **about the mass center** G as

$$\mathbf{H}'_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}'_i) \quad (14.17)$$

We now differentiate both members of Eq. (14.17) with respect to t . This operation is similar to that performed earlier on Eq. (14.7), so we can write immediately

$$\dot{\mathbf{H}}'_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{a}'_i) \quad (14.18)$$

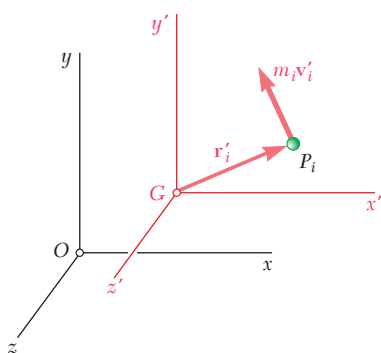


Fig. 14.5 A centroidal frame of reference $Gx'y'z'$ moving in translation with respect to a newtonian frame of reference $Oxyz$.

where $\mathbf{a}'_i$ denotes the acceleration of P_i relative to the moving frame of reference. Referring to Sec. 11.4D, we have

$$\mathbf{a}_i = \bar{\mathbf{a}} + \mathbf{a}'_i$$

where $\mathbf{a}_i$ and $\bar{\mathbf{a}}$ denote, respectively, the accelerations of P_i and G relative to the frame $Oxyz$. Solving for $\mathbf{a}'_i$ and substituting into Eq. (14.18), we have

$$\dot{\mathbf{H}}'_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{a}_i) - \left(\sum_{i=1}^n m_i \mathbf{r}'_i \right) \times \bar{\mathbf{a}} \quad (14.19)$$

However, by Eq. (14.12), the second sum in Eq. (14.19) is equal to $m\bar{\mathbf{r}}'$ and thus to zero, since the position vector $\bar{\mathbf{r}}'$ of G relative to the frame $Gx'y'z'$ is clearly zero. On the other hand, since $\mathbf{a}_i$ represents the acceleration of P_i relative to a newtonian frame, we can use Eq. (14.1) and replace $m_i \mathbf{a}_i$ by the sum of the internal forces $\mathbf{f}_{ij}$ and of the resultant $\mathbf{F}_i$ of the external forces acting on P_i . But a reasoning similar to that used in Sec. 14.1A shows that the moment resultant about G of the internal forces $\mathbf{f}_{ij}$ of the entire system is zero. The first sum in Eq. (14.19) therefore reduces to the resultant moment about G of the external forces acting on the particles of the system, and we have

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}'_G \quad (14.20)$$

This equation states:

The resultant moment about G of the external forces is equal to the rate of change of the angular momentum about G of the system of particles.

Note that in Eq. (14.17) we defined the angular momentum $\mathbf{H}'_G$ as the sum of the moments about G of the momenta of the particles $m_i \mathbf{v}'_i$ in their motion relative to the centroidal frame of reference $Gx'y'z'$. We may sometimes want to compute the sum $\mathbf{H}_G$ of the moments about G of the momenta of the particles $m_i \mathbf{v}_i$ in their absolute motion, i.e., in their motion as observed from the newtonian frame of reference $Oxyz$ (Fig. 14.6):

$$\mathbf{H}_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}_i) \quad (14.21)$$

Remarkably, the angular momenta $\mathbf{H}'_G$ and $\mathbf{H}_G$ are identically equal. This can be verified by referring to Sec. 11.4D and writing

$$\mathbf{v}_i = \bar{\mathbf{v}} + \mathbf{v}'_i \quad (14.22)$$

Substituting for $\mathbf{v}_i$ from Eq. (14.22) into Eq. (14.21), we have

$$\mathbf{H}_G = \left(\sum_{i=1}^n m_i \mathbf{r}'_i \right) \times \bar{\mathbf{v}} + \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}'_i)$$

But, as observed earlier, the first sum is equal to zero. Thus, $\mathbf{H}_G$ reduces to the second sum, which by definition is equal to $\mathbf{H}'_G$.[†]

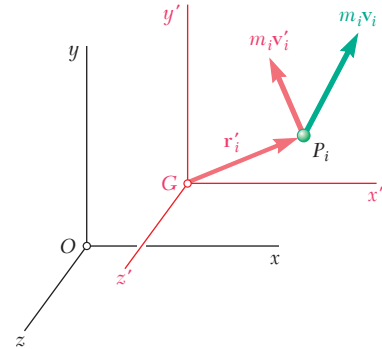


Fig. 14.6 The linear momentum of particle P_i with respect to the centroidal frame ($m_i \mathbf{v}'_i$) and with respect to a newtonian frame ($m_i \mathbf{v}_i$).

[†]Note that this property is peculiar to the centroidal frame $Gx'y'z'$ and does not, in general, hold for other frames of reference (see Prob. 14.29).

Taking advantage of the property we have just established, we simplify our notation by dropping the prime (') from Eq. (14.20) and writing

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad (14.23)$$

Here we can compute the angular momentum $\mathbf{H}_G$ by taking the moments about G of the momenta of the particles with respect to either the Newtonian frame $Oxyz$ or the centroidal frame $Gx'y'z'$:

$$\mathbf{H}_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}_i) = \sum_{i=1}^n (\mathbf{r}'_i \times m_i \mathbf{v}'_i) \quad (14.24)$$

14.1E Conservation of Momentum for a System of Particles

If no external force acts on the particles of a system, the left-hand sides of Eqs. (14.10) and (14.11) are equal to zero. These equations then reduce to $\dot{\mathbf{L}} = 0$ and $\dot{\mathbf{H}}_O = 0$. We conclude that

$$\mathbf{L} = \text{constant} \quad \mathbf{H}_O = \text{constant} \quad (14.25)$$

These equations state that the linear momentum of the system of particles and its angular momentum about the fixed point O are conserved.

In some applications, such as problems involving central forces, the moment about a fixed point O of each of the external forces can be zero without any of the forces being zero. In such cases, the second of Eqs. (14.25) still holds; the angular momentum of the system of particles about O is conserved.

We can also apply the concept of conservation of momentum to the analysis of the motion of the mass center G of a system of particles and to the analysis of the motion of the system about G . For example, if the sum of the external forces is zero, the first of Eqs. (14.25) applies. Recalling Eq. (14.14), we have

$$\bar{\mathbf{v}} = \text{constant} \quad (14.26)$$

This equation says that the mass center G of the system moves in a straight line and at a constant speed. On the other hand, if the sum of the moments about G of the external forces is zero, it follows from Eq. (14.23) that the angular momentum of the system about its mass center is conserved:

$$\mathbf{H}_G = \text{constant} \quad (14.27)$$



Photo 14.1 No external impulsive forces act on a fireworks as it explodes, so linear and angular momenta of the system are conserved.

Sample Problem 14.1

A 200-kg space vehicle passes through the origin of a newtonian reference frame $Oxyz$ at time $t = 0$ with velocity $\mathbf{v}_0 = (150 \text{ m/s})\mathbf{i}$ relative to the frame. Following the detonation of explosive charges, the vehicle separates into three parts A , B , and C , each with a mass of 100 kg, 60 kg, and 40 kg, respectively. Knowing that at $t = 2.5$ s, the positions of parts A and B are observed to be $A(555, -180, 240)$ and $B(255, 0, -120)$, where the coordinates are expressed in meters. Determine the position of part C at that time.

STRATEGY: There are no external forces, so the linear momentum of the system is conserved. Use kinematics to relate the motion of the center of mass of the spacecraft and the rectangular coordinates of its position.

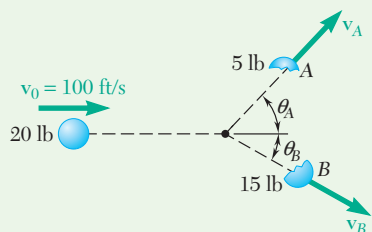
MODELING and ANALYSIS: The system is the space vehicle. After the explosion, the system is composed of all three parts: A , B , and C . The mass center G of the system moves with the constant velocity $\mathbf{v}_0 = (150 \text{ m/s})\mathbf{i}$. At $t = 2.5$ s, its position is

$$\bar{\mathbf{r}} = \mathbf{v}_0 t = (150 \text{ m/s})\mathbf{i}(2.5 \text{ s}) = (375 \text{ m})\mathbf{i}$$

Recalling Eq. (14.12), you have

$$\begin{aligned} m\bar{\mathbf{r}} &= m_A \mathbf{r}_A + m_B \mathbf{r}_B + m_C \mathbf{r}_C \\ (200 \text{ kg})(375 \text{ m})\mathbf{i} &= (100 \text{ kg})[(555 \text{ m})\mathbf{i} - (180 \text{ m})\mathbf{j} + (240 \text{ m})\mathbf{k}] \\ &\quad + (60 \text{ kg})[(255 \text{ m})\mathbf{i} - (120 \text{ m})\mathbf{k}] + (40 \text{ kg})\mathbf{r}_C \\ \mathbf{r}_C &= (105 \text{ m})\mathbf{i} + (450 \text{ m})\mathbf{j} - (420 \text{ m})\mathbf{k} \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: This kind of calculation can serve as a model for any situation involving fragmentation of a projectile with no external forces present.



Sample Problem 14.2

A 20-lb projectile is moving with a velocity of 100 ft/s when it explodes into two fragments A and B , weighing 5 lb and 15 lb, respectively. Knowing that immediately after the explosion, fragments A and B travel in directions defined respectively by $\theta_A = 45^\circ$ and $\theta_B = 30^\circ$, determine the velocity of each fragment.

STRATEGY: There are no external forces, so apply the conservation of linear momentum to the system.

(continued)

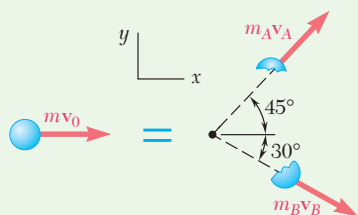


Fig. 1 Impulse-momentum diagram for the projectile.

MODELING and ANALYSIS: The system is the projectile. After the explosion, the system is composed of the two fragments. The impulse-momentum diagram for this system is shown in Fig. 1. There are no external impulses acting on this system, so linear momentum is conserved and:

$$m_A \mathbf{v}_A + m_B \mathbf{v}_B = m \mathbf{v}_0$$

$$(5/\text{g})\mathbf{v}_A + (15/\text{g})\mathbf{v}_B = (20/\text{g})\mathbf{v}_0$$

Applying this equation in the x and y directions gives you two scalar equations. Thus,

$$\rightarrow x \text{ components: } 5v_A \cos 45^\circ + 15v_B \cos 30^\circ = 20(100)$$

$$+\uparrow y \text{ components: } 5v_A \sin 45^\circ - 15v_B \sin 30^\circ = 0$$

Solving the two equations for v_A and v_B simultaneously gives

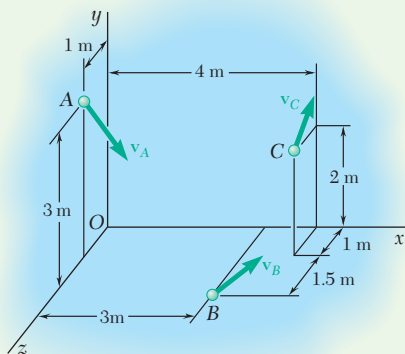
$$v_A = 207 \text{ ft/s} \quad v_B = 97.6 \text{ ft/s}$$

$$\mathbf{v}_A = 207 \text{ ft/s} \nearrow 45^\circ \quad \mathbf{v}_B = 97.6 \text{ ft/s} \searrow 30^\circ \quad \blacktriangleleft$$

REFLECT and THINK: As you might have predicted, the less massive fragment winds up with a larger magnitude of velocity and departs the original trajectory at a larger angle.

Sample Problem 14.3

A system consists of three particles A , B , and C , with masses $m_A = 1 \text{ kg}$, $m_B = 2 \text{ kg}$, and $m_C = 3 \text{ kg}$. The velocities of the particles expressed in m/s are, respectively, $\mathbf{v}_A = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$, $\mathbf{v}_B = 4\mathbf{i} + 3\mathbf{j}$, and $\mathbf{v}_C = 2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$. Determine (a) the angular momentum $\mathbf{H}_O$ of the system about O , (b) the position vector $\bar{\mathbf{r}}$ of the mass center G of the system, (c) the angular momentum $\mathbf{H}_G$ of the system about G .



STRATEGY: You have a system of particles, so use the definitions of angular momentum and center of mass.

MODELING: Choose the three particles as your system.

ANALYSIS: The linear momentum of each particle expressed in $\text{kg}\cdot\text{m/s}$ is

$$m_A \mathbf{v}_A = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$$

$$m_B \mathbf{v}_B = 8\mathbf{i} + 6\mathbf{j}$$

$$m_C \mathbf{v}_C = 6\mathbf{i} + 15\mathbf{j} - 9\mathbf{k}$$

The position vectors (in meters) are

$$\mathbf{r}_A = 3\mathbf{j} + \mathbf{k} \quad \mathbf{r}_B = 3\mathbf{i} + 2.5\mathbf{k} \quad \mathbf{r}_C = 4\mathbf{i} + 2\mathbf{j} + \mathbf{k}$$

a. Angular Momentum About O . Using the definition of angular momentum about O (in $\text{kg}\cdot\text{m}^2/\text{s}$) you find

$$\begin{aligned}\mathbf{H}_O &= \mathbf{r}_A \times (m_A \mathbf{v}_A) + \mathbf{r}_B \times (m_B \mathbf{v}_B) + \mathbf{r}_C \times (m_C \mathbf{v}_C) \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 3 & 1 \\ 3 & -2 & 4 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & 2.5 \\ 8 & 6 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 2 & 1 \\ 6 & 15 & -9 \end{vmatrix} \\ &= (14\mathbf{i} + 3\mathbf{j} - 9\mathbf{k}) + (-15\mathbf{i} + 20\mathbf{j} + 18\mathbf{k}) + (-33\mathbf{i} + 42\mathbf{j} + 48\mathbf{k}) \\ &= 34\mathbf{i} + 65\mathbf{j} + 57\mathbf{k} \\ \mathbf{H}_O &= -(34 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{i} + (65 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{j} + (57 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{k} \quad \blacktriangleleft\end{aligned}$$

b. Mass Center. Using the definition of mass center, you find

$$\begin{aligned}(m_A + m_B + m_C)\bar{\mathbf{r}} &= m_A \mathbf{r}_A + m_B \mathbf{r}_B + m_C \mathbf{r}_C \\ 6\bar{\mathbf{r}} &= (1)(3\mathbf{j} + \mathbf{k}) + (2)(3\mathbf{i} + 2.5\mathbf{k}) + (3)(4\mathbf{i} + 2\mathbf{j} + \mathbf{k}) \\ \bar{\mathbf{r}} &= 3\mathbf{i} + 1.5\mathbf{j} + 1.5\mathbf{k} \\ \bar{\mathbf{r}} &= (3.00 \text{ m})\mathbf{i} + (1.500 \text{ m})\mathbf{j} + (1.500 \text{ m})\mathbf{k} \quad \blacktriangleleft\end{aligned}$$

c. Angular Momentum About G . The angular momentum of the system about G is

$$\mathbf{H}_G = \mathbf{r}'_A \times m_A \mathbf{v}_A + \mathbf{r}'_B \times m_B \mathbf{v}_B + \mathbf{r}'_C \times m_C \mathbf{v}_C$$

where $\mathbf{r}'_A$, $\mathbf{r}'_B$, and $\mathbf{r}'_C$ are the position vectors from the particles to the center of mass; that is

$$\begin{aligned}\mathbf{r}'_A &= \mathbf{r}_A - \bar{\mathbf{r}} = -3\mathbf{i} + 1.5\mathbf{j} - 0.5\mathbf{k} \\ \mathbf{r}'_B &= \mathbf{r}_B - \bar{\mathbf{r}} = -1.5\mathbf{j} + \mathbf{k} \\ \mathbf{r}'_C &= \mathbf{r}_C - \bar{\mathbf{r}} = \mathbf{i} + 0.5\mathbf{j} - 0.5\mathbf{k}\end{aligned}$$

Therefore, you can calculate the angular momentum as

$$\begin{aligned}\mathbf{H}_G &= \mathbf{r}'_A \times m_A \mathbf{v}_A + \mathbf{r}'_B \times m_B \mathbf{v}_B + \mathbf{r}'_C \times m_C \mathbf{v}_C \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 1.5 & -0.5 \\ 3 & -2 & 4 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1.5 & 1 \\ 8 & 6 & 0 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4 & 0.5 & -0.5 \\ 6 & 15 & -9 \end{vmatrix} \\ &= (5\mathbf{i} + 10.5\mathbf{j} + 1.5\mathbf{k}) + (-6\mathbf{i} + 8\mathbf{j} + 12\mathbf{k}) + (3\mathbf{i} + 6\mathbf{j} + 12\mathbf{k}) \\ &= 2\mathbf{i} + 24.5\mathbf{j} + 25.5\mathbf{k} \\ \mathbf{H}_G &= (2.00 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{i} + (24.5 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{j} + (25.5 \text{ kg}\cdot\text{m}^2/\text{s})\mathbf{k} \quad \blacktriangleleft\end{aligned}$$

REFLECT and THINK: You should be able to verify that the answers to this problem satisfy the equations given in Prob. 14.27; that is, $\mathbf{H}_O = \bar{\mathbf{r}} \times m\bar{\mathbf{v}} + \mathbf{H}_G$. Because no impulses act on the system, the linear momentum of the overall system is constant; the location of the center of mass of the system, however, changes with time.

SOLVING PROBLEMS ON YOUR OWN

This chapter dealt with the motion of **systems of particles** where the motion of a large number of particles is considered together, rather than separately. In this first section, you learned to compute the **linear momentum** and the **angular momentum** of a system of particles. We defined the linear momentum $\mathbf{L}$ of a system of particles as the sum of the linear momenta of the particles, and we defined the angular momentum $\mathbf{H}_O$ of the system as the sum of the angular momenta of the particles about O :

$$\mathbf{L} = \sum_{i=1}^n m_i \mathbf{v}_i \quad \mathbf{H}_O = \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{v}_i) \quad (14.6, 14.7)$$

In this section, you will be asked to solve several problems of practical interest, either by observing that the linear momentum of a system of particles is conserved or by considering the motion of the mass center of a system of particles.

1. Conservation of the linear momentum of a system of particles. This occurs *when the resultant of the external forces acting on the particles of the system is zero*. You may encounter such a situation in the following types of problems.

a. Problems involving the rectilinear motion of objects, such as colliding automobiles and railroad cars. After you have checked that the resultant of the external forces is zero, equate the algebraic sums of the initial momenta and final momenta to obtain an equation that you can solve for one unknown.

b. Problems involving the two-dimensional or three-dimensional motion of objects, such as exploding shells or colliding aircraft, automobiles, or billiard balls. After you have checked that the resultant of the external forces is zero, add the initial momenta of the objects vectorially, add their final momenta vectorially, and equate the two sums to obtain a vector equation expressing that the linear momentum of the system is conserved.

In the case of two-dimensional motion, you can replace this equation with two scalar equations that you can solve for two unknowns. In the case of three-dimensional motion, you can replace the equation with three scalar equations that you can solve for three unknowns.

2. Motion of the mass center of a system of particles. You saw in Sec. 14.1C that *the mass center of a system of particles moves as if the entire mass of the system and all of the external forces were concentrated at that point*.

a. In the case of a body exploding while in motion, it follows that the mass center of the resulting fragments moves as the body itself would have moved if the explosion had not occurred. You can solve problems of this type by writing the equation of motion of the mass center of the system in vector form and expressing the position vector of the mass center in terms of the position vectors of the various fragments [Eq. (14.12) and Sample Prob. 14.1]. You can then rewrite the vector equation as two or three scalar equations and solve the equations for an equivalent number of unknowns.

b. In the case of the collision of several moving bodies, it follows that the motion of the mass center of the various bodies is unaffected by the collision. You can solve problems of this type by writing the equation of motion of the mass center of the system in vector form and expressing its position vector before and after the collision in terms of the position vectors of the relevant bodies [Eq. (14.12)]. You can then rewrite the vector equation as two or three scalar equations and solve these equations for an equivalent number of unknowns.

Problems

- 14.1** A 30-g bullet is fired with a horizontal velocity of 450 m/s and becomes embedded in block *B*, which has a mass of 3 kg. After the impact, block *B* slides on 30-kg carrier *C* until it impacts the end of the carrier. Knowing the impact between *B* and *C* is perfectly plastic and the coefficient of kinetic friction between *B* and *C* is 0.2, determine (a) the velocity of the bullet and *B* after the first impact, (b) the final velocity of the carrier.

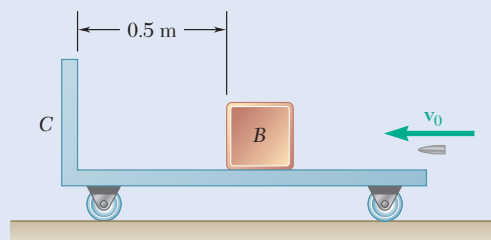


Fig. P14.1

- 14.2** Two identical 1350-kg automobiles *A* and *B* are at rest with their brakes released when *B* is struck by a 5400-kg truck *C* that is moving to the left at 8 km/h. A second collision then occurs when *B* strikes *A*. Assuming the first collision is perfectly plastic and the second collision is perfectly elastic, determine the velocities of the three vehicles just after the second collision.

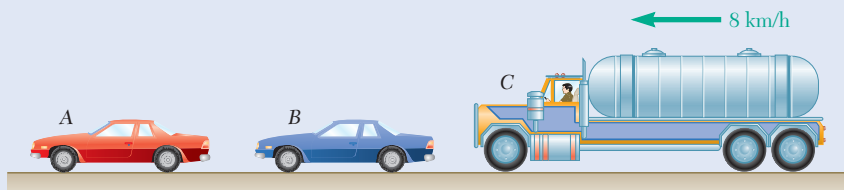


Fig. P14.2

- 14.3** An airline employee tosses two suitcases with weights of 30 lb and 40 lb, respectively, onto a 50-lb baggage carrier in rapid succession. Knowing that the carrier is initially at rest and that the employee imparts a 9-ft/s horizontal velocity to the 30-lb suitcase and a 6-ft/s horizontal velocity to the 40-lb suitcase, determine the final velocity of the baggage carrier if the first suitcase tossed onto the carrier is (a) the 30-lb suitcase, (b) the 40-lb suitcase.



Fig. P14.3

- 14.4** A bullet is fired with a horizontal velocity of 1500 ft/s through a 6-lb block *A* and becomes embedded in a 4.95-lb block *B*. Knowing that blocks *A* and *B* start moving with velocities of 5 ft/s and 9 ft/s, respectively, determine (a) the weight of the bullet, (b) its velocity as it travels from block *A* to block *B*.

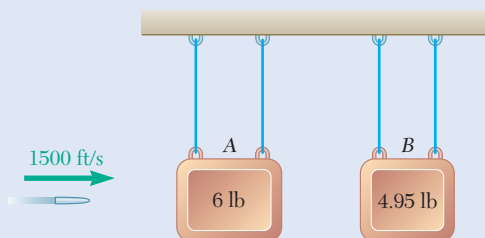


Fig. P14.4



Fig. P14.5

14.5 Two swimmers *A* and *B*, of weight 190 lb and 125 lb, respectively, are at diagonally opposite corners of a floating raft when they realize that the raft has broken away from its anchor. Swimmer *A* immediately starts walking toward *B* at a speed of 2 ft/s relative to the raft. Knowing that the raft weighs 300 lb, determine (a) the speed of the raft if *B* does not move, (b) the speed with which *B* must walk toward *A* if the raft is not to move.

14.6 A 180-lb man and a 120-lb woman stand side by side at the same end of a 300-lb boat, ready to dive, each with a 16-ft/s velocity relative to the boat. Determine the velocity of the boat after they have both dived, if (a) the woman dives first, (b) the man dives first.

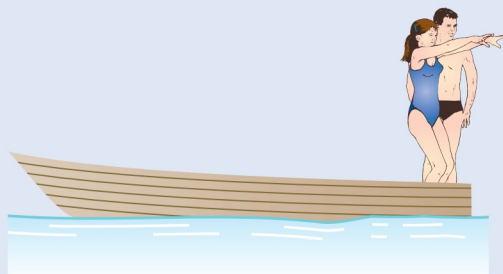


Fig. P14.6

14.7 A 40-Mg boxcar *A* is moving in a railroad switchyard with a velocity of 9 km/h toward cars *B* and *C*, which are both at rest with their brakes off at a short distance from each other. Car *B* is a 25-Mg flatcar supporting a 30-Mg container, and car *C* is a 35-Mg boxcar. As the cars hit each other they get automatically and tightly coupled. Determine the velocity of car *A* immediately after each of the two couplings, assuming that the container (a) does not slide on the flatcar, (b) slides after the first coupling but hits a stop before the second coupling occurs, (c) slides and hits the stop only after the second coupling has occurred.

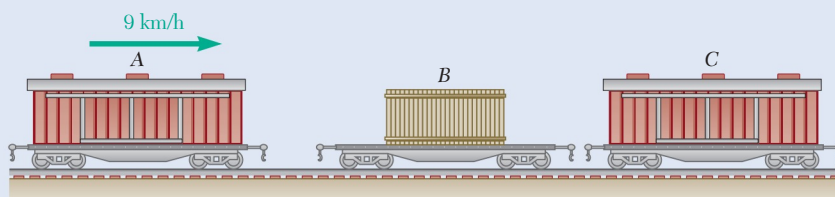


Fig. P14.7

14.8 Two identical cars *A* and *B* are at rest on a loading dock with brakes released. Car *C*, of a slightly different style but of the same weight, has been pushed by dockworkers and hits car *B* with a velocity of 1.5 m/s. Knowing that the coefficient of restitution is 0.8 between *B* and *C* and 0.5 between *A* and *B*, determine the velocity of each car after all collisions have taken place.



Fig. P14.8

14.9 A 20-kg base satellite deploys three sub-satellites, each which has its own thrust capabilities, to perform research on tether propulsion. The masses of sub-satellites A , B , and C are 4 kg, 6 kg, and 8 kg, respectively, and their velocities expressed in m/s are given by $\mathbf{v}_A = 4\mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$, $\mathbf{v}_B = \mathbf{i} + 4\mathbf{j}$, $\mathbf{v}_C = 2\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$. At the instant shown, what is the angular momentum $\mathbf{H}_O$ of the system about the base satellite?

14.10 For the satellite system of Prob. 14.9, assuming that the velocity of the base satellite is zero, determine (a) the position vector $\bar{\mathbf{r}}$ of the mass center G of the system, (b) the linear momentum $\mathbf{L}$ of the system, (c) the angular momentum $\mathbf{H}_G$ of the system about G . Also, verify that the answers to this problem and to Prob. 14.9 satisfy the equation given in Prob. 14.27.

14.11 A system consists of three identical 19.32-lb particles A , B , and C . The velocities of the particles are, respectively, $\mathbf{v}_A = v_A\mathbf{j}$, $\mathbf{v}_B = v_B\mathbf{i}$, and $\mathbf{v}_C = v_C\mathbf{k}$. Knowing that the angular momentum of the system about O expressed in ft·lb·s is $\mathbf{H}_O = -1.2\mathbf{k}$, determine (a) the velocities of the particles, (b) the angular momentum of the system about its mass center G .

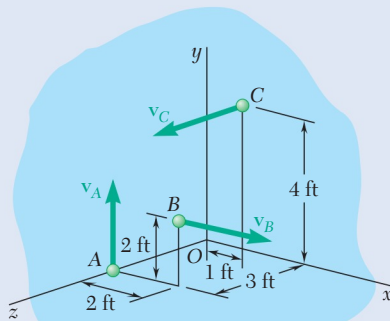


Fig. P14.11 and P14.12

14.12 A system consists of three identical 19.32-lb particles A , B , and C . The velocities of the particles are, respectively, $\mathbf{v}_A = v_A\mathbf{j}$, $\mathbf{v}_B = v_B\mathbf{i}$, and $\mathbf{v}_C = v_C\mathbf{k}$, and the magnitude of the linear momentum $\mathbf{L}$ of the system is 9 lb·s. Knowing that $\mathbf{H}_G = \mathbf{H}_O$, where $\mathbf{H}_G$ is the angular momentum of the system about its mass center G and $\mathbf{H}_O$ is the angular momentum of the system about O , determine (a) the velocities of the particles, (b) the angular momentum of the system about O .

14.13 A system consists of three particles A , B , and C . We know that $m_A = 3$ kg, $m_B = 2$ kg, and $m_C = 4$ kg and that the velocities of the particles expressed in m/s are, respectively, $\mathbf{v}_A = 4\mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, $\mathbf{v}_B = 4\mathbf{i} + 3\mathbf{j}$, and $\mathbf{v}_C = -2\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$. Determine the angular momentum $\mathbf{H}_O$ of the system about O .

14.14 For the system of particles of Prob. 14.13, determine (a) the position vector $\bar{\mathbf{r}}$ of the mass center G of the system, (b) the linear momentum $m\bar{\mathbf{v}}$ of the system, (c) the angular momentum $\mathbf{H}_G$ of the system about G . Also verify that the answers to this problem and to Problem 14.13 satisfy the equation given in Prob. 14.27.

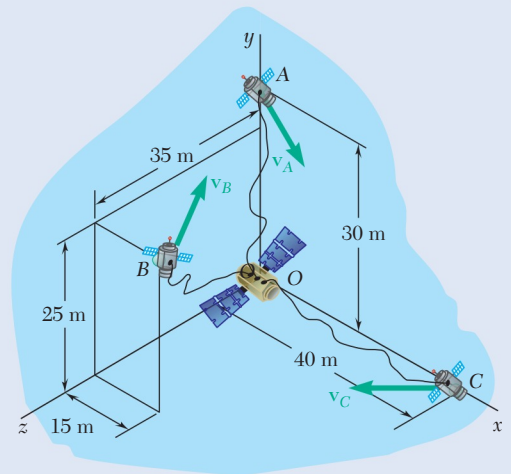


Fig. P14.9 and P14.10

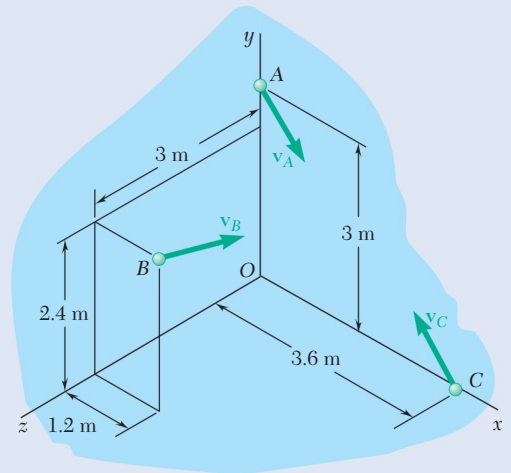


Fig. P14.13

14.15 A 13-kg projectile is passing through the origin O with a velocity $\mathbf{v}_0 = (35 \text{ m/s})\mathbf{i}$ when it explodes into two fragments A and B , of mass 5 kg and 8 kg, respectively. Knowing that 3 s later the position of fragment A is (90 m, 7 m, -14 m), determine the position of fragment B at the same instant. Assume $a_y = -g = -9.81 \text{ m/s}^2$ and neglect air resistance.

14.16 A 300-kg space vehicle traveling with a velocity $\mathbf{v}_0 = (360 \text{ m/s})\mathbf{i}$ passes through the origin O at $t = 0$. Explosive charges then separate the vehicle into three parts A , B , and C , with mass, respectively, 150 kg, 100 kg, and 50 kg. Knowing that at $t = 4 \text{ s}$, the positions of parts A and B are observed to be A (1170 m, -290 m, -585 m) and B (1975 m, 365 m, 800 m), determine the corresponding position of part C . Neglect the effect of gravity.

14.17 A 2-kg model rocket is launched vertically and reaches an altitude of 70 m with a speed of 30 m/s at the end of powered flight, time $t = 0$. As the rocket approaches its maximum altitude it explodes into two parts of masses $m_A = 0.7 \text{ kg}$ and $m_B = 1.3 \text{ kg}$. Part A is observed to strike the ground 80 m west of the launch point at $t = 6 \text{ s}$. Determine the position of part B at that time.

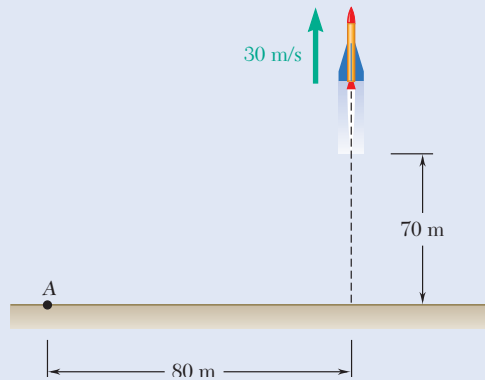


Fig. P14.17

14.18 An 18-kg cannonball and a 12-kg cannonball are chained together and fired horizontally with a velocity of 165 m/s from the top of a 15-m wall. The chain breaks during the flight of the cannonballs and the 12-kg cannonball strikes the ground at $t = 1.5 \text{ s}$, at a distance of 240 m from the foot of the wall, and 7 m to the right of the line of fire. Determine the position of the other cannonball at that instant. Neglect the resistance of the air.

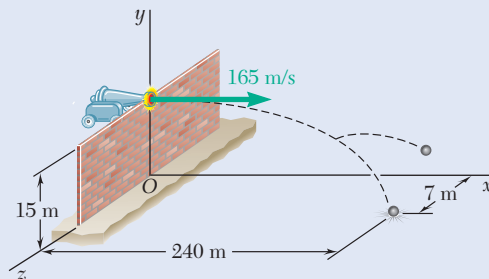


Fig. P14.18

14.19 and 14.20 Car A was traveling east at high speed when it collided at point O with car B, which was traveling north at 45 mi/h. Car C, which was traveling west at 60 mi/h, was 32 ft east and 10 ft north of point O at the time of the collision. Because the pavement was wet, the driver of car C could not prevent his car from sliding into the other two cars, and the three cars, stuck together, kept sliding until they hit the utility pole P . Knowing that the weights of cars A, B, and C are, respectively, 3000 lb, 2600 lb, and 2400 lb, and neglecting the forces exerted on the cars by the wet pavement, solve the problems indicated.

14.19 Knowing that the speed of car A was 75 mi/h and that the time elapsed from the first collision to the stop at P was 2.4 s, determine the coordinates of the utility pole P .

14.20 Knowing that the coordinates of the utility pole are $x_p = 46$ ft and $y_p = 59$ ft, determine (a) the time elapsed from the first collision to the stop at P , (b) the speed of car A.

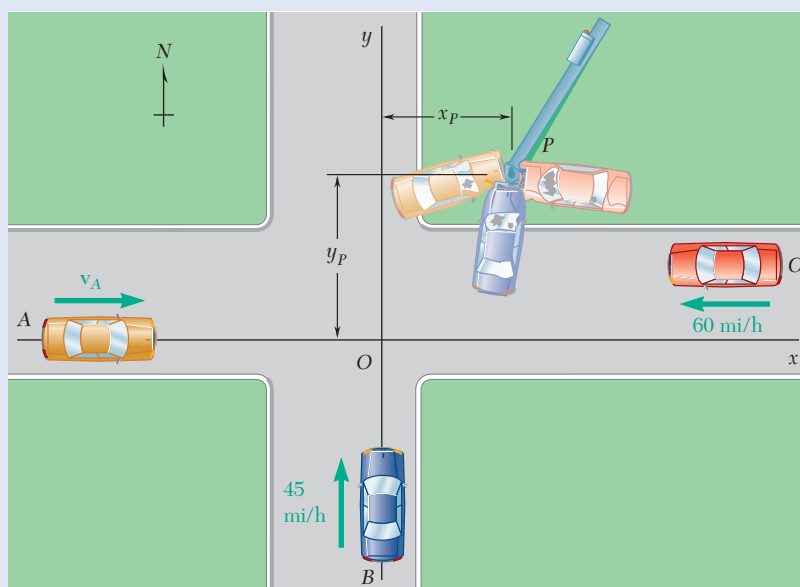


Fig. P14.19 and P14.20

14.21 An expert archer demonstrates his ability by hitting tennis balls thrown by an assistant. A 2-oz tennis ball has a velocity of $(32 \text{ ft/s})\mathbf{i} - (7 \text{ ft/s})\mathbf{j}$ and is 33 ft above the ground when it is hit by a 1.2-oz arrow traveling with a velocity of $(165 \text{ ft/s})\mathbf{j} + (230 \text{ ft/s})\mathbf{k}$ where $\mathbf{j}$ is directed upwards. Determine the position P where the ball and arrow will hit the ground, relative to point O located directly under the point of impact.

14.22 Two spheres, each of mass m , can slide freely on a frictionless, horizontal surface. Sphere A is moving at a speed $v_0 = 16 \text{ ft/s}$ when it strikes sphere B, which is at rest, and the impact causes sphere B to break into two pieces, each of mass $m/2$. Knowing that 0.7 s after the collision one piece reaches point C and 0.9 s after the collision the other piece reaches point D, determine (a) the velocity of sphere A after the collision, (b) the angle θ and the speeds of the two pieces after the collision.

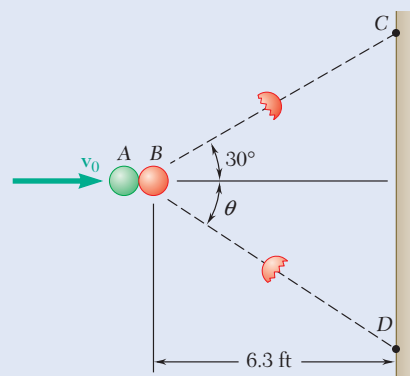


Fig. P14.22

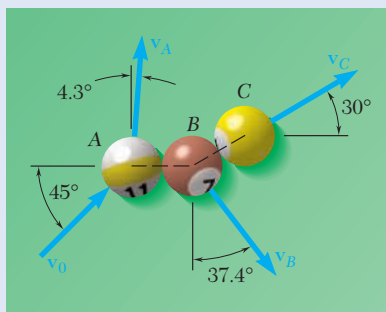


Fig. P14.23

14.23 In a game of pool, ball A is moving with a velocity $\mathbf{v}_0$ when it strikes balls B and C which are at rest and aligned as shown. Knowing that after the collision the three balls move in the directions indicated and that $v_0 = 12$ ft/s and $v_C = 6.29$ ft/s, determine the magnitude of the velocity of (a) ball A , (b) ball B .

14.24 A 6-kg shell moving with a velocity $\mathbf{v}_0 = (12 \text{ m/s})\mathbf{i} - (9 \text{ m/s})\mathbf{j} - (360 \text{ m/s})\mathbf{k}$ explodes at point D into three fragments A , B , and C of mass, respectively, 3 kg, 2 kg, and 1 kg. Knowing that the fragments hit the vertical wall at the points indicated, determine the speed of each fragment immediately after the explosion. Assume that elevation changes due to gravity may be neglected.

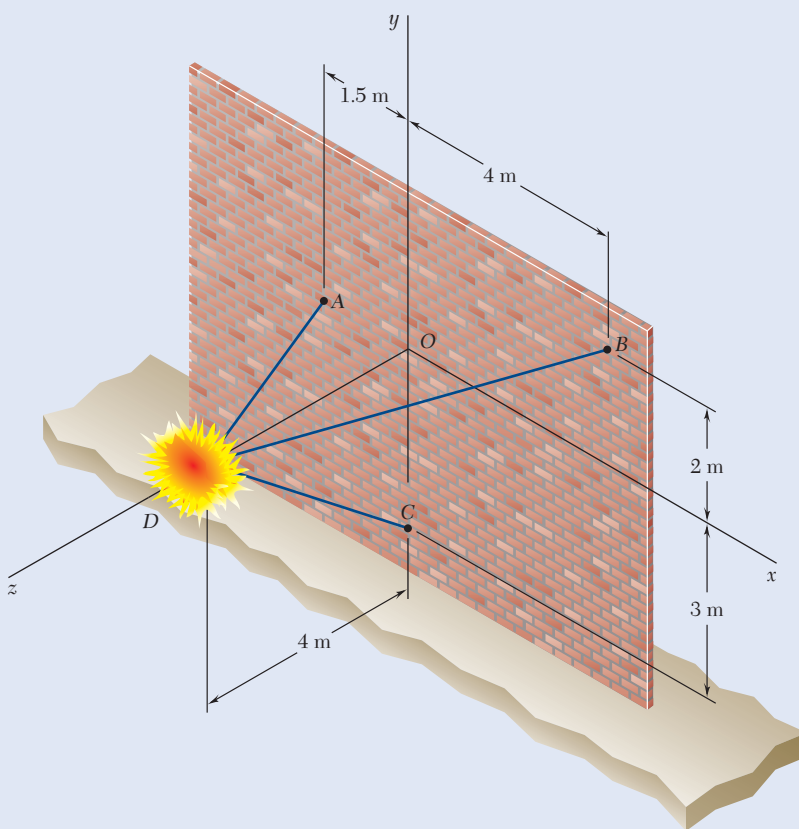


Fig. P14.24 and P14.25

14.25 A 6-kg shell moving with a velocity $\mathbf{v}_0 = (12 \text{ m/s})\mathbf{i} - (9 \text{ m/s})\mathbf{j} - (360 \text{ m/s})\mathbf{k}$ explodes at point D into three fragments A , B , and C of mass, respectively, 2 kg, 1 kg, and 3 kg. Knowing that the fragments hit the vertical wall at the points indicated, determine the speed of each fragment immediately after the explosion. Assume that elevation changes due to gravity may be neglected.

- 14.26** In a scattering experiment, an alpha particle A is projected with the velocity $\mathbf{u}_0 = -(600 \text{ m/s})\mathbf{i} + (750 \text{ m/s})\mathbf{j} - (800 \text{ m/s})\mathbf{k}$ into a stream of oxygen nuclei moving with a common velocity $\mathbf{v}_0 = (600 \text{ m/s})\mathbf{j}$. After colliding successively with the nuclei B and C , particle A is observed to move along the path defined by the points A_1 (280, 240, 120) and A_2 (360, 320, 160), while nuclei B and C are observed to move along paths defined, respectively, by B_1 (147, 220, 130) and B_2 (114, 290, 120), and by C_1 (240, 232, 90) and C_2 (240, 280, 75). All paths are along straight lines and all coordinates are expressed in millimeters. Knowing that the mass of an oxygen nucleus is four times that of an alpha particle, determine the speed of each of the three particles after the collisions.

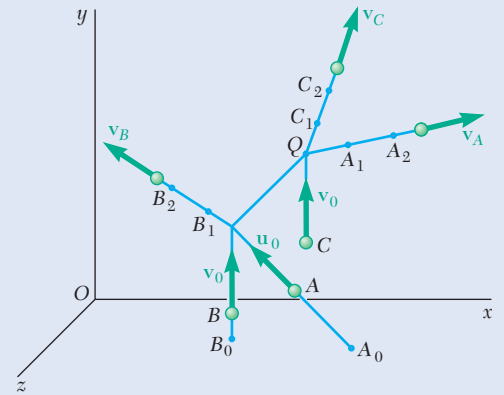


Fig. P14.26

- 14.27** Derive the relation

$$\mathbf{H}_O = \bar{\mathbf{r}} \times m\bar{\mathbf{v}} + H_G$$

between the angular momenta $\mathbf{H}_O$ and $\mathbf{H}_G$ defined in Eqs. (14.7) and (14.24), respectively. The vectors $\bar{\mathbf{r}}$ and $\bar{\mathbf{v}}$ define, respectively, the position and velocity of the mass center G of the system of particles relative to the newtonian frame of reference $Oxyz$, and m represents the total mass of the system.

- 14.28** Show that Eq. (14.23) may be derived directly from Eq. (14.11) by substituting for $\mathbf{H}_O$ the expression given in Prob. 14.27.

- 14.29** Consider the frame of reference $Ax'y'z'$ in translation with respect to the newtonian frame of reference $Oxyz$. We define the angular momentum $\mathbf{H}'_A$ of a system of n particles about A as the sum

$$\mathbf{H}'_A = \sum_{i=1}^n \mathbf{r}'_i \times m_i \mathbf{v}'_i \quad (1)$$

of the moments about A of the momenta $m_i \mathbf{v}'_i$ of the particles in their motion relative to the frame $Ax'y'z'$. Denoting by $\mathbf{H}_A$ the sum

$$\mathbf{H}_A = \sum_{i=1}^n \mathbf{r}'_i \times m_i \mathbf{v}_i$$

of the moments about A of the momenta $m_i \mathbf{v}_i$ of the particles in their motion relative to the newtonian frame $Oxyz$, show that $\mathbf{H}_A = \mathbf{H}'_A$ at a given instant if, and only if, one of the following conditions is satisfied at that instant: (a) A has zero velocity with respect to the frame $Oxyz$, (b) A coincides with the mass center G of the system, (c) the velocity $\mathbf{v}_A$ relative to $Oxyz$ is directed along the line AG .

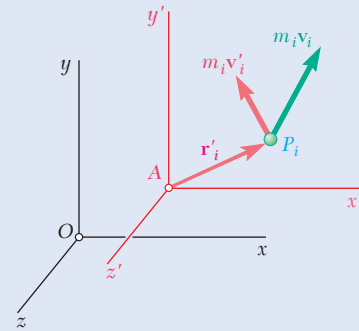


Fig. P14.29

- 14.30** Show that the relation $\Sigma \mathbf{M}_A = \dot{\mathbf{H}}'_A$, where $\mathbf{H}'_A$ is defined by Eq. (1) of Prob. 14.29 and where $\Sigma \mathbf{M}_A$ represents the sum of the moments about A of the external forces acting on the system of particles, is valid if, and only if, one of the following conditions is satisfied: (a) the frame $Ax'y'z'$ is itself a newtonian frame of reference, (b) A coincides with the mass center G , (c) the acceleration $\mathbf{a}_A$ of A relative to $Oxyz$ is directed along the line AG .

14.2 ENERGY AND MOMENTUM METHODS FOR A SYSTEM OF PARTICLES

Solving problems involving a system of particles is often made easier by applying energy and momentum methods, just as it was as for a single particle in Chapter 13. Definitions of terms and statements of the work–energy and impulse–momentum principles are very similar to the single-particle versions, especially when you take into account the mass center of the particles.

14.2A Kinetic Energy of a System of Particles

We define the kinetic energy T of a system of particles as the sum of the kinetic energies of the various particles of the system. Referring to Sec. 13.1B, we have

**Kinetic energy,
system of particles**

$$T = \frac{1}{2} \sum_{i=1}^n m_i v_i^2 \quad (14.28)$$

Using a Centroidal Frame of Reference. It is often convenient when computing the kinetic energy of a system comprised of a large number of particles (as in the case of a rigid body) to consider the motion of the mass center G of the system and the motion of the system relative to a moving frame attached to G separately.

Let P_i be a particle of the system, $\mathbf{v}_i$ be its velocity relative to the newtonian frame of reference $Oxyz$, and $\mathbf{v}'_i$ be its velocity relative to the moving frame $Gx'y'z'$ that is in translation with respect to $Oxyz$ (Fig. 14.7). Recall from Sec. 14.1D that

$$\mathbf{v}_i = \bar{\mathbf{v}} + \mathbf{v}'_i \quad (14.22)$$

where $\bar{\mathbf{v}}$ denotes the velocity of the mass center G relative to the newtonian frame $Oxyz$. Observing that v_i^2 is equal to the scalar product $\mathbf{v}_i \cdot \mathbf{v}_i$, we can express the kinetic energy T of the system relative to the newtonian frame $Oxyz$ as

$$T = \frac{1}{2} \sum_{i=1}^n m_i v_i^2 = \frac{1}{2} \sum_{i=1}^n (m_i \mathbf{v}_i \cdot \mathbf{v}_i)$$

or, substituting for $\mathbf{v}_i$ from Eq. (14.22),

$$\begin{aligned} T &= \frac{1}{2} \sum_{i=1}^n [m_i (\bar{\mathbf{v}} + \mathbf{v}'_i) \cdot (\bar{\mathbf{v}} + \mathbf{v}'_i)] \\ &= \frac{1}{2} \left(\sum_{i=1}^n m_i \right) \bar{v}^2 + \bar{\mathbf{v}} \cdot \sum_{i=1}^n m_i \mathbf{v}'_i + \frac{1}{2} \sum_{i=1}^n m_i v_i'^2 \end{aligned}$$

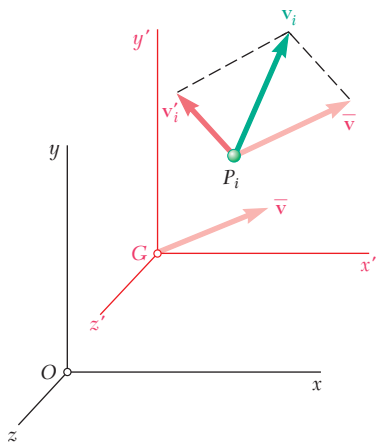


Fig. 14.7 A centroidal frame of reference $Gx'y'z'$ moving in translation with velocity $\bar{\mathbf{v}}$ with respect to a newtonian reference frame $Oxyz$.

In this equation, the first sum represents the total mass m of the system. Recalling Eq. (14.13), we note that the second sum is equal to $m\bar{\mathbf{v}}'$ and thus to zero, since $\bar{\mathbf{v}}'$, which represents the velocity of G relative to the frame $Gx'y'z'$, is clearly zero. We therefore have

$$T = \frac{1}{2}m\bar{\mathbf{v}}^2 + \frac{1}{2}\sum_{i=1}^n m_i v_i'^2 \quad (14.29)$$

This equation states that we can obtain the kinetic energy T of a system of particles **by adding the kinetic energy of the mass center G and the kinetic energy of the system in its motion relative to the frame $Gx'y'z'$.**

14.2B Work-Energy Principle and Conservation of Energy for a System of Particles

We can apply the principle of work and energy to each particle P_i of a system of particles, obtaining for each particle P_i

$$(T_1)_i + (U_{1 \rightarrow 2})_i = (T_2)_i$$

where $(U_{1 \rightarrow 2})_i$ represents the work done by the internal forces $\mathbf{f}_{ij}$ and the resultant external force $\mathbf{F}_i$ acting on P_i . Adding the kinetic energies of the various particles of the system and considering the work of all the forces involved, we obtain an expression for the entire system as

**Work-energy principle,
system of particles**

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (14.30)$$

The quantities T_1 and T_2 now represent the kinetic energy of the entire system and can be computed from either Eq. (14.28) or Eq. (14.29). The quantity $U_{1 \rightarrow 2}$ represents the work of all the forces acting on the particles of the system. Note that although the internal forces $\mathbf{f}_{ij}$ and $\mathbf{f}_{ji}$ are equal and opposite, the work of these forces does not, in general, cancel out, since the particles P_i and P_j on which they act generally undergo different displacements. Therefore, in computing $U_{1 \rightarrow 2}$, **we must consider the work of the internal forces $\mathbf{f}_{ij}$ as well as the work of the external forces $\mathbf{F}_i$.** An alternative way of writing Eq. (14.30) is

$$T_1 + V_{g_1} + V_{e_1} + U_{1 \rightarrow 2}^{\text{NC}} = T_2 + V_{g_2} + V_{e_2} \quad (14.30')$$

where V_g is the gravitational potential energy of the system, V_e is the elastic potential energy, and $U_{1 \rightarrow 2}^{\text{NC}}$ is the work due to non-conservative forces.

If all of the forces acting on the particles of the system are conservative, we can replace Eq. (14.30) by

**Conservation of energy,
system of particles**

$$T_1 + V_1 = T_2 + V_2 \quad (14.31)$$

where V represents the potential energy associated with the internal and external forces acting on the particles of the system.



Photo 14.2 When a golf ball is hit out of a sand trap, some of the momentum of the club is transferred to the golf ball and any sand that is hit.

14.2C Impulse-Momentum Principle and Conservation of Momentum for a System of Particles

Integrating Eqs. (14.10) and (14.11) with respect to t from t_1 to t_2 , we have

$$\sum \int_{t_1}^{t_2} \mathbf{F} dt = \mathbf{L}_2 - \mathbf{L}_1 \quad (14.32)$$

$$\sum \int_{t_1}^{t_2} \mathbf{M}_O dt = (\mathbf{H}_O)_2 - (\mathbf{H}_O)_1 \quad (14.33)$$

From the definition of the linear impulse of a force given in Sec. 13.3A, the integrals in Eq. (14.32) represent the linear impulses of the external forces acting on the particles of the system. In a similar way, we shall refer to the integrals in Eq. (14.33) as the **angular impulses** about O of the external forces. Thus, Eq. (14.32) states that the sum of the linear impulses of the external forces acting on the system is equal to the change in linear momentum of the system. Similarly, Eq. (14.33) says that the sum of the angular impulses about O of the external forces is equal to the change in angular momentum about O of the system.

To clarify the physical significance of Eqs. (14.32) and (14.33), we rearrange the terms in these equations, obtaining

$$\mathbf{L}_1 + \sum \int_{t_1}^{t_2} \mathbf{F} dt = \mathbf{L}_2 \quad (14.34)$$

$$(\mathbf{H}_O)_1 + \sum \int_{t_1}^{t_2} \mathbf{M}_O dt = (\mathbf{H}_O)_2 \quad (14.35)$$

In parts *a* and *c* of Fig. 14.8, we have sketched the momenta of the particles of the system at times t_1 and t_2 , respectively. In part *b*, we show terms equal to the sum of the linear impulses of the external forces and the sum of the angular impulses about O of the external forces.

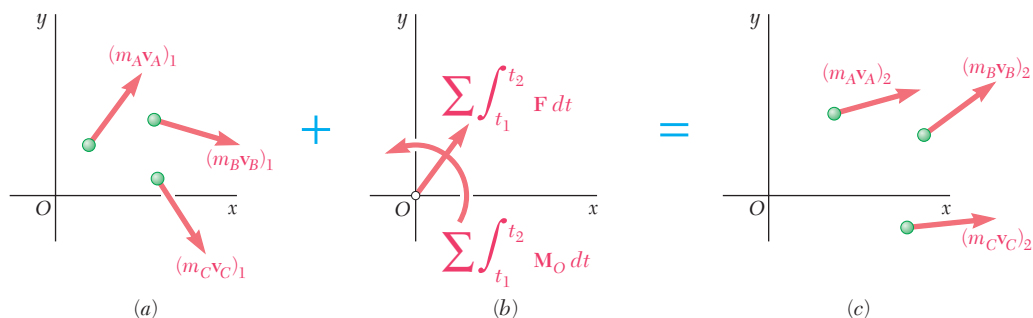


Fig. 14.8 The impulse-momentum diagram for a system of particles contains (a) momenta of particles at time t_1 ; (b) impulses of the external forces and moments about O ; (c) momenta of the particles at time t_2 .

For simplicity, we have assumed the particles move in the plane of the figure, but the present discussion remains valid in the case of particles moving in space. Recall from Eq. (14.6) that $\mathbf{L}$, by definition, is the resultant of the momenta $m_i \mathbf{v}_i$. Then Eq. (14.34) says that the resultant of the vectors shown in parts *a* and *b* of Fig. 14.8 is equal to the resultant of the vectors shown in part *c*. Recalling from Eq. (14.7) that $\mathbf{H}_O$ is the angular momentum, we note that Eq. (14.35) similarly says that the angular momentum of the vectors in parts *a* added to the angular impulses in part *b* of Fig. 14.8 is equal to the angular momentum of the vectors in part *c*. Together, Eqs. (14.34) and (14.35) state:

The momenta of the particles at time t_1 and the impulses of the external forces from t_1 to t_2 form a system of vectors equipollent to the system of the momenta of the particles at time t_2 .

This is indicated in Fig. 14.8 by the use of blue plus and equal signs.

If no external force acts on the particles of the system, the integrals in Eqs. (14.34) and (14.35) are zero, and these equations yield

Conservation of linear and angular momentum

$$\mathbf{L}_1 = \mathbf{L}_2 \quad (14.36)$$

$$(\mathbf{H}_O)_1 = (\mathbf{H}_O)_2 \quad (14.37)$$

We thus check the result obtained in Sec. 14.1E: If no external force acts on the particles of a system, the linear momentum and the angular momentum about O of the system of particles are conserved. The system of the initial momenta is equipollent to the system of the final momenta, and it follows that the angular momentum of the system of particles about *any* fixed point is conserved.

Sample Problem 14.4

For the 200-kg space vehicle of Sample Prob. 14.1, it is known that at $t = 2.5$ s, the velocity of part A is $\mathbf{v}_A = (270 \text{ m/s})\mathbf{i} - (120 \text{ m/s})\mathbf{j} + (160 \text{ m/s})\mathbf{k}$, and the velocity of part B is parallel to the xz plane. Determine (a) the velocity of part C, (b) the energy gained during the detonation.

STRATEGY: Since there are no external forces, use the conservation of linear momentum. Although it is not immediately apparent, you will also need to use the conservation of angular momentum to solve this problem.

MODELING and ANALYSIS: Choose the space vehicle as your system. After the explosion, the system is composed of three parts: A, B, and C. Figure 1 shows the momenta of the system before and after the explosion. From the conservation of linear momentum, you have

$$\mathbf{L}_1 = \mathbf{L}_2: \quad m\mathbf{v}_0 = m_A\mathbf{v}_A + m_B\mathbf{v}_B + m_C\mathbf{v}_C \quad (1)$$

From conservation of angular momentum about point O you have

$$(\mathbf{H}_O)_1 = (\mathbf{H}_O)_2: \quad \mathbf{0} = \mathbf{r}_A \times m_A\mathbf{v}_A + \mathbf{r}_B \times m_B\mathbf{v}_B + \mathbf{r}_C \times m_C\mathbf{v}_C \quad (2)$$

Recall from Sample Prob. 14.1 that $\mathbf{v}_0 = (150 \text{ m/s})\mathbf{i}$ and

$$\begin{aligned} m_A &= 100 \text{ kg} & m_B &= 60 \text{ kg} & m_C &= 40 \text{ kg} \\ \mathbf{r}_A &= (555 \text{ m})\mathbf{i} - (180 \text{ m})\mathbf{j} + (240 \text{ m})\mathbf{k} \\ \mathbf{r}_B &= (255 \text{ m})\mathbf{i} - (120 \text{ m})\mathbf{k} \\ \mathbf{r}_C &= (105 \text{ m})\mathbf{i} + (450 \text{ m})\mathbf{j} - (420 \text{ m})\mathbf{k} \end{aligned}$$

Then, using the information given in the statement of this problem, rewrite Eqs. (1) and (2) as

$$200(150\mathbf{i}) = 100(270\mathbf{i} - 120\mathbf{j} + 160\mathbf{k}) + 60[(v_B)_x\mathbf{i} + (v_B)_z\mathbf{k}] + 40[(v_C)_x\mathbf{i} + (v_C)_y\mathbf{j} + (v_C)_z\mathbf{k}] \quad (1')$$

$$\begin{aligned} 0 &= 100 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 555 & -180 & 240 \\ 270 & -120 & 160 \end{vmatrix} + 60 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 255 & 0 & -120 \\ (v_B)_x & 0 & (v_B)_z \end{vmatrix} \\ &\quad + 40 \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 105 & 450 & -420 \\ (v_C)_x & (v_C)_y & (v_C)_z \end{vmatrix} \quad (2') \end{aligned}$$

Equate the coefficient of $\mathbf{j}$ in Eq. (1') and the coefficients of $\mathbf{i}$ and $\mathbf{k}$ in Eq. (2'). After reductions, you obtain the three scalar equations of

$$\begin{aligned} (v_C)_y - 300 &= 0 \\ 450(v_C)_z + 420(v_C)_y &= 0 \\ 105(v_C)_y - 450(v_C)_x - 45\,000 &= 0 \end{aligned}$$

which yield, respectively,

$$(v_C)_y = 300 \quad (v_C)_z = -280 \quad (v_C)_x = -30$$

The velocity of part C is thus

$$\mathbf{v}_C = -(30 \text{ m/s})\mathbf{i} + (300 \text{ m/s})\mathbf{j} - (280 \text{ m/s})\mathbf{k} \quad \blacktriangleleft$$

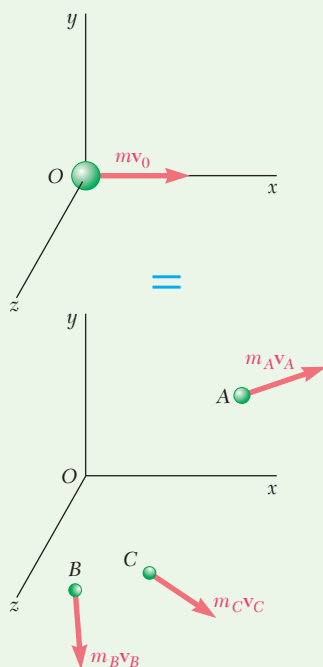


Fig. 1 Impulse-momentum diagram for the system.

Equating the coefficients of the **i** and **k** terms on each side of Eq. (1') and solving for the unknown components of the velocity of *B* gives

$$(v_B)_x = 70 \text{ m/s} \quad (v_B)_z = -80 \text{ m/s}$$

So

$$v_A = \sqrt{(270 \text{ m/s})^2 + (-120 \text{ m/s})^2 + (160 \text{ m/s})^2} = 336.0 \text{ m/s}$$

$$v_B = \sqrt{(70 \text{ m/s})^2 + (0)^2 + (-80 \text{ m/s})^2} = 106.3 \text{ m/s}$$

$$v_C = \sqrt{(-30 \text{ m/s})^2 + (300)^2 + (-280 \text{ m/s})^2} = 411.5 \text{ m/s}$$

The initial kinetic energy is

$$T_1 = \frac{1}{2}mv_0^2 = \frac{1}{2}(200 \text{ kg})(150 \text{ m/s})^2 = 2250 \text{ kJ}$$

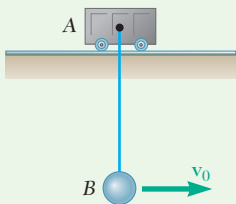
The final kinetic energy is

$$\begin{aligned} T_2 &= \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2 + \frac{1}{2}m_C v_C^2 \\ &= \frac{1}{2}(100 \text{ kg})(336.0 \text{ m/s})^2 + \frac{1}{2}(60 \text{ kg})(106.3 \text{ m/s})^2 + \frac{1}{2}(40 \text{ kg})(411.5 \text{ m/s})^2 \\ &= 9370 \text{ kJ} \end{aligned}$$

So

$$\Delta T = T_2 - T_1 = 9370 \text{ kJ} - 2250 \text{ kJ} \quad \Delta T = 7120 \text{ kJ} \quad \blacktriangleleft$$

REFLECT and THINK: The negative signs for $(v_C)_x$ and $(v_C)_z$ indicate that the velocity is not directed as shown in Fig. 1. We also notice that the directions of the components of $\mathbf{v}_C$ are opposite to those of $\mathbf{v}_A$. Given the lack of external forces, it seems reasonable to expect a more symmetric spread of velocities in all directions. You should also notice that the explosion added a lot of energy to the system.



Sample Problem 14.5

Ball *B*, with a mass of m_B , is suspended from a cord with a length l attached to cart *A*, with a mass of m_A , that can roll freely on a frictionless horizontal track. If the ball is given an initial horizontal velocity $\mathbf{v}_0$ while the cart is at rest, determine (a) the velocity of *B* as it reaches its maximum elevation, (b) the maximum vertical distance h through which *B* will rise. (Assume $v_0^2 < 2gl$.)

STRATEGY: You are asked about the velocity of the system at two different positions, so use the principle of work and energy for the cart–ball system. You will also use the impulse–momentum principle, since momentum is conserved in the x -direction.

(continued)

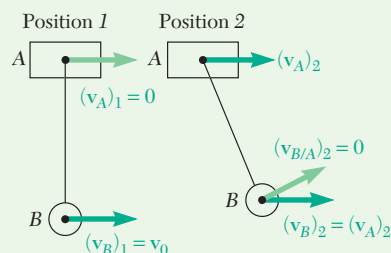


Fig. 1 Velocity vectors at the two positions.

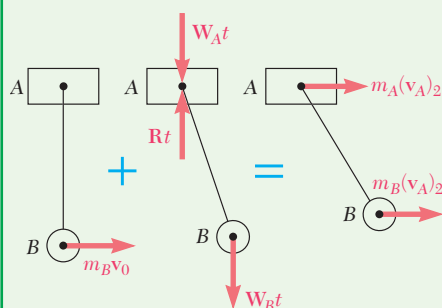


Fig. 2 Impulse-momentum diagram for the system.

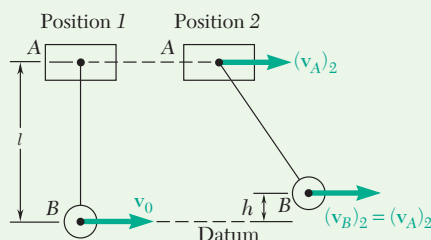


Fig. 3 The system drawn in position 1 and position 2.

MODELING and ANALYSIS: For your system, choose the ball and the cart and model them as particles.

Velocities.

Position 1: $(\mathbf{v}_A)_1 = 0$ $(\mathbf{v}_B)_1 = \mathbf{v}_0$ (1)

Position 2: When ball B reaches its maximum elevation, its velocity $(\mathbf{v}_{B/A})_2$ relative to its support A is zero (Fig. 1). Thus, at that instant, its absolute velocity is

$$(\mathbf{v}_B)_2 = (\mathbf{v}_A)_2 + (\mathbf{v}_{B/A})_2 = (\mathbf{v}_A)_2 \quad (2)$$

Impulse-Momentum Principle. The external impulses consist of $\mathbf{W}_A t$, $\mathbf{W}_B t$, and $\mathbf{R} t$, where $\mathbf{R}$ is the reaction of the track on the cart. Recalling Eqs. (1) and (2), draw the impulse-momentum diagram (Fig. 2) and write

$$\Sigma m \mathbf{v}_1 + \Sigma \text{Ext Imp}_{1 \rightarrow 2} = \Sigma m \mathbf{v}_2$$

$\rightarrow x$ components:

$$m_B v_0 = (m_A + m_B)(v_A)_2$$

This expresses that the linear momentum of the system is conserved in the horizontal direction. Solving for $(v_A)_2$, you have

$$(v_A)_2 = \frac{m_B}{m_A + m_B} v_0 \quad (\mathbf{v}_B)_2 = (\mathbf{v}_A)_2 = \frac{m_B}{m_A + m_B} v_0 \rightarrow \blacktriangleleft$$

Conservation of Energy. The system is shown in Fig. 3 in the two positions. Define your datum at the location of B in position 1 (although you could also choose to place it at A). You can now calculate the kinetic and potential energies in the two positions:

Position 1. Potential Energy:	$V_1 = m_A g l$
Kinetic Energy:	$T_1 = \frac{1}{2} m_B v_0^2$
Position 2. Potential Energy:	$V_2 = m_A g l + m_B g h$
Kinetic Energy:	$T_2 = \frac{1}{2} (m_A + m_B) (v_A)_2^2$

Substituting these into the conservation of energy gives

$$T_1 + V_1 = T_2 + V_2: \quad \frac{1}{2} m_B v_0^2 + m_A g l = \frac{1}{2} (m_A + m_B) (v_A)_2^2 + m_A g l + m_B g h$$

Solving for h , you have

$$h = \frac{v_0^2}{2g} - \frac{m_A + m_B}{m_B} \frac{(v_A)_2^2}{2g}$$

or substituting $(v_A)_2$ from above, you have

$$h = \frac{v_0^2}{2g} - \frac{m_B}{m_A + m_B} \frac{v_0^2}{2g} \quad h = \frac{m_A}{m_A + m_B} \frac{v_0^2}{2g} \quad \blacktriangleleft$$

REFLECT and THINK: Recalling that $v_0^2 < 2gl$, it follows from the last equation that $h < l$; this verifies that B stays below A , as assumed in the solution. For $m_A \gg m_B$, the answers reduce to $(\mathbf{v}_B)_2 = (\mathbf{v}_A)_2 = 0$ and $h = v_0^2/2g$; B oscillates as a simple pendulum with A fixed. For $m_A \ll m_B$, they reduce to $(\mathbf{v}_B)_2 = (\mathbf{v}_A)_2 = \mathbf{v}_0$ and $h = 0$; A and B move with the same constant velocity $\mathbf{v}_0$.

Sample Problem 14.6

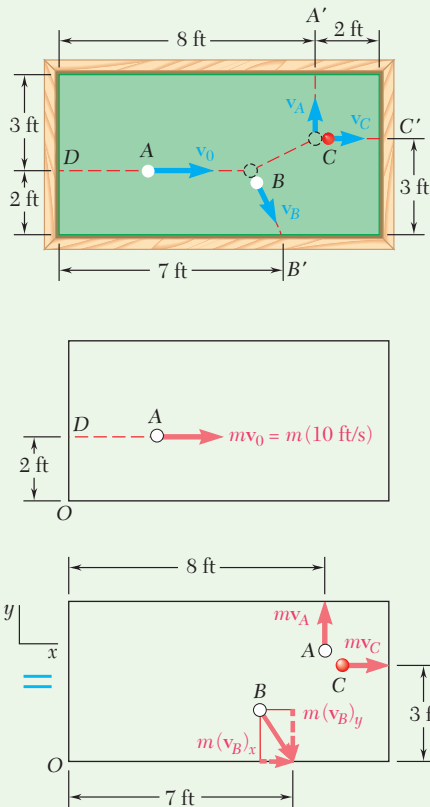


Fig. 1 Impulse-momentum diagram for the system.

In a game of billiards, ball A is given an initial velocity $\mathbf{v}_0$ with a magnitude of $v_0 = 10$ ft/s along line DA parallel to the axis of the table. It hits ball B and then ball C , which are both at rest. Balls A and C hit the sides of the table squarely at points A' and C' , respectively, and B hits the side obliquely at B' . Assuming frictionless surfaces and perfectly elastic impacts, determine the velocities $\mathbf{v}_A$, $\mathbf{v}_B$, and $\mathbf{v}_C$ with which the balls hit the sides of the table. (Remark: In this sample problem and in several of the problems that follow, we assume the billiard balls are particles moving freely in a horizontal plane, rather than the rolling and sliding spheres they actually are.)

STRATEGY: Since there are no externally applied forces, use the conservation of linear and angular momentum. Because you are told that the impacts are perfectly elastic, you can also use the conservation of energy (but note that in general, energy is lost in an impact).

MODELING and ANALYSIS: Choose the system to be all three billiard balls and model them as particles.

Conservation of Momentum. There is no external force, so the initial momentum $m\mathbf{v}_0$ is equipollent to the system of momenta after the two collisions (and before any of the balls hit the sides of the table). Referring to Fig. 1, you have

$$+\rightarrow x \text{ components:} \quad m(10 \text{ ft/s}) = m(v_B)_x + mv_C \quad (1)$$

$$+\uparrow y \text{ components:} \quad 0 = mv_A - m(v_B)_y \quad (2)$$

$$+\curvearrowright \text{ moments about } O: -(2 \text{ ft})m(10 \text{ ft/s}) = (8 \text{ ft})mv_A - (7 \text{ ft})m(v_B)_y - (3 \text{ ft})mv_C \quad (3)$$

Solving the three equations for v_A , $(v_B)_x$, and $(v_B)_y$ in terms of v_C gives

$$v_A = (v_B)_y = 3v_C - 20 \quad (v_B)_x = 10 - v_C \quad (4)$$

Conservation of Energy. The surfaces are frictionless and the impacts are perfectly elastic, so the initial kinetic energy $\frac{1}{2}mv_0^2$ is equal to the final kinetic energy of the system:

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv_A^2 + \frac{1}{2}mv_B^2 + \frac{1}{2}mv_C^2$$

$$v_A^2 + (v_B)_x^2 + (v_B)_y^2 + v_C^2 = (10 \text{ ft/s})^2 \quad (5)$$

Substituting for v_A , $(v_B)_x$, and $(v_B)_y$ from Eq. (4) into Eq. (5), you have

$$2(3v_C - 20)^2 + (10 - v_C)^2 + v_C^2 = 100$$

$$20v_C^2 - 260v_C + 800 = 0$$

Solving for v_C , you find $v_C = 5$ ft/s and $v_C = 8$ ft/s. Since only the second root yields a positive value for v_A after substitution into Eqs. (4), then $v_C = 8$ ft/s and

$$v_A = (v_B)_y = 3(8) - 20 = 4 \text{ ft/s} \quad (v_B)_x = 10 - 8 = 2 \text{ ft/s}$$

$$\mathbf{v}_A = 4 \text{ ft/s} \uparrow \quad \mathbf{v}_B = 4.47 \text{ ft/s} \searrow 63.4^\circ \quad \mathbf{v}_C = 8 \text{ ft/s} \rightarrow \blacktriangleleft$$

REFLECT and THINK: In a real situation, energy would not be conserved, and you would need to know the coefficient of restitution between the balls to solve this problem. We also neglected friction and the rotation of the balls in our analysis, which is often a poor assumption in pool or billiards. We discuss rigid-body impacts in Chapter 17.

SOLVING PROBLEMS ON YOUR OWN

In Sec. 14.1, we defined the linear momentum and the angular momentum of a system of particles. In this section, we defined the **kinetic energy** T of a system of particles as

$$T = \frac{1}{2} \sum_{i=1}^n m_i v_i^2 \quad (14.28)$$

The solutions of the problems in Sec. 14.1 were based on the conservation of linear momentum of a system of particles or on the observation of the motion of the mass center of a system of particles. In this section, you will solve problems involving the following concepts.

1. Computation of the kinetic energy lost in collisions. You can compute the kinetic energy T_1 of the system of particles before the collisions and its kinetic energy T_2 after the collisions from Eq. (14.28) and subtract one from the other. Keep in mind that although linear momentum and angular momentum are vector quantities, kinetic energy is a *scalar* quantity.

2. Conservation of linear momentum and conservation of energy. As you saw in Sec. 14.1, when the resultant of the external forces acting on a system of particles is zero, the linear momentum of the system is conserved. In problems involving two-dimensional motion, expressing that the initial linear momentum and the final linear momentum of the system are equipollent yields two algebraic equations. Equating the initial total energy of the system of particles (including potential energy as well as kinetic energy) to its final total energy yields an additional equation. Thus, you can write three equations that you can solve for three unknowns [Sample Prob. 14.6]. Note that if the resultant of the external forces is not zero but has a fixed direction, the component of the linear momentum in a direction perpendicular to the resultant is still conserved; the number of equations that you can use is then reduced to two [Sample Prob. 14.5].

3. Conservation of linear and angular momentum. When no external forces act on a system of particles, both the linear momentum of the system and its angular momentum about some arbitrary point are conserved. In the case of three-dimensional motion, this enables you to write as many as six equations, although you may need to solve only some of them to obtain the desired answers [Sample Prob. 14.4]. In the case of two-dimensional motion, you will be able to write three equations that you can solve for three unknowns.

4. Conservation of linear and angular momentum and conservation of energy. In the case of the two-dimensional motion of a system of particles that is not subjected to any external forces, you can obtain two algebraic equations by expressing that the linear momentum of the system is conserved; one equation by writing that the angular momentum of the system about some arbitrary point is conserved; and a fourth equation by expressing that the total energy of the system is conserved. These equations can be solved for four unknowns.

Problems

- 14.31** Determine the energy lost due to friction and the impacts for Prob. 14.1.
- 14.32** Assuming that the airline employee of Prob. 14.3 first tosses the 30-lb suitcase on the baggage carrier, determine the energy lost (a) as the first suitcase hits the carrier, (b) as the second suitcase hits the carrier.
- 14.33** In Prob. 14.6, determine the work done by the woman and by the man as each dives from the boat, assuming that the woman dives first.
- 14.34** Determine the energy lost as a result of the series of collisions described in Prob. 14.8.
- 14.35** Two automobiles A and B , of mass m_A and m_B , respectively, are traveling in opposite directions when they collide head on. The impact is assumed perfectly plastic, and it is further assumed that the energy absorbed by each automobile is equal to its loss of kinetic energy with respect to a moving frame of reference attached to the mass center of the two-vehicle system. Denoting by E_A and E_B , respectively, the energy absorbed by automobile A and by automobile B , (a) show that $E_A/E_B = m_B/m_A$, that is, the amount of energy absorbed by each vehicle is inversely proportional to its mass, (b) compute E_A and E_B , knowing that $m_A = 1600$ kg and $m_B = 900$ kg and that the speeds of A and B are, respectively, 90 km/h and 60 km/h.

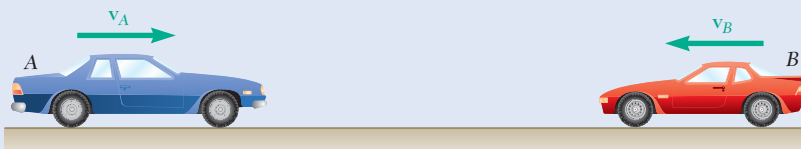


Fig. P14.35

- 14.36** It is assumed that each of the two automobiles involved in the collision described in Prob. 14.35 had been designed to safely withstand a test in which it crashed into a solid, immovable wall at the speed v_0 . The severity of the collision of Prob. 14.35 may then be measured for each vehicle by the ratio of the energy it absorbed in the collision to the energy it absorbed in the test. On that basis, show that the collision described in Prob. 14.35 is $(m_A/m_B)^2$ times more severe for automobile B than for automobile A .
- 14.37** Solve Sample Problem 14.5, assuming that cart A is given an initial horizontal velocity v_0 while ball B is at rest.
- 14.38** Two hemispheres are held together by a cord which maintains a spring under compression (the spring is not attached to the hemispheres). The potential energy of the compressed spring is 120 J and the assembly has an initial velocity v_0 of magnitude $v_0 = 8$ m/s. Knowing that the cord is severed when $\theta = 30^\circ$, causing the hemispheres to fly apart, determine the resulting velocity of each hemisphere.

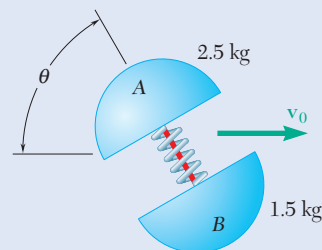


Fig. P14.38

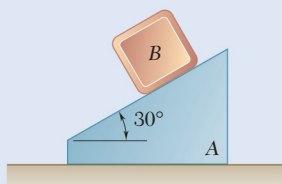


Fig. P14.39

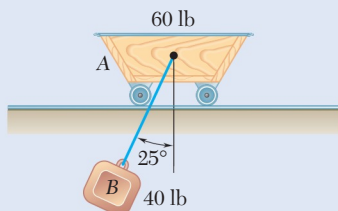


Fig. P14.40

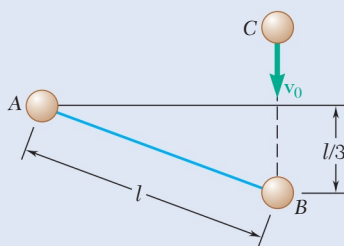


Fig. P14.43

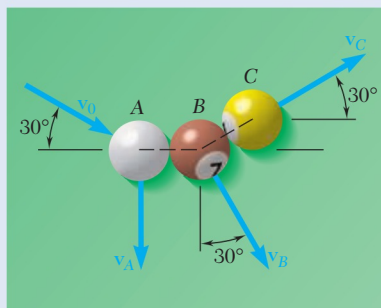


Fig. P14.41

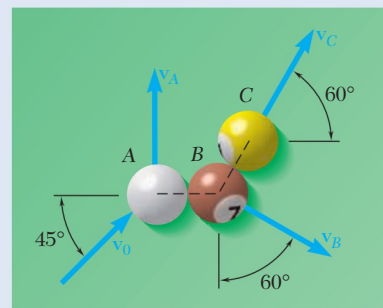


Fig. P14.42

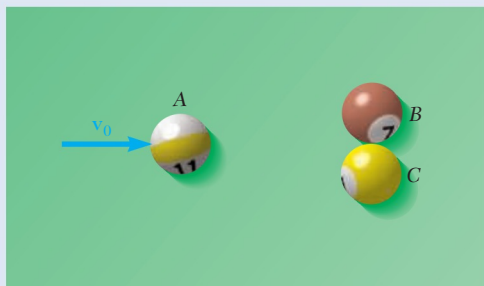


Fig. P14.44

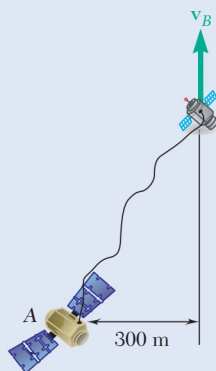


Fig. P14.45

14.39 A 15-lb block B starts from rest and slides on the 25-lb wedge A , which is supported by a horizontal surface. Neglecting friction, determine (a) the velocity of B relative to A after it has slid 3 ft down the inclined surface of the wedge, (b) the corresponding velocity of A .

14.40 A 40-lb block B is suspended from a 6-ft cord attached to a 60-lb cart A , which may roll freely on a frictionless, horizontal track. If the system is released from rest in the position shown, determine the velocities of A and B as B passes directly under A .

14.41 and 14.42 In a game of pool, ball A is moving with a velocity $\mathbf{v}_0$ with a magnitude of $v_0 = 15$ ft/s when it strikes balls B and C , which are at rest and aligned as shown. Knowing that after the collision the three balls move in the directions indicated and assuming frictionless surfaces and perfectly elastic impact (that is, conservation of energy), determine the magnitudes of the velocities $\mathbf{v}_A$, $\mathbf{v}_B$, and $\mathbf{v}_C$.

14.43 Three spheres, each with a mass of m , can slide freely on a frictionless, horizontal surface. Spheres A and B are attached to an inextensible, inelastic cord with a length l and are at rest in the position shown when sphere B is struck squarely by sphere C , which is moving with a velocity $\mathbf{v}_0$. Knowing that the cord is taut when sphere B is struck by sphere C and assuming perfectly elastic impact between B and C , and thus the conservation of energy for the entire system, determine the velocity of each sphere immediately after impact.

14.44 In a game of pool, ball A is moving with the velocity $\mathbf{v}_0 = v_0 \mathbf{i}$ when it strikes balls B and C , which are at rest side by side. Assuming frictionless surfaces and perfectly elastic impact (i.e., conservation of energy), determine the final velocity of each ball, assuming that the path of A is (a) perfectly centered and that A strikes B and C simultaneously, (b) not perfectly centered and that A strikes B slightly before it strikes C .

14.45 The 2-kg sub-satellite B has an initial velocity $\mathbf{v}_B = (3 \text{ m/s})\mathbf{j}$. It is connected to the 20-kg base-satellite A by a 500-m space tether. Determine the velocity of the base satellite and sub-satellite immediately after the tether becomes taut (assuming no rebound).

14.46 A 900-lb space vehicle traveling with a velocity $\mathbf{v}_0 = (1500 \text{ ft/s})\mathbf{k}$ passes through the origin O . Explosive charges then separate the vehicle into three parts A , B , and C , with masses of 150 lb, 300 lb, and 450 lb, respectively. Knowing that shortly thereafter the positions of the three parts are, respectively, $A(250, 250, 2250)$, $B(600, 1300, 3200)$, and $C(-475, -950, 1900)$, where the coordinates are expressed in feet, that the velocity of B is $\mathbf{v}_B = (500 \text{ ft/s})\mathbf{i} + (1100 \text{ ft/s})\mathbf{j} + (2100 \text{ ft/s})\mathbf{k}$, and that the x component of the velocity of C is -400 ft/s , determine the velocity of part A .

14.47 Four small disks A , B , C , and D can slide freely on a frictionless horizontal surface. Disks B , C , and D are connected by light rods and are at rest in the position shown when disk B is struck squarely by disk A which is moving to the right with a velocity $\mathbf{v}_0 = (38.5 \text{ ft/s})\mathbf{i}$. The weights of the disks are $W_A = W_B = W_C = 15 \text{ lb}$, and $W_D = 30 \text{ lb}$. Knowing that the velocities of the disks immediately after the impact are $\mathbf{v}_A = \mathbf{v}_B = (8.25 \text{ ft/s})\mathbf{i}$, $\mathbf{v}_C = v_C\mathbf{i}$, and $\mathbf{v}_D = v_D\mathbf{i}$, determine (a) the speeds v_C and v_D , (b) the fraction of the initial kinetic energy of the system which is dissipated during the collision.

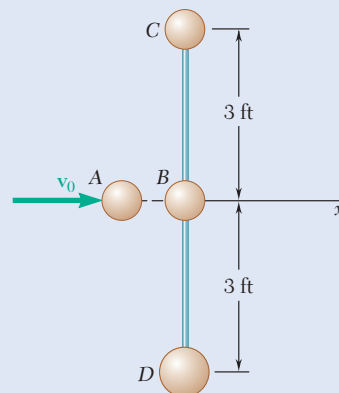


Fig. P14.47

14.48 In the scattering experiment of Prob. 14.26, it is known that the alpha particle is projected from $A_0(300, 0, 300)$ and that it collides with the oxygen nucleus C at $Q(240, 200, 100)$, where all coordinates are expressed in millimeters. Determine the coordinates of point B_0 where the original path of nucleus B intersects the zx plane. (Hint: Express that the angular momentum of the three particles about Q is conserved.)

14.49 Three identical small spheres, each of weight 2 lb, can slide freely on a horizontal frictionless surface. Spheres B and C are connected by a light rod and are at rest in the position shown when sphere B is struck squarely by sphere A which is moving to the right with a velocity $\mathbf{v}_0 = (8 \text{ ft/s})\mathbf{i}$. Knowing that $\theta = 45^\circ$ and that the velocities of spheres A and B immediately after the impact are $\mathbf{v}_A = 0$ and $\mathbf{v}_B = (6 \text{ ft/s})\mathbf{i} + (v_B)_y\mathbf{j}$, determine $(v_B)_y$ and the velocity of C immediately after impact.

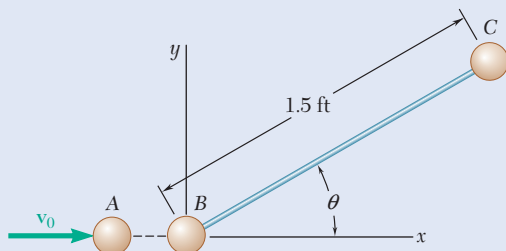


Fig. P14.49

14.50 Three small spheres A , B , and C , each of mass m , are connected to a small ring D of negligible mass by means of three inextensible, inelastic cords of length l . The spheres can slide freely on a frictionless horizontal surface and are rotating initially at a speed v_0 about ring D which is at rest. Suddenly the cord CD breaks. After the other two cords have again become taut, determine (a) the speed of ring D , (b) the relative speed at which spheres A and B rotate about D , (c) the fraction of the original energy of spheres A and B that is dissipated when cords AD and BD again became taut.

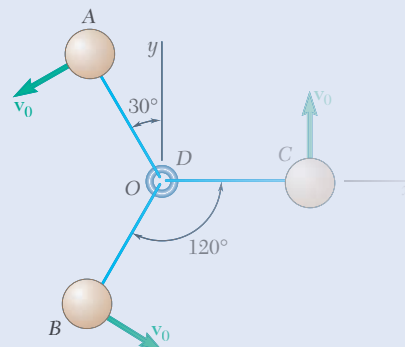


Fig. P14.50

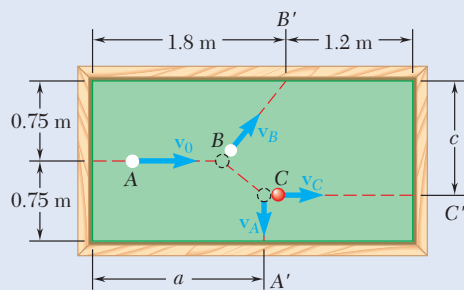


Fig. P14.51

14.51 In a game of billiards, ball A is given an initial velocity v_0 along the longitudinal axis of the table. It hits ball B and then ball C , which are both at rest. Balls A and C are observed to hit the sides of the table squarely at A' and C' , respectively, and ball B is observed to hit the side obliquely at B' . Knowing that $v_0 = 4$ m/s, $v_A = 1.92$ m/s, and $a = 1.65$ m, determine (a) the velocities v_B and v_C of balls B and C , (b) the point C' where ball C hits the side of the table. Assume frictionless surfaces and perfectly elastic impacts (i.e., conservation of energy).

14.52 For the game of billiards of Prob. 14.51, it is now assumed that $v_0 = 5$ m/s, $v_C = 3.2$ m/s, and $c = 1.22$ m. Determine (a) the velocities v_A and v_B of balls A and B , (b) the point A' where ball A hits the side of the table.

14.53 Two small disks A and B , of mass 3 kg and 1.5 kg, respectively, may slide on a horizontal, frictionless surface. They are connected by a cord, 600 mm long, and spin counterclockwise about their mass center G at the rate of 10 rad/s. At $t = 0$, the coordinates of G are $\bar{x}_0 = 0$, $\bar{y}_0 = 2$ m, and its velocity $\bar{v}_0 = (1.2 \text{ m/s})\mathbf{i} + (0.96 \text{ m/s})\mathbf{j}$. Shortly thereafter the cord breaks; disk A is then observed to move along a path parallel to the y axis and disk B along a path which intersects the x axis at a distance $b = 7.5$ m from O . Determine (a) the velocities of A and B after the cord breaks, (b) the distance a from the y axis to the path of A .

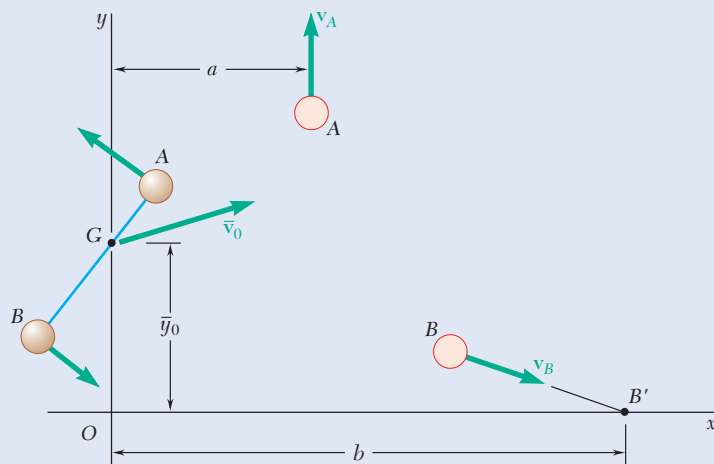


Fig. P14.53 and P14.54

14.54 Two small disks A and B , of mass 2 kg and 1 kg, respectively, may slide on a horizontal and frictionless surface. They are connected by a cord of negligible mass and spin about their mass center G . At $t = 0$, G is moving with the velocity $\bar{v}_0$ and its coordinates are $\bar{x}_0 = 0$, $\bar{y}_0 = 1.89$ m. Shortly thereafter, the cord breaks and disk A is observed to move with a velocity $v_A = (5 \text{ m/s})\mathbf{j}$ in a straight line and at a distance $a = 2.56$ m from the y axis, while B moves with a velocity $v_B = (7.2 \text{ m/s})\mathbf{i} - (4.6 \text{ m/s})\mathbf{j}$ along a path intersecting the x axis at a distance $b = 7.48$ m from the origin O . Determine (a) the initial velocity $\bar{v}_0$ of the mass center G of the two disks, (b) the length of the cord initially connecting the two disks, (c) the rate in rad/s at which the disks were spinning about G .

- 14.55** Three small identical spheres A , B , and C , which can slide on a horizontal, frictionless surface, are attached to three 9-in.-long strings, which are tied to a ring G . Initially the spheres rotate clockwise about the ring with a relative velocity of 2.6 ft/s and the ring moves along the x axis with a velocity $\mathbf{v}_0 = (1.3 \text{ ft/s})\mathbf{i}$. Suddenly the ring breaks and the three spheres move freely in the xy plane with A and B following paths parallel to the y axis at a distance $a = 1.0 \text{ ft}$ from each other and C following a path parallel to the x axis. Determine (a) the velocity of each sphere, (b) the distance d .

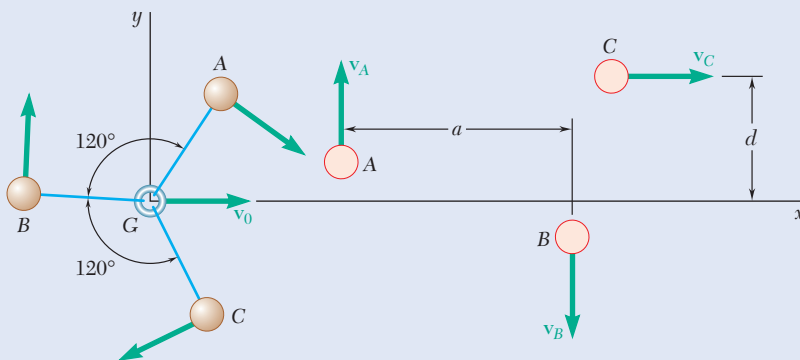


Fig. P14.55 and P14.56

- 14.56** Three small identical spheres A , B , and C , which can slide on a horizontal, frictionless surface, are attached to three strings of length l which are tied to a ring G . Initially the spheres rotate clockwise about the ring which moves along the x axis with a velocity $\mathbf{v}_0$. Suddenly the ring breaks and the three spheres move freely in the xy plane. Knowing that $\mathbf{v}_A = (3.5 \text{ ft/s})\mathbf{j}$, $\mathbf{v}_C = (6.0 \text{ ft/s})\mathbf{i}$, $a = 16 \text{ in.}$, and $d = 9 \text{ in.}$, determine (a) the initial velocity of the ring, (b) the length l of the strings, (c) the rate in rad/s at which the spheres were rotating about G .

*14.3 VARIABLE SYSTEMS OF PARTICLES

All of the systems considered so far consisted of well-defined particles. These systems did not gain or lose any particles during their motion. In a large number of engineering applications, however, it is necessary to consider **variable systems of particles**, i.e., systems that are continually gaining or losing particles, or doing both at the same time. Consider, for example, a hydraulic turbine. Its analysis involves determining the forces exerted by a stream of water on rotating blades, and the particles of water in contact with the blades form an ever-changing system that continually acquires and loses particles. Rockets furnish another example of variable systems, since their propulsion depends upon the continual ejection of fuel particles.

To analyze variable systems of particles, we must find a way to reduce the analysis to that of an auxiliary constant system. We indicate the procedure to follow in Secs. 14.3A and 14.3B for two broad categories of applications: a steady stream of particles and a system that is gaining or losing mass.

*14.3A Steady Stream of Particles

Consider a steady stream of particles, such as a stream of water diverted by a fixed vane or a flow of air through a duct or through a blower. In order to determine the resultant of the forces exerted on the particles in contact with the vane, duct, or blower, we isolate these particles and define them to be a system S (Fig. 14.9). Note that S is a variable system of particles, since it continually gains particles flowing in and loses an equal number of particles flowing out. Therefore, the kinetics principles that we have established so far do not apply directly to S .

However, we can easily define an auxiliary system of particles that does remain constant for a short interval of time Δt . Consider at time t the system S plus the particles that will enter S during the interval of time Δt (Fig. 14.10a). Next, consider at time $t + \Delta t$ the system S plus the particles

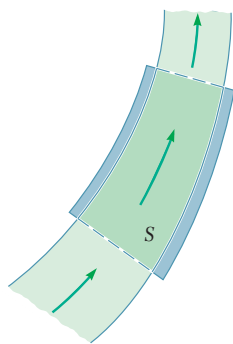


Fig. 14.9 A system of particles in a steady stream.

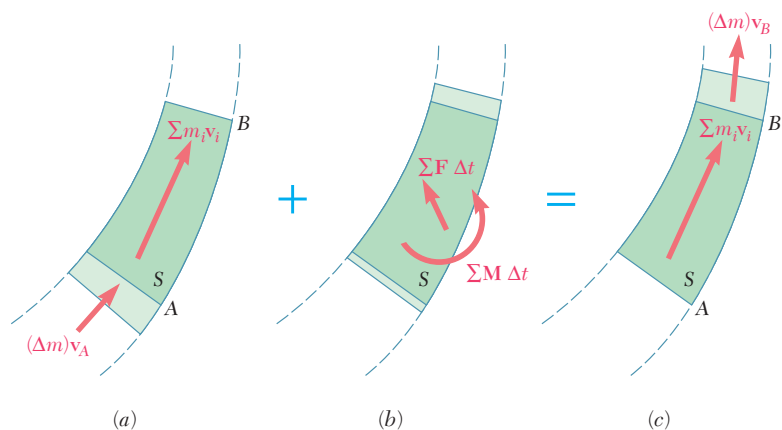


Fig. 14.10 The impulse-momentum diagram for a stream of particles contains (a) momenta of particles entering and in the system S plus (b) impulses during the time interval Δt , and (c) momenta of the particles in and leaving the system.

that have left S during the interval Δt (Fig. 14.10c). Clearly, *the same particles are involved in both cases*, and we can apply the principle of impulse and momentum to those particles. Since the total mass m of the system S remains constant, the particles entering the system and those leaving the system in the time Δt must have the same mass Δm . Suppose we denote the velocities of the particles entering S at A and leaving S at B by $\mathbf{v}_A$ and $\mathbf{v}_B$, respectively. Then we can represent the momentum of the particles entering S by $(\Delta m)\mathbf{v}_A$ (Fig. 14.10a) and the momentum of the particles leaving S by $(\Delta m)\mathbf{v}_B$ (Fig. 14.10c). We also represent the momenta $m_i\mathbf{v}_i$ of the particles forming S and the impulses of the forces exerted on S by appropriate vectors. Then we indicate by blue plus and equals signs that the system of the momenta and impulses in parts a and b of Fig. 14.10 is equipollent to the system of the momenta in part c of the same figure.

The resultant $\Sigma m_i\mathbf{v}_i$ of the momenta of the particles of S is found on both sides of the equals sign and thus can be omitted. We conclude:

The system formed by the momentum $(\Delta m)\mathbf{v}_A$ of the particles entering S in the time Δt and the impulses of the forces exerted on S during that time is equipollent to the momentum $(\Delta m)\mathbf{v}_B$ of the particles leaving S in the same time Δt .

Mathematically, we have

$$(\Delta m)\mathbf{v}_A + \Sigma \mathbf{F} \Delta t = (\Delta m)\mathbf{v}_B \quad (14.38)$$

We can obtain a similar equation by taking the moments of the vectors involved (see Sample Prob. 14.7). Dividing all terms of Eq. (14.38) by Δt and letting Δt approach zero, we obtain at the limit

$$\Sigma \mathbf{F} = \frac{dm}{dt}(\mathbf{v}_B - \mathbf{v}_A) \quad (14.39)$$

where $\mathbf{v}_B - \mathbf{v}_A$ represents the difference between the *vector* $\mathbf{v}_B$ and the *vector* $\mathbf{v}_A$.

If we use SI units, dm/dt is expressed in kg/s and the velocities in m/s; we check that both sides of Eq. (14.39) are expressed in the same units (newtons). If we use U.S. customary units, dm/dt must be expressed in slugs/s and the velocities in ft/s; we check again that both sides of the equation are expressed in the same units (pounds).[†]

We can use this principle to analyze a large number of engineering applications. Let's look at some of the more common of these applications.

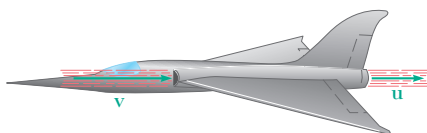
Fluid Stream Diverted by a Vane. If the vane is fixed, we can apply directly the method of analysis given here to find the force $\mathbf{F}$ exerted by the vane on the stream. Note that $\mathbf{F}$ is the only force we need to consider, because the pressure in the stream is constant (atmospheric pressure). The force exerted by the stream on the vane is equal and opposite to $\mathbf{F}$.

[†]It is often convenient to express the mass rate of flow dm/dt as the product ρQ , where ρ is the density of the stream (mass per unit volume) and Q is its volume rate of flow (volume per unit time). If you use SI units, ρ is in kg/m^3 (for instance, $\rho = 1000 \text{ kg/m}^3$ for water) and Q is in m^3/s . However, if you use U.S. customary units, ρ generally has to be computed from the corresponding specific weight γ (weight per unit volume), $\rho = \gamma/g$. Since γ is expressed in lb/ft^3 (for instance, $\gamma = 62.4 \text{ lb/ft}^3$ for water), we obtain ρ in slug/ft^3 . The volume rate of flow Q is expressed in ft^3/s .

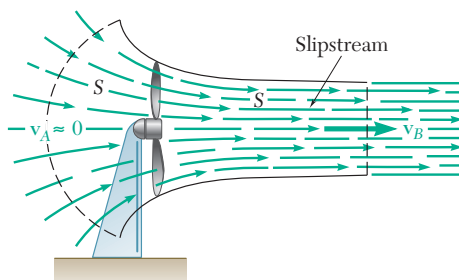
If the vane moves with a constant velocity, the stream is not steady. However, it will appear steady to an observer moving with the vane. We should therefore choose a system of axes moving with the vane. Since this system of axes is not accelerated, we can still use Eq. (14.38), but we must replace $\mathbf{v}_A$ and $\mathbf{v}_B$ by the *relative velocities* of the stream with respect to the vane (see Sample Prob. 14.8).

Fluid Flowing Through a Pipe. We can determine the force exerted by the fluid on a pipe transition, such as a bend or a contraction, by considering the system of particles S in contact with the transition. Since, in general, the pressure in the flow will vary, we should also consider the forces exerted on S by the adjoining portions of the fluid.

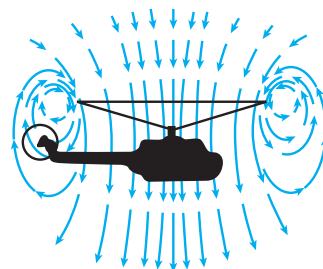
Jet Engine. In a jet engine, air enters the front of the engine with no velocity and leaves through the rear with a high velocity. The energy required to accelerate the air particles is obtained by burning fuel. The mass of the burned fuel in the exhaust gases is usually small enough compared with the mass of the air flowing through the engine that it can be neglected. Thus, the analysis of a jet engine reduces to that of an airstream. We can consider this stream as a steady stream if we measure all velocities with respect to the airplane. We assume, therefore, that the airstream enters the engine with a velocity $\mathbf{v}$ of magnitude equal to the speed of the airplane and leaves with a velocity $\mathbf{u}$ equal to the relative velocity of the exhaust gases (Fig. 14.11a). Since the intake and exhaust pressures are nearly atmospheric, the only external force we need to



(a) Jet plane



(b) Fan



(c) Helicopter

Fig. 14.11 Applications of a steady stream of particles.

consider is the force exerted by the engine on the airstream. This force is equal and opposite to the thrust.[†]

Fan. Consider the system of particles S shown in Fig. 14.11*b*. We assume the velocity $\mathbf{v}_A$ of the particles entering the system is equal to zero, and the velocity $\mathbf{v}_B$ of the particles leaving the system is the velocity of the *slipstream*. We can obtain the rate of flow by multiplying v_B by the cross-sectional area of the slipstream. Since the pressure all around S is atmospheric, the only external force acting on S is the thrust of the fan.

Helicopter. Determining the thrust created by the rotating blades of a hovering helicopter is similar to the determination of the thrust of a fan (Fig. 14.11*c*). We assume the velocity $\mathbf{v}_A$ of the air particles as they approach the blades is zero, and we obtain the rate of flow by multiplying the magnitude of the velocity $\mathbf{v}_B$ of the slipstream by its cross-sectional area.

*14.3B Systems Gaining or Losing Mass

Let us now analyze a different type of variable system of particles; namely, a system that gains mass by continually absorbing particles or loses mass by continually expelling particles. Consider the system S shown in Fig. 14.12. Its mass, equal to m at the instant t , increases by Δm in the time interval Δt . In order to apply the principle of impulse and momentum to this system, we must consider at time t the system S *plus* the particles of mass Δm that S absorbs during the time interval Δt . The velocity of S at time t is denoted by $\mathbf{v}$, the velocity of S at time $t + \Delta t$ is denoted by $\mathbf{v} + \Delta \mathbf{v}$, and the absolute velocity of the particles absorbed is denoted by $\mathbf{v}_a$. Applying the principle of impulse and momentum, we have

$$m\mathbf{v} + (\Delta m)\mathbf{v}_a + \Sigma \mathbf{F} \Delta t = (m + \Delta m)(\mathbf{v} + \Delta \mathbf{v}) \quad (14.40)$$

Solving for the sum $\Sigma \mathbf{F} \Delta t$ of the impulses of the external forces acting on S (excluding the forces exerted by the particles being absorbed), we obtain

$$\Sigma \mathbf{F} \Delta t = m\Delta \mathbf{v} + \Delta m(\mathbf{v} - \mathbf{v}_a) + (\Delta m)(\Delta \mathbf{v}) \quad (14.41)$$

Now we introduce the *relative velocity* $\mathbf{u}$ with respect to S of the particles that are absorbed. We have $\mathbf{u} = \mathbf{v}_a - \mathbf{v}$ and note, since $v_a < v$, that the relative velocity $\mathbf{u}$ is directed to the left, as shown in Fig. 14.12. Neglecting the last term in Eq. (14.41), which is of the second order, we have

$$\Sigma \mathbf{F} \Delta t = m \Delta \mathbf{v} - (\Delta m)\mathbf{u}$$

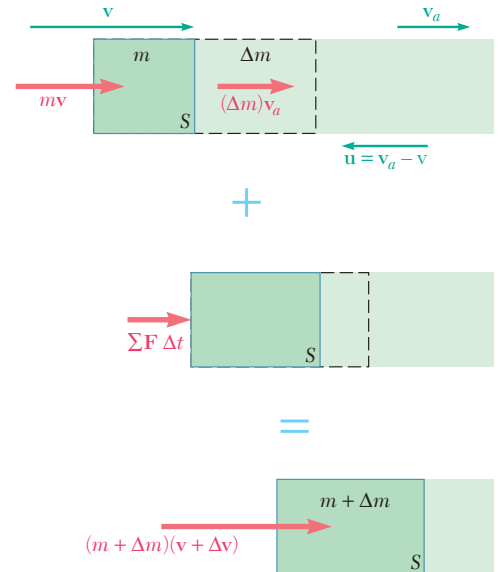


Fig. 14.12 Impulse-momentum diagram for a system that gains mass.

[†]Note that if the airplane is accelerating, we cannot use it as a newtonian frame of reference. However, we can obtain the same result for the thrust by using a reference frame at rest with respect to the atmosphere. In this frame, the air particles enter the engine with no velocity and leave it with a velocity of magnitude $u - v$.



Photo 14.3 As booster rockets are fired, the gas particles they eject provide the thrust required for liftoff.

Now we divide through by Δt and let Δt approach zero. In the limit we obtain[†]

$$\Sigma \mathbf{F} = m \frac{d\mathbf{v}}{dt} - \frac{dm}{dt} \mathbf{u} \quad (14.42)$$

Rearranging terms and recalling that $d\mathbf{v}/dt = \mathbf{a}$, where $\mathbf{a}$ is the acceleration of the system S , we have

$$\Sigma \mathbf{F} + \frac{dm}{dt} \mathbf{u} = m\mathbf{a} \quad (14.43)$$

This equation states that the action on S of the particles being absorbed is equivalent to a thrust

$$\mathbf{P} = \frac{dm}{dt} \mathbf{u} \quad (14.44)$$

that tends to slow down the motion of S , since the relative velocity $\mathbf{u}$ of the particles is directed to the left. If we use SI units, dm/dt is expressed in kg/s, the relative velocity u is in m/s, and the corresponding thrust is in newtons. If we use U.S. customary units, dm/dt must be expressed in slug/s, u in ft/s, and the corresponding thrust in pounds.

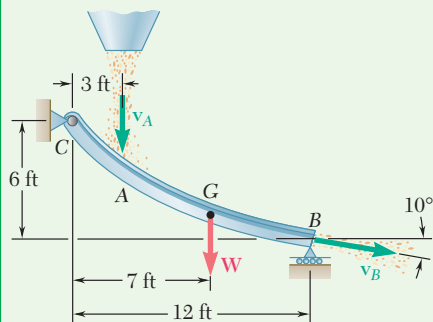
We can also use these equations to determine the motion of a system S losing mass. In this case, the rate of change of mass is negative, and the action on S of the particles being expelled is equivalent to a thrust in the direction of $-\mathbf{u}$; that is, in the direction opposite to that in which the particles are being expelled. A *rocket* represents a typical case of a system continually losing mass (see Sample Prob. 14.9).

[†]When the absolute velocity $\mathbf{v}_a$ of the particles absorbed is zero, $\mathbf{u} = -\mathbf{v}$ and formula (14.42) becomes

$$\Sigma \mathbf{F} = \frac{d}{dt}(m\mathbf{v})$$

Comparing this formula with Eq. (12.3) of Sec. 12.1B, we see that this is Newton's second law applied to a system gaining mass, *provided that the particles absorbed are initially at rest*. We can also apply it to a system losing mass, *provided that the velocity of the particles expelled is zero* with respect to the chosen frame of reference.

Sample Problem 14.7



Grain falls from a hopper onto a chute CB at the rate of 240 lb/s. It hits the chute at A with a velocity of 20 ft/s and leaves at B with a velocity of 15 ft/s, forming an angle of 10° with the horizontal. Knowing that the combined weight of the chute and of the grain it supports is a force $\mathbf{W}$ with a magnitude of 600 lb applied at G , determine the reaction at the roller support B and the components of the reaction at the hinge C .

STRATEGY: Since we have a steady stream of particles, apply the principle of impulse and momentum for the time interval Δt .

MODELING: Choose a system that consists of the chute, the grain it supports, and the amount of grain that hits the chute in the interval Δt . The impulse-momentum diagram for this system is shown in Fig. 1. Since the chute does not move, it has no momentum. Note that the sum $\sum m_i \mathbf{v}_i$ of the momenta of the particles supported by the chute is the same at t and $t + \Delta t$ and thus can be omitted.

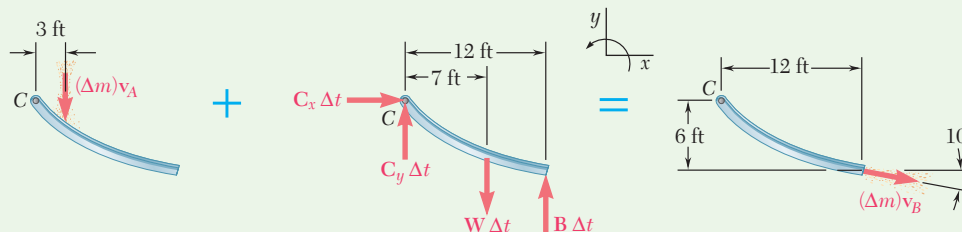


Fig. 1 Impulse-momentum diagram for the system.

ANALYSIS: You can use the impulse-momentum diagram to obtain scalar equations for the x and y directions and for moments about point C .

$$\rightarrow x \text{ components:} \quad C_x \Delta t = (\Delta m)v_B \cos 10^\circ \quad (1)$$

$$+\uparrow y \text{ components:} \quad -(\Delta m)v_A + C_y \Delta t - W \Delta t + B \Delta t = -(\Delta m)v_B \sin 10^\circ \quad (2)$$

$$+\curvearrowright \text{ moments about } C: \quad -3(\Delta m)v_A - 7(W \Delta t) + 12(B \Delta t) = 6(\Delta m)v_B \cos 10^\circ - 12(\Delta m)v_B \sin 10^\circ \quad (3)$$

Using the given data, $W = 600$ lb, $v_A = 20$ ft/s, $v_B = 15$ ft/s, and $\Delta m/\Delta t = 240/32.2 = 7.45$ slug/s, and solving Eq. (3) for B and Eq. (1) for C_x , you obtain

$$\begin{aligned} 12B &= 7(600) + 3(7.45)(20) + 6(7.45)(15)(\cos 10^\circ - 2 \sin 10^\circ) \\ 12B &= 5075 \quad B = 423 \text{ lb} \quad \mathbf{B = 423 \text{ lb} \uparrow} \end{aligned}$$

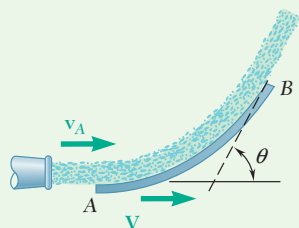
$$C_x = (7.45)(15) \cos 10^\circ = 110.1 \text{ lb} \quad \mathbf{C_x = 110.1 \text{ lb} \rightarrow}$$

Substituting for B and solving Eq. (2) for C_y , you end up with

$$C_y = 600 - 423 + (7.45)(20 - 15 \sin 10^\circ) = 307 \text{ lb} \quad \mathbf{C_y = 307 \text{ lb} \uparrow}$$

REFLECT and THINK: This kind of situation is common in factory and storage settings. Being able to determine the reactions is essential for designing a proper chute that will support the stream safely. We can compare this situation to the case when there is no mass flow, which results in reactions of $B_y = 350$ lb, $C_y = 250$ lb, and $C_x = 0$ lb.

Sample Problem 14.8



A nozzle discharges a stream of water of cross-sectional area A with a velocity $\mathbf{v}_A$. The stream is deflected by a *single* blade that moves to the right with a constant velocity $\mathbf{V}$. Assuming that the water moves along the blade at constant speed, determine (a) the components of the force $\mathbf{F}$ exerted by the blade on the stream, (b) the velocity $\mathbf{V}$ for which maximum power is developed.

STRATEGY: Since you have a steady stream of particles, apply the principle of impulse and momentum.

MODELING: Choose the system to be the particles in contact with the blade and the particles striking the blade in the time Δt , and use a coordinate system that moves with the blade at a constant velocity $\mathbf{V}$. The particles of water strike the blade with a relative velocity $\mathbf{u}_A = \mathbf{v}_A - \mathbf{V}$ and leave the blade with a relative velocity $\mathbf{u}_B$, as shown in Fig. 1. Since the particles move along the blade at a constant speed, the relative velocities $\mathbf{u}_A$ and $\mathbf{u}_B$ have the same magnitude u . Denoting the density of water by ρ , the mass of the particles striking the blade during the time interval Δt is $\Delta m = A\rho(v_A - V)\Delta t$; an equal mass of particles leaves the blade during Δt . The impulse-momentum diagram for this system is shown in Fig. 2.

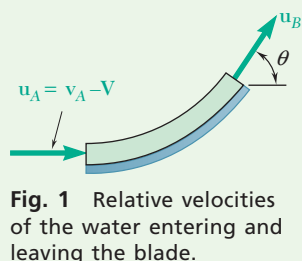


Fig. 1 Relative velocities of the water entering and leaving the blade.

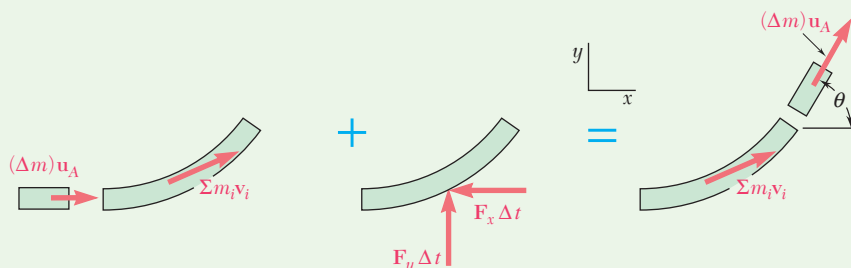


Fig. 2 Impulse-momentum diagram for the system.

ANALYSIS:

a. Components of Force Exerted on Stream. Recalling that $\mathbf{u}_A$ and $\mathbf{u}_B$ have the same magnitude u and omitting the momentum $\sum m_i \mathbf{v}_i$ that appears on both sides, applying the principle of impulse and momentum gives you

$$\pm x \text{ components:} \quad (\Delta m)u - F_x \Delta t = (\Delta m)u \cos \theta$$

$$+\uparrow y \text{ components:} \quad +F_y \Delta t = (\Delta m)u \sin \theta$$

Substituting $\Delta m = A\rho(v_A - V)\Delta t$ and $u = v_A - V$, you obtain

$$\mathbf{F}_x = A\rho(v_A - V)^2(1 - \cos \theta) \leftarrow \quad \mathbf{F}_y = A\rho(v_A - V)^2 \sin \theta \uparrow \blacktriangleleft$$

b. Velocity of Blade for Maximum Power. You can obtain the power by multiplying the velocity V of the blade by the component F_x of the force exerted by the stream on the blade.

$$\text{Power} = F_x V = A\rho(v_A - V)^2(1 - \cos \theta)V$$

Differentiating the power with respect to V and setting the derivative equal to zero, you have

$$\frac{d(\text{power})}{dV} = A\rho(v_A^2 - 4v_A V + 3V^2)(1 - \cos \theta) = 0$$

$$V = v_A \quad V = \frac{1}{3}v_A \quad \text{For maximum power } \mathbf{V} = \frac{1}{3}v_A \rightarrow \blacktriangleleft$$

REFLECT and THINK: These results are valid only when a *single* blade deflects the stream. Different results appear when a series of blades deflects the stream, as in a Pelton-wheel turbine (see Prob. 14.81).

Sample Problem 14.9



A rocket of initial mass m_0 (including shell and fuel) is fired vertically at time $t = 0$. The fuel is consumed at a constant rate $q = dm/dt$ and is expelled at a constant speed u relative to the rocket. Derive an expression for the magnitude of the velocity of the rocket at time t , neglecting the resistance of the air.

STRATEGY: Since you have a system that is losing mass, apply the principle of impulse and momentum. This gives you an equation you can integrate to obtain the velocity.

MODELING: Choose the rocket shell and its fuel as your system. At time t , the mass of the rocket shell and remaining fuel is $m = m_0 - qt$, and the velocity is $\mathbf{v}$. During the time interval Δt , a mass of fuel $\Delta m = q \Delta t$ is expelled with a speed u relative to the rocket. The impulse–momentum diagram for this system is shown in Fig. 1, where $\mathbf{v}_e$ is the absolute velocity of the expelled fuel.

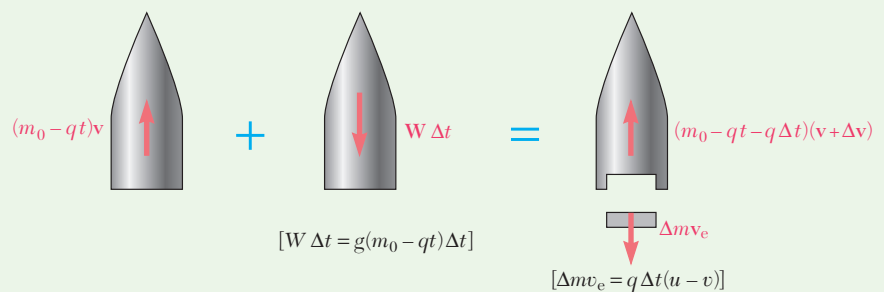


Fig. 1 Impulse–momentum diagram for the system.

(continued)

ANALYSIS: Apply the principle of impulse and momentum between time t and time $t + \Delta t$ to find

$$(m_0 - qt)v - g(m_0 - qt) \Delta t = (m_0 - qt - q \Delta t)(v + \Delta v) - q \Delta t(u - v)$$

Divide through by Δt and let Δt approach zero for

$$-g(m_0 - qt) = (m_0 - qt) \frac{dv}{dt} - qu$$

Separating variables and integrating from $t = 0, v = 0$ to $t = t, v = v$, you have

$$dv = \left(\frac{qu}{m_0 - qt} - g \right) dt$$

$$\int_0^v dv = \int_0^t \left(\frac{qu}{m_0 - qt} - g \right) dt$$

$$v = [-u \ln(m_0 - qt) - gt]_0^t \quad v = u \ln \frac{m_0}{m_0 - qt} - gt \quad \blacktriangleleft$$

REFLECT and THINK: The mass remaining at time t_f , after all of the fuel has been expended, is equal to the mass of the rocket shell $m_s = m_0 - qt_f$, and the maximum velocity attained by the rocket is $v_m = u \ln(m_0/m_s) - gt_f$. Assuming that the fuel is expelled in a relatively short period of time, the term gt_f is small, and we have $v_m \approx u \ln(m_0/m_s)$. In order to escape the gravitational field of Earth, a rocket must reach a velocity of 11.18 km/s. Assuming $u = 2200$ m/s and $v_m = 11.18$ km/s, we obtain $m_0/m_s = 161$. Thus, to project each kilogram of the rocket shell into space, it is necessary to consume more than 161 kg of fuel if we use a propellant yielding $u = 2200$ m/s.

SOLVING PROBLEMS ON YOUR OWN

This section was devoted to the motion of **variable systems of particles**, i.e., systems that are continually *gaining or losing particles* or doing both at the same time. The problems you will be asked to solve will involve (1) **steady streams of particles** and (2) **systems gaining or losing mass**.

1. To solve problems involving a steady stream of particles [Sample Probs. 14.7 and 14.8], consider a portion S of the stream and express mathematically that the system formed by the momentum of the particles entering S at A in the time Δt and that the impulses of the forces exerted on S during that time is equipollent to the momentum of the particles leaving S at B in the same time Δt (Fig. 14.10). Considering only the resultants of the vector systems involved, you can write the vector equation

$$(\Delta m)\mathbf{v}_A + \Sigma \mathbf{F} \Delta t = (\Delta m)\mathbf{v}_B \quad (14.38)$$

You may also want to consider the angular momentum of the particle systems to obtain an additional equation [Sample Prob. 14.7]. However, many problems can be solved using Eq. (14.38) or the equation obtained by dividing all terms by Δt and letting Δt approach zero,

$$\Sigma \mathbf{F} = \frac{dm}{dt}(\mathbf{v}_B - \mathbf{v}_A) \quad (14.39)$$

Here $\mathbf{v}_B - \mathbf{v}_A$ represents a *vector subtraction*, and the mass rate of flow dm/dt can be expressed as the product ρQ of the density ρ of the stream (mass per unit volume) and the volume rate of flow Q (volume per unit time). In U.S. customary units, ρ is expressed as the ratio γ/g , where γ is the specific weight of the stream and g is the acceleration due to gravity.

Typical problems involving a steady stream of particles have been described in Sec. 14.3A. You may be asked to determine the following,

a. Thrust caused by a diverted flow. Equation (14.39) is applicable, but you will get a better understanding of the problem if you use a solution based on Eq. (14.38).

b. Reactions at supports of vanes or conveyor belts. First draw a diagram showing on one side of the equals sign the momentum $(\Delta m)\mathbf{v}_A$ of the particles impacting the vane or belt in the time Δt , as well as the impulses of the loads and reactions at the supports during that time. On the other side, show the momentum $(\Delta m)\mathbf{v}_B$ of the particles leaving the vane or belt in the time Δt [Sample Prob. 14.7]. Equating the x components, y components, and moments of the quantities on both sides of the equals sign will yield three scalar equations that you can solve for three unknowns.

c. Thrust developed by a jet engine, a propeller, or a fan. In most cases, a single unknown is involved, and you can obtain that unknown by solving the scalar equation derived from Eq. (14.38) or Eq. (14.39).

2. To solve problems involving systems gaining mass, consider the system S , which has a mass m and is moving with a velocity $\mathbf{v}$ at time t , and the particles of mass Δm with velocity $\mathbf{v}_a$ that S absorbs in the time interval Δt (Fig. 14.12). You will then express that the total momentum of S and of the particles absorbed, *plus* the impulse of the external forces exerted on S , are equipollent to the momentum of S at time $t + \Delta t$. Noting that the mass of S and its velocity at that time are, respectively, $m + \Delta m$ and $\mathbf{v} + \Delta \mathbf{v}$, you will write the vector equation

$$m\mathbf{v} + (\Delta m)\mathbf{v}_a + \Sigma \mathbf{F} \Delta t = (m + \Delta m)(\mathbf{v} + \Delta \mathbf{v}) \quad (14.40)$$

As we showed in Sec. 14.3B, if you introduce the relative velocity $\mathbf{u} = \mathbf{v}_a - \mathbf{v}$ of the particles being absorbed, you obtain the following expression for the resultant of the external forces applied to S as

$$\Sigma \mathbf{F} = m \frac{d\mathbf{v}}{dt} - \frac{dm}{dt} \mathbf{u} \quad (14.42)$$

Furthermore, the action on S of the particles being absorbed is equivalent to a thrust

$$\mathbf{P} = \frac{dm}{dt} \mathbf{u} \quad (14.44)$$

exerted in the direction of the relative velocity of the particles being absorbed.

Examples of systems gaining mass are conveyor belts, moving railroad cars being loaded with gravel or sand, and chains being pulled out of a pile.

3. To solve problems involving systems losing mass, such as rockets and rocket engines, you can use Eqs. (14.40) through (14.44)—provided that you give negative values to the increment of mass Δm and to the rate of change of mass dm/dt [Sample Prob. 14.9]. It follows that the thrust defined by Eq. (14.44) is exerted in a direction opposite to the direction of the relative velocity of the particles being ejected.

Problems

- 14.57** A stream of water with a density of $\rho = 1000 \text{ kg/m}^3$ is discharged from a nozzle at the rate of $0.06 \text{ m}^3/\text{s}$. Using Bernoulli's equation, the gage pressure P in the pipe just upstream from the nozzle is $P = 0.5 \rho (v_2^2 - v_1^2)$. Knowing the nozzle is held to the pipe by six flange bolts, determine the tension in each bolt, neglecting the initial tension caused by the tightening of the nuts.

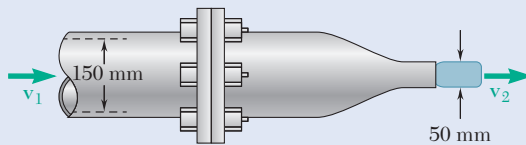


Fig. P14.57

- 14.58** A jet ski is placed in a channel and is tethered so that it is stationary. Water enters the jet ski with velocity $\mathbf{v}_1$ and exits with velocity $\mathbf{v}_2$. Knowing the inlet area is A_1 and the exit area is A_2 , determine the tension in the tether.

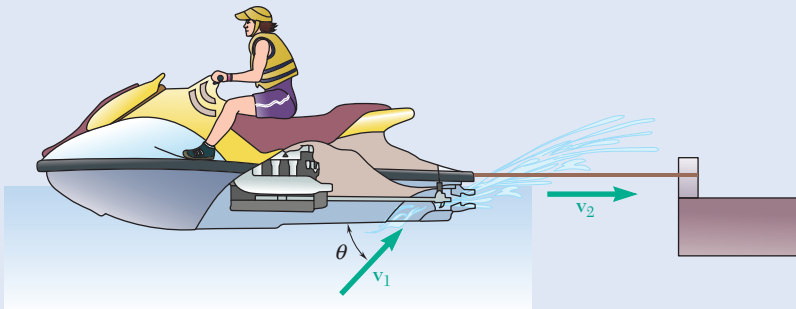
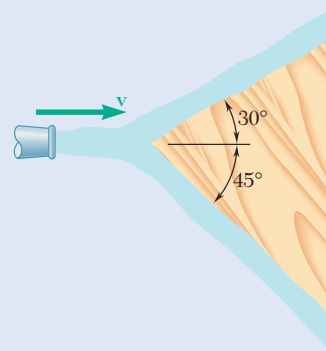


Fig. P14.58

- 14.59** The nozzle shown discharges a stream of water at a flow rate of $Q = 475 \text{ gal/min}$ with a velocity $\mathbf{v}$ and a magnitude of 60 ft/s . The stream is split into two streams with equal flow rates by a wedge that is kept in a fixed position. Determine the components (drag and lift) of the force exerted by the stream on the wedge. (Note: $1 \text{ ft}^3 = 7.48 \text{ gal.}$)



- 14.60** The nozzle shown discharges a stream of water at a flow rate of $Q = 500 \text{ gal/min}$ with a velocity $\mathbf{v}$ and a magnitude of 48 ft/s . The stream is split into two streams of equal flow rates by a wedge that is moving to the left at a constant speed of 12 ft/s . Determine the components (drag and lift) of the force exerted by the stream on the wedge. (Note: $1 \text{ ft}^3 = 7.48 \text{ gal.}$)

Fig. P14.59 and P14.60

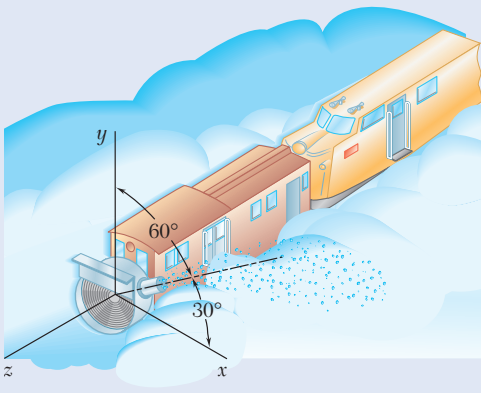


Fig. P14.61

14.61 A rotary power plow is used to remove snow from a level section of railroad track. The plow car is placed ahead of an engine that propels it at a constant speed of 20 km/h. The plow car clears 160 Mg of snow per minute, projecting it in the direction shown with a velocity of 12 m/s relative to the plow car. Neglecting friction, determine (a) the force exerted by the engine on the plow car, (b) the lateral force exerted by the track on the plow car.

14.62 Tree limbs and branches are being fed at A at the rate of 5 kg/s into a shredder which spews the resulting wood chips at C with a velocity of 20 m/s. Determine the horizontal component of the force exerted by the shredder on the truck hitch at D.

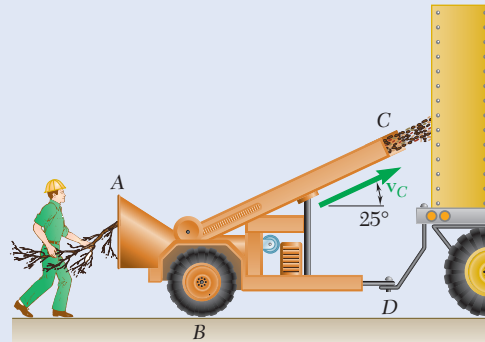


Fig. P14.62

14.63 Sand falls from three hoppers onto a conveyor belt at a rate of 90 lb/s for each hopper. The sand hits the belt with a vertical velocity $v_1 = 10$ ft/s and is discharged at A with a horizontal velocity $v_2 = 13$ ft/s. Knowing that the combined mass of the beam, belt system, and the sand it supports is 1300 lb with a mass center at G, determine the reaction at E.

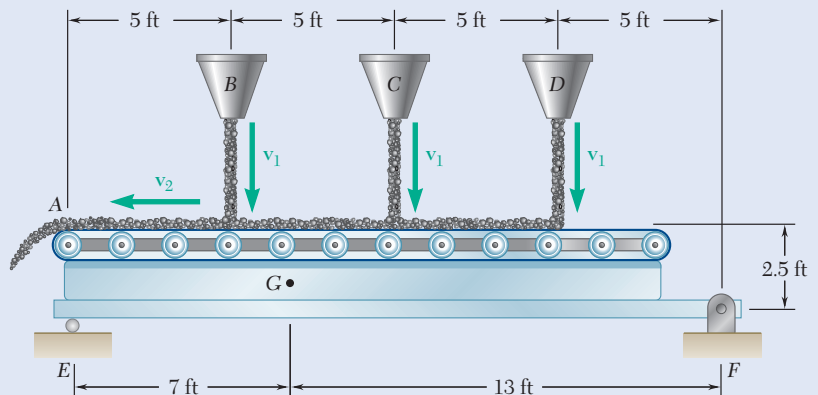


Fig. P14.63

14.64 The stream of water shown flows at a rate of 550 L/min and moves with a velocity of magnitude 18 m/s at both A and B. The vane is supported by a pin and bracket at C and by a load cell at D that can exert only a horizontal force. Neglecting the weight of the vane, determine the components of the reactions at C and D.

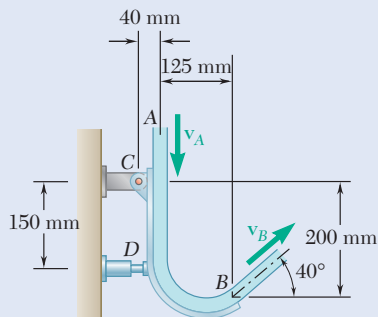


Fig. P14.64

14.65 The nozzle shown discharges water at the rate of $40 \text{ ft}^3/\text{min}$. Knowing that at both A and B the stream of water moves with a velocity of magnitude 75 ft/s and neglecting the weight of the vane, determine the components of the reactions at C and D .

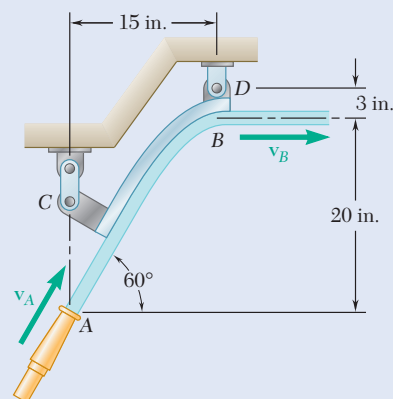


Fig. P14.65

14.66 A stream of water flowing at a rate of $1.2 \text{ m}^3/\text{min}$ and moving with a speed of 30 m/s at both A and B is deflected by a vane welded to a hinged plate. Knowing that the combined mass of the vane and plate is 20 kg with the mass center at point G , determine (a) the angle θ , (b) the reaction at C .

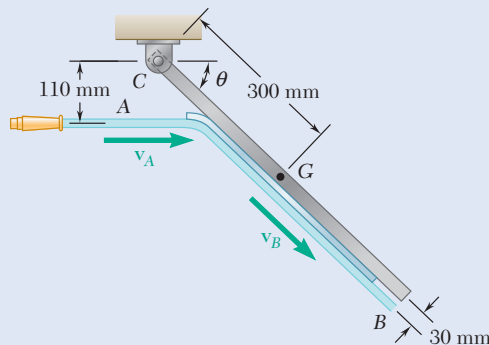


Fig. P14.66 and P14.67

14.67 A stream of water flowing at a rate of $1.2 \text{ m}^3/\text{min}$ and moving with a speed of v at both A and B is deflected by a vane welded to a hinged plate. The combined mass of the vane and plate is 20 kg with the mass center at point G . Knowing that $\theta = 45^\circ$, determine (a) the speed v of the flow, (b) the reaction at C .

14.68 Coal is being discharged from a first conveyor belt at the rate of 120 kg/s . It is received at A by a second belt that discharges it again at B . Knowing that $v_1 = 3 \text{ m/s}$ and $v_2 = 4.25 \text{ m/s}$ and that the second belt assembly and the coal it supports have a total mass of 472 kg , determine the components of the reactions at C and D .

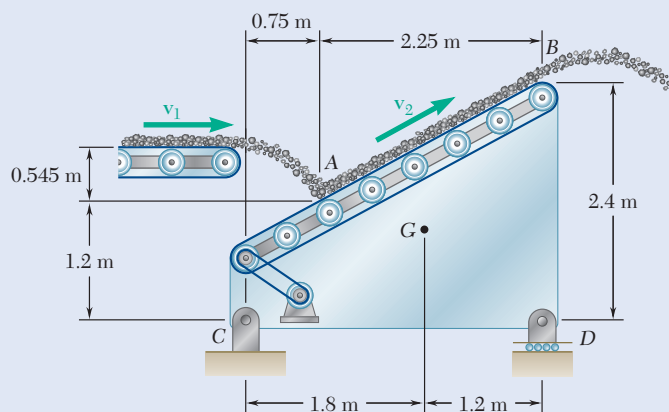


Fig. P14.68

14.69 The total drag due to air friction on a jet airplane traveling at 900 km/h is 35 kN . Knowing that the exhaust velocity is 600 m/s relative to the airplane, determine the mass of air that must pass through the engine per second to maintain the speed of 900 km/h in level flight.

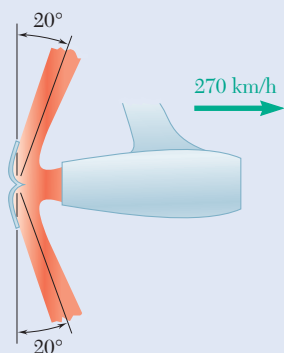


Fig. P14.71

14.70 While cruising in level flight at a speed of 600 mi/h, a jet plane scoops in air at the rate of 200 lb/s and discharges it with a velocity of 2100 ft/s relative to the airplane. Determine the total drag due to air friction on the airplane.

14.71 In order to shorten the distance required for landing, a jet airplane is equipped with movable vanes that partially reverse the direction of the air discharged by each of its engines. Each engine scoops in the air at a rate of 120 kg/s and discharges it with a velocity of 600 m/s relative to the engine. At an instant when the speed of the airplane is 270 km/h, determine the reverse thrust provided by each of the engines.

14.72 The helicopter shown can produce a maximum downward air speed of 80 ft/s in a 30-ft-diameter slipstream. Knowing that the weight of the helicopter and its crew is 3500 lb and assuming $\gamma = 0.076 \text{ lb/ft}^3$ for air, determine the maximum load that the helicopter can lift while hovering in midair.

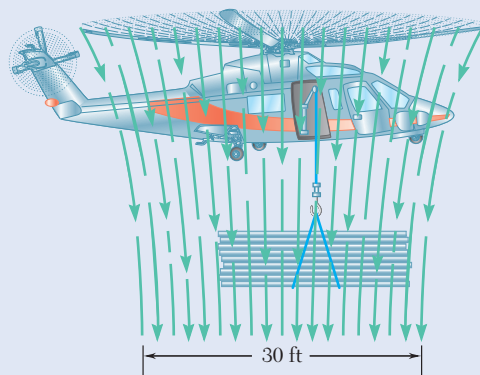


Fig. P14.72

14.73 Prior to takeoff, the pilot of a 3000-kg twin-engine airplane tests the reversible-pitch propellers by increasing the reverse thrust with the brakes at point *B* locked. Knowing that point *G* is the center of gravity of the airplane, determine the velocity of the air in the two 2.2-m-diameter slipstreams when the nose wheel *A* begins to lift off the ground. Assume $\rho = 1.21 \text{ kg/m}^3$ and neglect the approach velocity of the air.

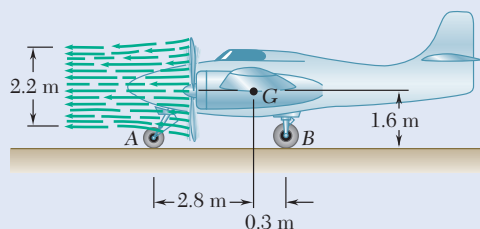


Fig. P14.73

14.74 The jet engine shown scoops in air at *A* at a rate of 200 lb/s and discharges it at *B* with a velocity of 2000 ft/s relative to the airplane. Determine the magnitude and line of action of the propulsive thrust developed by the engine when the speed of the airplane is (a) 300 mi/h, (b) 600 mi/h.

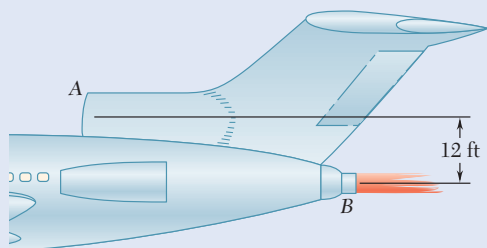


Fig. P14.74

14.75 A jet airliner is cruising at a speed of 900 km/h with each of its three engines discharging air with a velocity of 800 m/s relative to the plane. Determine the speed of the airliner after it has lost the use of (a) one of its engines, (b) two of its engines. Assume that the drag due to air friction is proportional to the square of the speed and that the remaining engines keep operating at the same rate.

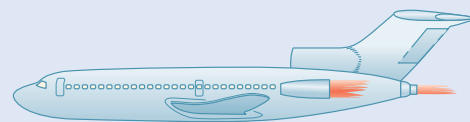


Fig. P14.75

14.76 A 16-Mg jet airplane maintains a constant speed of 774 km/h while climbing at an angle $\alpha = 18^\circ$. The airplane scoops in air at a rate of 300 kg/s and discharges it with a velocity of 665 m/s relative to the airplane. If the pilot changes to a horizontal flight while maintaining the same engine setting, determine (a) the initial acceleration of the plane, (b) the maximum horizontal speed that will be attained. Assume that the drag due to air friction is proportional to the square of the speed.

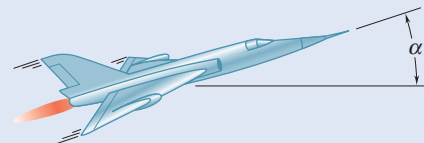


Fig. P14.76

14.77 The propeller of a small airplane has a 2-m-diameter slipstream and produces a thrust of 3600 N when the airplane is at rest on the ground. Assuming $\rho = 1.225 \text{ kg/m}^3$ for air, determine (a) the speed of the air in the slipstream, (b) the volume of air passing through the propeller per second, (c) the kinetic energy imparted per second to the air in the slipstream.

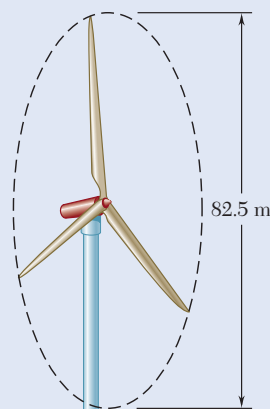


Fig. P14.78 and P14.79

14.78 The wind turbine generator shown has an output-power rating of 1.5 MW for a wind speed of 36 km/h. For the given wind speed, determine (a) the kinetic energy of the air particles entering the 82.5-m-diameter circle per second, (b) the efficiency of this energy conversion system. Assume $\rho = 1.21 \text{ kg/m}^3$ for air.

14.79 A wind turbine generator system having a diameter of 82.5 m produces 1.5 MW at a wind speed of 12 m/s. Determine the diameter of blade necessary to produce 10 MW of power assuming the efficiency is the same for both designs and $\rho = 1.21 \text{ kg/m}^3$ for air.

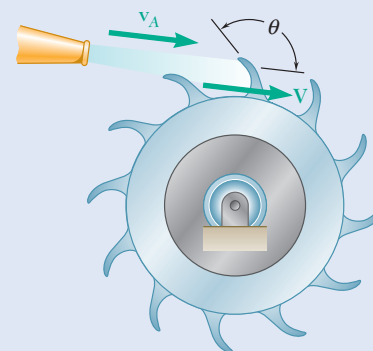


Fig. P14.81

14.80 While cruising in level flight at a speed of 570 mi/h, a jet airplane scoops in air at a rate of 240 lb/s and discharges it with a velocity of 2200 ft/s relative to the airplane. Determine (a) the power actually used to propel the airplane, (b) the total power developed by the engine, (c) the mechanical efficiency of the airplane.

14.81 In a Pelton-wheel turbine, a stream of water is deflected by a series of blades so that the rate at which water is deflected by the blades is equal to the rate at which water issues from the nozzle ($\Delta m/\Delta t = A\rho v_A$). Using the same notation as in Sample Prob. 14.8, (a) determine the velocity V of the blades for which maximum power is developed, (b) derive an expression for the maximum power, (c) derive an expression for the mechanical efficiency.

14.82 A circular reentrant orifice (also called Borda's mouthpiece) of diameter D is placed at a depth h below the surface of a tank. Knowing that the speed of the issuing stream is $v = \sqrt{2gh}$ and assuming that the speed of approach v_1 is zero, show that the diameter of the stream is $d = D/\sqrt{2}$. (Hint: Consider the section of water indicated, and note that P is equal to the pressure at a depth h multiplied by the area of the orifice.)

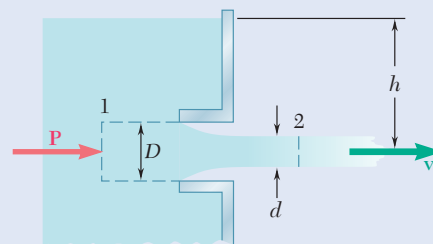


Fig. P14.82

- 14.83** A railroad car with length L and mass m_0 when empty is moving freely on a horizontal track while being loaded with sand from a stationary chute at a rate $dm/dt = q$. Knowing that the car was approaching the chute at a speed v_0 , determine (a) the mass of the car and its load after the car has cleared the chute, (b) the speed of the car at that time.

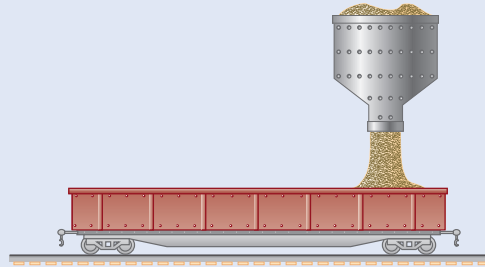


Fig. P14.83

- *14.84** The depth of water flowing in a rectangular channel of width b at a speed v_1 and a depth d_1 increases to a depth d_2 at a *hydraulic jump*. Express the rate of flow Q in terms of b , d_1 , and d_2 .



Fig. P14.84

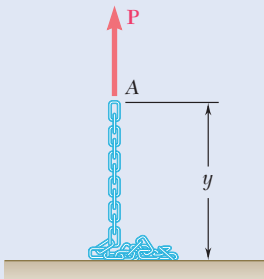


Fig. P14.86

- *14.85** Determine the rate of flow in the channel of Prob. 14.84, knowing that $b = 12$ ft, $d_1 = 4$ ft, and $d_2 = 5$ ft.
- 14.86** A chain of length l and mass m lies in a pile on the floor. If its end A is raised vertically at a constant speed v , express in terms of the length y of chain that is off the floor at any given instant (a) the magnitude of the force $\mathbf{P}$ applied to A , (b) the reaction of the floor.
- 14.87** Solve Prob. 14.86, assuming that the chain is being *lowered* to the floor at a constant speed v .
- 14.88** The ends of a chain lie in piles at A and C . When released from rest at time $t = 0$, the chain moves over the pulley at B , which has a negligible mass. Denoting by L the length of chain connecting the two piles and neglecting friction, determine the speed v of the chain at time t .

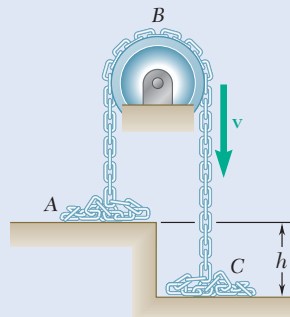


Fig. P14.88

- 14.89** A toy car is propelled by water that squirts from an internal tank at a constant 6 ft/s relative to the car. The weight of the empty car is 0.4 lb and it holds 2 lb of water. Neglecting other tangential forces, determine the top speed of the car.

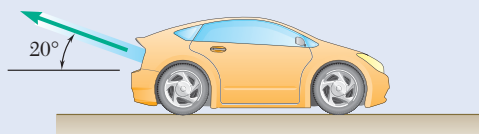


Fig. P14.89 and P14.90

- 14.90** A toy car is propelled by water that squirts from an internal tank. The weight of the empty car is 0.4 lb and it holds 2 lb of water. Knowing the top speed of the car is 8 ft/s, determine the relative velocity of the water that is being ejected.

- 14.91** The main propulsion system of a space shuttle consists of three identical rocket engines that provide a total thrust of 6 MN. Determine the rate at which the hydrogen-oxygen propellant is burned by each of the three engines, knowing that it is ejected with a relative velocity of 3750 m/s.

- 14.92** The main propulsion system of a space shuttle consists of three identical rocket engines, each of which burns the hydrogen-oxygen propellant at the rate of 750 lb/s and ejects it with a relative velocity of 12,000 ft/s. Determine the total thrust provided by the three engines.

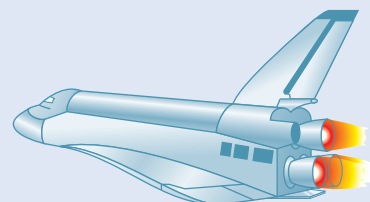


Fig. P14.91 and P14.92

- 14.93** A rocket sled burns fuel at the constant rate of 120 lb/s. The initial weight of the sled is 1800 lb, including 360 lb of fuel. Assume that the track is lubricated and the sled is aerodynamically designed so that air resistance and friction are negligible. (a) Derive a formula for the acceleration a of the sled as a function of time t and the exhaust velocity v_{ex} of the burned fuel relative to the sled. Plot the ratio a/v_{ex} versus time t for the range $0 < t < 4$ s, and check the slope of the graph at $t = 0$ and $t = 4$ s using the formula for a . (b) Determine the ratio of the velocity v_b of the sled at burnout to the exhaust velocity v_{ex} .

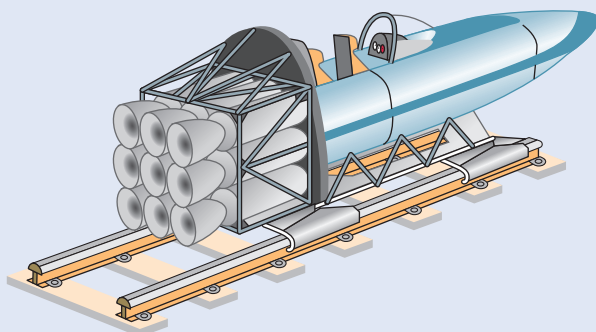


Fig. P14.93

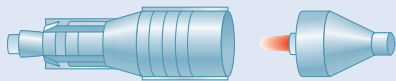


Fig. P14.94

14.94 A space vehicle describing a circular orbit about the earth at a speed of 24×10^3 km/h releases at its front end a capsule that has a gross mass of 600 kg, including 400 kg of fuel. If the fuel is consumed at the rate of 18 kg/s and ejected with a relative velocity of 3000 m/s, determine (a) the tangential acceleration of the capsule as its engine is fired, (b) the maximum speed attained by the capsule.

14.95 A 540-kg spacecraft is mounted on top of a rocket with a mass of 19 Mg, including 17.8 Mg of fuel. Knowing that the fuel is consumed at a rate of 225 kg/s and ejected with a relative velocity of 3600 m/s, determine the maximum speed imparted to the spacecraft if the rocket is fired vertically from the ground.



Fig. P14.95



Fig. P14.96

14.96 The rocket used to launch the 540-kg spacecraft of Prob. 14.95 is redesigned to include two stages *A* and *B*, each of mass 9.5 Mg, including 8.9 Mg of fuel. The fuel is again consumed at a rate of 225 kg/s and ejected with a relative velocity of 3600 m/s. Knowing that when stage *A* expels its last particle of fuel, its casing is released and jettisoned, determine (a) the speed of the rocket at that instant, (b) the maximum speed imparted to the spacecraft.

14.97 The weight of a spacecraft, including fuel, is 11,600 lb when the rocket engines are fired to increase its velocity by 360 ft/s. Knowing that 1000 lb of fuel is consumed, determine the relative velocity of the fuel ejected.

14.98 The rocket engines of a spacecraft are fired to increase its velocity by 450 ft/s. Knowing that 1200 lb of fuel is ejected at a relative velocity of 5400 ft/s, determine the weight of the spacecraft after the firing.

14.99 Determine the distance traveled by the spacecraft of Prob. 14.97 during the rocket engine firing, knowing that its initial speed was 7500 ft/s and the duration of the firing was 60 s.

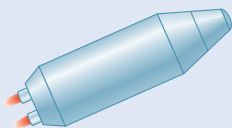


Fig. P14.97 and P14.98

- 14.100** A rocket weighs 2600 lb, including 2200 lb of fuel, which is consumed at the rate of 25 lb/s and ejected with a relative velocity of 13,000 ft/s. Knowing that the rocket is fired vertically from the ground, determine (a) its acceleration as it is fired, (b) its acceleration as the last particle of fuel is being consumed, (c) the altitude at which all the fuel has been consumed, (d) the velocity of the rocket at that time.
- 14.101** Determine the altitude reached by the spacecraft of Prob. 14.95 when all the fuel of its launching rocket has been consumed.
- 14.102** For the spacecraft and the two-stage launching rocket of Prob. 14.96, determine the altitude at which (a) stage A of the rocket is released, (b) the fuel of both stages has been consumed.
- 14.103** In a jet airplane, the kinetic energy imparted to the exhaust gases is wasted as far as propelling the airplane is concerned. The useful power is equal to the product of the force available to propel the airplane and the speed of the airplane. If v is the speed of the airplane and u is the relative speed of the expelled gases, show that the mechanical efficiency of the airplane is $\eta = 2v/(u + v)$. Explain why $\eta = 1$ when $u = v$.
- 14.104** In a rocket, the kinetic energy imparted to the consumed and ejected fuel is wasted as far as propelling the rocket is concerned. The useful power is equal to the product of the force available to propel the rocket and the speed of the rocket. If v is the speed of the rocket and u is the relative speed of the expelled fuel, show that the mechanical efficiency of the rocket is $\eta = 2uv/(u^2 + v^2)$. Explain why $\eta = 1$ when $u = v$.

Review and Summary

In this chapter, we analyzed the motion of **systems of particles**, i.e., the motion of a large number of particles considered together. In the first part of the chapter, we considered systems consisting of well-defined particles, whereas in the second part, we analyzed systems that are continually gaining or losing particles or doing both at the same time.

Newton's Second Law for a System of Particles

We showed that **the system of the external forces acting on the particles and the system of the $m_i \mathbf{a}_i$ terms of the particles are equipollent**; i.e., both systems have the *same resultant* and the *same moment resultant* about O :

$$\sum_{i=1}^n \mathbf{F}_i = \sum_{i=1}^n m_i \mathbf{a}_i \quad (14.4)$$

$$\sum_{i=1}^n (\mathbf{r}_i \times \mathbf{F}_i) = \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{a}_i) \quad (14.5)$$

Linear and Angular Momentum of a System of Particles

We defined the *linear momentum* $\mathbf{L}$ and the *angular momentum* $\mathbf{H}_O$ about point O of the system of particles [Sec. 14.1B] as

$$\mathbf{L} = \sum_{i=1}^n m_i \mathbf{v}_i \quad \mathbf{H}_O = \sum_{i=1}^n (\mathbf{r}_i \times m_i \mathbf{v}_i) \quad (14.6, 14.7)$$

Then we showed that we can replace Eqs. (14.4) and (14.5) with the equations

$$\Sigma \mathbf{F} = \dot{\mathbf{L}} \quad \Sigma \mathbf{M}_O = \dot{\mathbf{H}}_O \quad (14.10, 14.11)$$

Together, these equations state that **the sum of external forces is equal to the rate of change of the linear momentum, and the sum of the moments about O is equal to the rate of change of the angular momentum about O .**

Motion of the Mass Center of a System of Particles

In Sec. 14.1C, we defined the mass center of a system of particles as the point G whose position vector $\bar{\mathbf{r}}$ satisfies the equation

$$m\bar{\mathbf{r}} = \sum_{i=1}^n m_i \mathbf{r}_i \quad (14.12)$$

where m represents the total mass $m = \sum_{i=1}^n m_i$ of the particles. Differentiating both sides of Eq. (14.12) twice with respect to t , we obtained the relations

$$\mathbf{L} = m\bar{\mathbf{v}} \quad \dot{\mathbf{L}} = m\bar{\mathbf{a}} \quad (14.14, 14.15)$$

where $\bar{\mathbf{v}}$ and $\bar{\mathbf{a}}$ represent, respectively, the velocity and the acceleration of the mass center G . Substituting for $\dot{\mathbf{L}}$ from Eq. (14.15) into Eq. (14.10), we obtained

$$\Sigma \mathbf{F} = m\bar{\mathbf{a}} \quad (14.16)$$

From this, we concluded that **the mass center of a system of particles moves as if the entire mass of the system and all of the external forces were concentrated at that point** [Sample Prob. 14.1].

Angular Momentum of a System of Particles about its Mass Center

In Sec. 14.1D, we considered the motion of the particles of a system with respect to a centroidal frame $Gx'y'z'$ attached to the mass center G of the system and in translation with respect to the newtonian frame $Oxyz$ (Fig. 14.13). We defined the *angular momentum* of the system *about its mass center* G as the sum of the moments about G of the momenta $m_i\mathbf{v}'_i$ of the particles relative to the frame $Gx'y'z'$. We also noted that we can obtain the same earlier result by considering the moments about G of the momenta $m_i\mathbf{v}_i$ of the particles in their absolute motion. We therefore wrote

$$\mathbf{H}_G = \sum_{i=1}^n (\mathbf{r}'_i \times m_i\mathbf{v}'_i) = \sum_{i=1}^n (\mathbf{r}'_i \times m_i\mathbf{v}_i) \quad (14.24)$$

and derived the relation

$$\Sigma \mathbf{M}_G = \dot{\mathbf{H}}_G \quad (14.23)$$

This equation states that **the resultant moment about G of the external forces is equal to the rate of change of the angular momentum about G of the system of particles**. As you will see later, this relation is fundamental to the study of the motion of rigid bodies.

Conservation of Momentum

When no external force acts on a system of particles [Sec. 14.1E], it follows from Eqs. (14.10) and (14.11) that the linear momentum $\mathbf{L}$ and the angular momentum $\mathbf{H}_O$ of the system are conserved [Sample Probs. 14.2 and 14.4]. In problems involving central forces, the angular momentum of the system about the center of force O is also conserved.

Kinetic Energy of a System of Particles

The kinetic energy T of a system of particles was defined as the sum of the kinetic energies of the particles [Sec. 14.2A]:

$$T = \frac{1}{2} \sum_{i=1}^n m_i v_i^2 \quad (14.28)$$

Using the centroidal frame of reference $Gx'y'z'$ of Fig. 14.13, we noted that we can also obtain the kinetic energy of the system by adding the kinetic energy $\frac{1}{2}m\bar{v}^2$ associated with the motion of the mass center G and the kinetic energy of the system relative to the frame $Gx'y'z'$. Thus,

$$T = \frac{1}{2}m\bar{v}^2 + \frac{1}{2} \sum_{i=1}^n m_i v_i'^2 \quad (14.29)$$

Principle of Work and Energy

We applied the **principle of work and energy** to a system of particles as well as to individual particles [Sec. 14.2B]. We have

$$T_1 + U_{1 \rightarrow 2} = T_2 \quad (14.30)$$

and noted that $U_{1 \rightarrow 2}$ represents the work of *all* of the forces acting on the particles of the system—internal as well as external.

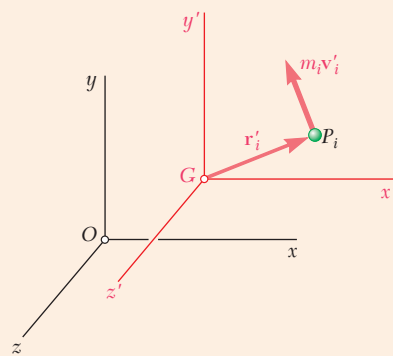


Fig. 14.13

Conservation of Energy

If all of the forces acting on the particles of a system are *conservative*, we can determine the potential energy V of the system and write

$$T_1 + V_1 = T_2 + V_2 \quad (14.31)$$

which expresses the **principle of conservation of energy** for a system of particles.

Principle of Impulse and Momentum

We saw in Sec. 14.2C that the **principle of impulse and momentum** for a system of particles can be expressed graphically, as shown in Fig. 14.14. The principle states that the momenta of the particles at time t_1 and the impulses of the external forces from t_1 to t_2 form a system of vectors equipollent to the system of the momenta of the particles at time t_2 .

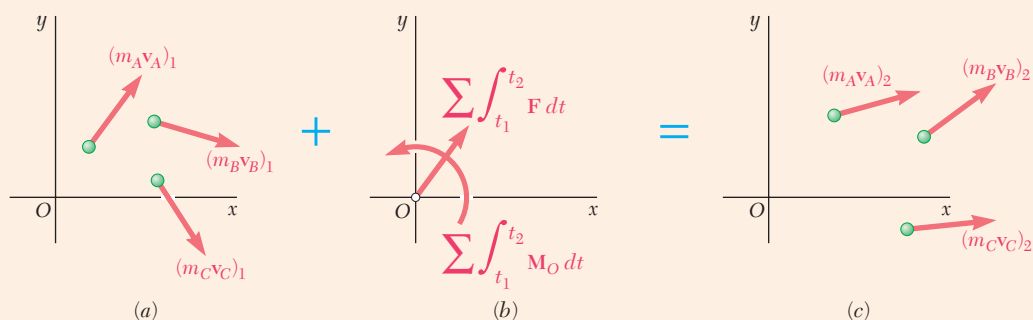


Fig. 14.14

If no external force acts on the particles of the system, the systems of momenta shown in parts a and c of Fig. 14.14 are equipollent, and we have

$$\mathbf{L}_1 = \mathbf{L}_2 \quad (\mathbf{H}_O)_1 = (\mathbf{H}_O)_2 \quad (14.36, 14.37)$$

Use of Conservation Principles in the Solution of Problems Involving Systems of Particles

We can solve many problems involving the motion of systems of particles by applying simultaneously the principle of impulse and momentum and the principle of conservation of energy [Sample Prob. 14.5] or by expressing that the linear momentum, angular momentum, and energy of the system are conserved [Sample Prob. 14.6].

Steady Stream of Particles

In the second part of the chapter, we considered **variable systems of particles**. First we considered a **steady stream of particles**, such as a stream of water diverted by a fixed vane or the flow of air through a jet engine [Sec. 14.3A]. We applied the principle of impulse and momentum to a system S of particles during a time interval Δt , including the particles that enter the system at A during that time interval and those (of the same mass Δm) that leave the system at B . We concluded that **the system formed by the momentum $(\Delta m)\mathbf{v}_A$ of the particles entering S in the time Δt and the impulses of the forces**

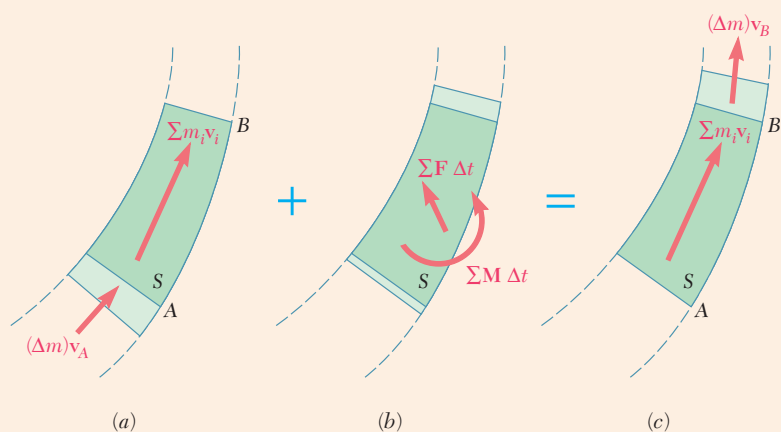


Fig. 14.15

exerted on S during that time is equipollent to the momentum $(\Delta m)\mathbf{v}_B$ of the particles leaving S in the same time Δt (Fig. 14.15). Equating the x components, y components, and moments about a fixed point of the vectors involved, we could obtain as many as three equations that you could solve for the desired unknowns [Sample Probs. 14.7 and 14.8]. From this result, we also derived the expression for the resultant $\Sigma \mathbf{F}$ of the forces exerted on S as

$$\Sigma \mathbf{F} = \frac{dm}{dt}(\mathbf{v}_B - \mathbf{v}_A) \quad (14.39)$$

where $\mathbf{v}_B - \mathbf{v}_A$ represents the difference between the *vectors* $\mathbf{v}_B$ and $\mathbf{v}_A$ and dm/dt is the mass rate of flow of the stream (see footnote, page 951).

Systems Gaining or Losing Mass

We considered next a system of particles gaining mass by continually absorbing particles or losing mass by continually expelling particles [Sec. 14.3B], as in the case of a rocket. We applied the principle of impulse and momentum to the system during a time interval Δt , being careful to include the particles gained or lost during that time interval [Sample Prob. 14.9]. We also noted that the action on a system S of the particles being *absorbed* by S was equivalent to a thrust

$$\mathbf{P} = \frac{dm}{dt} \mathbf{u} \quad (14.44)$$

where dm/dt is the rate at which mass is being absorbed and $\mathbf{u}$ is the velocity of the particles *relative to* S . In the case of particles being *expelled* by S , the rate dm/dt is negative, and the thrust $\mathbf{P}$ is exerted in a direction opposite to that in which the particles are being expelled.

Review Problems

- 14.105** Three identical cars are being unloaded from an automobile carrier. Cars B and C have just been unloaded and are at rest with their brakes off when car A leaves the unloading ramp with a velocity of 5.76 ft/s and hits car B , which hits car C . Car A then again hits car B . Knowing that the velocity of car B is 5.04 ft/s after the first collision, 0.630 ft/s after the second collision, and 0.709 ft/s after the third collision, determine (a) the final velocities of cars A and C , (b) the coefficient of restitution for each of the collisions.

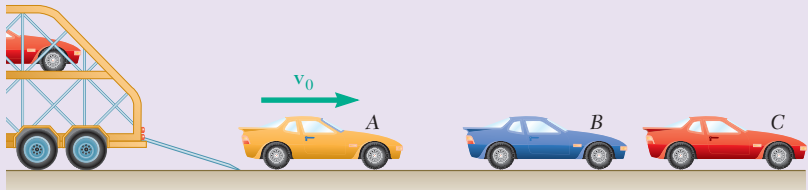


Fig. P14.105

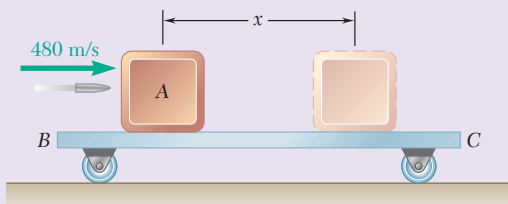


Fig. P14.106

- 14.106** A 30-g bullet is fired with a velocity of 480 m/s into block A , which has a mass of 5 kg . The coefficient of kinetic friction between block A and cart BC is 0.50 . Knowing that the cart has a mass of 4 kg and can roll freely, determine (a) the final velocity of the cart and block, (b) the final position of the block on the cart.

- 14.107** An 80-Mg railroad engine A coasting at 6.5 km/h strikes a 20-Mg flatcar C carrying a 30-Mg load B that can slide along the floor of the car ($\mu_k = 0.25$). Knowing that the car was at rest with its brakes released and that it automatically coupled with the engine upon impact, determine the velocity of the car (a) immediately after impact, (b) after the load has slid to a stop relative to the car.

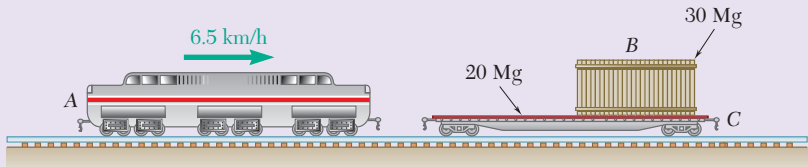


Fig. P14.107

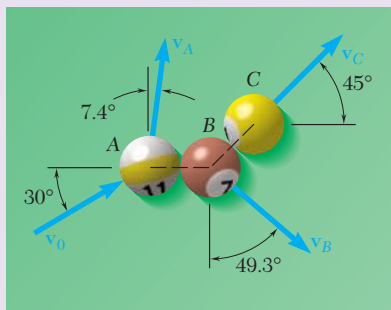


Fig. P14.108

- 14.108** In a game of pool, ball A is moving with a velocity v_0 when it strikes balls B and C , which are at rest and aligned as shown. Knowing that after the collision the three balls move in the directions indicated and that $v_0 = 12 \text{ ft/s}$ and $v_C = 6.29 \text{ ft/s}$, determine the magnitude of the velocity of (a) ball A , (b) ball B .

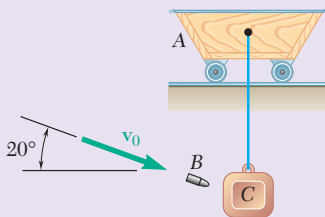


Fig. P14.109

- 14.109** Mass C , which has a mass of 4 kg , is suspended from a cord attached to cart A , which has a mass of 5 kg and can roll freely on a frictionless horizontal track. A 60-g bullet is fired with a speed $v_0 = 500 \text{ m/s}$ and gets lodged in block C . Determine (a) the velocity of C as it reaches its maximum elevation, (b) the maximum vertical distance h through which C will rise.

- 14.110** A 15-lb block B is at rest and a spring of constant $k = 72 \text{ lb/in.}$ is held compressed 3 in. by a cord. After 5-lb block A is placed against the end of the spring, the cord is cut causing A and B to move. Neglecting friction, determine the velocities of blocks A and B immediately after A leaves B .

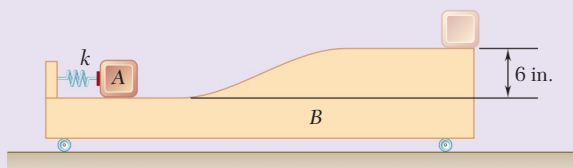


Fig. P14.110

- 14.111** Car A of mass 1800 kg and car B of mass 1700 kg are at rest on a 20-Mg flatcar which is also at rest. Cars A and B then accelerate and quickly reach constant speeds relative to the flatcar of 2.35 m/s and 1.175 m/s, respectively, before decelerating to a stop at the opposite end of the flatcar. Neglecting friction and rolling resistance, determine the velocity of the flatcar when the cars are moving at constant speeds.

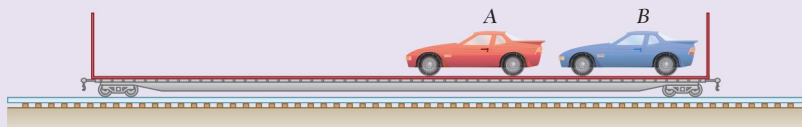


Fig. P14.111

- 14.112** The nozzle shown discharges water at the rate of 200 gal/min. Knowing that at both B and C the stream of water moves with a velocity of magnitude 100 ft/s, and neglecting the weight of the vane, determine the force-couple system that must be applied at A to hold the vane in place ($1 \text{ ft}^3 = 7.48 \text{ gal}$).
- 14.113** An airplane with a weight W and a total wing span b flies horizontally at a constant speed v . Use the airplane as a reference frame; that is, consider the airplane to be motionless and the air to flow past it with speed v . Suppose that a cylinder of air with diameter b is deflected downward by the wing (the cross section of the cylinder is the dashed circle in the figure). Show that the angle through which the cylinder stream is deflected (called the *downwash angle*) is determined by the formula $\sin \theta = 4W/(\pi b^2 \rho v^2)$, where ρ is the mass density of the air.

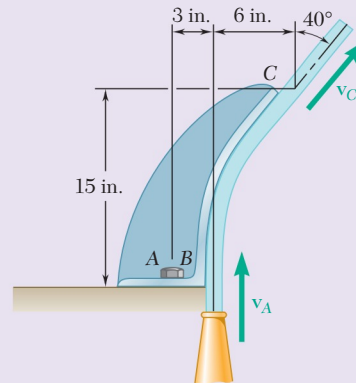


Fig. P14.112

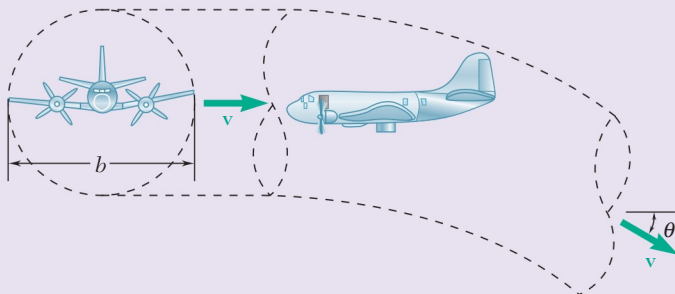


Fig. P14.113

- 14.114** The final component of a conveyor system receives sand at a rate of 100 kg/s at A and discharges it at B . The sand is moving horizontally at A and B with a velocity of magnitude $v_A = v_B = 4.5$ m/s. Knowing that the combined weight of the component and of the sand it supports is $W = 4$ kN, determine the reactions at C and D .

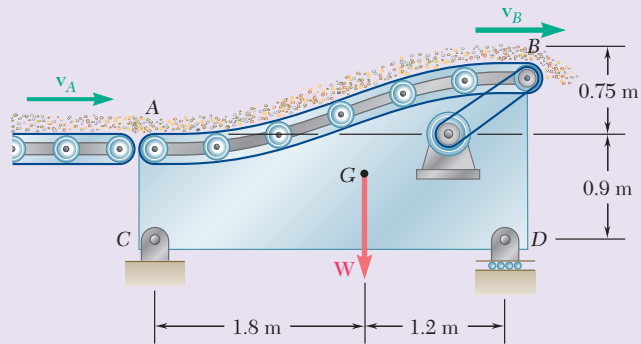


Fig. P14.114

- 14.115** A garden sprinkler has four rotating arms, each of which consists of two horizontal straight sections of pipe forming an angle of 120° with each other. Each arm discharges water at a rate of 20 L/min with a velocity of 18 m/s relative to the arm. Knowing that the friction between the moving and stationary parts of the sprinkler is equivalent to a couple of magnitude $M = 0.375$ N·m, determine the constant rate at which the sprinkler rotates.

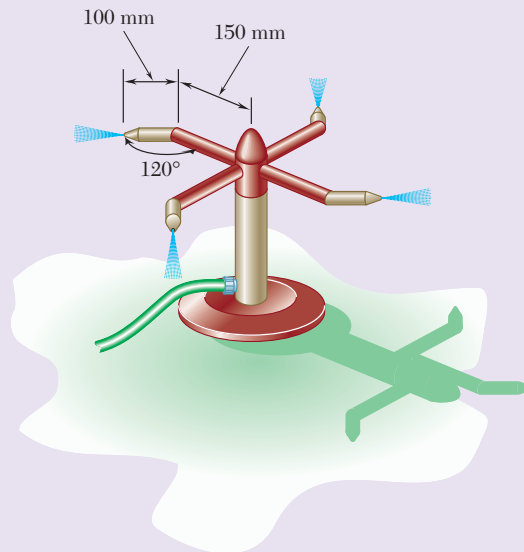


Fig. P14.115

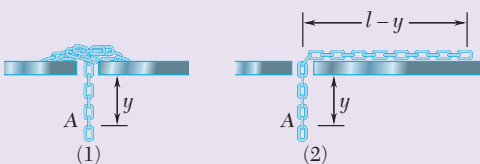


Fig. P14.116

- 14.116** A chain of length l and mass m falls through a small hole in a plate. Initially, when y is very small, the chain is at rest. In each case shown, determine (a) the acceleration of the first link A as a function of y , (b) the velocity of the chain as the last link passes through the hole. In case 1, assume that the individual links are at rest until they fall through the hole; in case 2, assume that at any instant all links have the same speed. Ignore the effect of friction.



15

Kinematics of Rigid Bodies

This huge crank belongs to a large diesel engine. In this chapter, you will learn to perform the *kinematic* analysis of rigid bodies that undergo *translation*, *fixed axis rotation*, and *general plane motion*.

Introduction

15.1 TRANSLATION AND FIXED-AXIS ROTATION

15.1A Translation

15.1B Rotation About a Fixed Axis

15.1C Equations Defining the Rotation of a Rigid Body About a Fixed Axis

15.2 GENERAL PLANE MOTION: VELOCITY

15.2A Analyzing General Plane Motion

15.2B Absolute and Relative Velocity in Plane Motion

15.3 INSTANTANEOUS CENTER OF ROTATION

15.4 GENERAL PLANE MOTION: ACCELERATION

15.4A Absolute and Relative Acceleration in Plane Motion

*15.4B Analysis of Plane Motion in Terms of a Parameter

15.5 ANALYZING MOTION WITH RESPECT TO A ROTATING FRAME

15.5A Rate of Change of a Vector with Respect to a Rotating Frame

15.5B Plane Motion of a Particle Relative to a Rotating Frame

*15.6 MOTION OF A RIGID BODY IN SPACE

15.6A Motion About a Fixed Point

15.6B General Motion

*15.7 MOTION RELATIVE TO A MOVING REFERENCE FRAME

15.7A Three-Dimensional Motion of a Particle Relative to a Rotating Frame

15.7B General Three-Dimensional Motion

Objectives

- **Describe** the five basic types of rigid body motion: translation, rotation about a fixed axis, general plane motion, motion about a fixed point, and general motion.
- **Use** angular kinematic relationships involving θ , ω , and α to determine the angular motion of a rigid body.
- **Identify** the directions of terms in the relative velocity and relative acceleration equations.
- **Calculate** the linear velocity and acceleration of any point on a rigid body undergoing translation, fixed axis rotation, or general plane motion.
- **Solve** planar rigid body kinematics problems using the relative velocity and relative acceleration equations.
- **Determine** the instantaneous center of rotation and use it to analyze the planar velocity kinematics of a rigid body.
- When appropriate, **define** a rotating coordinate frame and use it to solve planar and three-dimensional kinematics problems.
- **Determine** the angular velocity and angular acceleration of a body undergoing three-dimensional motion.
- **Calculate** the linear velocity and acceleration of any point on a rigid body undergoing three-dimensional motion.

Introduction

In this chapter, we consider the kinematics of **rigid bodies**. We will investigate the relations between the time, the positions, the velocities, and the accelerations of the various particles forming a rigid body. As you will see, the various types of rigid-body motion can be conveniently grouped as follows:

1. **Translation.** A motion is said to be a translation if any straight line inside the body maintains the same orientation during the motion. In a translation, all of the particles forming the body move along parallel paths. If these paths are straight lines, the motion is called **rectilinear translation** (Fig. 15.1); if the paths are curved lines, the motion is called **curvilinear translation** (Fig. 15.2).
2. **Rotation About a Fixed Axis.** In this motion, the particles forming the rigid body move in parallel planes along circles centered on the same fixed axis (Fig. 15.3). If this axis, called the **axis of rotation**, intersects the rigid body, the particles located on the axis have zero velocity and zero acceleration.

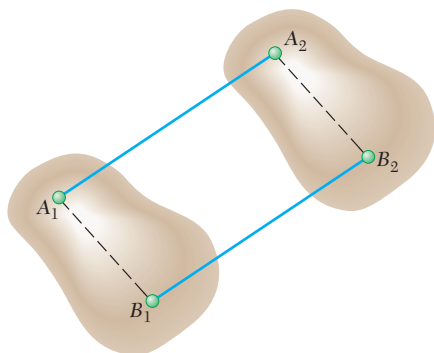


Fig. 15.1 A rigid body in rectilinear translation.

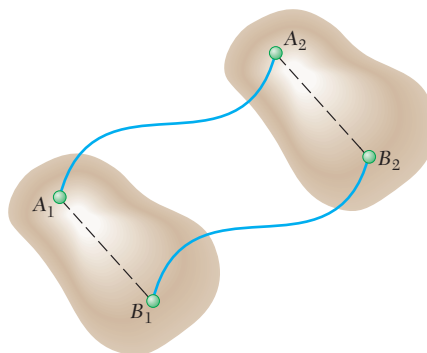


Fig. 15.2 A rigid body in curvilinear translation.

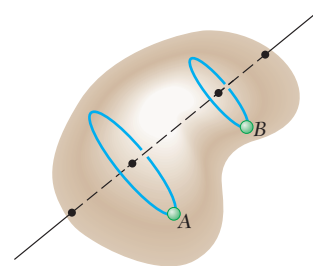


Fig. 15.3 A rigid body rotating about a fixed axis.

Be careful not to confuse rotation with certain types of curvilinear translation. For example, the plate shown in Fig. 15.4a is in curvilinear translation, with all of its particles moving along *parallel* circles, whereas the plate shown in Fig. 15.4b is in rotation, with all of its particles moving along *concentric* circles. In the first case, any given straight line drawn on the plate maintains the same direction, whereas in the second case, the orientation of the plate changes throughout the rotation. Because each particle moves in a given plane, the rotation of a body about a fixed axis is said to be a **plane motion**.

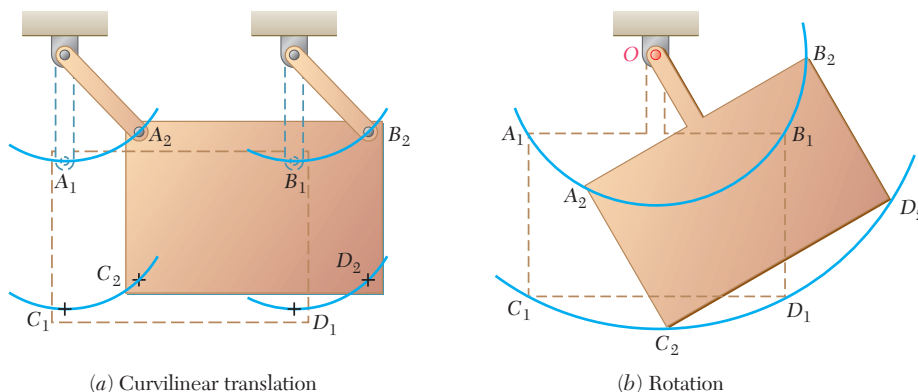


Fig. 15.4 (a) In curvilinear motion, particles move along parallel circles, whereas (b) in fixed-axis rotation, particles move along concentric circles.

3. **General Plane Motion.** Many other types of plane motion can occur, i.e., motions in which all the particles of the body move in a single plane. Any plane motion that is neither a rotation nor a translation is referred to as general plane motion. Figure 15.5 shows two examples of general plane motion.
4. **Motion About a Fixed Point.** The three-dimensional motion of a rigid body attached at a fixed point O , such as the motion of a top on a rough floor (Fig. 15.6), is known as motion about a fixed point.
5. **General Motion.** Any motion of a rigid body that does not fall in any of these categories is referred to as a general motion.

After a brief discussion of the motion of translation, we consider the rotation of a rigid body about a fixed axis. We define the *angular velocity* and the *angular acceleration* of a rigid body rotating about a fixed axis,

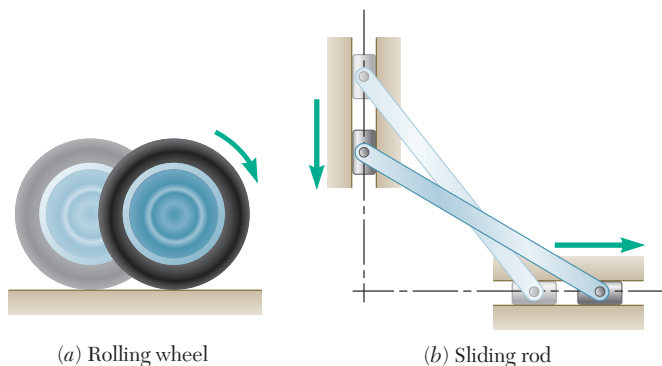


Fig. 15.5 (a) A rolling wheel and (b) a sliding rod are common examples of general plane motion.

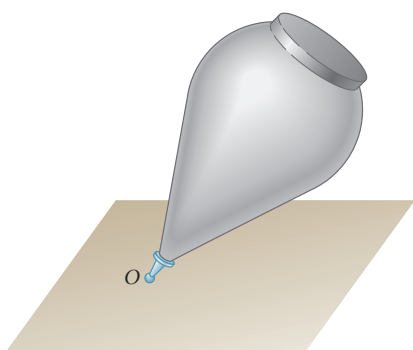


Fig. 15.6 The motion of a spinning top on a rough surface is an example of three-dimensional motion about a fixed point.

and you will see how to express the velocity and acceleration of a given point of the body in terms of its position vector and the angular velocity and angular acceleration of the body.

Afterwards, we study the general plane motion of a rigid body and apply the results to the analysis of mechanisms such as gears, connecting rods, and pin-connected linkages. If we resolve the plane motion of a rigid body into a translation and a rotation, we can then express the velocity of a point B of the body as the sum of the velocity of a reference point A and of the velocity of B relative to a frame of reference translating with A (i.e., moving with A but not rotating). We use the same approach later in Sec. 15.4 to express the acceleration of B in terms of the acceleration of A and of the acceleration of B relative to a frame translating with A . We also present an alternative method for analyzing velocities in plane motion based on the concept of the *instantaneous center of rotation*, and we discuss still another method of analysis based on the use of parametric expressions for the coordinates of a given point.

The motion of a particle relative to a rotating frame of reference and the concept of *Coriolis acceleration* are discussed in Sec. 15.5. We apply the results to the analysis of the plane motion of mechanisms containing parts that slide on each other.

In the remainder of this chapter, we analyze the three-dimensional motion of a rigid body, specifically, the motion of a rigid body with a fixed point and the general motion of a rigid body. We use a fixed frame of reference or a frame of reference in translation to carry out this analysis, then we consider the motion of the body relative to a rotating frame or to a frame in general motion. Again, we use the concept of Coriolis acceleration.

15.1 TRANSLATION AND FIXED AXIS ROTATION

We noted in the introduction that we can resolve a general plane motion into a translation and a rotation. Thus, our first step is to formulate the mathematical descriptions of simple translations and rotations.

15.1A Translation

Consider a rigid body in translation (either rectilinear or curvilinear translation), and let A and B be any two of its particles (Fig. 15.7a). Denoting the position vectors of A and B with respect to a fixed frame of

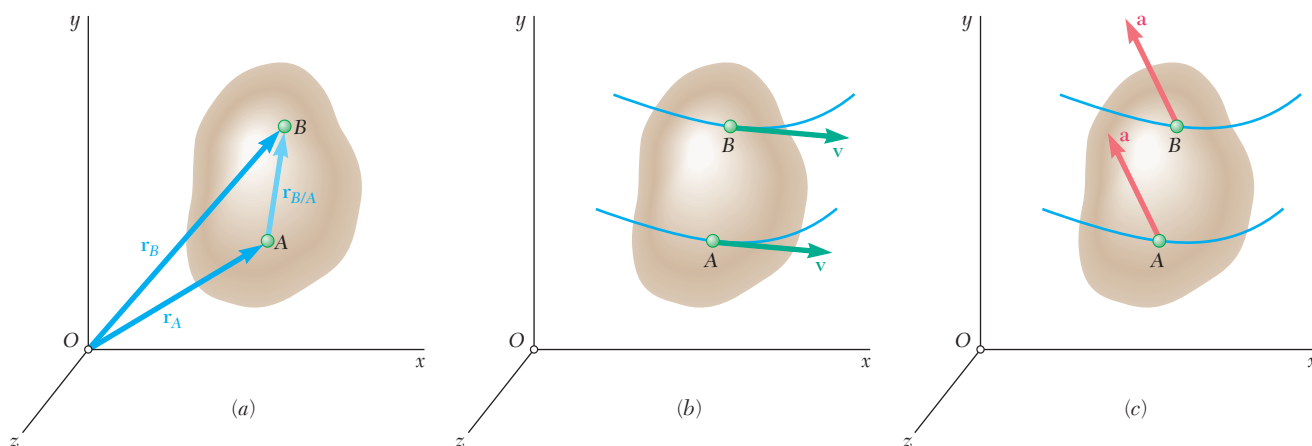


Fig. 15.7 For a rigid body in translation: (a) the position vector between any two points is constant in magnitude and direction; (b) every point has the same velocity; (c) every point has the same acceleration.

reference by $\mathbf{r}_A$ and $\mathbf{r}_B$, respectively, and the vector from A to B by $\mathbf{r}_{B/A}$, we have

$$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A} \quad (15.1)$$

To obtain the relationship between the velocities of A and B , we differentiate this expression with respect to t . Note that, from the very definition of a translation, the vector $\mathbf{r}_{B/A}$ must maintain a constant direction; its magnitude must also be constant, since A and B belong to the same rigid body. Thus, the derivative of $\mathbf{r}_{B/A}$ is zero, and we have

$$\mathbf{v}_B = \mathbf{v}_A \quad (15.2)$$

Differentiating once more, we obtain the relationship between the accelerations of A and B as

$$\mathbf{a}_B = \mathbf{a}_A \quad (15.3)$$

Thus, **when a rigid body is in translation, all the points of the body have the same velocity and the same acceleration at any given instant** (Fig. 15.7*b* and *c*). In the case of curvilinear translation, the velocity and acceleration change in direction as well as in magnitude at every instant. In the case of rectilinear translation, all particles of the body move along parallel straight lines, and their velocity and acceleration keep the same direction during the entire motion.



Photo 15.1 The horizontal linkage of a locomotive undergoes curvilinear translation.

15.1B Rotation About a Fixed Axis

Consider a rigid body that rotates about a fixed axis AA' . Let P be a point of the body and $\mathbf{r}$ be its position vector with respect to a fixed frame of reference. For convenience, let us assume that the frame is centered at point O on AA' and that the z axis coincides with AA' (Fig. 15.8). Let B be the projection of P on AA' . Since P must remain at a constant distance from B , it describes a circle with a center B and radius $r \sin \phi$, where ϕ denotes the angle formed by $\mathbf{r}$ and AA' .

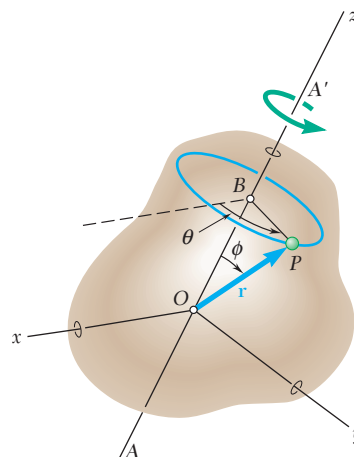


Fig. 15.8 For a rigid body in rotation about a fixed axis, each point of the body moves in a circular path centered on the axis.



Photo 15.2 For the central gear rotating about a fixed axis, the angular velocity and angular acceleration of that gear are vectors directed along the vertical axis of rotation.

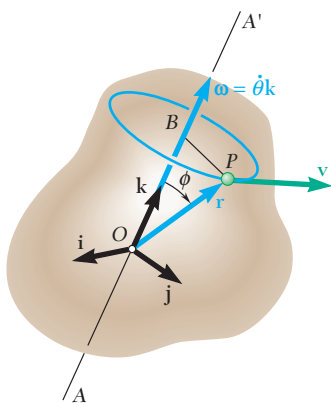


Fig. 15.9 For a rigid body in rotation about a fixed axis, the velocity of a particle is the vector product of the angular velocity of the body and the position vector of the particle.

The position both of P and of the entire body is completely defined by the angle θ that the line BP forms with the zx plane. The angle θ is known as the **angular coordinate** of the body and is defined as positive when viewed as counterclockwise from A' . The angular coordinate is expressed in radians (rad) or, occasionally, in degrees ($^\circ$) or revolutions (rev). Recall that

$$1 \text{ rev} = 2\pi \text{ rad} = 360^\circ$$

Recall from Sec. 11.4A that the velocity $\mathbf{v} = d\mathbf{r}/dt$ of a particle P is a vector tangent to the path of P and with a magnitude of $v = ds/dt$. The length Δs of the arc described by P when the body rotates through $\Delta\theta$ is

$$\Delta s = (BP) \Delta\theta = (r \sin \phi) \Delta\theta$$

Then dividing both members by Δt , we obtain in the limit, as Δt approaches zero,

$$v = \frac{ds}{dt} = r\dot{\theta} \sin \phi \quad (15.4)$$

where $\dot{\theta}$ denotes the time derivative of θ . (Note that the angle θ depends on the position of P within the body, but the rate of change $\dot{\theta}$ is itself independent of P .) We conclude that the velocity $\mathbf{v}$ of P is a vector perpendicular to the plane containing AA' and $\mathbf{r}$, and of magnitude v defined by Eq. (15.4). But this is precisely the result we would obtain if we drew a vector $\boldsymbol{\omega} = \dot{\theta}\mathbf{k}$ along AA' and formed the vector product $\boldsymbol{\omega} \times \mathbf{r}$ (Fig. 15.9). We thus have

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \boldsymbol{\omega} \times \mathbf{r} \quad (15.5)$$

The vector

$$\boldsymbol{\omega} = \omega\mathbf{k} = \dot{\theta}\mathbf{k} \quad (15.6)$$

is directed along the axis of rotation. It is called the **angular velocity** of the body and is equal in magnitude to the rate of change $\dot{\theta}$ of the angular coordinate. You can obtain the sense of the vector by using the right hand rule (Sec. 3.1E); using your right hand, curl your fingers in the direction of the angular velocity, and your thumb will point in the direction of the vector.[†]

Now we can determine the acceleration $\mathbf{a}$ of particle P . Differentiating Eq. (15.5) and recalling the rule for the differentiation of a vector product (Sec. 11.4B), we have

$$\begin{aligned} \mathbf{a} &= \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(\boldsymbol{\omega} \times \mathbf{r}) \\ &= \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{r} + \boldsymbol{\omega} \times \frac{d\mathbf{r}}{dt} \\ &= \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{r} + \boldsymbol{\omega} \times \mathbf{v} \end{aligned} \quad (15.7)$$

[†]We will show in Sec. 15.6 the more general case of a rigid body, rotating simultaneously about axes having different directions, where angular velocities obey the parallelogram law of addition and thus are actually vector quantities.

The vector $d\boldsymbol{\omega}/dt$ is denoted by $\boldsymbol{\alpha}$ and is called the **angular acceleration** of the body. Substituting for $\mathbf{v}$ from Eq. (15.5), we have

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad (15.8)$$

Differentiating Eq. (15.6) and recalling that $\mathbf{k}$ is constant in magnitude and direction, we have

$$\boldsymbol{\alpha} = \alpha \mathbf{k} = \dot{\omega} \mathbf{k} = \ddot{\theta} \mathbf{k} \quad (15.9)$$

Thus, the angular acceleration of a body rotating about a fixed axis is a vector directed along the axis of rotation and is equal in magnitude to the rate of change $\dot{\omega}$ of the angular velocity.

Returning to Eq. (15.8), we note that the acceleration of P is the sum of two vectors. The first vector is equal to the vector product $\boldsymbol{\alpha} \times \mathbf{r}$; it is tangent to the circle described by P and therefore represents the tangential component of the acceleration. The second vector is equal to the *vector triple product* $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$ obtained by forming the vector product of $\boldsymbol{\omega}$ and $\boldsymbol{\omega} \times \mathbf{r}$. Since $\boldsymbol{\omega} \times \mathbf{r}$ is tangent to the circle described by P , the vector triple product is directed toward the center B of the circle and therefore represents the normal component of the acceleration.

Rotation of a Representative Slab. We can express the rotation of a rigid body about a fixed axis by examining the motion of a representative slab in a reference plane perpendicular to the axis of rotation. We choose the xy plane as the reference plane and assume that it coincides with the plane of the figure with the z axis pointing out of the page (Fig. 15.10). Recalling from Eq. (15.6) that $\boldsymbol{\omega} = \omega \mathbf{k}$, we note that a positive value of the scalar ω corresponds to a counterclockwise rotation of the representative slab, and a negative value corresponds to a clockwise rotation. Substituting $\omega \mathbf{k}$ for $\boldsymbol{\omega}$ in Eq. (15.5), we express the velocity of any given point P of the slab as

$$\mathbf{v} = \omega \mathbf{k} \times \mathbf{r} \quad (15.10)$$

Since the vectors $\mathbf{k}$ and $\mathbf{r}$ are mutually perpendicular, the magnitude of the velocity $\mathbf{v}$ is

$$v = r\omega \quad (15.10')$$

We can obtain its direction by rotating $\mathbf{r}$ through 90° in the sense of rotation of the slab.

If we substitute $\omega \mathbf{k}$ into Eq. (15.8), we obtain $\omega \mathbf{k} \times (\omega \mathbf{k} \times \mathbf{r})$, which simplifies to $-\omega^2 \mathbf{r}$. This indicates that the direction of the normal acceleration is $-\mathbf{r}$, or toward the center of rotation, which is exactly what we expect. Using this expression and $\boldsymbol{\alpha} = \alpha \mathbf{k}$ in Eq. (15.8), we obtain

$$\mathbf{a} = \alpha \mathbf{k} \times \mathbf{r} - \omega^2 \mathbf{r} \quad (15.11)$$

Resolving $\mathbf{a}$ into tangential and normal components (Fig. 15.11) gives

$$\begin{aligned} a_t &= \alpha \mathbf{k} \times \mathbf{r} & a_t &= r\alpha \\ a_n &= -\omega^2 \mathbf{r} & a_n &= r\omega^2 \end{aligned} \quad (15.11')$$

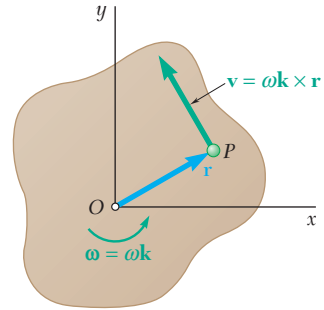


Fig. 15.10 For an object undergoing fixed-axis rotation, the velocity of a point P equals the vector product of the angular velocity vector and the position vector of P . A positive value of the scalar ω corresponds to counterclockwise motion.

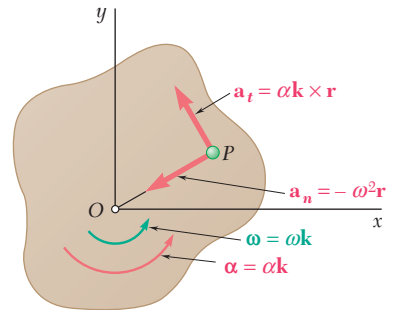


Fig. 15.11 For an object undergoing fixed-axis rotation, the acceleration of a point P has a tangential component that depends on angular acceleration and a normal component that depends on angular velocity.

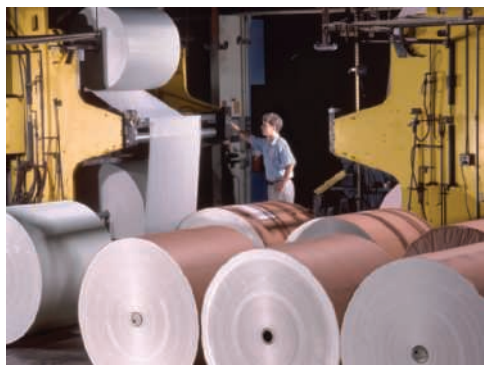


Photo 15.3 If the lower roll has a constant angular velocity, the speed of the paper being wound onto it increases as the radius of the roll increases.

The tangential component $\mathbf{a}_t$ points in the counterclockwise direction if the scalar α is positive and in the clockwise direction if α is negative. The normal component $\mathbf{a}_n$ always points in the direction opposite to that of $\mathbf{r}$, that is, toward O .

15.1C Equations Defining the Rotation of a Rigid Body About a Fixed Axis

The motion of a rigid body rotating about a fixed axis AA' is said to be *known* when we can express its angular coordinate θ as a known function of t . In practice, however, we can seldom describe the rotation of a rigid body by a relation between θ and t . More often, the conditions of motion are specified by the angular acceleration of the body. For example, α may be given as a function of t , as a function of θ , or as a function of ω . From the relations in Eqs. (15.6) and (15.9), we have

$$\omega = \frac{d\theta}{dt} \quad (15.12)$$

$$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} \quad (15.13)$$

or solving Eq. (15.12) for dt and substituting into Eq. (15.13), we have

$$\alpha = \omega \frac{d\omega}{d\theta} \quad (15.14)$$

These equations are similar to those obtained in Chap. 11 for the rectilinear motion of a particle, so we can integrate them by following the procedures outlined in Sec. 11.1B.

Two particular cases of rotation occur frequently:

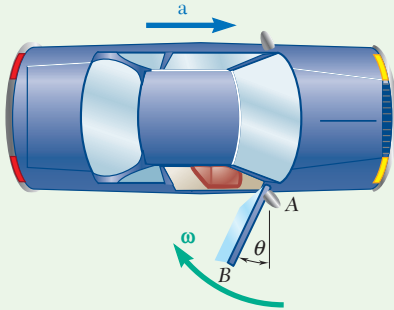
1. *Uniform Rotation.* This case is characterized by the fact that the angular acceleration is zero, therefore the angular velocity is constant and the angular position is given by

$$\theta = \theta_0 + \omega t$$

2. *Uniformly Accelerated Rotation.* In this case, the angular acceleration is constant. We can derive the following formulas relating angular velocity, angular position, and time in a manner similar to that described in Sec. 11.2B. The similarity between the formulas derived here and those obtained for the rectilinear uniformly accelerated motion of a particle is apparent.

$$\begin{aligned} \omega &= \omega_0 + \alpha t \\ \theta &= \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2 \\ \omega^2 &= \omega_0^2 + 2\alpha(\theta - \theta_0) \end{aligned} \quad (15.16)$$

We emphasize that you can use formula (15.15) only when $\alpha = 0$, and formulas (15.16) only when $\alpha = \text{constant}$. In any other case, you need to use the general Eq. (15.12) through Eq. (15.14).



Sample Problem 15.1

A driver starts his car with the door on the passenger's side wide open ($\theta = 0$). As the car moves forward with constant acceleration, the angular acceleration of the door is $\alpha = 2.5 \cos \theta$, where α is in rad/s^2 . Determine the angular velocity of the door as it slams shut ($\theta = 90^\circ$).

STRATEGY: You are given the angular acceleration as a function of θ , so use the kinematic relationships between angular acceleration, angular velocity, angular position, and time.

MODELING and ANALYSIS: Model the door as a rigid body. Using the basic kinematic relationship gives

$$\alpha = \frac{d\omega}{dt} = \omega \frac{d\omega}{d\theta} = 2.5 \cos \theta$$

Separating variables gives

$$\omega d\omega = 2.5 \cos \theta d\theta$$

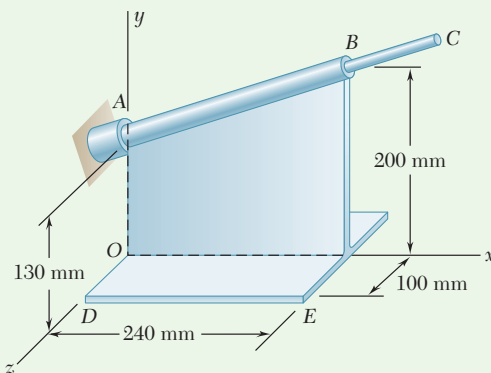
Integrating, using $\omega = 0$ when $\theta = 0$, you have

$$\int_0^\omega \omega d\omega = \int_0^\theta 2.5 \cos \theta d\theta$$

$$\frac{1}{2} \omega^2 = 2.5 \sin \theta \Big|_0^{\pi/2} = 2.5$$

$$\omega = 2.24 \text{ rad/s} \quad \blacktriangleleft$$

REFLECT and THINK: If the angular acceleration of the door had been a constant 2.5 rad/s^2 , you would have found $\frac{1}{2} \omega^2 = 2.5 \Big|_0^{\pi/2}$ or $\omega = 2.80 \text{ rad/s}$. Since $\alpha = 2.5 \cos \theta$ decreases as θ increases, it makes sense that the answer you found in this case is smaller than the case for constant angular acceleration.



Sample Problem 15.2

The assembly shown rotates about the rod AC . At the instant shown, the assembly has an angular velocity of 5 rad/s that is increasing with an angular acceleration of 25 rad/s^2 . Knowing that the y -component of the velocity of corner D is negative at this instant in time, determine the velocity and acceleration of corner E .

STRATEGY: You are interested in determining the velocity and acceleration of a point on a body undergoing fixed-axis rotation, so use rigid body kinematics.

(continued)

MODELING and ANALYSIS: Model the assembly as a rigid body. You can find the velocity and acceleration of E using

$$\mathbf{v}_E = \boldsymbol{\omega} \times \mathbf{r}_{E/B} \quad (1)$$

$$\mathbf{a}_E = \boldsymbol{\alpha} \times \mathbf{r}_{E/B} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{E/B}) = \boldsymbol{\alpha} \times \mathbf{r}_{E/B} + \boldsymbol{\omega} \times \mathbf{v}_E \quad (2)$$

To use these equations, you need the angular velocity vector, the angular acceleration vector, and the position vector. The direction of the angular velocity and acceleration vectors are along the axis of rotation. Since the corner D is moving downward and using the right-hand rule, you know $\boldsymbol{\omega}$ is in the direction shown in Fig. 1. Therefore, to write the angular velocity vector, you need a unit vector in this direction. You know that

$$\mathbf{AB} = (0.24 \text{ m})\mathbf{i} + (0.07 \text{ m})\mathbf{j}$$

so the unit vector from A to B is

$$\boldsymbol{\lambda}_{AB} = \frac{(0.24 \text{ m})\mathbf{i} + (0.07 \text{ m})\mathbf{j}}{\sqrt{(0.24 \text{ m})^2 + (0.07 \text{ m})^2}} = 0.960\mathbf{i} + 0.280\mathbf{j}$$

Thus, the angular velocity and angular acceleration are

$$\boldsymbol{\omega} = \omega \boldsymbol{\lambda}_{AB} = (5 \text{ rad/s})(0.960\mathbf{i} + 0.280\mathbf{j}) = (4.80 \text{ rad/s})\mathbf{i} + (1.40 \text{ rad/s})\mathbf{j}$$

$$\boldsymbol{\alpha} = \alpha \boldsymbol{\lambda}_{AB} = (25 \text{ rad/s})(0.960\mathbf{i} + 0.280\mathbf{j}) = (24.0 \text{ rad/s}^2)\mathbf{i} + (7.00 \text{ rad/s}^2)\mathbf{j}$$

The position vector of E with respect to B is

$$\mathbf{r}_{E/B} = (-0.20 \text{ m})\mathbf{j} + (0.10 \text{ m})\mathbf{k}$$

Substituting these expressions into Eqs. (1) and (2) gives

$$\mathbf{v}_E = \boldsymbol{\omega} \times \mathbf{r}_{E/B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4.80 & 1.40 & 0 \\ 0 & -0.20 & 0.10 \end{vmatrix} = 0.140\mathbf{i} - 0.480\mathbf{j} - 0.960\mathbf{k}$$

$$\mathbf{v}_E = (0.140 \text{ m/s})\mathbf{i} - (0.480 \text{ m/s})\mathbf{j} - (0.960 \text{ m/s})\mathbf{k} \quad \blacktriangleleft$$

$$\begin{aligned} \mathbf{a}_E &= \boldsymbol{\alpha} \times \mathbf{r}_{E/B} + \boldsymbol{\omega} \times \mathbf{v}_E = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 24.0 & 7.00 & 0 \\ 0 & -0.20 & 0.10 \end{vmatrix} \\ &\quad + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 4.80 & 1.40 & 0 \\ 0.140 & -0.480 & -0.960 \end{vmatrix} \\ &= 0.70\mathbf{i} - 2.40\mathbf{j} - 4.80\mathbf{k} - 1.344\mathbf{i} + 4.608\mathbf{j} + (-2.304 - 0.196)\mathbf{k} \\ \mathbf{a}_E &= -(0.644 \text{ m/s}^2)\mathbf{i} + (2.21 \text{ m/s}^2)\mathbf{j} - (7.30 \text{ m/s}^2)\mathbf{k} \quad \blacktriangleleft \end{aligned}$$

REFLECT and THINK: The first term of Eq. (2) represents the tangential acceleration of point E . The second term of Eq. (2) represents the normal acceleration of point E and points toward the bar AB . Note that you could have chosen any point along the axis of rotation to define your position vector.

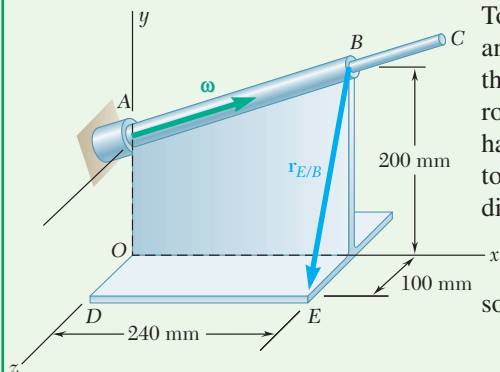


Fig. 1 Direction of the angular velocity and the position vector to point E .

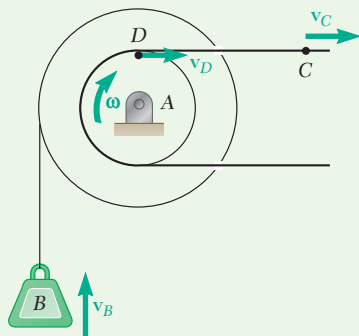
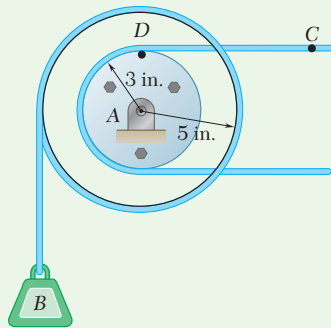


Fig. 1 The velocity of two point on an inextensible cable are equal.

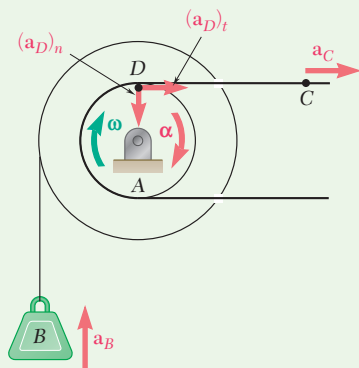


Fig. 2 Acceleration of B , C , and D .

Sample Problem 15.3

Load B is connected to a double pulley by one of the two inextensible cables shown. The motion of the pulley is controlled by cable C , which has a constant acceleration of 9 in./s^2 and an initial velocity of 12 in./s , both directed to the right. Determine (a) the number of revolutions executed by the pulley in 2 s , (b) the velocity and change in position of the load B after 2 s , and (c) the acceleration of point D on the rim of the inner pulley at $t = 0$.

STRATEGY: This is a case of uniformly accelerated rotation, so you can use the kinematic relationships between angular acceleration, angular velocity, angular position, and time. You also need to use the kinematic relationships for the velocity and acceleration of a point on an object undergoing fixed axis rotation.

MODELING and ANALYSIS:

a. Motion of Pulley. You can model the pulley as a rigid body rotating about a fixed axis A . Since the cable is inextensible, the velocity of point D is equal to the velocity of point C (Fig. 1), and the tangential component of the acceleration of D is equal to the acceleration of C (Fig. 2).

$$(\mathbf{v}_D)_0 = (\mathbf{v}_C)_0 = 12 \text{ in./s} \rightarrow \quad (\mathbf{a}_D)_t = \mathbf{a}_C = 9 \text{ in./s}^2 \rightarrow$$

The distance from D to the center of the pulley is 3 in. , so you have

$$\begin{aligned} (v_D)_0 &= r\omega_0 & 12 \text{ in./s} &= (3 \text{ in.})\omega_0 & \omega_0 &= 4 \text{ rad/s} \downarrow \\ (a_D)_t &= r\alpha & 9 \text{ in./s}^2 &= (3 \text{ in.})\alpha & \alpha &= 3 \text{ rad/s}^2 \downarrow \end{aligned}$$

Using the equations of uniformly accelerated motion, for $t = 2 \text{ s}$ you obtain

$$\begin{aligned} \omega &= \omega_0 + \alpha t = 4 \text{ rad/s} + (3 \text{ rad/s}^2)(2 \text{ s}) = 10 \text{ rad/s} \\ \omega &= 10 \text{ rad/s} \downarrow \\ \theta &= \omega_0 t + \frac{1}{2}\alpha t^2 = (4 \text{ rad/s})(2 \text{ s}) + \frac{1}{2}(3 \text{ rad/s}^2)(2 \text{ s})^2 = 14 \text{ rad} \\ \theta &= 14 \text{ rad} \downarrow \end{aligned}$$

$$\text{Number of revolutions} = (14 \text{ rad}) \left(\frac{1 \text{ rev}}{2\pi \text{ rad}} \right) = 2.23 \text{ rev} \quad \blacktriangleleft$$

b. Motion of Load B . The motion of load B is the same as a point on the outer rim of the double pulley. Using $r = 5 \text{ in.}$, you have

$$\begin{aligned} v_B &= r\omega = (5 \text{ in.})(10 \text{ rad/s}) = 50 \text{ in./s} & \mathbf{v}_B &= 50 \text{ in./s} \uparrow \quad \blacktriangleleft \\ \Delta y_B &= r\theta = (5 \text{ in.})(14 \text{ rad}) = 70 \text{ in.} & \Delta y_B &= 70 \text{ in. upward} \quad \blacktriangleleft \end{aligned}$$

c. Acceleration of Point D at $t = 0$. The acceleration of point D has a tangential and a normal component (Fig. 2). The tangential component of the acceleration is

$$(\mathbf{a}_D)_t = \mathbf{a}_C = 9 \text{ in./s}^2 \rightarrow$$

Since, at $t = 0$, $\omega_0 = 4 \text{ rad/s}$, the normal component of the acceleration is

$$(a_D)_n = r_D\omega_0^2 = (3 \text{ in.})(4 \text{ rad/s})^2 = 48 \text{ in./s}^2 \quad (\mathbf{a}_D)_n = 48 \text{ in./s}^2 \downarrow$$

(continued)

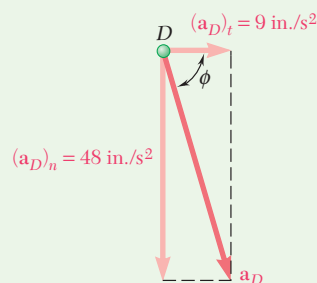


Fig. 3 Vector triangle for resolving the acceleration vector into a magnitude and direction.

You can obtain the magnitude and direction of the total acceleration from Fig. 3.

$$\begin{aligned}\tan \phi &= (48 \text{ in./s}^2)/(9 \text{ in./s}^2) & \phi &= 79.4^\circ \\ a_D \sin 79.4^\circ &= 48 \text{ in./s}^2 & a_D &= 48.8 \text{ in./s}^2 \\ \mathbf{a_D} &= 48.8 \text{ in./s}^2 \searrow 79.4^\circ\end{aligned}$$

REFLECT and THINK: A double pulley acts similarly to a system of gears; for every 3 inches that point C moves to the right, point B moves 5 inches upward. This is also similar to how your bicycle works; the size ratio of the front chainring to the rear sprocket controls the rotation of the rear tire.

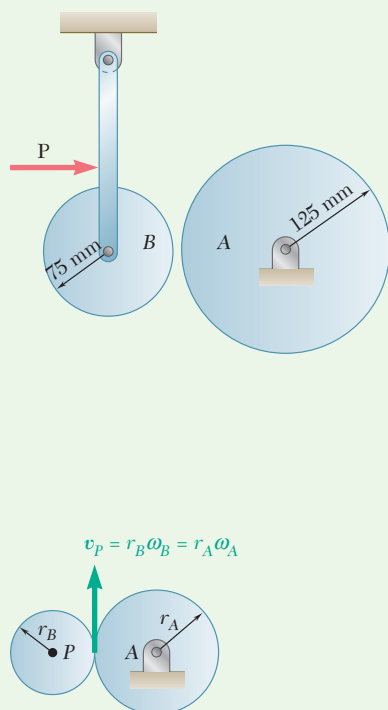


Fig. 1 The wheels will stop slipping when the velocities of the points of contact are equal.

Sample Problem 15.4

Two friction wheels A and B are both rotating freely at 300 rpm clockwise when they are brought into contact. After 6 s of slippage, during which each wheel has a constant angular acceleration, wheel A reaches a final angular velocity of 60 rpm clockwise. Determine the angular acceleration of each wheel during the period of slippage.

STRATEGY: You are not given any masses or forces, so you can use kinematics to solve this problem.

MODELING and ANALYSIS: Model each wheel as a rigid body.

Initial Data. The initial angular velocities of the wheels are $(\omega_A)_0 = (\omega_B)_0 = 300 \text{ rpm} = 31.42 \text{ rad/s}$, both clockwise. After 6 s of slippage, the final angular velocity of A is $\omega_A = 60 \text{ rpm} = 6.28 \text{ rad/s}$ clockwise.

Wheel A. You are told the angular accelerations of the wheels are constant, so

$$\begin{aligned}\omega_A &= (\omega_A)_0 + \alpha_A t: & 6.28 \text{ rad/s} &= 31.42 \text{ rad/s} + \alpha_A(6 \text{ s}) \\ \alpha_A &= -4.19 \text{ rad/s}^2 & \mathbf{\alpha_A} &= 4.19 \text{ rad/s}^2 \nwarrow\end{aligned}$$

Wheel B. At $t = 6 \text{ s}$, the wheels stop slipping and the two points in contact have the same velocity (Fig. 1). Thus,

$$r_A \omega_A = r_B \omega_B$$

so

$$\omega_B = \frac{r_A \omega_A}{r_B} = \frac{(125 \text{ mm})(6.28 \text{ rad/s})}{(75 \text{ mm})} = 10.47 \text{ rad/s} \nearrow$$

The angular acceleration of B is constant, so

$$\begin{aligned}\omega_B &= (\omega_B)_0 + \alpha_B t: & -10.47 \text{ rad/s} &= 31.42 \text{ rad/s} + \alpha_B(6 \text{ s}) \\ \alpha_B &= -6.98 \text{ rad/s}^2\end{aligned}$$

$$\mathbf{\alpha_B} = 6.98 \text{ rad/s}^2 \nwarrow$$

REFLECT and THINK: The initial angular velocity of B is clockwise, and its final angular velocity is counterclockwise. There must be some time when this wheel has an angular velocity of zero and changes direction from rotating clockwise to rotating counterclockwise.

SOLVING PROBLEMS ON YOUR OWN

In this section, we began the study of the motion of rigid bodies by considering two particular types of motion: **translation** and **rotation about a fixed axis**.

1. Rigid body in translation. At any given instant, all the points of a rigid body in translation have the *same velocity* and the *same acceleration* (Fig. 15.7).

2. Rigid body rotating about a fixed axis. The position of a rigid body rotating about a fixed axis is defined at any given instant by the **angular position** θ , which is usually measured in radians. Selecting the unit vector $\mathbf{k}$ along the fixed axis in such a way that the rotation of the body appears counterclockwise as seen from the tip of $\mathbf{k}$, we define the **angular velocity** $\boldsymbol{\omega}$ and the **angular acceleration** $\boldsymbol{\alpha}$ of the body as

$$\boldsymbol{\omega} = \dot{\theta}\mathbf{k} \quad \boldsymbol{\alpha} = \ddot{\theta}\mathbf{k} \quad (15.6, 15.9)$$

In solving problems, keep in mind that the vectors $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ are both directed along the fixed axis of rotation and that their sense can be obtained by the right-hand rule [Sample Prob. 15.2].

a. The velocity of a point P of a body rotating about a fixed axis is

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} \quad (15.5)$$

where $\boldsymbol{\omega}$ is the angular velocity of the body and $\mathbf{r}$ is the position vector drawn from any point on the axis of rotation to point P (Fig. 15.9).

b. The acceleration of point P of a body rotating about a fixed axis is

$$\mathbf{a} = \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \quad (15.8)$$

Since vector products are not commutative, *be sure to write the vectors in the order shown* when using either of the above two equations.

3. Rotation of a representative slab. In many problems, you will be able to reduce the analysis of the rotation of a three-dimensional body about a fixed axis to the case of the rotation of a representative slab in a plane perpendicular to the fixed axis. The z axis should be directed along the axis of rotation and point out of the page. Thus, the representative slab rotates in the xy plane about the origin O of the coordinate system (Fig. 15.10).

To solve problems of this type, you should do the following steps.

a. Draw a diagram of the representative slab showing its dimensions, its angular velocity and angular acceleration, and the vectors representing the velocities and accelerations of the points of the slab.

b. Relate the rotation of the slab and the motion of points of the slab by writing

$$v = r\omega \quad (15.10')$$

$$a_t = r\alpha \quad a_n = r\omega^2 \quad (15.11')$$

Remember that the velocity $\mathbf{v}$ and the component $\mathbf{a}_t$ of the acceleration of a point P of the slab are tangent to the circular path described by P [Sample Probs. 15.3 and 15.4]. You can find the directions of $\mathbf{v}$ and $\mathbf{a}_t$ by rotating the position vector $\mathbf{r}$ through 90° in the sense indicated by $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$, respectively. The normal component $\mathbf{a}_n$ of the acceleration of P is always directed toward the axis of rotation.

4. Equations defining the rotation of a rigid body. Note the similarity between the equations defining the rotation of a rigid body about a fixed axis [Eqs. (15.12) through (15.16)] and those in Chap. 11 defining the rectilinear motion of a particle [Eqs. (11.1) through (11.8)]. All you have to do to obtain the new set of equations is to substitute θ , ω , and α for x , v , and a , respectively, in the equations of Chap. 11 [Sample Prob. 15.1].

Problems

CONCEPT QUESTIONS

- 15.CQ1** A rectangular plate swings from arms of equal length as shown. What is the magnitude of the angular velocity of the plate?
- 0 rad/s
 - 1 rad/s
 - 2 rad/s
 - 3 rad/s
 - Need to know the location of the center of gravity.

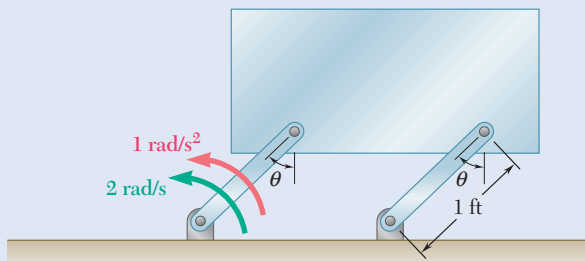


Fig. P15.CQ1

- 15.CQ2** Knowing that wheel A rotates with a constant angular velocity and that no slipping occurs between ring C and wheel A and wheel B, which of the following statements concerning the angular speeds of the three objects is true?

- $\omega_a = \omega_b$
- $\omega_a > \omega_b$
- $\omega_a < \omega_b$
- $\omega_a = \omega_c$
- The contact points between A and C have the same acceleration.

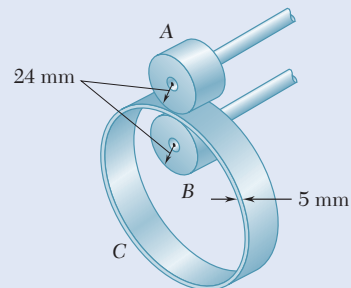


Fig. P15.CQ2

END-OF-SECTION PROBLEMS

- 15.1** The brake drum is attached to a larger flywheel that is not shown. The motion of the brake drum is defined by the relation $\theta = 36t - 1.6t^2$, where θ is expressed in radians and t in seconds. Determine (a) the angular velocity at $t = 2$ s, (b) the number of revolutions executed by the brake drum before coming to rest.
- 15.2** The motion of an oscillating flywheel is defined by the relation $\theta = \theta_0 e^{-3\pi t} \cos 4\pi t$, where θ is expressed in radians and t in seconds. Knowing that $\theta_0 = 0.5$ rad, determine the angular coordinate, the angular velocity, and the angular acceleration of the flywheel when (a) $t = 0$, (b) $t = 0.125$ s.
- 15.3** The motion of an oscillating flywheel is defined by the relation $\theta = \theta_0 e^{-7\pi t/6} \sin 4\pi t$, where θ is expressed in radians and t in seconds. Knowing that $\theta_0 = 0.4$ rad, determine the angular coordinate, the angular velocity, and the angular acceleration of the flywheel when (a) $t = 0.125$ s, (b) $t = \infty$.

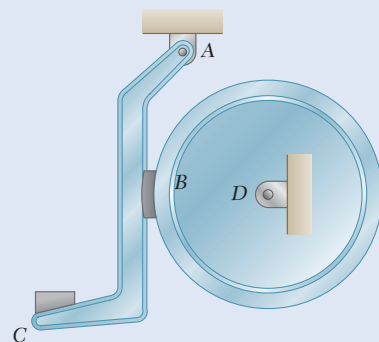


Fig. P15.1

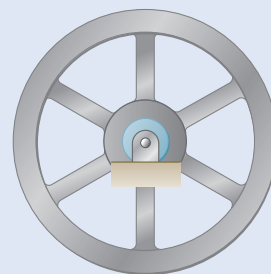


Fig. P15.2 and P15.3

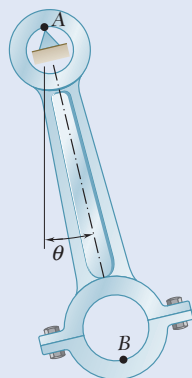


Fig. P15.6

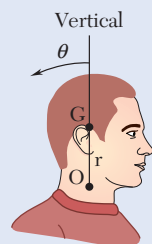


Fig. P15.7

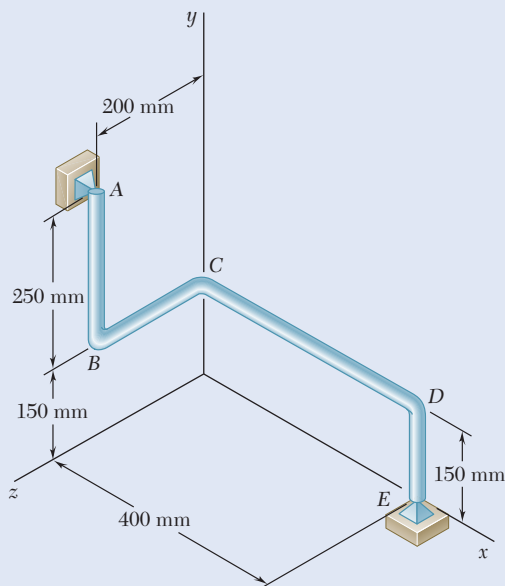


Fig. P15.10

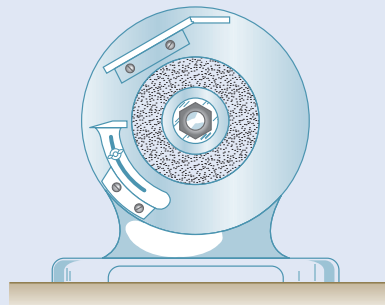


Fig. P15.5

- 15.4** The rotor of a gas turbine is rotating at a speed of 6900 rpm when the turbine is shut down. It is observed that 4 min is required for the rotor to coast to rest. Assuming uniformly accelerated motion, determine (a) the angular acceleration, (b) the number of revolutions that the rotor executes before coming to rest.
- 15.5** A small grinding wheel is attached to the shaft of an electric motor which has a rated speed of 3600 rpm. When the power is turned on, the unit reaches its rated speed in 5 s, and when the power is turned off, the unit coasts to rest in 70 s. Assuming uniformly accelerated motion, determine the number of revolutions that the motor executes (a) in reaching its rated speed, (b) in coasting to rest.

- 15.6** A connecting rod is supported by a knife-edge at point A. For small oscillations the angular acceleration of the connecting rod is governed by the relation $\alpha = -6\theta$ where α is expressed in rad/s^2 and θ in radians. Knowing that the connecting rod is released from rest when $\theta = 20^\circ$, determine (a) the maximum angular velocity, (b) the angular position when $t = 2$ s.
- 15.7** When studying whiplash resulting from rear-end collisions, the rotation of the head is of primary interest. An impact test was performed, and it was found that the angular acceleration of the head is defined by the relation $\alpha = 700 \cos \theta + 70 \sin \theta$, where α is expressed in rad/s^2 and θ in radians. Knowing that the head is initially at rest, determine the angular velocity of the head when $\theta = 30^\circ$.
- 15.8** The angular acceleration of an oscillating disk is defined by the relation $\alpha = -k\theta$, where α is expressed in rad/s^2 and θ in radians. Determine (a) the value of k for which $\omega = 12 \text{ rad/s}$ when $\theta = 0$ and $\theta = 6 \text{ rad}$ when $\omega = 0$, (b) the angular velocity of the disk when $\theta = 3 \text{ rad}$.
- 15.9** The angular acceleration of a shaft is defined by the relation $\alpha = -0.5\omega$, where α is expressed in rad/s^2 and ω in rad/s . Knowing that at $t = 0$ the angular velocity of the shaft is 30 rad/s , determine (a) the number of revolutions the shaft will execute before coming to rest, (b) the time required for the shaft to come to rest, (c) the time required for the angular velocity of the shaft to reduce to 2 percent of its initial value.
- 15.10** The bent rod ABCDE rotates about a line joining points A and E with a constant angular velocity of 9 rad/s . Knowing that the rotation is clockwise as viewed from E, determine the velocity and acceleration of corner C.
- 15.11** In Prob. 15.10, determine the velocity and acceleration of corner B, assuming that the angular velocity is 9 rad/s and increases at the rate of 45 rad/s^2 .

- 15.12** The rectangular block shown rotates about the diagonal OA with a constant angular velocity of 6.76 rad/s . Knowing that the rotation is counterclockwise as viewed from A , determine the velocity and acceleration of point B at the instant shown.
- 15.13** The rectangular block shown rotates about the diagonal OA with an angular velocity of 3.38 rad/s that is decreasing at the rate of 5.07 rad/s^2 . Knowing that the rotation is counterclockwise as viewed from A , determine the velocity and acceleration of point B at the instant shown.
- 15.14** A circular plate of 120-mm radius is supported by two bearings A and B as shown. The plate rotates about the rod joining A and B with a constant angular velocity of 26 rad/s . Knowing that, at the instant considered, the velocity of point C is directed to the right, determine the velocity and acceleration of point E .

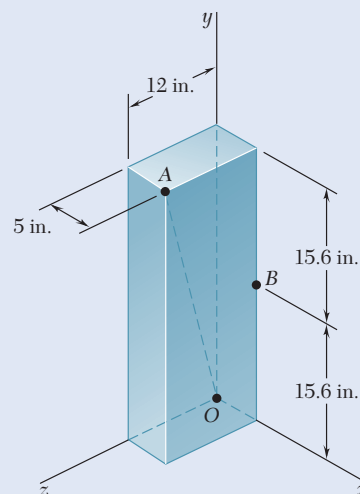


Fig. P15.12 and P15.13

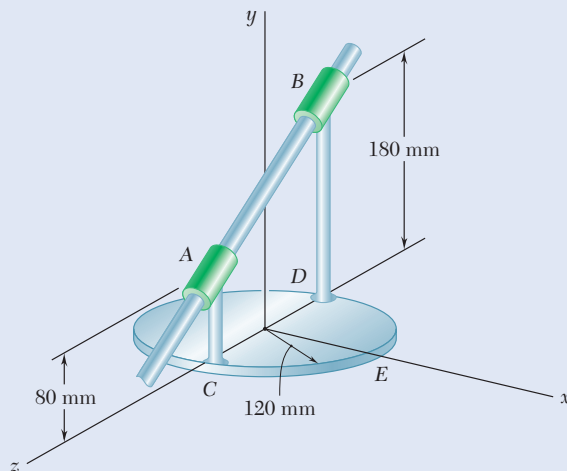


Fig. P15.14

- 15.15** In Prob. 15.14, determine the velocity and acceleration of point E , assuming that the angular velocity is 26 rad/s and increases at the rate of 65 rad/s^2 .
- 15.16** The earth makes one complete revolution around the sun in 365.24 days. Assuming that the orbit of the earth is circular and has a radius of $93,000,000 \text{ mi}$, determine the velocity and acceleration of the earth.
- 15.17** The earth makes one complete revolution on its axis in 23 h 56 min. Knowing that the mean radius of the earth is 3960 mi , determine the linear velocity and acceleration of a point on the surface of the earth (a) at the equator, (b) at Philadelphia, latitude 40° north, (c) at the North Pole.
- 15.18** A series of small machine components being moved by a conveyor belt pass over a 120-mm -radius idler pulley. At the instant shown, the velocity of point A is 300 mm/s to the left and its acceleration is 180 mm/s^2 to the right. Determine (a) the angular velocity and angular acceleration of the idler pulley, (b) the total acceleration of the machine component at B .
- 15.19** A series of small machine components being moved by a conveyor belt pass over a 120-mm -radius idler pulley. At the instant shown, the angular velocity of the idler pulley is 4 rad/s clockwise. Determine the angular acceleration of the pulley for which the magnitude of the total acceleration of the machine component at B is 2400 mm/s^2 .

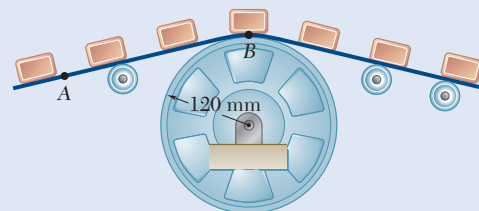


Fig. P15.18 and P15.19

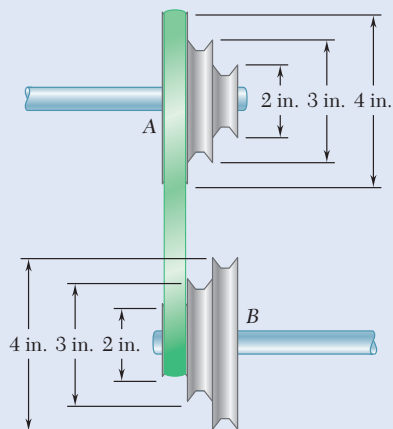


Fig. P15.22

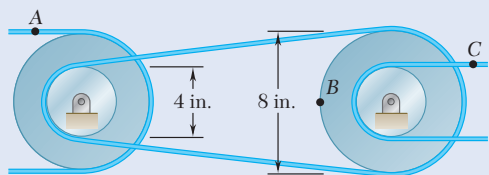


Fig. P15.23

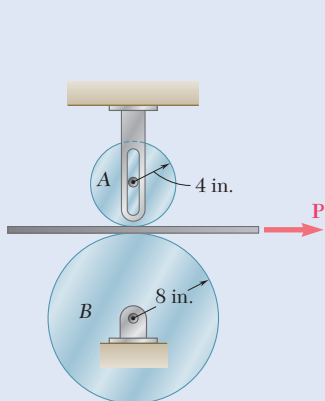


Fig. P15.25

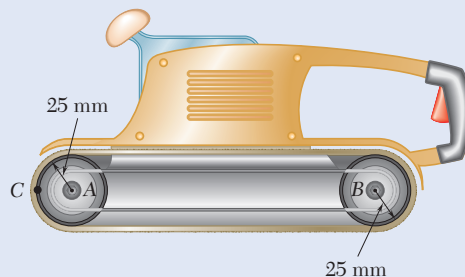


Fig. P15.20 and P15.21

15.20 The belt sander shown is initially at rest. If the driving drum B has a constant angular acceleration of 120 rad/s^2 counterclockwise, determine the magnitude of the acceleration of the belt at point C when (a) $t = 0.5 \text{ s}$, (b) $t = 2 \text{ s}$.

15.21 The rated speed of drum B of the belt sander shown is 2400 rpm. When the power is turned off, it is observed that the sander coasts from its rated speed to rest in 10 s. Assuming uniformly decelerated motion, determine the velocity and acceleration of point C of the belt, (a) immediately before the power is turned off, (b) 9 s later.

15.22 The two pulleys shown may be operated with the V belt in any of three positions. If the angular acceleration of shaft A is 6 rad/s^2 and if the system is initially at rest, determine the time required for shaft B to reach a speed of 400 rpm with the belt in each of the three positions.

15.23 Three belts move over two pulleys without slipping in the speed reduction system shown. At the instant shown, the velocity of point A on the input belt is 2 ft/s to the right, decreasing at the rate of 6 ft/s^2 . Determine, at this instant, (a) the velocity and acceleration of point C on the output belt, (b) the acceleration of point B on the output pulley.

15.24 A gear reduction system consists of three gears A , B , and C . Knowing that gear A rotates clockwise with a constant angular velocity $\omega_A = 600 \text{ rpm}$, determine (a) the angular velocities of gears B and C , (b) the accelerations of the points on gears B and C which are in contact.

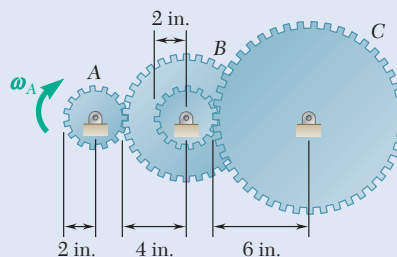


Fig. P15.24

15.25 A belt is pulled to the right between cylinders A and B . Knowing that the speed of the belt is a constant 5 ft/s and no slippage occurs, determine (a) the angular velocities of A and B , (b) the accelerations of the points which are in contact with the belt.

- 15.26** Ring C has an inside radius of 55 mm and an outside radius of 60 mm and is positioned between two wheels A and B , each of 24-mm outside radius. Knowing that wheel A rotates with a constant angular velocity of 300 rpm and that no slipping occurs, determine (a) the angular velocity of ring C and of wheel B , (b) the acceleration of the points on A and B that are in contact with C .

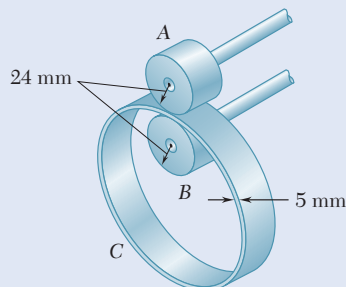


Fig. P15.26

- 15.27** Ring B has an inside radius r_2 and hangs from the horizontal shaft A as shown. Shaft A rotates with a constant angular velocity of 25 rad/s and no slipping occurs. Knowing that $r_1 = 12$ mm, $r_2 = 30$ mm, and $r_3 = 40$ mm, determine (a) the angular velocity of ring B , (b) the accelerations of the points of shaft A and ring B which are in contact, (c) the magnitude of the acceleration of point D .

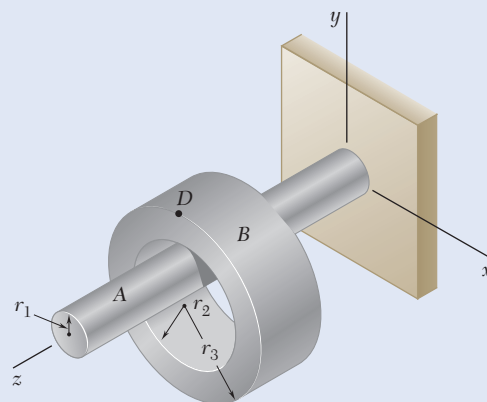


Fig. P15.27

- 15.28** A plastic film moves over two drums. During a 4-s interval the speed of the tape is increased uniformly from $v_0 = 2$ ft/s to $v_1 = 4$ ft/s. Knowing that the tape does not slip on the drums, determine (a) the angular acceleration of drum B , (b) the number of revolutions executed by drum B during the 4-s interval.

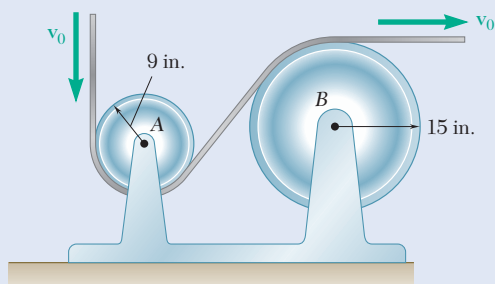


Fig. P15.28

- 15.29** Cylinder A is moving downward with a velocity of 3 m/s when the brake is suddenly applied to the drum. Knowing that the cylinder moves 6 m downward before coming to rest and assuming uniformly accelerated motion, determine (a) the angular acceleration of the drum, (b) the time required for the cylinder to come to rest.

- 15.30** The system shown is held at rest by the brake-and-drum system shown. After the brake is partially released at $t = 0$ it is observed that the cylinder moves 5 m in 4.5 s. Assuming uniformly accelerated motion, determine (a) the angular acceleration of the drum, (b) the angular velocity of the drum at $t = 3.5$ s.

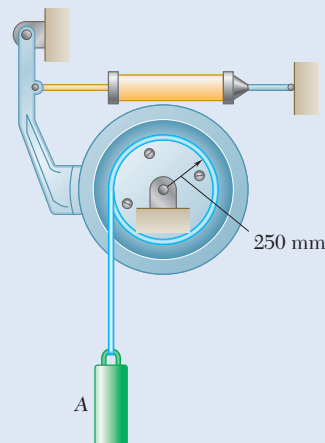


Fig. P15.29 and P15.30

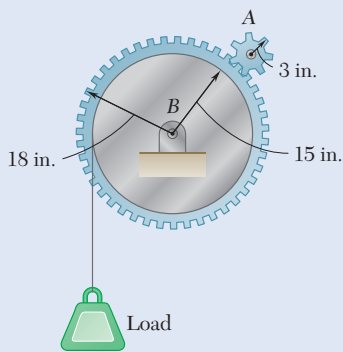


Fig. P15.31

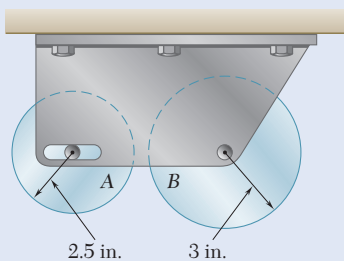


Fig. P15.32 and P15.33

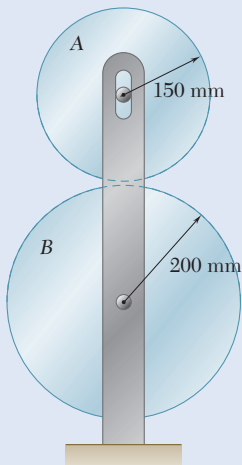


Fig. P15.34 and P15.35

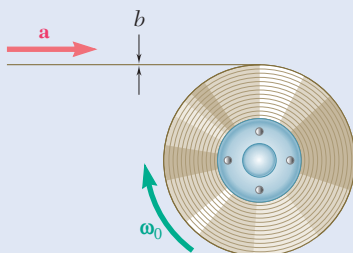


Fig. P15.36

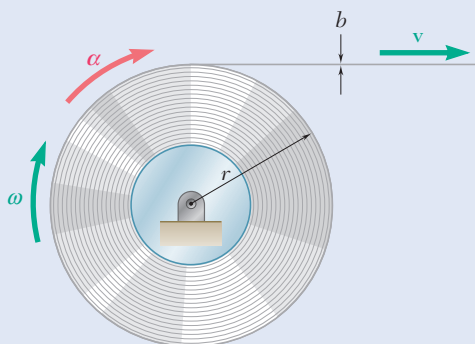


Fig. P15.37

- 15.31** A load is to be raised 20 ft by the hoisting system shown. Assuming gear A is initially at rest, accelerates uniformly to a speed of 120 rpm in 5 s, and then maintains a constant speed of 120 rpm, determine (a) the number of revolutions executed by gear A in raising the load, (b) the time required to raise the load.
- 15.32** A simple friction drive consists of two disks A and B. Initially, disk B has a clockwise angular velocity of 500 rpm, and disk A is at rest. It is known that disk B will coast to rest in 60 s. However, rather than waiting until both disks are at rest to bring them together, disk A is given a constant angular acceleration of 3 rad/s^2 counterclockwise. Determine (a) at what time the disks can be brought together if they are not to slip, (b) the angular velocity of each disk as contact is made.
- 15.33** Two friction wheels A and B are both rotating freely at 300 rpm counterclockwise when they are brought into contact. After 12 s of slippage, during which time each wheel has a constant angular acceleration, wheel B reaches a final angular velocity of 75 rpm counterclockwise. Determine (a) the angular acceleration of each wheel during the period of slippage, (b) the time at which the angular velocity of wheel A is equal to zero.
- 15.34** Two friction disks A and B are to be brought into contact without slipping when the angular velocity of disk A is 240 rpm counterclockwise. Disk A starts from rest at time $t = 0$ and is given a constant angular acceleration with a magnitude α . Disk B starts from rest at time $t = 2 \text{ s}$ and is given a constant clockwise angular acceleration, also with a magnitude α . Determine (a) the required angular acceleration magnitude α , (b) the time at which the contact occurs.
- 15.35** Two friction disks A and B are brought into contact when the angular velocity of disk A is 240 rpm counterclockwise and disk B is at rest. A period of slipping follows and disk B makes two revolutions before reaching its final angular velocity. Assuming that the angular acceleration of each disk is constant and inversely proportional to the cube of its radius, determine (a) the angular acceleration of each disk, (b) the time during which the disks slip.
- *15.36** Steel tape is being wound onto a spool that rotates with a constant angular velocity ω_0 . Denoting by r the radius of the spool and tape at any given time and by b the thickness of the tape, derive an expression for the acceleration of the tape as it approaches the spool.
- *15.37** In a continuous printing process, paper is drawn into the presses at a constant speed v . Denoting by r the radius of the paper roll at any given time and by b the thickness of the paper, derive an expression for the angular acceleration of the paper roll.

15.2 GENERAL PLANE MOTION: VELOCITY

As indicated in the chapter introduction, general plane motion describes a plane motion that is neither a pure translation nor a pure rotation. As you will presently see, however, **a general plane motion can always be considered as the sum of a translation and a rotation.**

15.2A Analyzing General Plane Motion

As an example of general plane motion, consider a wheel rolling on a straight track (Fig. 15.12). Over some interval of time, two given points A and B will have moved, respectively, from A_1 to A_2 and from B_1 to B_2 . However, we could obtain the same result through a translation that would bring A_1 and B_1 into A_2 and B'_1 (the line AB remaining vertical), followed by a rotation about A , bringing B into B_2 . The original rolling motion differs from the combination of translation and rotation when these motions are taken in succession, but we can duplicate the original motion exactly using a combination of simultaneous translation and rotation.

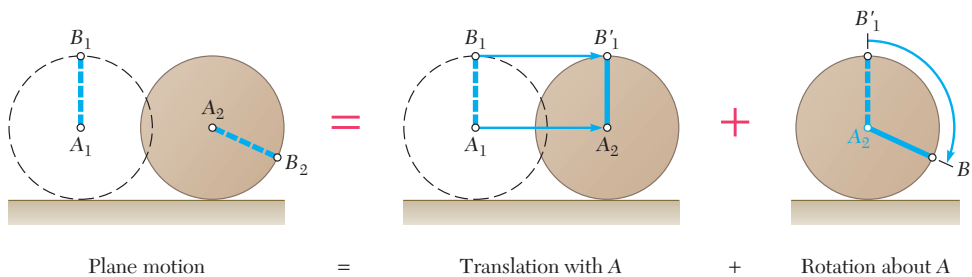


Fig. 15.12 The general plane motion of a rolling wheel can be analyzed as a combination of translation plus a fixed-axis rotation.

Another example of plane motion is shown in Fig. 15.13, which represents a rod whose ends slide along a horizontal and a vertical track. We can replace this motion using a horizontal translation and a rotation about A (Fig. 15.13a) or using a vertical translation and a rotation about B (Fig. 15.13b).

In the general case of plane motion, we consider a small displacement that brings two particles A and B of a representative rigid body, respectively, from A_1 and B_1 into A_2 and B_2 (Fig. 15.14). We can divide this displacement into two parts: in one, the particles move into A_2 and B'_1 while the line AB maintains the same direction; in the other, B moves into B_2 while A remains fixed. The first part of the motion is clearly a translation, and the second part is clearly a rotation about A .

Recall from Sec. 11.4D the definition of the relative motion of a particle with respect to a moving frame of reference—as opposed to its absolute motion with respect to a fixed frame of reference. With that definition in mind, we can restate our results: Given two particles A and B of a rigid body in plane motion, the relative motion of B with respect

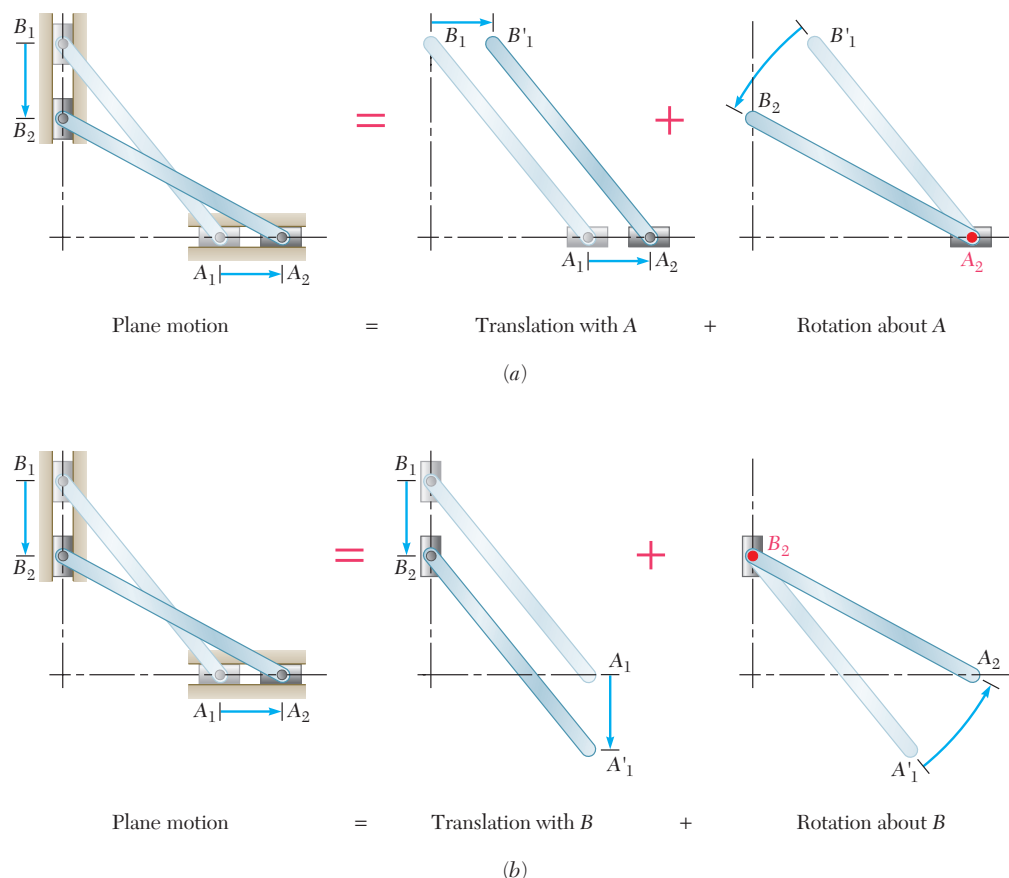


Fig. 15.13 The general plane motion of this sliding rod can be analyzed as (a) a horizontal translation plus a fixed-axis rotation about A or (b) a vertical translation and a fixed-axis rotation about B. The results are the same either way.

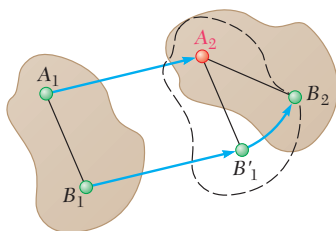


Fig. 15.14 General plane motion is a combination of a translation plus a fixed-axis rotation. To an observer moving with A but not rotating, particle B appears to travel in a circle centered at A.

to a frame attached to A and of fixed orientation is a rotation. To an observer moving with A but not rotating, particle B appears to describe an arc of a circle centered at A.

15.2B Absolute and Relative Velocity in Plane Motion

We have just seen that any plane motion of a rigid body can be replaced by a translation of an arbitrary reference point A and a simultaneous rotation about A. We can obtain the absolute velocity $\mathbf{v}_B$ of a particle B of the rigid body from the relative velocity formula derived in Sec. 11.4D, as

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad (15.17)$$

where the right-hand side represents a vector sum. The velocity $\mathbf{v}_A$ corresponds to the translation of the rigid body with A, whereas the relative velocity $\mathbf{v}_{B/A}$ is associated with the rotation of the rigid body about A and is measured with respect to axes centered at A and of fixed orientation (Fig. 15.15). Denoting the position vector of B relative to A by $\mathbf{r}_{B/A}$ (which points from A to B) and the angular velocity of the rigid body with respect to axes of fixed orientation by $\omega \mathbf{k}$, we have from Eqs. (15.10) and (15.10')

$$\mathbf{v}_{B/A} = \omega \mathbf{k} \times \mathbf{r}_{B/A} \quad v_{B/A} = r\omega \quad (15.18)$$

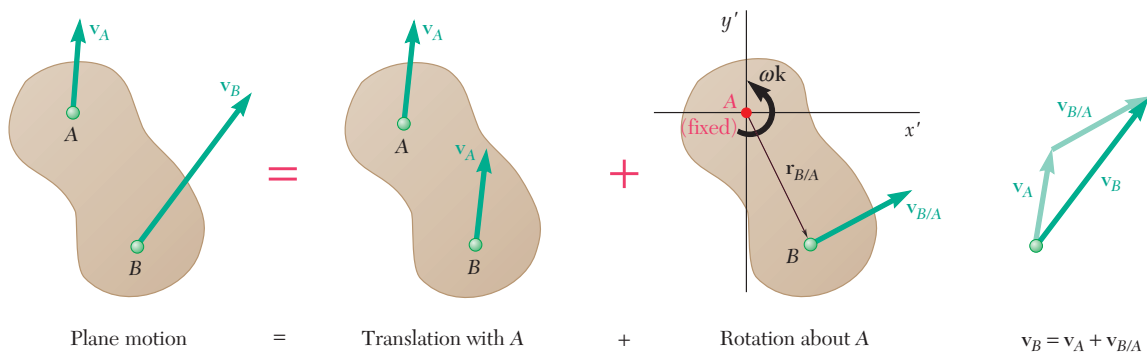


Fig. 15.15 A pictorial representation of the vector equation relating the velocity of two points on a rigid body undergoing general plane motion.

where r is the distance from A to B . Substituting for $\mathbf{v}_{B/A}$ from Eq. (15.18) into Eq. (15.17), we also have

Relative velocity for two points on a rigid body

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A} \quad (15.17')$$

As an example, let us again consider rod AB of Fig. 15.13. Assuming that we know the velocity $\mathbf{v}_A$ of end A , we propose to find the velocity $\mathbf{v}_B$ of end B and the angular velocity $\boldsymbol{\omega}$ of the rod in terms of the velocity $\mathbf{v}_A$, the length l , and the angle θ . Choosing A as a reference point, the given motion is equivalent to a translation of A and a simultaneous rotation about A (Fig. 15.16). The absolute velocity of B therefore must be equal to the vector sum

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad (15.17)$$

Note that although we know the direction of $\mathbf{v}_{B/A}$, its magnitude $l\omega$ is unknown. However, this is compensated for by the fact that the direction of $\mathbf{v}_B$ is known. We can therefore complete the vector diagram of Fig. 15.16. Solving for the magnitudes v_B and ω , we obtain

$$v_B = v_A \tan \theta \quad \omega = \frac{v_{B/A}}{l} = \frac{v_A}{l \cos \theta} \quad (15.19)$$

Alternatively, we can also solve this problem by using the vector relationship in Eq. (15.17'). Recognizing that point A is constrained to

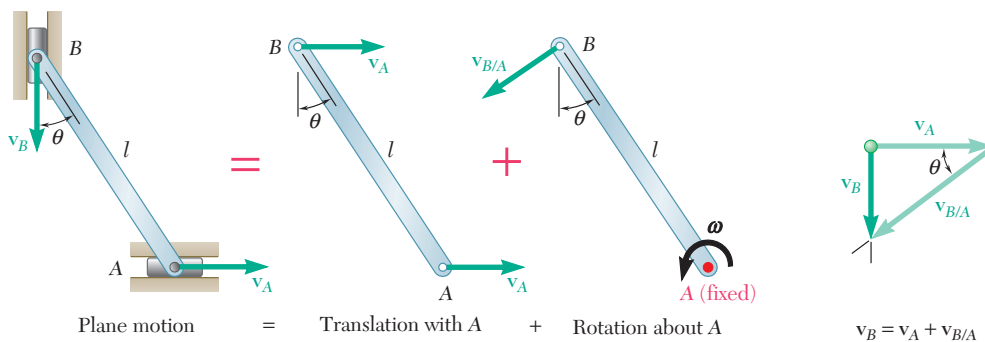


Fig. 15.16 Pictorial representation of Eq. (15.17) for a sliding rod. The relative velocity $\mathbf{v}_{B/A}$ is perpendicular to the line connecting A and B .



Photo 15.4 Planetary gear systems are used in applications requiring a large reduction ratio and a high torque-to-weight ratio. The small gears undergo general plane motion.

move only in the x direction and B moves only in the y direction (assume it moves down), we can write

$$-v_B \mathbf{j} = v_A \mathbf{i} + \omega \mathbf{k} \times (-l \sin \theta \mathbf{i} + l \cos \theta \mathbf{j}) = (v_A - \omega l \cos \theta) \mathbf{i} - \omega l \sin \theta \mathbf{j}$$

Equating components in the x direction, we obtain

$$v_A - \omega l \cos \theta = 0 \quad \omega = \frac{v_A}{l \cos \theta}$$

Equating components in the y direction, we obtain

$$v_B = \omega l \sin \theta = \left(\frac{v_A}{l \cos \theta} \right) l \sin \theta = v_A \tan \theta$$

These are the same results as we obtained in Eq. 15.19. We obtain the same result by using B as a point of reference. Resolving the given motion into a translation of B and a simultaneous rotation about B (Fig. 15.17), we have the equation

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B} = \mathbf{v}_B + \omega \mathbf{k} \times \mathbf{r}_{A/B} \quad (15.20)$$

which is represented graphically in Fig. 15.17. Note that $\mathbf{v}_{A/B}$ and $\mathbf{v}_{B/A}$ have the same magnitude $l\omega$ but opposite sense. The sense of the relative velocity depends, therefore, upon the point of reference that we have selected and should be carefully ascertained from the appropriate diagram (Fig. 15.16 or 15.17).

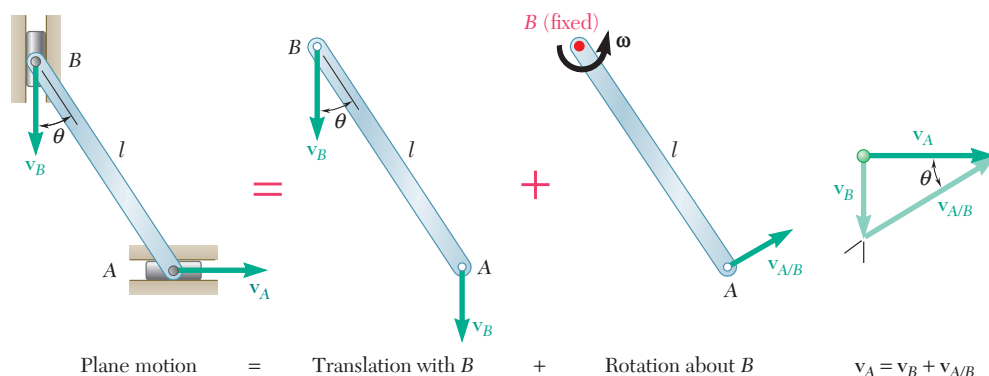


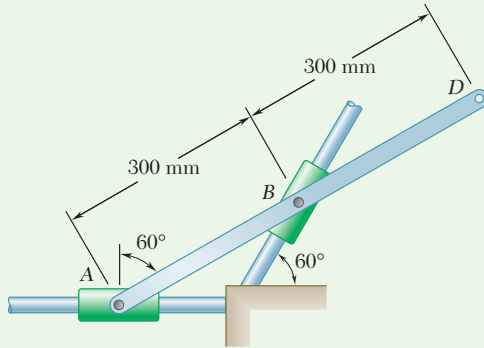
Fig. 15.17 Pictorial representation of Eq. (15.20) for a sliding rod. The relative velocity $\mathbf{v}_{A/B}$ is perpendicular to the line connecting A and B .

Finally, observe that the angular velocity ω of the rod in its rotation about B is the same as in its rotation about A . It is measured in both cases by the rate of change of the angle θ . This result is quite general; you should therefore bear in mind that

The angular velocity ω of a rigid body in plane motion is independent of the reference point.

Most mechanisms consist not of one but of *several* moving parts. When the various parts of a mechanism are connected by pins, we can analyze the mechanism by considering each part as a rigid body, keeping in mind that the points where two parts are connected must have the same absolute velocity (see Sample Probs. 15.7 and 15.8). We can use a similar analysis when gears are involved, since the teeth in contact also must have the same absolute velocity. However, when a mechanism contains parts that slide on each other, the relative velocity of the parts in contact must be taken into account (see Sec. 15.5).

Sample Problem 15.5



Collars A and B are pin-connected to bar ABD and can slide along fixed rods. Knowing that at the instant shown the velocity of A is 0.9 m/s to the right, determine (a) the angular velocity of ABD , (b) the velocity of point D .

STRATEGY: Use the kinematic equation that relates the velocity of two points on the same rigid body. Because you know the directions of the velocities of points A and B , choose these two points to relate.

MODELING and ANALYSIS: Model bar ABD as a rigid body. From kinematics you know

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

Substituting in known values (Fig. 1) and assuming $\boldsymbol{\omega} = \omega \mathbf{k}$ gives you

$$\begin{aligned} v_B \cos 60^\circ \mathbf{i} + v_B \sin 60^\circ \mathbf{j} &= (0.9)\mathbf{i} + \\ &\quad \omega \mathbf{k} \times [(0.3 \cos 30^\circ)\mathbf{i} + (0.3 \sin 30^\circ)\mathbf{j}] \\ 0.500v_B \mathbf{i} + 0.866v_B \mathbf{j} &= (0.9 - 0.15\omega)\mathbf{i} + 0.260\omega \mathbf{j} \end{aligned}$$

Equating components,

$$\mathbf{i}: 0.500v_B = 0.9 - 0.15\omega$$

$$\mathbf{j}: 0.866v_B = 0.260\omega$$

Solving these equations gives you $v_B = 0.900\text{ m/s}$ and $\omega = 3.00\text{ rad/s}$

$$\omega = 3.00\text{ rad/s} \uparrow \quad \blacktriangleleft$$

Velocity of D . The relationship between the velocities of A and D is

$$\mathbf{v}_D = \mathbf{v}_A + \mathbf{v}_{D/A} = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{D/A}$$

Substituting in values from above gives

$$\mathbf{v}_D = (0.9)\mathbf{i} + 3.00\mathbf{k} \times [(0.6 \cos 30^\circ)\mathbf{i} + (0.6 \sin 30^\circ)\mathbf{j}]$$

$$\mathbf{v}_D = (0.9 - 0.9)\mathbf{i} + 1.559\mathbf{j}$$

$$\mathbf{v}_D = 1.559\text{ m/s} \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: The velocity of point D is straight up at this instant in time, but as the bar continues to rotate counter-clockwise, the direction of the velocity of D will continuously change.

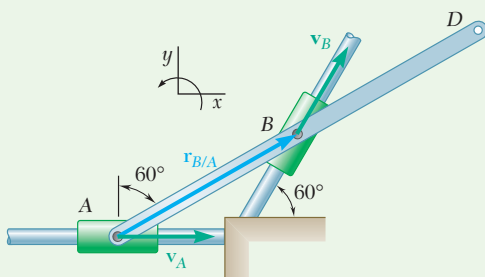
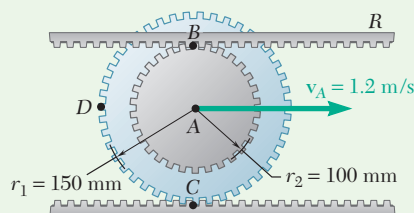


Fig. 1 Position vector and directions of the velocities of A and B .

Sample Problem 15.6



The double gear shown rolls on the stationary lower rack; the velocity of its center A is 1.2 m/s directed to the right. Determine (a) the angular velocity of the gear, (b) the velocities of the upper rack R and of point D of the gear.

STRATEGY: The double gear is undergoing general motion, so use rigid body kinematics. Resolve the rolling motion into two component motions: a translation of point A and a rotation about the center A (Fig. 1). In the translation, all points of the gear move with the same velocity $\mathbf{v}_A$. In the rotation, each point P of the gear moves about A with a relative velocity $\mathbf{v}_{P/A} = \boldsymbol{\omega} \times \mathbf{r}_{P/A}$, where $\mathbf{r}_{P/A}$ is the position vector of P relative to A .

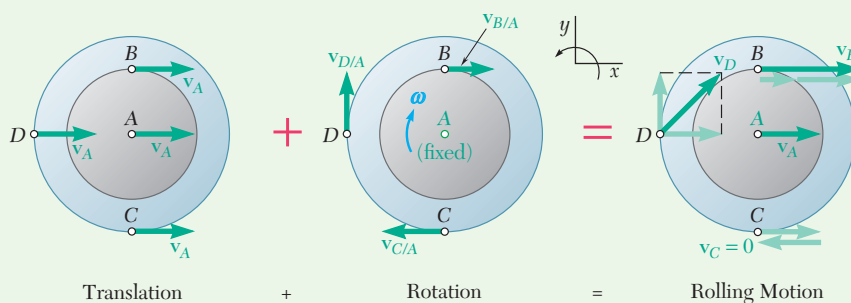


Fig. 1 The gear motion can be modeled as a translation plus a rotation.

MODELING and ANALYSIS:

a. Angular Velocity of the Gear. Since the gear rolls on the lower rack, its center A moves through a distance equal to the outer circumference $2\pi r_1$ for each full revolution of the gear. Noting that 1 rev = 2π rad, and that when A moves to the right ($x_A > 0$), the gear rotates clockwise ($\theta < 0$), you have

$$\frac{x_A}{2\pi r_1} = -\frac{\theta}{2\pi} \quad x_A = -r_1 \theta$$

Differentiating with respect to the time t and substituting the known values $v_A = 1.2$ m/s and $r_1 = 150$ mm = 0.150 m, you obtain

$$v_A = -r_1 \omega \quad 1.2 \text{ m/s} = -(0.150 \text{ m})\omega \quad \omega = -8 \text{ rad/s} \\ \boldsymbol{\omega} = \omega \mathbf{k} = -(8 \text{ rad/s})\mathbf{k} \quad \blacktriangleleft$$

where $\mathbf{k}$ is a unit vector pointing out of the page.

b. Velocity of Upper Rack. The velocity of the upper rack is equal to the velocity of point B ; you have

$$\begin{aligned} \mathbf{v}_R = \mathbf{v}_B &= \mathbf{v}_A + \mathbf{v}_{B/A} = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A} \\ &= (1.2 \text{ m/s})\mathbf{i} - (8 \text{ rad/s})\mathbf{k} \times (0.100 \text{ m})\mathbf{j} \\ &= (1.2 \text{ m/s})\mathbf{i} + (0.8 \text{ m/s})\mathbf{i} = (2 \text{ m/s})\mathbf{i} \\ \mathbf{v}_R &= 2 \text{ m/s} \rightarrow \quad \blacktriangleleft \end{aligned}$$

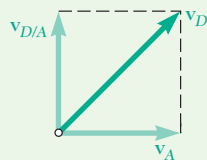


Fig. 2 The two components of the velocity of D .

Velocity of Point D . The velocity of point D has two components (Fig. 2):

$$\begin{aligned}\mathbf{v}_D &= \mathbf{v}_A + \mathbf{v}_{D/A} = \mathbf{v}_A + \omega \mathbf{k} \times \mathbf{r}_{D/A} \\ &= (1.2 \text{ m/s})\mathbf{i} - (8 \text{ rad/s})\mathbf{k} \times (-0.150 \text{ m})\mathbf{i} \\ &= (1.2 \text{ m/s})\mathbf{i} + (1.2 \text{ m/s})\mathbf{j}\end{aligned}$$

$$\mathbf{v}_D = 1.697 \text{ m/s} \nearrow 45^\circ \quad \blacktriangleleft$$

REFLECT and THINK: The principles involved in this problem are similar to those that you used in Sample Prob. 15.3, but in this problem, point A was free to translate. Point C , since it is in contact with the fixed lower rack, has a velocity of zero. Every point along diameter CAB has a velocity vector directed to the right (Fig. 1) and the magnitude of the velocity increases linearly as the distance from point C increases.

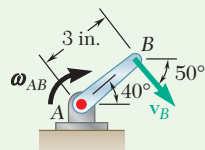
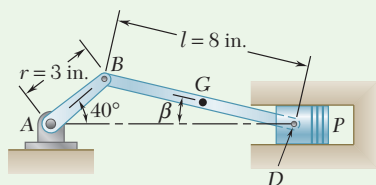


Fig. 1 Crank AB is undergoing fixed axis rotation.

Sample Problem 15.7

In the engine system shown, the crank AB has a constant clockwise angular velocity of 2000 rpm. For the crank position shown, determine (a) the angular velocity of the connecting rod BD , (b) the velocity of the piston P .

STRATEGY: Connecting rod BD is undergoing general motion, so use rigid-body kinematics. Crank AB is undergoing fixed axis rotation, and piston P is translating. The motion of the piston is the same as the end D of the connecting rod.

MODELING and ANALYSIS:

Motion of Crank AB . The crank AB rotates about point A . Expressing ω_{AB} in rad/s and writing $v_B = r\omega_{AB}$, you have (Fig. 1)

$$\omega_{AB} = \left(2000 \frac{\text{rev}}{\text{min}} \right) \left(\frac{1 \text{ min}}{60 \text{ s}} \right) \left(\frac{2\pi \text{ rad}}{1 \text{ rev}} \right) = 209.4 \text{ rad/s}$$

$$v_B = (AB)\omega_{AB} = (3 \text{ in.})(209.4 \text{ rad/s}) = 628.3 \text{ in./s}$$

$$\mathbf{v}_B = 628.3 \text{ in./s} \searrow 50^\circ$$

Motion of Connecting Rod BD . Consider this as a general plane motion. Using the law of sines, compute the angle β between the connecting rod and the horizontal as

$$\frac{\sin 40^\circ}{8 \text{ in.}} = \frac{\sin \beta}{3 \text{ in.}} \quad \beta = 13.95^\circ$$

The velocity $\mathbf{v}_D$ of point D where the rod is attached to the piston must be horizontal, while the velocity of point B is equal to the velocity $\mathbf{v}_B$ obtained previously. Expressing the relation between the velocities $\mathbf{v}_D$, $\mathbf{v}_B$, and $\mathbf{v}_{D/B}$, you have

$$\mathbf{v}_D = \mathbf{v}_B + \mathbf{v}_{D/B}$$

(continued)

This equation is shown pictorially in Fig. 2 where the motion of BD is resolved into a translation of B and a rotation about B .

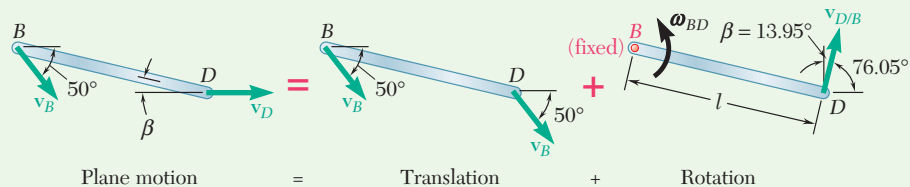


Fig. 2 The general plane motion of the connecting rod can be modelled as a translation plus a rotation.

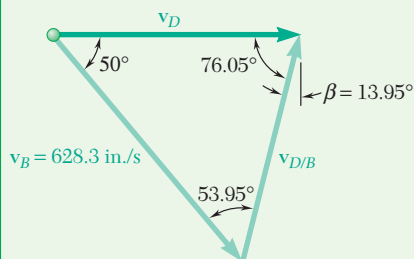


Fig. 3 Vector triangle showing the relationship between the velocities of B and D .

Draw the vector diagram corresponding to this equation (Fig. 3). Recalling that $\beta = 13.95^\circ$, you can determine the angles of the triangle and write

$$\frac{v_D}{\sin 53.95^\circ} = \frac{v_{D/B}}{\sin 50^\circ} = \frac{628.3 \text{ in./s}}{\sin 76.05^\circ}$$

$$v_{D/B} = 495.9 \text{ in./s} \quad v_{D/B} = 495.9 \text{ in./s} \angle 76.05^\circ$$

$$v_D = 523.4 \text{ in./s} = 43.6 \text{ ft/s} \quad v_D = 43.6 \text{ ft/s} \rightarrow$$

$$v_P = v_D = 43.6 \text{ ft/s} \rightarrow \blacktriangleleft$$

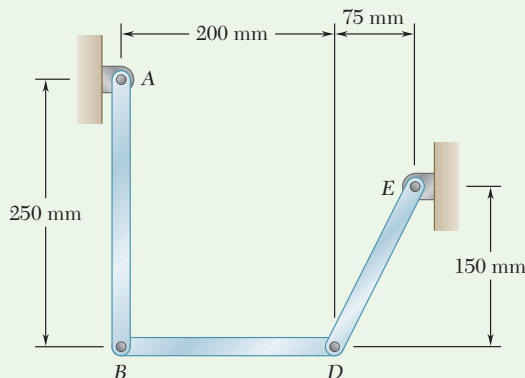
Since $v_{D/B} = l\omega_{BD}$, you have

$$495.9 \text{ in./s} = (8 \text{ in.})\omega_{BD} \quad \omega_{BD} = 62.0 \text{ rad/s} \uparrow \blacktriangleleft$$

REFLECT and THINK: Note that as the crank continues to move clockwise below the center line, the piston changes direction and starts to move to the left. Can you see what happens to the motion of the connecting rod at that point? You can also solve this problem using the vector relationship expressed in Eq. (15.17'); this type of approach is shown in Sample Prob. 15.8.

Sample Problem 15.8

In the position shown, bar AB has an angular velocity of 4 rad/s clockwise. Determine the angular velocity of bars BD and DE .



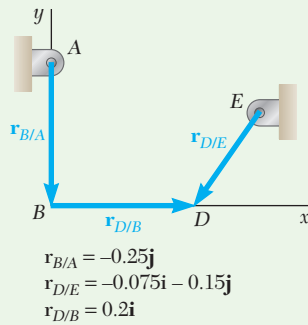


Fig. 1 Relative position vectors for points B and D .

STRATEGY: The bars AB and DE are undergoing fixed-axis rotation, whereas bar BD is undergoing general plane motion. You will need to use rigid-body kinematics to analyze the motion.

MODELING and ANALYSIS: Model the bars as rigid bodies. The angular velocity of AB is given and is equal to $\omega_{AB} = -(4 \text{ rad/s})\mathbf{k}$. You can use vector algebra to relate the velocities of points B and D on bar BD after you find the velocities of B and D from the connecting bars. Position vectors are defined in Fig. 1.

Bar AB. (Rotation about A)

$$\mathbf{v}_B = \omega_{AB} \times \mathbf{r}_{B/A} = (-4\mathbf{k}) \times (-0.25\mathbf{j}) = -(1.00 \text{ m/s})\mathbf{i} \quad (1)$$

Bar ED. (Rotation about E) Assuming ω_{DE} is positive, you have

$$\mathbf{v}_D = \omega_{DE}\mathbf{k} \times \mathbf{r}_{D/E} = \omega_{DE}\mathbf{k} \times (-0.075\mathbf{i} - 0.15\mathbf{j}) = 0.15\omega_{DE}\mathbf{i} - 0.075\omega_{DE}\mathbf{j} \quad (2)$$

Bar BD. (Translation with B and rotation about B .)

$$\mathbf{v}_D = \mathbf{v}_B + \mathbf{v}_{D/B} \quad (3)$$

where you assume ω_{BD} is positive. The relative velocity is

$$\mathbf{v}_{D/B} = \omega_{BD}\mathbf{k} \times \mathbf{r}_{D/B} = \omega_{BD}\mathbf{k} \times 0.2\mathbf{i} = 0.2\omega_{BD}\mathbf{j} \quad (4)$$

Substituting Eqs. (1), (2), and (4) into Eq. (3) gives

$$0.15\omega_{DE}\mathbf{i} - 0.075\omega_{DE}\mathbf{j} = -1.00\mathbf{i} + 0.2\omega_{BD}\mathbf{j}$$

Equating components allows you to solve for the unknown angular velocities:

$$\mathbf{i}: 0.15\omega_{DE} = -1.00, \quad \omega_{DE} = -6.667 \text{ rad/s} \quad \omega_{DE} = 6.67 \text{ rad/s} \downarrow \blacktriangleleft$$

$$\mathbf{j}: -0.075\omega_{DE} = 0.2\omega_{BD} \quad \omega_{BD} = \frac{-(0.075)(-6.667)}{0.2} \quad \omega_{BD} = 2.50 \text{ rad/s} \uparrow \blacktriangleleft$$

REFLECT and THINK: The vector algebra approach is very straightforward for problems like this. It makes sense that if AB is rotating clockwise, BD is rotating counterclockwise and DE is rotating clockwise.

SOLVING PROBLEMS ON YOUR OWN

In this section, you learned how to analyze the velocity of bodies in **general plane motion**. You found that you can always consider a general plane motion to be the sum of the two motions you studied in the Sec. 15.1, namely, *a translation and a rotation*.

To solve a problem involving the velocity of a body in plane motion, you should take the following steps.

1. Whenever possible, determine the velocity of the points of the body where it is connected to another body whose motion is known [Sample Prob. 15.6]. That other body may be an arm or crank rotating with a given angular velocity [Sample Probs. 15.7 and 15.8].

2. Next, draw a diagram to use in your solution (Figs. 15.15 and 15.16) if you are not using the vector algebra approach. This diagram consists of the following diagrams.

a. Plane motion diagram: Draw a diagram of the body including all dimensions and showing those points for which you know or seek the velocity.

b. Translation diagram: Select a reference point A for which you know the direction and/or the magnitude of the velocity $\mathbf{v}_A$, and draw a second diagram showing the body in translation with all of its points having the same velocity $\mathbf{v}_A$.

c. Rotation diagram: Consider point A as a fixed point and draw a diagram showing the body in rotation about A . Show the angular velocity $\boldsymbol{\omega} = \omega \mathbf{k}$ of the body and the relative velocities with respect to A of the other points, such as the velocity $\mathbf{v}_{B/A}$ of B relative to A .

3. Write the relative velocity formula as

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad (15.17)$$

or for plane motion as

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \mathbf{k} \times \mathbf{r}_{B/A} \quad (15.17')$$

You can solve this vector equation analytically by writing the corresponding scalar equations, or you can solve it by using a vector triangle (Fig. 15.16).

4. Use a different reference point to obtain an equivalent solution. For example, if you select point B as the reference point, the relative velocity of point A is

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B} = \mathbf{v}_B + \boldsymbol{\omega} \mathbf{k} \times \mathbf{r}_{A/B} \quad (15.20)$$

Note that the relative velocities $\mathbf{v}_{B/A}$ and $\mathbf{v}_{A/B}$ have the same magnitude but opposite sense. Relative velocities, therefore, depend upon the reference point that you select. The angular velocity, however, is independent of the choice of reference point.

5. Write additional relative velocity equations if you are analyzing a multi-body linkage. For problems such as the crankshaft-piston in Sample Prob. 15.7, you may have to write multiple relative velocity equations. In that problem, you can express the velocity of P with respect to B and then the velocity of B with respect to A . Generally, the ends of the linkages will have some type of constraint (e.g., the piston moving only in the x direction).

Problems

CONCEPT QUESTIONS

15.CQ3 The ball rolls without slipping on the fixed surface as shown. What is the direction of the velocity of point A?

- a. $\rightarrow$ b. $\nearrow$ c. $\uparrow$ d. $\downarrow$ e. $\searrow$

15.CQ4 Three uniform rods— ABC , DCE , and FGH —are connected as shown. Which of the following statements concerning the angular speed of the three objects is true?

- a. $\omega_{ABC} = \omega_{DCE} = \omega_{FGH}$
 b. $\omega_{DCE} > \omega_{ABC} > \omega_{FGH}$
 c. $\omega_{DCE} < \omega_{ABC} < \omega_{FGH}$
 d. $\omega_{ABC} > \omega_{DCE} > \omega_{FGH}$
 e. $\omega_{FGH} = \omega_{DCE} < \omega_{ABC}$

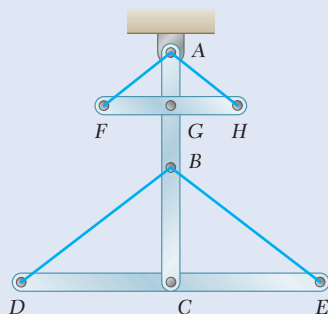


Fig. P15.CQ4

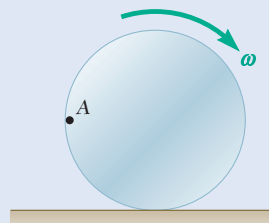


Fig. P15.CQ3

END-OF-SECTION PROBLEMS

15.38 An automobile travels to the right at a constant speed of 48 mi/h. If the diameter of a wheel is 22 in., determine the velocities of points B, C, D, and E on the rim of the wheel.

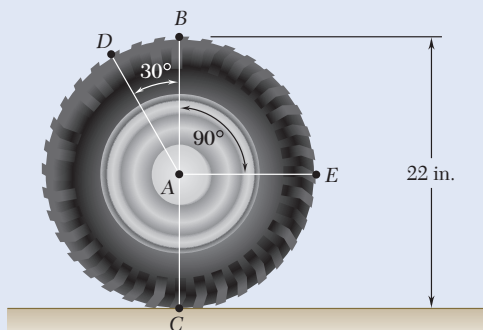


Fig. P15.38

15.39 The motion of rod AB is guided by pins attached at A and B that slide in the slots shown. At the instant shown, $\theta = 40^\circ$ and the pin at B moves upward to the left with a constant velocity of 6 in./s. Determine (a) the angular velocity of the rod, (b) the velocity of the pin at end A .

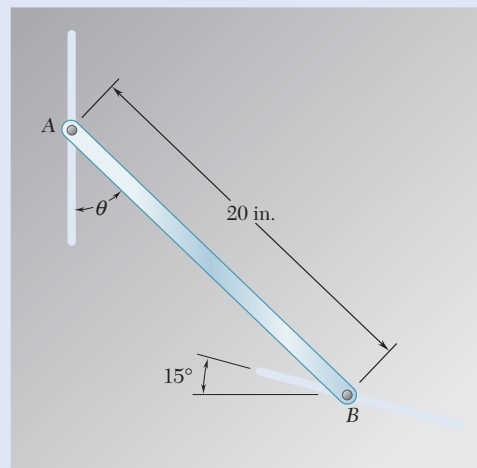


Fig. P15.39

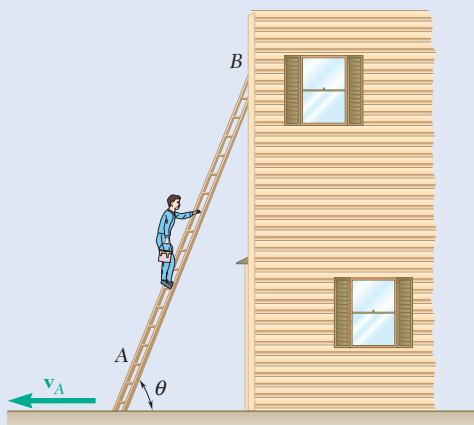


Fig. P15.40

15.40 A painter is halfway up a 10-m ladder when the bottom starts sliding out from under him. Knowing that point A has a velocity $\mathbf{v}_A = 2 \text{ m/s}$ directed to the left when $\theta = 60^\circ$, determine (a) the angular velocity of the ladder, (b) the velocity of the painter.

15.41 Rod AB can slide freely along the floor and the inclined plane. At the instant shown, the velocity of end A is 1.4 m/s to the left. Determine (a) the angular velocity of the rod, (b) the velocity of end B of the rod.

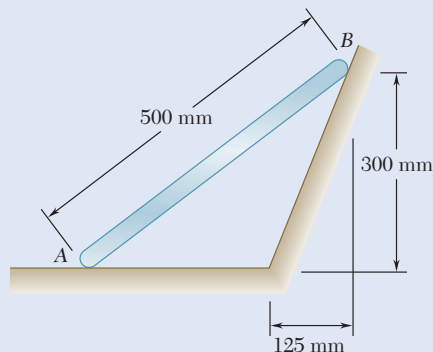


Fig. P15.41 and P15.42

15.42 Rod AB can slide freely along the floor and the inclined plane. At the instant shown, the angular velocity of the rod is 4.2 rad/s counterclockwise. Determine (a) the velocity of end A of the rod, (b) the velocity of end B of the rod.

15.43 Rod AB moves over a small wheel at C while end A moves to the right with a constant velocity of 25 in./s. At the instant shown, determine (a) the angular velocity of the rod, (b) the velocity of end B of the rod.

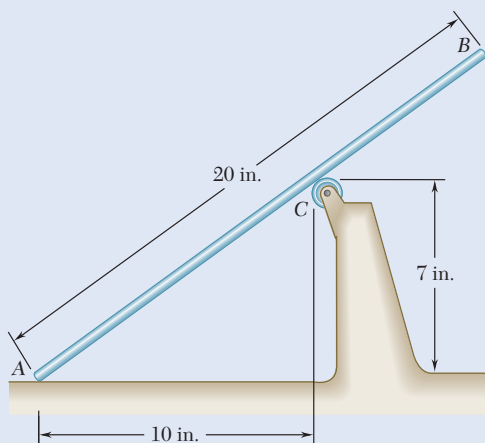


Fig. P15.43

15.44 The disk shown moves in the xy plane. Knowing that $(v_A)_y = -7 \text{ m/s}$, $(v_B)_x = -7.4 \text{ m/s}$, and $(v_C)_x = -1.4 \text{ m/s}$, determine (a) the angular velocity of the disk, (b) the velocity of point B.

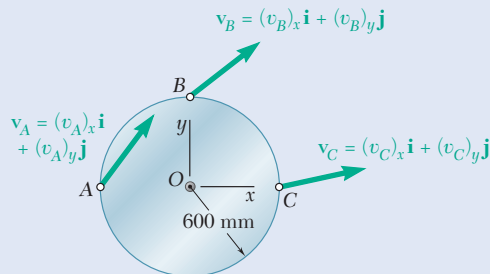


Fig. P15.44 and P15.45

15.45 The disk shown moves in the xy plane. Knowing that $(v_A)_y = -7 \text{ m/s}$, $(v_B)_x = -7.4 \text{ m/s}$, and $(v_C)_x = -1.4 \text{ m/s}$, determine (a) the velocity of point O, (b) the point of the disk with zero velocity.

- 15.46** The plate shown moves in the xy plane. Knowing that $(v_A)_x = 250$ mm/s, $(v_B)_y = -450$ mm/s, and $(v_C)_x = -500$ mm/s, determine (a) the angular velocity of the plate, (b) the velocity of point A.

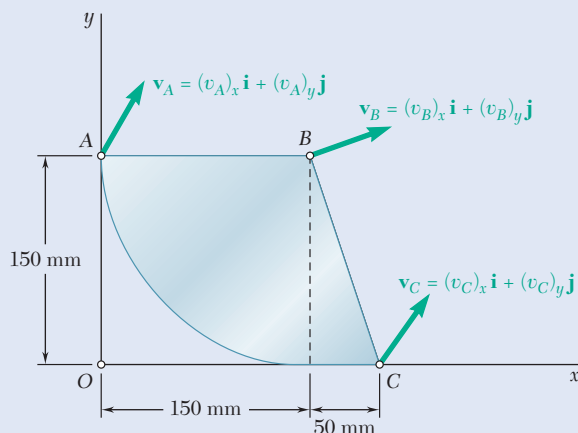


Fig. P15.46

- 15.47** Velocity sensors are placed on a satellite that is moving only in the xy plane. Knowing that at the instant shown the unidirectional sensors measure $(v_A)_x = 2$ ft/s, $(v_B)_x = -0.333$ ft/s, and $(v_C)_y = -2$ ft/s, determine (a) the angular velocity of the satellite, (b) the velocity of point B.

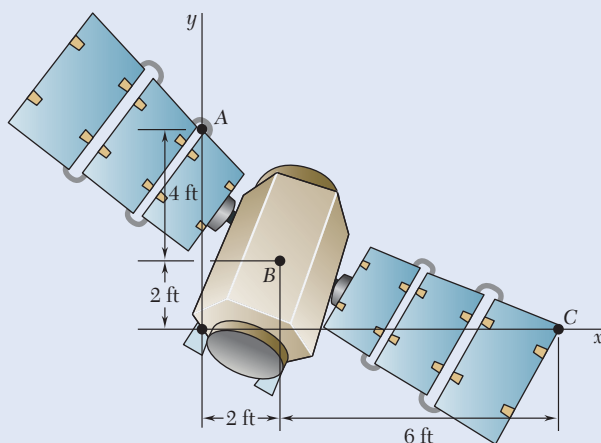


Fig. P15.47

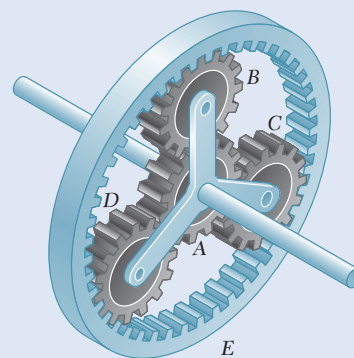


Fig. P15.48 and P15.49

- 15.48** In the planetary gear system shown, the radius of gears A, B, C, and D is a and the radius of the outer gear E is $3a$. Knowing that the angular velocity of gear A is ω_A clockwise and that the outer gear E is stationary, determine (a) the angular velocity of each planetary gear, (b) the angular velocity of the spider connecting the planetary gears.
- 15.49** In the planetary gear system shown, the radius of gears A, B, C, and D is 30 mm and the radius of the outer gear E is 90 mm. Knowing that gear E has an angular velocity of 180 rpm clockwise and that the central gear A has an angular velocity of 240 rpm clockwise, determine (a) the angular velocity of each planetary gear, (b) the angular velocity of the spider connecting the planetary gears.
- 15.50** Arm AB rotates with an angular velocity of 20 rad/s counterclockwise. Knowing that the outer gear C is stationary, determine (a) the angular velocity of gear B, (b) the velocity of the gear tooth located at point D.

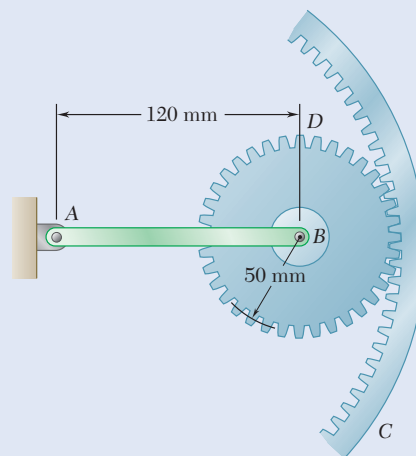


Fig. P15.50

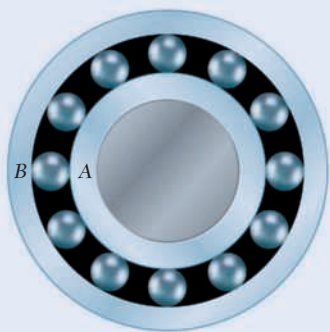


Fig. P15.51

15.51 In the simplified sketch of a ball bearing shown, the diameter of the inner race *A* is 60 mm and the diameter of each ball is 12 mm. The outer race *B* is stationary while the inner race has an angular velocity of 3600 rpm. Determine (a) the speed of the center of each ball, (b) the angular velocity of each ball, (c) the number of times per minute each ball describes a complete circle.

15.52 A simplified gear system for a mechanical watch is shown. Knowing that gear *A* has a constant angular velocity of 1 rev/h and gear *C* has a constant angular velocity of 1 rpm, determine (a) the radius *r*, (b) the magnitudes of the accelerations of the points on gear *B* that are in contact with gears *A* and *C*.

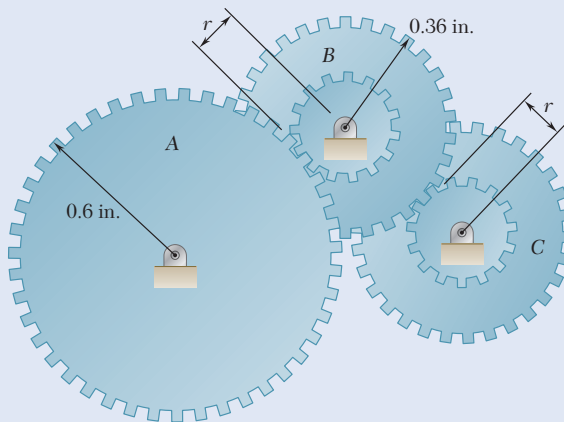


Fig. P15.52

15.53 and 15.54 Arm *ACB* rotates about point *C* with an angular velocity of 40 rad/s counterclockwise. Two friction disks *A* and *B* are pinned at their centers to arm *ACB* as shown. Knowing that the disks roll without slipping at surfaces of contact, determine the angular velocity of (a) disk *A*, (b) disk *B*.

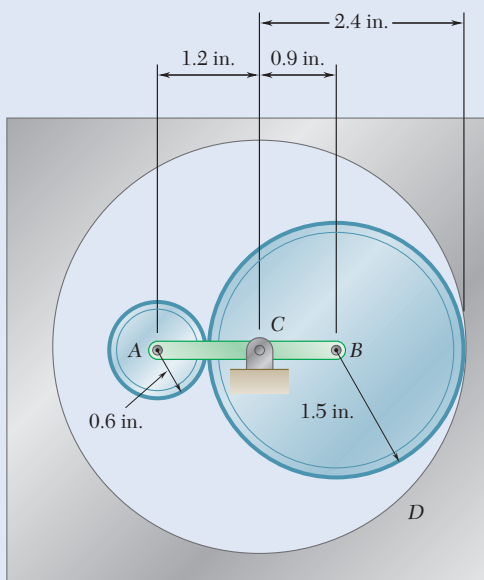


Fig. P15.53

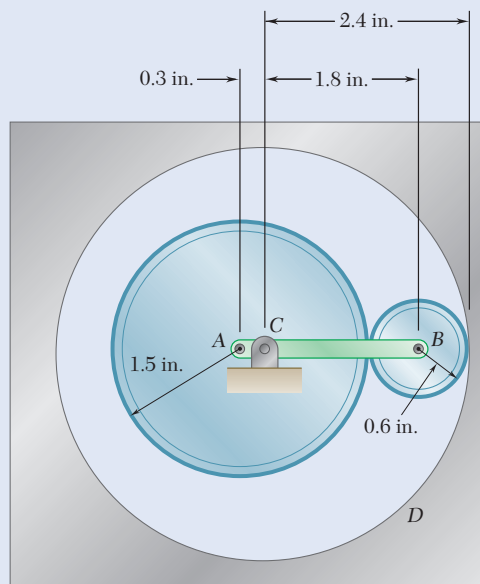


Fig. P15.54

15.55 Knowing that at the instant shown the velocity of collar *A* is 900 mm/s to the left, determine (a) the angular velocity of rod *ADB*, (b) the velocity of point *B*.

15.56 Knowing that at the instant shown the angular velocity of rod *DE* is 2.4 rad/s clockwise, determine (a) the velocity of collar *A*, (b) the velocity of point *B*.

15.57 Knowing that the disk has a constant angular velocity of 15 rad/s clockwise, determine the angular velocity of bar *BD* and the velocity of collar *D* when (a) $\theta = 0^\circ$, (b) $\theta = 90^\circ$, (c) $\theta = 180^\circ$.

15.58 The disk has a constant angular velocity of 20 rad/s clockwise. (a) Determine the two values of the angle θ for which the velocity of collar *D* is zero. (b) For each of these values of θ , determine the corresponding value of the angular velocity of bar *BD*.

15.59 The test rig shown was developed to perform fatigue testing on fitness trampolines. A motor drives the 9-in.-radius flywheel *AB*, which is pinned at its center point *A*, in a counterclockwise direction. The flywheel is attached to slider *CD* by the 18-in. connecting rod *BC*. Knowing that the “feet” at *D* should hit the trampoline twice every second, at the instant when $\theta = 0^\circ$, determine (a) the angular velocity of the connecting rod *BC*, (b) the velocity of *D*, (c) the velocity of midpoint *CB*.

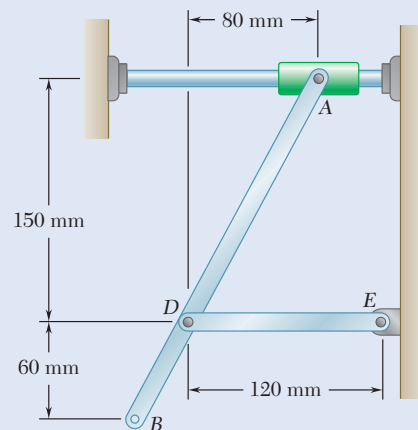


Fig. P15.55 and P15.56

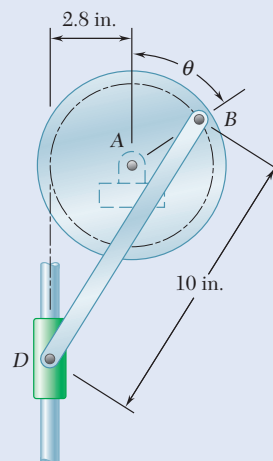


Fig. P15.57 and P15.58

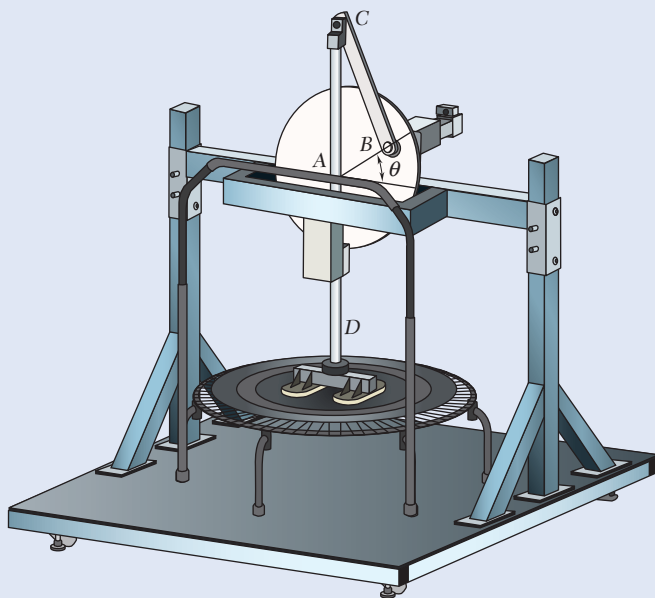


Fig. P15.59

15.60 In the eccentric shown, a disk of 2-in. radius revolves about shaft *O* that is located 0.5 in. from the center *A* of the disk. The distance between the center *A* of the disk and the pin at *B* is 8 in. Knowing that the angular velocity of the disk is 900 rpm clockwise, determine the velocity of the block when $\theta = 30^\circ$.

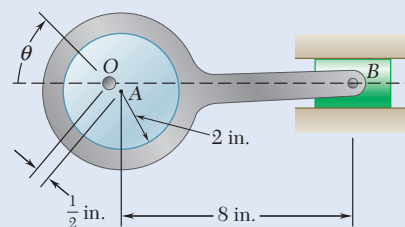


Fig. P15.60

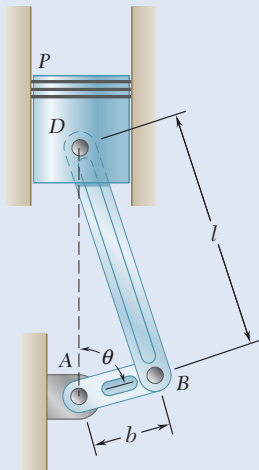


Fig. P15.61 and P15.62

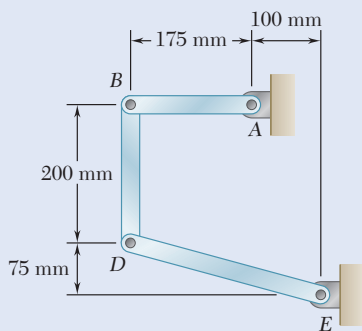


Fig. P15.64

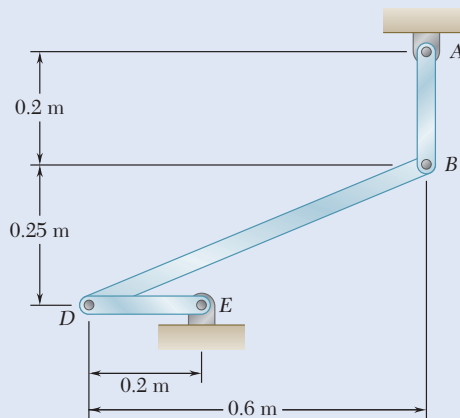


Fig. P15.63

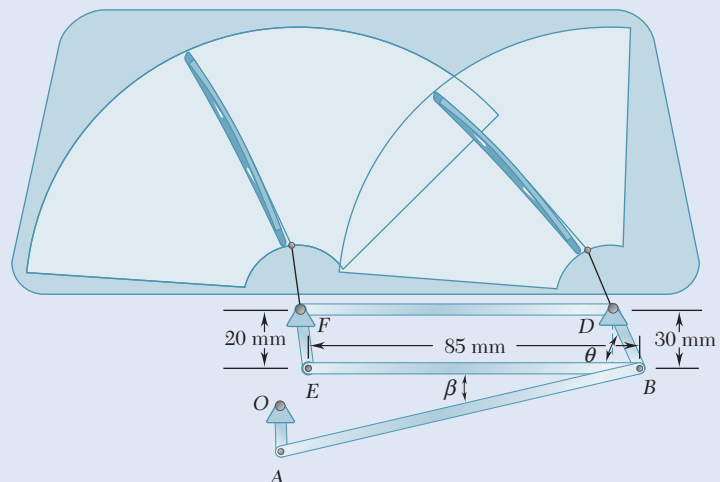


Fig. P15.65

- 15.61** In the engine system shown, $l = 160$ mm and $b = 60$ mm. Knowing that the crank AB rotates with a constant angular velocity of 1000 rpm clockwise, determine the velocity of the piston P and the angular velocity of the connecting rod when (a) $\theta = 0^\circ$, (b) $\theta = 90^\circ$.
- 15.62** In the engine system shown, $l = 160$ mm and $b = 60$ mm. Knowing that crank AB rotates with a constant angular velocity of 1000 rpm clockwise, determine the velocity of the piston P and the angular velocity of the connecting rod when $\theta = 60^\circ$.
- 15.63** Knowing that at the instant shown the angular velocity of rod AB is 15 rad/s clockwise, determine (a) the angular velocity of rod BD , (b) the velocity of the midpoint of rod BD .
- 15.64** In the position shown, bar AB has an angular velocity of 4 rad/s clockwise. Determine the angular velocity of bars BD and DE .
- 15.65** Linkage $DBEF$ is part of a windshield wiper mechanism, where points O , F , and D are fixed pin connections. At the position shown, $\theta = 30^\circ$ and link EB is horizontal. Knowing that link EF has a counterclockwise angular velocity of 4 rad/s at the instant shown, determine the angular velocity of links EB and DB .

15.66 Roberts linkage is named after Richard Roberts (1789–1864) and can be used to draw a close approximation to a straight line by locating a pen at point F . The distance AB is the same as BF , DF , and DE . Knowing that the angular velocity of bar AB is 5 rad/s clockwise in the position shown, determine (a) the angular velocity of bar DE , (b) the velocity of point F .

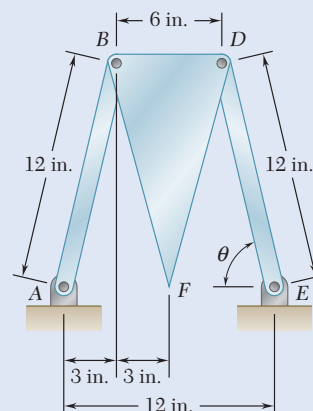


Fig. P15.66 and P15.67

15.67 Roberts linkage is named after Richard Roberts (1789–1864) and can be used to draw a close approximation to a straight line by locating a pen at point F . The distance AB is the same as BF , DF , and DE . Knowing that the angular velocity of plate BDF is 2 rad/s counterclockwise when $\theta = 90^\circ$, determine (a) the angular velocities of bars AB and DE , (b) the velocity of point F . When $\theta = 90^\circ$, point F may be assumed to coincide with point E , with negligible error in the velocity analysis.

15.68 In the position shown, bar DE has a constant angular velocity of 10 rad/s clockwise. Knowing that $h = 500$ mm, determine (a) the angular velocity of bar FBD , (b) the velocity of point F .

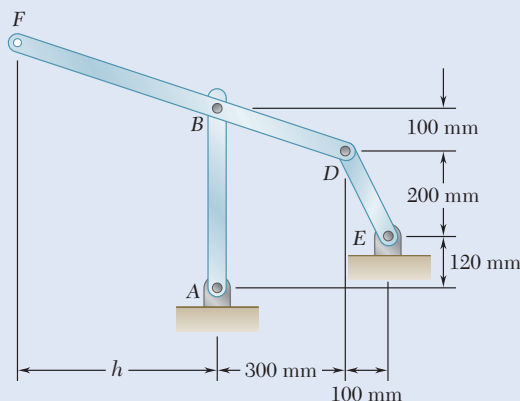


Fig. P15.68 and P15.69

15.69 In the position shown, bar DE has a constant angular velocity of 10 rad/s clockwise. Determine (a) the distance h for which the velocity of point F is vertical, (b) the corresponding velocity of point F .

15.70 Both 6-in.-radius wheels roll without slipping on the horizontal surface. Knowing that the distance AD is 5 in., the distance BE is 4 in., and D has a velocity of 6 in./s to the right, determine the velocity of point E .

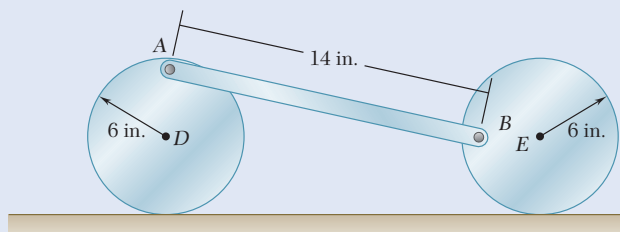


Fig. P15.70

- 15.71** The 80-mm-radius wheel shown rolls to the left with a velocity of 900 mm/s. Knowing that the distance AD is 50 mm, determine the velocity of the collar and the angular velocity of rod AB when (a) $\beta = 0$, (b) $\beta = 90^\circ$.

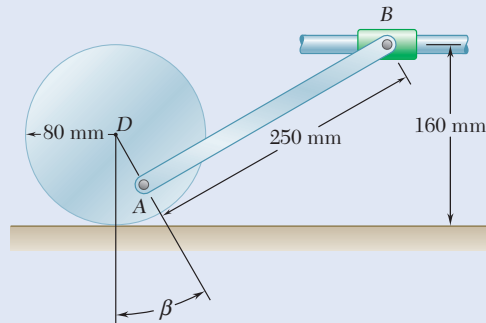


Fig. P15.71

- *15.72** For the gearing shown, derive an expression for the angular velocity ω_C of gear C and show that ω_C is independent of the radius of gear B. Assume that point A is fixed and denote the angular velocities of rod ABC and gear A by ω_{ABC} and ω_A , respectively.

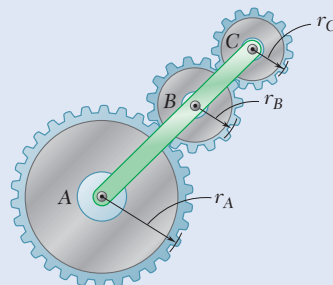


Fig. P15.72

15.3 INSTANTANEOUS CENTER OF ROTATION

Consider the general plane motion of a rigid body. We will show that, at any given instant, the velocities of the various particles of the rigid body are the same as if the body were rotating about an axis perpendicular to the plane of the body, called the **instantaneous axis of rotation**. This axis intersects the plane of the rigid body at a point C , called the **instantaneous center of rotation** of the body or the **instantaneous center of zero velocity**. This gives us an alternative method for solving problems involving the velocities of points on an object in plane motion, and it is sometimes simpler than using the equations in Sec. 15.2.

Recall that we can always replace the plane motion of a rigid body by a translation defined by the motion of an arbitrary reference point A and by a rotation about A . As far as the velocities are concerned, the translation is characterized by the velocity $\mathbf{v}_A$ of the reference point A and the rotation is characterized by the angular velocity $\boldsymbol{\omega}$ of the body (which is independent of the choice of A). Thus, the velocity $\mathbf{v}_A$ of point A and the angular velocity $\boldsymbol{\omega}$ of the rigid body define completely the velocities of all the other particles of the body (Fig. 15.18a).

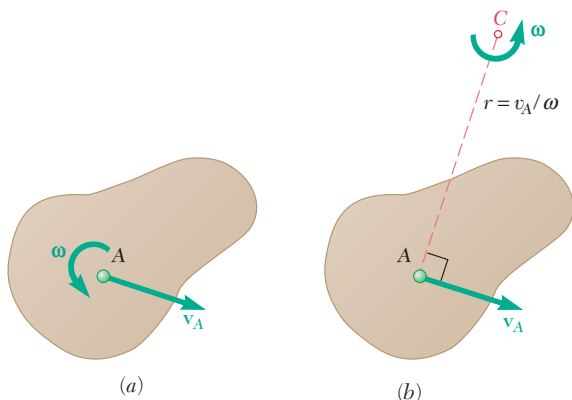


Fig. 15.18 As far as velocities are concerned, at every instant in time the rigid body seems to rotate about a point called the instantaneous center C .

Now let us assume that $\mathbf{v}_A$ and $\boldsymbol{\omega}$ are known and that they are both different from zero. (If $\mathbf{v}_A = 0$, point A is itself the instantaneous center of rotation, and if $\boldsymbol{\omega} = 0$, you have rigid body translation where all of the particles have the same velocity $\mathbf{v}_A$.) We could obtain these velocities by letting the rigid body rotate with the angular velocity $\boldsymbol{\omega}$ about a point C located on the perpendicular to $\mathbf{v}_A$ at a distance $r = v_A/\omega$ from A , as shown in Fig. 15.18b. We check that the velocity of A would be perpendicular to AC and that its magnitude would be $r\omega = (v_A/\omega)\omega = v_A$. Thus, the velocities of all the other particles of the body are the same as originally defined. Therefore, *as far as the velocities are concerned, the rigid body seems to rotate about the instantaneous center C at the instant considered.*



Photo 15.5 If the tires of this car are rolling without sliding, the instantaneous center of rotation of each tire is the point of contact between the road and the tire.

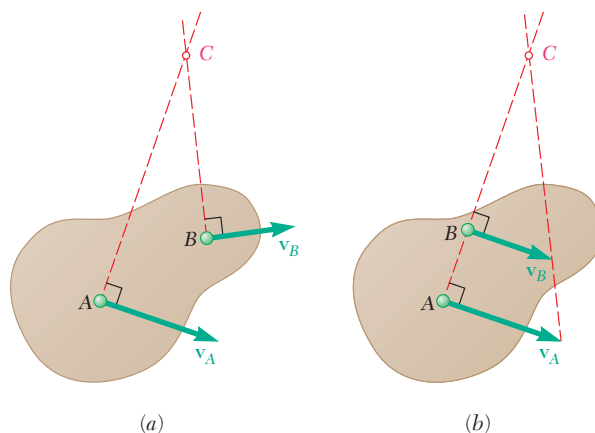


Fig. 15.19 Locating the instantaneous center of rotation C
 (a) when you know the directions of the velocities of two points;
 (b) when the velocities of two points are perpendicular to line AB .

We can define the position of the instantaneous center in two other ways. If we know the directions of the velocities of two particles A and B of the rigid body and if they are different, we can obtain the instantaneous center C by drawing the perpendicular to $\mathbf{v}_A$ through A and the perpendicular to $\mathbf{v}_B$ through B . The point C is where these two lines intersect (Fig. 15.19a). If the velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ of two particles A and B are perpendicular to line AB and we know their magnitudes, we can find the instantaneous center by intersecting line AB with the line joining the ends of the vectors $\mathbf{v}_A$ and $\mathbf{v}_B$ (Fig. 15.19b). Note that if $\mathbf{v}_A$ and $\mathbf{v}_B$ were parallel in Fig. 15.19a or if $\mathbf{v}_A$ and $\mathbf{v}_B$ had the same magnitude in Fig. 15.19b, the instantaneous center C would be at an infinite distance and ω would be zero; all points of the rigid body would have the same velocity.

To see how we can use the concept of the instantaneous center of rotation, let us consider again the sliding rod of Sec. 15.2. Drawing the perpendicular to $\mathbf{v}_A$ through A and the perpendicular to $\mathbf{v}_B$ through B (Fig. 15.20), we obtain the instantaneous center C . At the instant

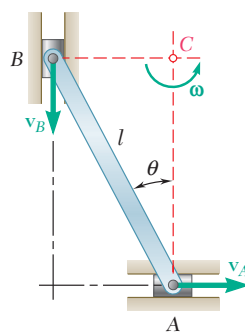


Fig. 15.20 Instantaneous center of rotation C for the sliding rod AB .

considered, the velocities of all the particles of the rod are thus the same as if the rod rotated about C . Now, if we know the magnitude v_A of the velocity of A , we can obtain the magnitude ω of the angular velocity of the rod from

$$\omega = \frac{v_A}{AC} = \frac{v_A}{l \cos \theta}$$

Then we obtain the magnitude of the velocity of B as

$$v_B = (BC)\omega = l \sin \theta \frac{v_A}{l \cos \theta} = v_A \tan \theta$$

Note that we used only *absolute* velocities in the computation.

The instantaneous center of a body in plane motion can be located either on the body or outside the body. If it is located on the rigid body, the particle C coinciding with the instantaneous center at a given instant t must have zero velocity at that instant. However, the instantaneous center of rotation is valid only at a given instant. Thus, particle C of the rigid body that coincides with the instantaneous center at time t generally does not coincide with the instantaneous center at time $t + \Delta t$. Its velocity is zero at time t , but it will probably be different from zero at time $t + \Delta t$. This means, in general, that particle C *does not have zero acceleration* and, therefore, that the accelerations of the various particles of the rigid body cannot be determined as if the body were rotating about C .

As the motion of the rigid body proceeds, the instantaneous center moves in space. However, we just pointed out that the position of the instantaneous center on the body keeps changing. Thus, the instantaneous center describes one curve in space, called the *space centrode*, and another curve on the rigid body, called the *body centrode* (Fig. 15.21). It can be shown that at any instant, these two curves are tangent at C and that as the rigid body moves, the body centrode appears to roll on the space centrode.

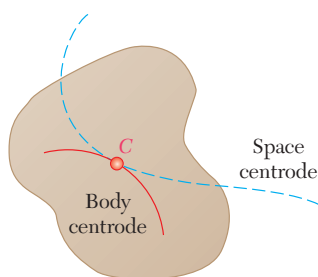


Fig. 15.21 The space centrode and the body centrode are tangent to each other.

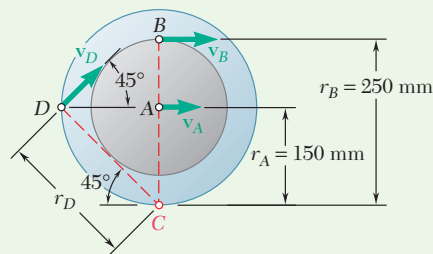
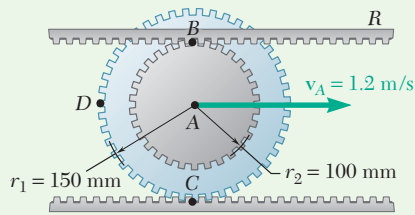


Fig. 1 Distances from the instantaneous center of rotation to A, B, and D.

Sample Problem 15.9

Solve Sample Prob. 15.6 using the method of the instantaneous center of rotation.

STRATEGY: You know the velocity direction of two points on the same rigid body, so you can find an instantaneous center of rotation. Since the gear rolls on the stationary lower rack, the point of contact C of the gear with the rack has no velocity; point C is therefore the instantaneous center of rotation.

MODELING and ANALYSIS:

a. Angular Velocity of the Gear. You can calculate the angular velocity directly from the data in Fig. 1.

$$v_A = r_A \omega \quad 1.2 \text{ m/s} = (0.150 \text{ m}) \omega$$

$$\omega = 8 \text{ rad/s} \downarrow$$

b. Velocities. As far as velocities are concerned, all points of the gear seem to rotate about the instantaneous center.

Velocity of Upper Rack. Recalling that $v_R = v_B$, you have

$$v_R = v_B = r_B \omega \quad v_R = (0.250 \text{ m})(8 \text{ rad/s}) = 2 \text{ m/s}$$

$$\mathbf{v}_R = 2 \text{ m/s} \rightarrow$$

Velocity of Point D. Since $r_D = (0.150 \text{ m})\sqrt{2} = 0.2121 \text{ m}$, you obtain

$$v_D = r_D \omega \quad v_D = (0.2121 \text{ m})(8 \text{ rad/s}) = 1.697 \text{ m/s}$$

$$\mathbf{v}_D = 1.697 \text{ m/s} \nearrow 45^\circ$$

REFLECT and THINK: The results are the same as in Sample Prob. 15.6, as you would expect, but it took much less computation to get them.

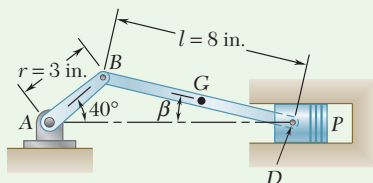
Sample Problem 15.10

Solve Sample Prob. 15.7 using the method of the instantaneous center of rotation.

STRATEGY: You know the velocity of point B from the motion of the crank (see Sample Prob. 15.7), and you know the direction of the velocity of point D. Therefore, you can find an instantaneous center of rotation.

MODELING and ANALYSIS:

Motion of Crank AB. Referring to Sample Prob. 15.7, you obtain the velocity of point B; $\mathbf{v}_B = 628.3 \text{ in./s} \searrow 50^\circ$.



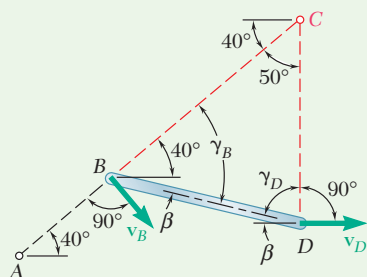


Fig. 1 Instantaneous center of rotation for bar BD .

Motion of the Connecting Rod BD . First locate the instantaneous center C by drawing lines perpendicular to the absolute velocities $\mathbf{v}_B$ and $\mathbf{v}_D$ (Fig. 1). Recalling from Sample Prob. 15.7 that $\beta = 13.95^\circ$ and that $BD = 8$ in., solve the triangle BCD .

$$\gamma_B = 40^\circ + \beta = 53.95^\circ \quad \gamma_D = 90^\circ - \beta = 76.05^\circ$$

$$\frac{BC}{\sin 76.05^\circ} = \frac{CD}{\sin 53.95^\circ} = \frac{8 \text{ in.}}{\sin 50^\circ}$$

$$BC = 10.14 \text{ in.} \quad CD = 8.44 \text{ in.}$$

Since the connecting rod BD seems to rotate about point C , you have

$$v_B = (BC)\omega_{BD}$$

$$628.3 \text{ in./s} = (10.14 \text{ in.})\omega_{BD}$$

$$\omega_{BD} = 62.0 \text{ rad/s} \rightarrow \blacktriangleleft$$

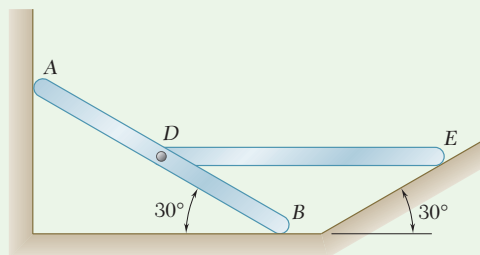
$$\begin{aligned} v_D &= (CD)\omega_{BD} = (8.44 \text{ in.})(62.0 \text{ rad/s}) \\ &= 523 \text{ in./s} = 43.6 \text{ ft/s} \end{aligned}$$

$$\mathbf{v}_P = \mathbf{v}_D = 43.6 \text{ ft/s} \rightarrow \blacktriangleleft$$

REFLECT and THINK: Often, the hardest part of solving a problem using the instantaneous center of rotation is the geometry. Remembering how to use the law of sines or the law of cosines is often helpful.

Sample Problem 15.11

Two 20-in. rods AB and DE are connected as shown. Point D is the midpoint of rod AB , and at the instant shown, rod DE is horizontal. Knowing that the velocity of point A is 1 ft/s downward, determine (a) the angular velocity of rod DE , (b) the velocity of point E .



STRATEGY: You know the directions of several points on these objects, so you can use instantaneous centers of rotation to solve this problem.

(continued)

MODELING and ANALYSIS: Locate the instantaneous center of rotation C of bar AB as the intersection of line AC perpendicular to $\mathbf{v}_A$ and line BC perpendicular to $\mathbf{v}_B$ (Fig. 1). Knowing the location of C , you can determine the direction of the velocity of D . From this direction, and the direction of E , you can find the instantaneous center, I , for bar DE (Fig. 1).

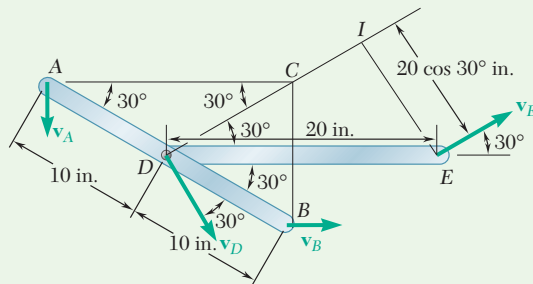


Fig. 1 The instantaneous centers of rotation for bar AB and DE are C and I , respectively.

a. Angular velocity of DE . From geometry, $r_{AC} = (20 \cos 30^\circ)$ in., so

$$\omega_{AB} = \frac{v_A}{r_{AC}} = \frac{12 \text{ in./s}}{20 \cos 30^\circ \text{ in.}} = 0.6928 \text{ rad/s } \curvearrowright$$

Now you can find v_D since $r_{D/C} = 10$ in.

$$v_D = \omega_{AB} r_{D/C} = (0.6928 \text{ rad/s})(10 \text{ in.}) = 6.928 \text{ in./s}$$

$$v_D = 6.928 \text{ in./s } \searrow 30^\circ$$

Now, since you know the directions of the velocities of D and E , $\mathbf{v}_E = v_E \nearrow 30^\circ$, you can find point I , which is the instantaneous center of bar DE . From geometry, $r_{DI} = 20 \cos 30^\circ$ in., and therefore

$$\omega_{DE} = \frac{v_D}{r_{DI}} = \frac{6.928 \text{ in./s}}{20 \cos 30^\circ \text{ in.}} = 0.400 \text{ rad/s} \quad \omega_{DE} = 0.400 \text{ rad/s } \curvearrowright \quad \blacktriangleleft$$

b. Velocity of E . Using this angular velocity, you can easily determine the velocity of E :

$$v_E = \omega_{DE} r_{EI} = (0.400 \text{ rad/s})(20 \sin 30^\circ \text{ in.}) = 4.00 \text{ in./s}$$

$$\mathbf{v}_E = 0.333 \text{ ft/s } \nearrow 30^\circ \quad \blacktriangleleft$$

REFLECT and THINK: The direction of ω_{DE} makes intuitive sense; you would expect it to be rotating counterclockwise at the instant shown. You could have also solved this problem using vector equations.

SOLVING PROBLEMS ON YOUR OWN

In this section, we introduced the **instantaneous center of rotation** in plane motion. This provides us with an alternative way of solving problems involving the *velocities* of the various points of a body in plane motion [Sample Probs. 15.9 through 15.11]. As its name suggests, the instantaneous center of rotation is the point about which you can assume a body is rotating at a given instant; you can use the instantaneous center to determine the velocity of any point on the body at that instant in time.

A. To determine the instantaneous center of rotation of a body in plane motion, you should use one of the following procedures.

1. If you know both the velocity $\mathbf{v}_A$ of a point A and the angular velocity ω of the body (Fig. 15.18):

a. Draw a sketch of the body, showing point A , its velocity $\mathbf{v}_A$, and the angular velocity ω of the body.

b. From A draw a line perpendicular to $\mathbf{v}_A$ on the side of $\mathbf{v}_A$ from which this velocity is viewed as having *the same sense as ω* .

c. Locate the instantaneous center C on this line at a distance $r = v_A/\omega$ from point A .

2. If you know the directions of the velocities of two points A and B and they are different (Fig. 15.19a):

a. Draw a sketch of the body showing points A and B and their velocities $\mathbf{v}_A$ and $\mathbf{v}_B$.

b. From A and B draw lines perpendicular to $\mathbf{v}_A$ and $\mathbf{v}_B$, respectively. The instantaneous center C is located at the point where the two lines intersect.

c. If you know the velocity of one of the two points, you can determine the angular velocity of the body at that instant in time. For example, if you know $\mathbf{v}_A$, you can write $\omega = v_A/AC$, where AC is the distance from point A to the instantaneous center C .

3. If you know the velocities of two points A and B and both are perpendicular to the line AB (Fig. 15.19b):

a. Draw a sketch of the body, showing points A and B with their velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ *drawn to scale*.

b. Draw a line through points A and B , and another line through the tips of the vectors $\mathbf{v}_A$ and $\mathbf{v}_B$. The instantaneous center C is located at the point where the two lines intersect.

c. Obtain the angular velocity of the body by either dividing $\mathbf{v}_A$ by AC or $\mathbf{v}_B$ by BC .

d. If the velocities $\mathbf{v}_A$ and $\mathbf{v}_B$ have the same magnitude, the two lines drawn in part *b* do not intersect; the instantaneous center C is at an infinite distance. The angular velocity ω is zero and the body is in translation.

B. Once you have determined the instantaneous center and the angular velocity of a body, you can determine the velocity $\mathbf{v}_P$ of any point P of the body in the following way.

1. Draw a sketch of the body, showing point P , the instantaneous center of rotation C , and the angular velocity ω .

2. Draw a line from P to the instantaneous center C and measure or calculate the distance from P to C .

3. The velocity $\mathbf{v}_P$ is a vector perpendicular to the line PC , of the same sense as ω , and with a magnitude of $v_P = (PC)\omega$.

Finally, keep in mind that the instantaneous center of rotation can be used *only* to determine velocities at a specific instant in time. It cannot be used to determine accelerations.

Problems

CONCEPT QUESTIONS

15.CQ5 The disk rolls without sliding on the fixed horizontal surface. At the instant shown, the instantaneous center of zero velocity for rod AB would be located in which region?

- Region 1
- Region 2
- Region 3
- Region 4
- Region 5
- Region 6

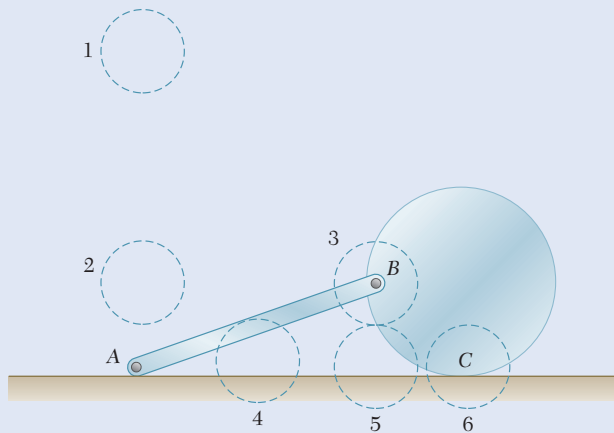


Fig. P15.CQ5

15.CQ6 Bar BDE is pinned to two links, AB and CD . At the instant shown, the angular velocities of link AB , link CD , and bar BDE are ω_{AB} , ω_{CD} , and ω_{BDE} , respectively. Which of the following statements concerning the angular speeds of the three objects is true at this instant?

- $\omega_{AB} = \omega_{CD} = \omega_{BDE}$
- $\omega_{BDE} > \omega_{AB} > \omega_{CD}$
- $\omega_{AB} = \omega_{CD} > \omega_{BDE}$
- $\omega_{AB} > \omega_{CD} > \omega_{BDE}$
- $\omega_{CD} > \omega_{AB} > \omega_{BDE}$

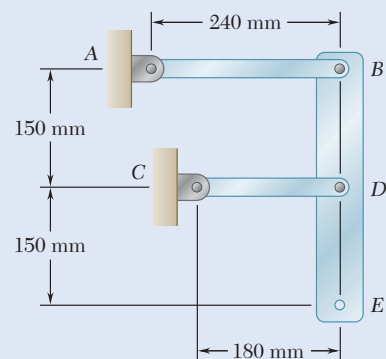


Fig. P15.CQ6

END-OF-SECTION PROBLEMS

15.73 A juggling club is thrown vertically into the air. The center of gravity G of the 20-in. club is located 12 in. from the knob. Knowing that at the instant shown, G has a velocity of 4 ft/s upwards and the club has an angular velocity of 30 rad/s counterclockwise, determine (a) the speeds of points A and B , (b) the location of the instantaneous center of rotation.

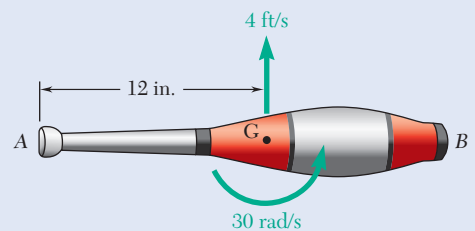


Fig. P15.73

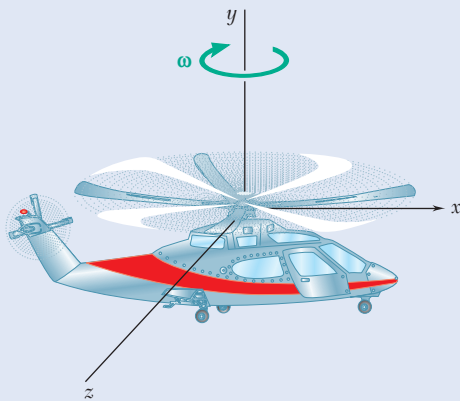


Fig. P15.75

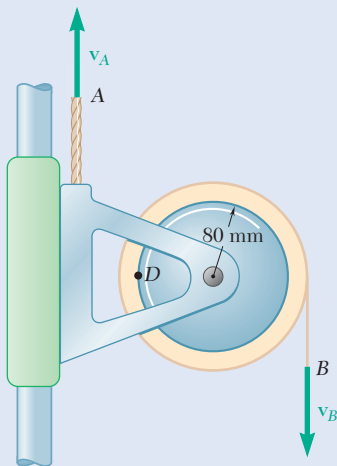


Fig. P15.78 and P15.79

- 15.74** At the instant shown during deceleration, the velocity of an automobile is 40 ft/s to the right. Knowing that the velocity of the contact point A of the wheel with the ground is 5 ft/s to the right, determine (a) the instantaneous center of rotation of the wheel, (b) the velocity of point B , (c) the velocity of point D .

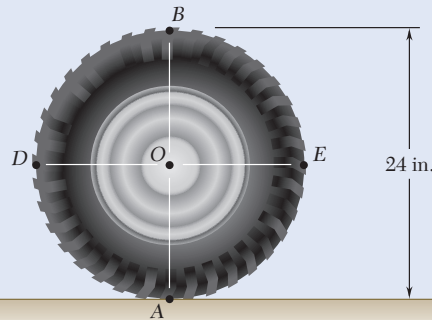


Fig. P15.74

- 15.75** A helicopter moves horizontally in the x direction at a speed of 120 mi/h. Knowing that the main blades rotate clockwise when viewed from above with an angular velocity of 180 rpm, determine the instantaneous axis of rotation of the main blades.

- 15.76 and 15.77** A 60-mm-radius drum is rigidly attached to a 100-mm-radius drum as shown. One of the drums rolls without sliding on the surface shown, and a cord is wound around the other drum. Knowing that end E of the cord is pulled to the left with a velocity of 120 mm/s, determine (a) the angular velocity of the drums, (b) the velocity of the center of the drums, (c) the length of cord wound or unwound per second.

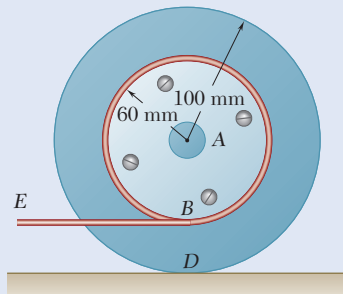


Fig. P15.76

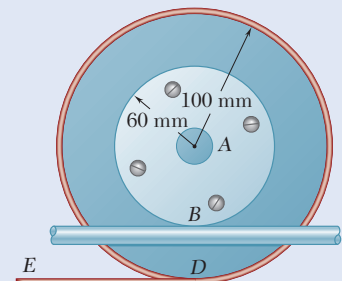


Fig. P15.77

- 15.78** The spool of tape shown and its frame assembly are pulled upward at a speed $v_A = 750$ mm/s. Knowing that the 80-mm-radius spool has an angular velocity of 15 rad/s clockwise and that at the instant shown the total thickness of the tape on the spool is 20 mm, determine (a) the instantaneous center of rotation of the spool, (b) the velocities of points B and D .

- 15.79** The spool of tape shown and its frame assembly are pulled upward at a speed $v_A = 100$ mm/s. Knowing that end B of the tape is pulled downward with a velocity of 300 mm/s and that at the instant shown the total thickness of the tape on the spool is 20 mm, determine (a) the instantaneous center of rotation of the spool, (b) the velocity of point D of the spool.

15.80 The arm ABC rotates with an angular velocity of 4 rad/s counterclockwise. Knowing that the angular velocity of the intermediate gear B is 8 rad/s counterclockwise, determine (a) the instantaneous centers of rotation of gears A and C , (b) the angular velocities of gears A and C .

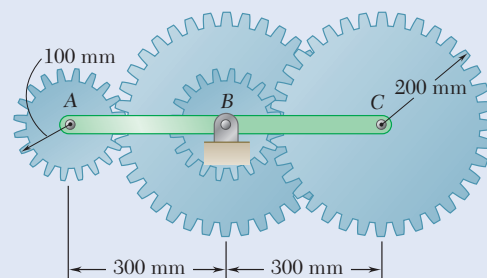


Fig. P15.80

15.81 The double gear rolls on the stationary left rack R . Knowing that the rack on the right has a constant velocity of 2 ft/s , determine (a) the angular velocity of the gear, (b) the velocities of points A and D .

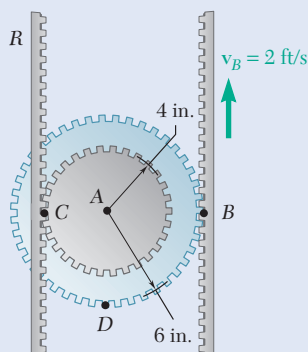


Fig. P15.81

15.82 An overhead door is guided by wheels at A and B that roll in horizontal and vertical tracks. Knowing that when $\theta = 40^\circ$ the velocity of wheel B is 1.5 ft/s upward, determine (a) the angular velocity of the door, (b) the velocity of end D of the door.

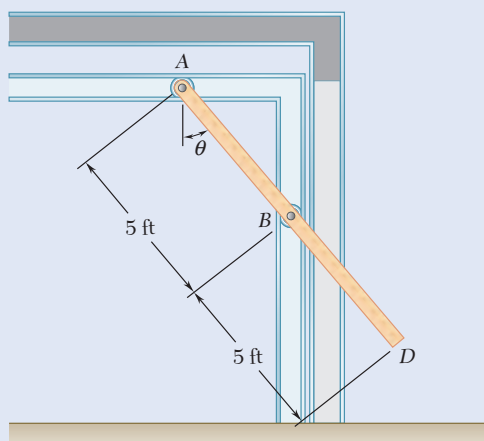


Fig. P15.82

15.83 Rod ABD is guided by wheels at A and B that roll in horizontal and vertical tracks. Knowing that at the instant $\beta = 60^\circ$ and the velocity of wheel B is 40 in./s downward, determine (a) the angular velocity of the rod, (b) the velocity of point D .

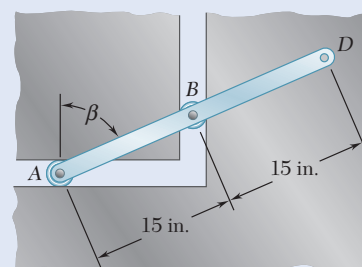


Fig. P15.83

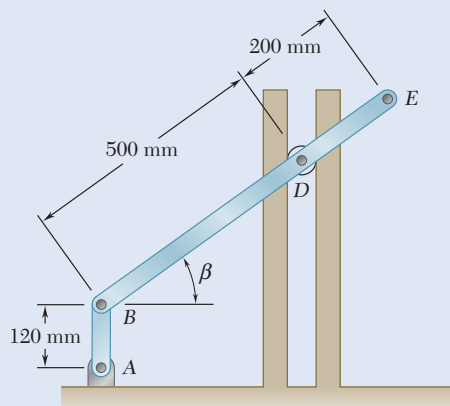


Fig. P15.84 and P15.85

15.84 Rod BDE is partially guided by a roller at D that moves in a vertical track. Knowing that at the instant shown the angular velocity of crank AB is 5 rad/s clockwise and that $\beta = 25^\circ$, determine (a) the angular velocity of the rod, (b) the velocity of point E .

15.85 Rod BDE is partially guided by a roller at D that moves in a vertical track. Knowing that at the instant shown $\beta = 30^\circ$, point E has a velocity of 2 m/s down and to the right, determine the angular velocities of rod BDE and crank AB .

15.86 A motor at O drives the windshield wiper mechanism so that OA has a constant angular velocity of 15 rpm . Knowing that at the instant shown linkage OA is vertical, $\theta = 30^\circ$, and $\beta = 15^\circ$, determine (a) the angular velocity of bar AB , (b) the velocity of the center of bar AB .

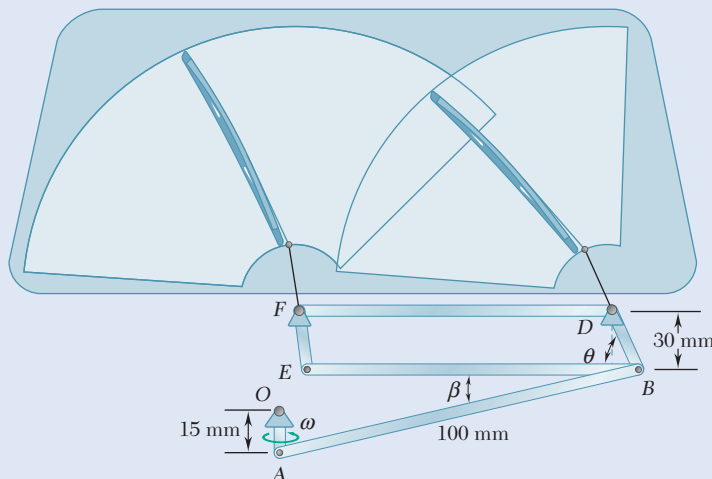


Fig. P15.86 and P15.87

15.87 A motor at O drives the windshield wiper mechanism so that point B has a speed of 2 m/s . Knowing that at the instant shown linkage OA is vertical, $\theta = 30^\circ$, and $\beta = 15^\circ$, determine (a) the angular velocity of bar OA , (b) the velocity of the center of bar AB .

15.88 Rod AB can slide freely along the floor and the inclined plane. Denoting the velocity of point A by $\mathbf{v}_A$, derive an expression for (a) the angular velocity of the rod, (b) the velocity of end B .

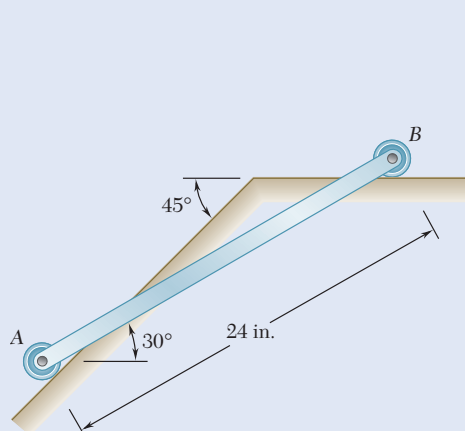


Fig. P15.89

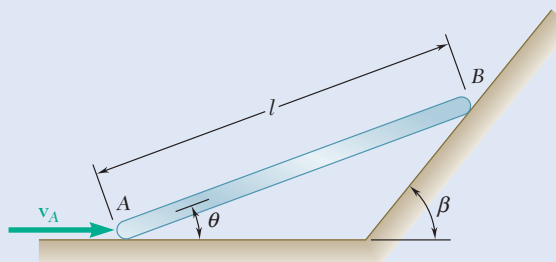


Fig. P15.88

15.89 Small wheels have been attached to the ends of bar AB and roll freely along the surfaces shown. Knowing that the velocity of wheel B is 7.5 ft/s to the right at the instant shown, determine (a) the velocity of end A of the bar, (b) the angular velocity of the bar, (c) the velocity of the midpoint of the bar.

- 15.90** Two slots have been cut in plate FG and the plate has been placed so that the slots fit two fixed pins A and B . Knowing that at the instant shown the angular velocity of crank DE is 6 rad/s clockwise, determine (a) the velocity of point F , (b) the velocity of point G .

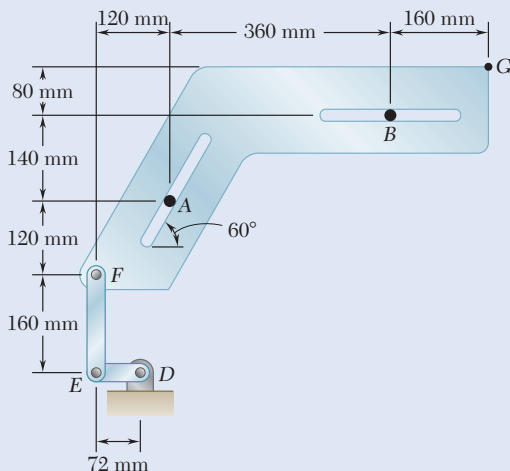


Fig. P15.90

- 15.91** The disk is released from rest and rolls down the incline. Knowing that the speed of A is 1.2 m/s when $\theta = 0^\circ$, determine at that instant (a) the angular velocity of the rod, (b) the velocity of B . (Only portions of the two tracks are shown.)

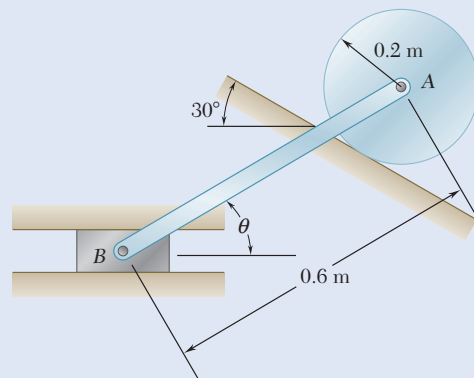


Fig. P15.91

- 15.92** The pin at B is attached to member ABD and can slide freely along the slot cut in the fixed plate. Knowing that at the instant shown the angular velocity of arm DE is 3 rad/s clockwise, determine (a) the angular velocity of member ABD , (b) the velocity of point A .

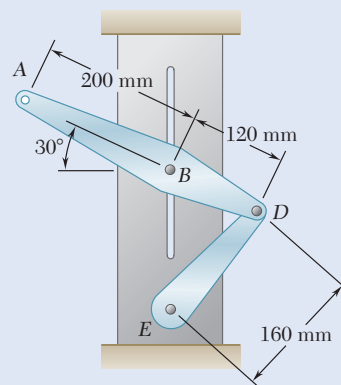


Fig. P15.92

- 15.93** Two identical rods ABF and DBE are connected by a pin at B . Knowing that at the instant shown the velocity of point D is 200 mm/s upward, determine the velocity of (a) point E , (b) point F .

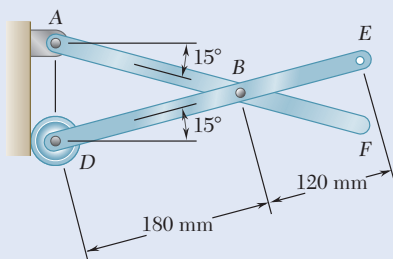


Fig. P15.93

- 15.94** Arm ABD is connected by pins to a collar at B and to crank DE . Knowing that the velocity of collar B is 16 in./s upward, determine (a) the angular velocity of arm ABD , (b) the velocity of point A .

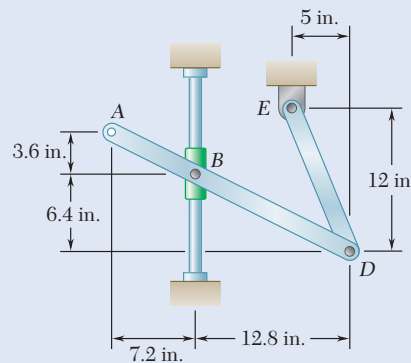


Fig. P15.94

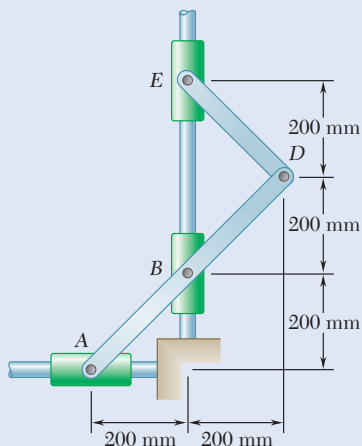


Fig. P15.96

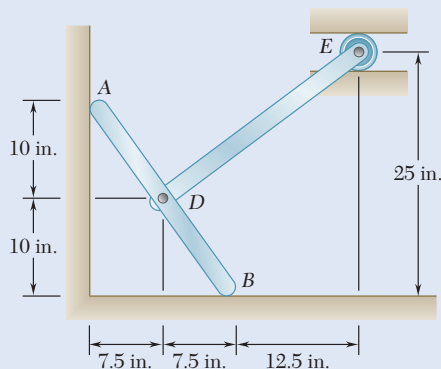


Fig. P15.95

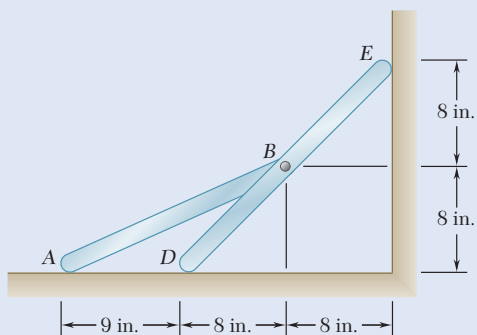


Fig. P15.98

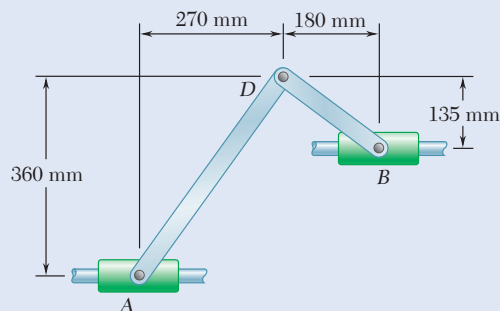


Fig. P15.97

15.95 Two 25-in. rods are pin-connected at D as shown. Knowing that B moves to the left with a constant velocity of 24 in./s, determine at the instant shown (a) the angular velocity of each rod, (b) the velocity of E .

15.96 Two rods ABD and DE are connected to three collars as shown. Knowing that the angular velocity of ABD is 5 rad/s clockwise, determine at the instant shown (a) the angular velocity of DE , (b) the velocity of collar E .

15.97 At the instant shown, the velocity of collar A is 0.4 m/s to the right and the velocity of collar B is 1 m/s to the left. Determine (a) the angular velocity of bar AD , (b) the angular velocity of bar BD , (c) the velocity of point D .

15.98 Two rods AB and DE are connected as shown. Knowing that point D moves to the left with a velocity of 40 in./s, determine (a) the angular velocity of each rod, (b) the velocity of point A .

15.99 Describe the space centroid and the body centroid of rod ABD of Prob. 15.83. (Hint: The body centroid need not lie on a physical portion of the rod.)

15.100 Describe the space centroid and the body centroid of the gear of Sample Prob. 15.6 as the gear rolls on the stationary horizontal rack.

15.101 Using the method of Sec. 15.3, solve Prob. 15.60.

15.102 Using the method of Sec. 15.3, solve Prob. 15.64.

15.103 Using the method of Sec. 15.3, solve Prob. 15.65.

15.104 Using the method of Sec. 15.3, solve Prob. 15.38.

15.4 GENERAL PLANE MOTION: ACCELERATION

We saw in Sec. 15.2A that any plane motion can be replaced by a translation defined by the motion of an arbitrary reference point A and a simultaneous rotation about A . We used this property in Sec. 15.2B to determine the velocity of the various points of a moving rigid body. We now use this same property to determine the acceleration of the points of the body.

15.4A Absolute and Relative Acceleration in Plane Motion

We first recall that the absolute acceleration $\mathbf{a}_B$ of a particle of the rigid body can be obtained from the relative-acceleration formula derived in Sec. 11.4D,

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A} \quad (15.21)$$

where the right-hand side represents a vector sum. The acceleration $\mathbf{a}_A$ corresponds to the translation of the rigid body with A . The relative acceleration $\mathbf{a}_{B/A}$ is associated with the rotation of the body about A and is measured with respect to axes centered at A and with fixed orientation.

Recall from Sec. 15.1B that we can resolve the relative acceleration $\mathbf{a}_{B/A}$ into two components: a **tangential component** $(\mathbf{a}_{B/A})_t$ perpendicular to the line AB and a **normal component** $(\mathbf{a}_{B/A})_n$ directed toward A (Fig. 15.22). We denote the position vector of B relative to A by $\mathbf{r}_{B/A}$ and the angular velocity and angular acceleration of the rigid body with respect to axes of fixed orientation by $\omega\mathbf{k}$ and $\alpha\mathbf{k}$, respectively. Then we have

$$\begin{aligned} (\mathbf{a}_{B/A})_t &= \alpha\mathbf{k} \times \mathbf{r}_{B/A} & (\mathbf{a}_{B/A})_t &= r\alpha \\ (\mathbf{a}_{B/A})_n &= -\omega^2\mathbf{r}_{B/A} & (\mathbf{a}_{B/A})_n &= r\omega^2 \end{aligned} \quad (15.22)$$

where r is the distance from A to B . Substituting the expressions obtained for the tangential and normal components of $\mathbf{a}_{B/A}$ into Eq. (15.21), we also have

Relative acceleration for two points on a rigid body

$$\mathbf{a}_B = \mathbf{a}_A + \alpha\mathbf{k} \times \mathbf{r}_{B/A} - \omega^2\mathbf{r}_{B/A} \quad (15.21')$$

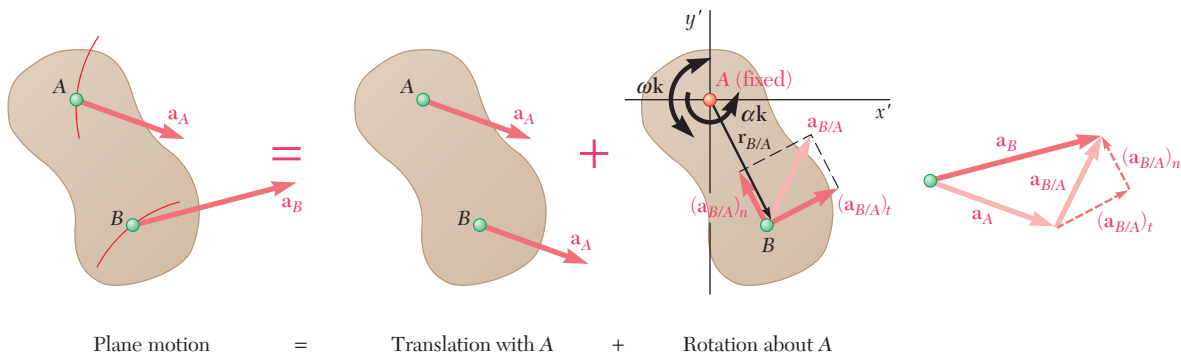


Fig. 15.22 Pictorial representation of the vector equation relating the acceleration of two points on a rigid body undergoing general plane motion.



Photo 15.6 The central gear rotates about a fixed axis and is pin-connected to three bars in general plane motion.

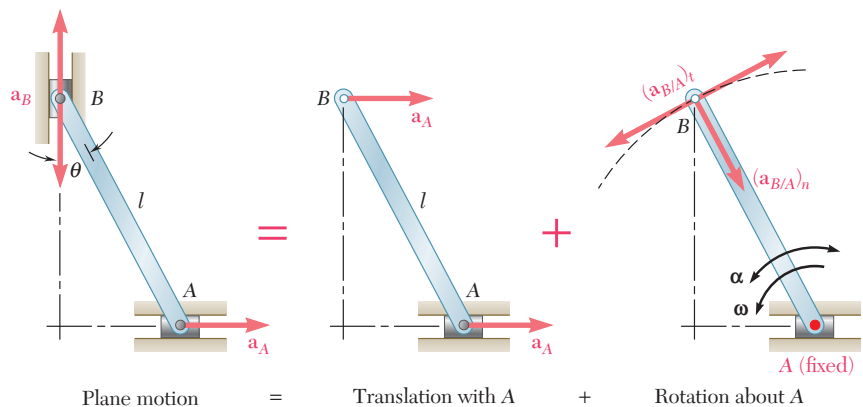


Fig. 15.23 For a sliding rod in general plane motion, the acceleration of point B relative to point A may have a tangential component in either direction perpendicular to the rod. The normal acceleration of B relative to A will always point toward A .

As an example, let us again consider the rod AB whose ends slide along a horizontal and a vertical track (Fig. 15.23). Assuming that we know the velocity $\mathbf{v}_A$ and the acceleration $\mathbf{a}_A$ of A , we propose to determine the acceleration $\mathbf{a}_B$ of B and the angular acceleration α of the rod. Choosing A as a reference point, the given motion is equivalent to a translation with A and a rotation about A . The absolute acceleration of B must be equal to the sum

$$\begin{aligned}\mathbf{a}_B &= \mathbf{a}_A + \mathbf{a}_{B/A} \\ &= \mathbf{a}_A + (\mathbf{a}_{B/A})_n + (\mathbf{a}_{B/A})_t\end{aligned}\quad (15.23)$$

where $(\mathbf{a}_{B/A})_n$ has magnitude $\omega^2 l$ and is directed toward A , while $(\mathbf{a}_{B/A})_t$ has the magnitude $l\alpha$ and is perpendicular to AB . Note that there is no way to tell whether the tangential component $(\mathbf{a}_{B/A})_t$ is directed to the left or to the right, and therefore, both possible directions for this component are indicated in Fig. 15.23. Similarly, both possible senses for $\mathbf{a}_B$ are indicated, since we do not know whether point B is accelerated upward or downward.

We can illustrate Eq. (15.23) geometrically. Figure 15.24 shows four different vector polygons, depending upon the sense of $\mathbf{a}_A$ and the relative magnitudes of $\mathbf{a}_A$ and $(\mathbf{a}_{B/A})_n$. To determine a_B and α from one of these diagrams, we must know not only a_A and θ but also ω . Therefore, we need to determine the angular velocity of the rod separately, by one of the methods indicated in Secs. 15.2 and 15.3. Then we can obtain the values of a_B and α by considering successively the x and y components of the vectors shown in Fig. 15.24. In the case of polygon a , we are assuming that α is in the counter-clockwise direction and $\mathbf{a}_B$ is down. Therefore, we have

$$\begin{aligned}\rightarrow x \text{ components:} & \quad 0 = a_A + l\omega^2 \sin \theta - l\alpha \cos \theta \\ \uparrow y \text{ components:} & \quad -a_B = -l\omega^2 \cos \theta - l\alpha \sin \theta\end{aligned}$$

and solve for a_B and α . An alternative approach to drawing Fig. 15.24 is to use a vector algebra solution; that is, you substitute the vector quantities into (15.21'), take the cross product, and equate components to obtain the two scalar equations shown previously.

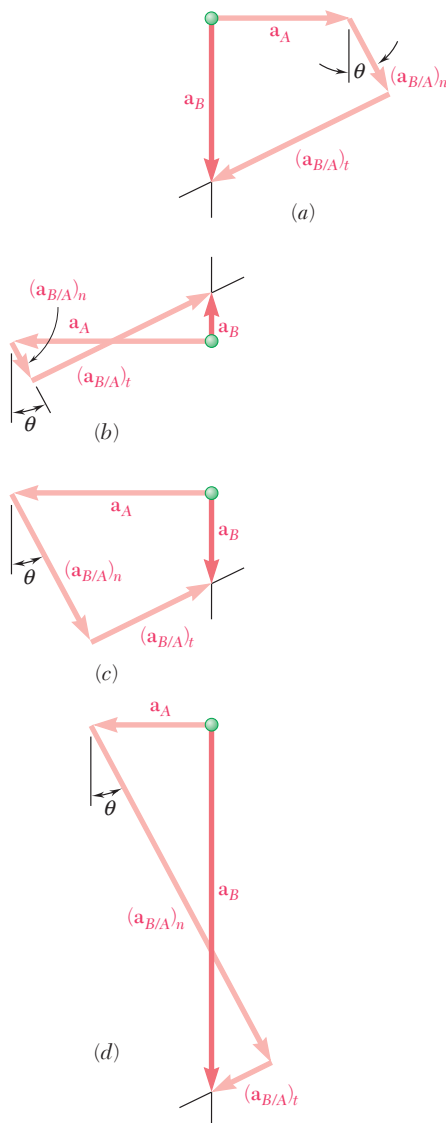


Fig. 15.24 Four possible vector polygons for the acceleration of the sliding rod.

Clearly, the determination of accelerations is considerably more involved than the determination of velocities. Yet in the example considered here, the ends A and B of the rod were moving along straight tracks, and the diagrams drawn were relatively simple. If A and B had moved along curved tracks, it would have been necessary to resolve the accelerations $\mathbf{a}_A$ and $\mathbf{a}_B$ into normal and tangential components and the solution of the problem would have involved six different vectors.

When a mechanism consists of several moving parts that are pin-connected, we can analyze the mechanism by considering each part to be a rigid body, keeping in mind that the points at which two parts are connected must have the same absolute acceleration (see Sample Prob. 15.15). In the case of meshed gears (see Sample Prob. 15.13), the tangential components of the accelerations of the teeth in contact are equal, but their normal components are different.

*15.4B Analysis of Plane Motion in Terms of a Parameter

In analyzing some mechanisms, it is possible to express the coordinates x and y of all the significant points of the mechanism by means of simple analytic expressions containing a single parameter. It is sometimes advantageous in such a case to determine the absolute velocity and the absolute acceleration of the various points of the mechanism directly, since we can obtain the components of the velocity and of the acceleration of a given point by differentiating the coordinates x and y of that point.

Let us consider again the rod AB whose ends slide, respectively, in a horizontal and a vertical track (Fig. 15.25). We can express the coordinates x_A and y_B of the ends of the rod in terms of the angle θ that the rod forms with the vertical:

$$x_A = l \sin \theta \quad y_B = l \cos \theta \quad (15.24)$$

Differentiating Eqs. (15.24) twice with respect to t , we have

$$\begin{aligned} v_A &= \dot{x}_A = l\dot{\theta} \cos \theta \\ a_A &= \ddot{x}_A = -l\dot{\theta}^2 \sin \theta + l\ddot{\theta} \cos \theta \end{aligned}$$

$$\begin{aligned} v_B &= \dot{y}_B = -l\dot{\theta} \sin \theta \\ a_B &= \ddot{y}_B = -l\dot{\theta}^2 \cos \theta - l\ddot{\theta} \sin \theta \end{aligned}$$

Recalling that $\dot{\theta} = \omega$ and $\ddot{\theta} = \alpha$, we obtain

$$v_A = l\omega \cos \theta \quad v_B = -l\omega \sin \theta \quad (15.25)$$

$$a_A = -l\omega^2 \sin \theta + l\alpha \cos \theta \quad a_B = -l\omega^2 \cos \theta - l\alpha \sin \theta \quad (15.26)$$

Note that a positive sign for v_A or a_A indicates that the velocity $\mathbf{v}_A$ or the acceleration $\mathbf{a}_A$ is directed to the right; a positive sign for v_B or a_B indicates that $\mathbf{v}_B$ or $\mathbf{a}_B$ is directed upward. We can use Eqs. (15.25) to determine, for example, v_B and ω when v_A and θ , are known. Substituting for ω in Eqs. (15.26), we can then determine a_B and α if we know a_A .

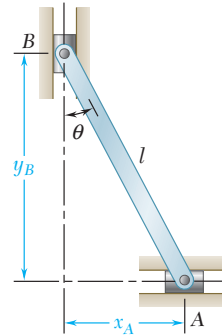
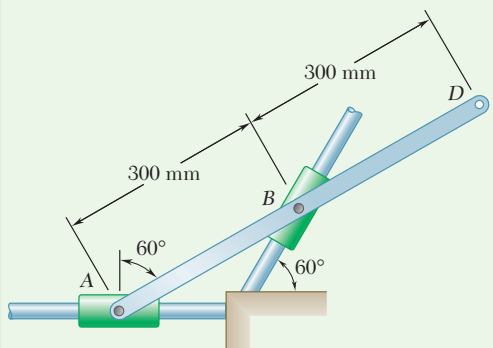


Fig. 15.25 The coordinates of the ends of the rod may be expressed in terms of the parameter θ .



Sample Problem 15.12

Collars A and B are pin-connected to bar ABD and can slide along fixed rods. Knowing that, at the instant shown, the velocity of A is a constant 0.9 m/s to the right, determine the angular acceleration of AB and the acceleration of B.

STRATEGY: Use the kinematic equation that relates the acceleration of two points on the same rigid body. Because you know that the directions of the accelerations of A and B must be along the fixed rods, choose these two points to relate.

MODELING and ANALYSIS: Model bar ABD as a rigid body. From Sample Prob. 15.5, you know $\omega = 3.00 \text{ rad/s} \uparrow$. The accelerations of A and B are related by

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A} = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$$

Substituting in known values (Fig. 1) and assuming $\boldsymbol{\alpha} = \alpha \mathbf{k}$ gives

$$\begin{aligned} a_B \cos 60^\circ \mathbf{i} + a_B \sin 60^\circ \mathbf{j} &= 0\mathbf{i} + \alpha \mathbf{k} \times [(0.3 \cos 30^\circ)\mathbf{i} + (0.3 \sin 30^\circ)\mathbf{j}] \\ &\quad - 3^2 [(0.3 \cos 30^\circ)\mathbf{i} + (0.3 \sin 30^\circ)\mathbf{j}] \\ 0.500a_B \mathbf{i} + 0.866a_B \mathbf{j} &= (0 - 0.15\alpha - 2.338)\mathbf{i} + (0.260\alpha - 1.350)\mathbf{j} \end{aligned}$$

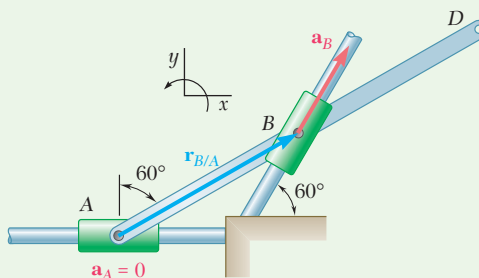


Fig. 1 Position vector and the assumed direction of the acceleration of point B.

Equating components, you have

$$\mathbf{i}: \quad 0.500a_B = -0.15\alpha - 2.338$$

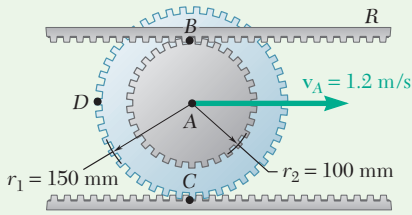
$$\mathbf{j}: \quad 0.866a_B = 0.260\alpha - 1.350$$

Solving these equations gives $a_B = -3.12 \text{ m/s}^2$ and $\alpha = -5.20 \text{ rad/s}^2$.

$$\boldsymbol{\alpha} = 5.20 \text{ rad/s}^2 \downarrow \quad \blacktriangleleft$$

$$\mathbf{a}_B = 3.12 \text{ m/s}^2 \nearrow 60^\circ \quad \blacktriangleleft$$

REFLECT and THINK: Even though A is traveling at a constant speed, bar AB still has an angular acceleration, and B has a linear acceleration. Just because one point on a body is moving at a constant speed doesn't mean the rest of the points on the body also have a constant speed.



Sample Problem 15.13

The center of the double gear of Sample Prob. 15.6 has a velocity of 1.2 m/s to the right and an acceleration of 3 m/s² to the right. Recalling that the lower rack is stationary, determine (a) the angular acceleration of the gear, (b) the acceleration of points B, C, and D of the gear.

STRATEGY: The double gear is a rigid body undergoing general plane motion, so use acceleration kinematics. You can also differentiate the equation for the gear's velocity and use that to find the gear's acceleration.

MODELING and ANALYSIS:

a. Angular Acceleration of the Gear. In Sample Prob. 15.6, you found that $x_A = -r_1\theta$ and $v_A = -r_1\omega$. Differentiating the second equation with respect to time, you obtain $a_A = -r_1\alpha$.

$$\begin{aligned} v_A &= -r_1\omega & 1.2 \text{ m/s} &= -(0.150 \text{ m})\omega & \omega &= -8 \text{ rad/s} \\ a_A &= -r_1\alpha & 3 \text{ m/s}^2 &= -(0.150 \text{ m})\alpha & \alpha &= -20 \text{ rad/s}^2 \end{aligned}$$

$$\alpha = \alpha\mathbf{k} = -(20 \text{ rad/s}^2)\mathbf{k} \quad \blacktriangleleft$$

b. Accelerations. The relationship between the acceleration of any two points on a rigid body undergoing general plane motion is

$$\begin{aligned} \mathbf{a}_B &= \mathbf{a}_A + \mathbf{a}_{B/A} = \mathbf{a}_A + (\mathbf{a}_{B/A})_t + (\mathbf{a}_{B/A})_n \\ &= \mathbf{a}_A + \alpha\mathbf{k} \times \mathbf{r}_{B/A} - \omega^2\mathbf{r}_{B/A} \end{aligned} \quad (1)$$

This equation indicates that the rolling motion of the gear can be thought of as a translation with A and a rotation about A (Fig. 1).

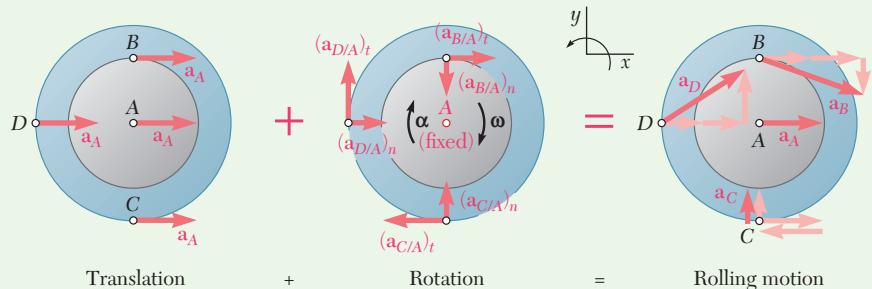


Fig. 1 A pictorial representation of Eq. 1.

Acceleration of Point B. Substituting values into Eq. (1) gives

$$\begin{aligned} \mathbf{a}_B &= \mathbf{a}_A + \mathbf{a}_{B/A} = \mathbf{a}_A + (\mathbf{a}_{B/A})_t + (\mathbf{a}_{B/A})_n \\ &= \mathbf{a}_A + \alpha\mathbf{k} \times \mathbf{r}_{B/A} - \omega^2\mathbf{r}_{B/A} \\ &= (3 \text{ m/s}^2)\mathbf{i} - (20 \text{ rad/s}^2)\mathbf{k} \times (0.100 \text{ m})\mathbf{j} - (8 \text{ rad/s})^2(0.100 \text{ m})\mathbf{j} \\ &= (3 \text{ m/s}^2)\mathbf{i} + (2 \text{ m/s}^2)\mathbf{i} - (6.40 \text{ m/s}^2)\mathbf{j} \end{aligned}$$

$$\mathbf{a}_B = 8.12 \text{ m/s}^2 \searrow 52.0^\circ \quad \blacktriangleleft$$

The vector triangle corresponding to this equation is shown in Fig. 2.

Acceleration of Point C Referring to Fig 3,

$$\begin{aligned} \mathbf{a}_C &= \mathbf{a}_A + \mathbf{a}_{C/A} = \mathbf{a}_A + \alpha\mathbf{k} \times \mathbf{r}_{C/A} - \omega^2\mathbf{r}_{C/A} \\ &= (3 \text{ m/s}^2)\mathbf{i} - (20 \text{ rad/s}^2)\mathbf{k} \times (-0.150 \text{ m})\mathbf{j} - (8 \text{ rad/s})^2(-0.150 \text{ m})\mathbf{j} \\ &= (3 \text{ m/s}^2)\mathbf{i} - (3 \text{ m/s}^2)\mathbf{i} + (9.60 \text{ m/s}^2)\mathbf{j} \end{aligned}$$

$$\mathbf{a}_C = 9.60 \text{ m/s}^2 \uparrow \quad \blacktriangleleft$$

(continued)

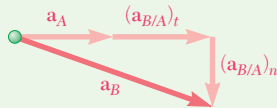


Fig. 2 Vector diagram relating the accelerations of A and B.

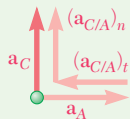


Fig. 3 Vector diagram of the equation relating the accelerations of A and C.

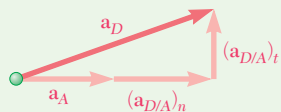


Fig. 4 Vector diagram relating the accelerations of A and D .

Acceleration of Point D (Fig. 4).

$$\begin{aligned}\mathbf{a}_D &= \mathbf{a}_A + \mathbf{a}_{D/A} = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{D/A} - \omega^2 \mathbf{r}_{D/A} \\ &= (3 \text{ m/s}^2)\mathbf{i} - (20 \text{ rad/s}^2)\mathbf{k} \times (-0.150 \text{ m})\mathbf{i} - (8 \text{ rad/s})^2(-0.150 \text{ m})\mathbf{i} \\ &= (3 \text{ m/s}^2)\mathbf{i} + (3 \text{ m/s}^2)\mathbf{j} + (9.60 \text{ m/s}^2)\mathbf{i}\end{aligned}$$

$$\mathbf{a}_D = 12.95 \text{ m/s}^2 \nearrow 13.4^\circ \quad \blacktriangleleft$$

REFLECT and THINK: It is interesting to note that the x -component of acceleration for point C is zero since it is in direct contact with the fixed lower rack. It does, however, have a normal acceleration pointed upward. This is also true for a wheel rolling without slip.

Sample Problem 15.14



Two adjacent identical wheels of a train can be modeled as rolling cylinders connected by a horizontal link. The distance between A and D is 10 in. Assume the wheels roll without sliding on the tracks. Knowing that the train is traveling at a constant 30 mph, determine the acceleration of the center of mass of DE .

STRATEGY: The connecting bar DE is undergoing curvilinear translation, so the acceleration of every point is identical; that is, $a_G = a_D$. Therefore, all you need to do is determine the acceleration of D using the kinematic relationship between A and D .

MODELING and ANALYSIS: Model the wheels and bar DE as rigid bodies. The speed of A is $v_A = 30 \text{ mph} = 44 \text{ ft/s}$. Since the wheel does not slip, the point of contact with the ground, C (Fig. 1), has a velocity of zero, so

$$\omega = \frac{v_A}{r_{A/C}} = \frac{44 \text{ ft/s}}{(20/12) \text{ ft}} = 26.4 \text{ rad/s}$$

Acceleration of D . The acceleration of D is

$$\mathbf{a}_D = \mathbf{a}_A + \mathbf{a}_{D/A} = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{D/A} - \omega^2 \mathbf{r}_{D/A} \quad (1)$$

The train is traveling at a constant speed, so $\mathbf{a}_A$ and $\boldsymbol{\alpha}$ are both zero. Substituting known quantities into Eq. (1) gives

$$\begin{aligned}\mathbf{a}_D &= 0 + 0 - (26.4 \text{ rad/s})^2 \left[\left(\frac{10}{12} \cos 60^\circ \text{ ft} \right) \mathbf{i} + \left(\frac{10}{12} \sin 60^\circ \text{ ft} \right) \mathbf{j} \right] \\ &= -(290.4 \text{ ft/s}^2)\mathbf{i} - (503.0 \text{ ft/s}^2)\mathbf{j}\end{aligned}$$

$$\mathbf{a}_G = \mathbf{a}_D = -(290 \text{ ft/s}^2)\mathbf{i} - (503 \text{ ft/s}^2)\mathbf{j} \quad \blacktriangleleft$$

REFLECT and THINK: Instead of using vector algebra, you could have recognized that the direction of $-\omega^2 \mathbf{r}_{D/A}$ is directed from D to A . So the final acceleration of D is simply $-\omega^2 \mathbf{r}_{D/A} \nearrow 60^\circ$.

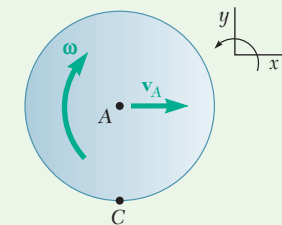
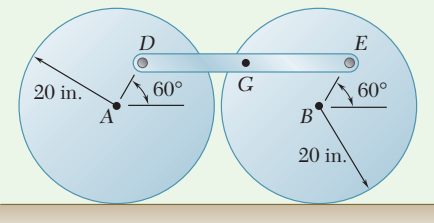


Fig. 1 Velocity and angular velocity of the wheel.

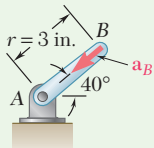
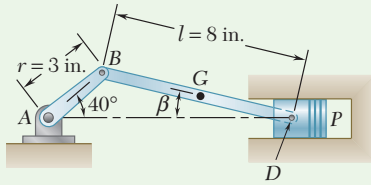


Fig. 1 The acceleration of B is only in the normal direction.

Sample Problem 15.15

Crank AB of the engine system of Sample Prob. 15.7 has a constant clockwise angular velocity of 2000 rpm. For the crank position shown, determine the angular acceleration of the connecting rod BD and the acceleration of point D .

STRATEGY: The linkage consists of two rigid bodies: crank AB is rotating about a fixed axis and connecting rod BD is undergoing general plane motion. Therefore, you need to use rigid-body kinematics.

MODELING and ANALYSIS:

Motion of Crank AB . Since the crank rotates about A with constant $\omega_{AB} = 2000 \text{ rpm} = 209.4 \text{ rad/s}$, you have $\alpha_{AB} = 0$. The acceleration of B is therefore directed toward A (Fig. 1) and has the magnitude of

$$a_B = r\omega_{AB}^2 = \left(\frac{3}{12} \text{ ft}\right)(209.4 \text{ rad/s})^2 = 10,962 \text{ ft/s}^2$$

$$\mathbf{a}_B = 10,962 \text{ ft/s}^2 \nearrow 40^\circ$$

Motion of Connecting Rod BD . The angular velocity ω_{BD} and the value of β were obtained in Sample Prob. 15.7 using relative velocity equations:

$$\omega_{BD} = 62.0 \text{ rad/s} \uparrow \quad \beta = 13.95^\circ$$

Resolve the motion of BD into a translation with B and a rotation about B (Fig. 2). Resolve the relative acceleration $\mathbf{a}_{D/B}$ into normal and tangential components:

$$(a_{D/B})_n = (BD)\omega_{BD}^2 = \left(\frac{8}{12} \text{ ft}\right)(62.0 \text{ rad/s})^2 = 2563 \text{ ft/s}^2$$

$$(\mathbf{a}_{D/B})_n = 2563 \text{ ft/s}^2 \searrow 13.95^\circ$$

$$(a_{D/B})_t = (BD)\alpha_{BD} = \left(\frac{8}{12}\right)\alpha_{BD} = 0.6667\alpha_{BD}$$

$$(\mathbf{a}_{D/B})_t = 0.6667\alpha_{BD} \swarrow 76.05^\circ$$

Although $(\mathbf{a}_{D/B})_t$ must be perpendicular to BD , its sense is not known.

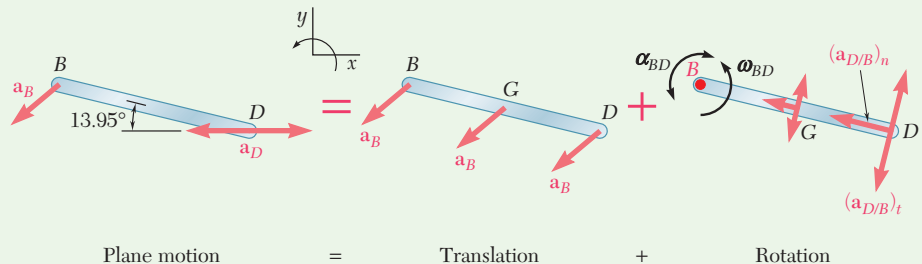


Fig. 2 General plane motion is a translation plus a rotation.

Noting that the acceleration $\mathbf{a}_D$ must be horizontal, you have

$$\mathbf{a}_D = \mathbf{a}_B + \mathbf{a}_{D/B} = \mathbf{a}_B + (\mathbf{a}_{D/B})_n + (\mathbf{a}_{D/B})_t \quad (1)$$

$$[a_D \leftrightarrow] = [10,962 \nearrow 40^\circ] + [2563 \searrow 13.95^\circ] + [0.6667\alpha_{BD} \swarrow 76.05^\circ]$$

Equating x and y components, you obtain the following scalar equations, as $\rightarrow x$ components:

$$-a_D = -10,962 \cos 40^\circ - 2563 \cos 13.95^\circ + 0.6667\alpha_{BD} \sin 13.95^\circ$$

+ $\uparrow y$ components:

$$0 = -10,962 \sin 40^\circ + 2563 \sin 13.95^\circ + 0.6667\alpha_{BD} \cos 13.95^\circ$$

(continued)

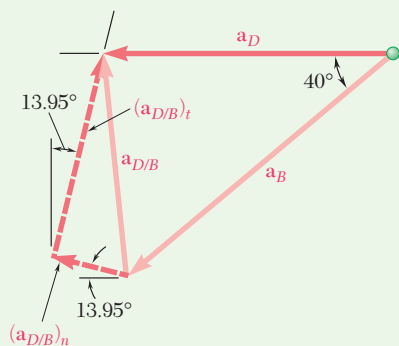


Fig. 3 Vector polygon relating the accelerations of *B* and *D*.

Solving the equations simultaneously gives $\alpha_{BD} = +9940 \text{ rad/s}^2$ and $a_D = +9290 \text{ ft/s}^2$. The positive signs indicate that the senses shown on the vector polygon (Fig. 3) are correct.

$$\alpha_{BD} = 9940 \text{ rad/s}^2 \nearrow$$

$$a_D = 9290 \text{ ft/s}^2 \leftarrow$$

REFLECT and THINK: In this solution, you looked at the magnitude and direction of each term in Eq. (1) and then found the *x* and *y* components. Alternatively, you could have assumed that a_D was to the left, α_{BD} was positive, and then substituted in the vector quantities to get

$$\mathbf{a}_D = \mathbf{a}_B + \alpha \mathbf{k} \times \mathbf{r}_{D/B} - \omega^2 \mathbf{r}_{D/B}$$

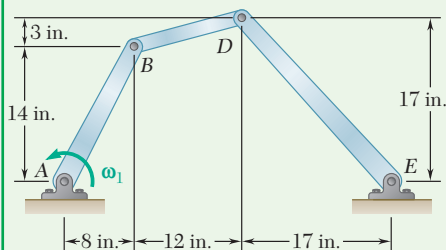
$$\begin{aligned} -a_D \mathbf{i} &= -a_B \cos 40^\circ \mathbf{i} - a_B \sin 40^\circ \mathbf{j} + \alpha_{BD} \mathbf{k} \times (l \cos \beta \mathbf{i} - l \sin \beta \mathbf{j}) \\ &\quad - \omega_{BD}^2 (l \cos \beta \mathbf{i} - l \sin \beta \mathbf{j}) \\ &= -a_B \cos 40^\circ \mathbf{i} - a_B \sin 40^\circ \mathbf{j} + \alpha_{BD} l \cos \beta \mathbf{j} + \alpha_{BD} l \sin \beta \mathbf{i} \\ &\quad - \omega_{BD}^2 l \cos \beta \mathbf{i} + \omega_{BD}^2 l \sin \beta \mathbf{j} \end{aligned}$$

Equating components gives

$$\mathbf{i}: -a_D = -a_B \cos 40^\circ + \alpha_{BD} l \sin \beta - \omega_{BD}^2 l \cos \beta$$

$$\mathbf{j}: 0 = -a_B \sin 40^\circ + \alpha_{BD} l \cos \beta + \omega_{BD}^2 l \sin \beta$$

These are identical to the previous equations if you substitute in the numbers.



Sample Problem 15.16

The linkage *ABDE* moves in the vertical plane. Knowing that, in the position shown, crank *AB* has a constant angular velocity ω_1 of 20 rad/s counterclockwise, determine the angular velocities and angular accelerations of the connecting rod *BD* and of the crank *DE*.

STRATEGY: The linkage consists of three interconnected rigid bodies. Use multiple velocity and acceleration kinematic equations to relate the motions of each body. You could solve this problem with the method used in Sample Prob. 15.15; however, we illustrate a vector approach, choosing position vectors $\mathbf{r}_B$, $\mathbf{r}_D$, and $\mathbf{r}_{D/B}$ as shown in Fig. 1.

MODELING and ANALYSIS:

Velocities. Assuming that the angular velocities of *BD* and *DE* are counterclockwise, you have

$$\omega_{AB} = \omega_{AB} \mathbf{k} = (20 \text{ rad/s}) \mathbf{k} \quad \omega_{BD} = \omega_{BD} \mathbf{k} \quad \omega_{DE} = \omega_{DE} \mathbf{k}$$

where $\mathbf{k}$ is a unit vector pointing out of the page. We can obtain the velocity of *D* by relating it to point *E*, as

$$\mathbf{v}_D = \mathbf{v}_E + \mathbf{v}_{D/E} = 0 + \omega_{AB} \mathbf{k} \times \mathbf{r}_D \quad (1)$$

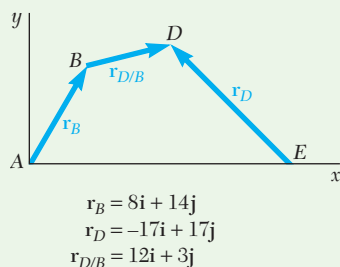


Fig. 1 Position vectors for points *B*, *D*, and *E*.

We can obtain the velocity of B by relating it to point A , as

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} = 0 + \omega_{AB}\mathbf{k} \times \mathbf{r}_B \quad (2)$$

The relationship between the velocities of D and B is

$$\mathbf{v}_D = \mathbf{v}_B + \mathbf{v}_{D/B} \quad (3)$$

Substituting Eqs. (1) and (2) into Eq. (3) and using $\mathbf{v}_{D/B} = \omega_{BD}\mathbf{k} \times \mathbf{r}_{D/B}$ gives

$$\begin{aligned} \omega_{DE}\mathbf{k} \times \mathbf{r}_D &= \omega_{AB}\mathbf{k} \times \mathbf{r}_B + \omega_{BD}\mathbf{k} \times \mathbf{r}_{D/B} \\ \omega_{DE}\mathbf{k} \times (-17\mathbf{i} + 17\mathbf{j}) &= 20\mathbf{k} \times (8\mathbf{i} + 14\mathbf{j}) + \omega_{BD}\mathbf{k} \times (12\mathbf{i} + 3\mathbf{j}) \\ -17\omega_{DE}\mathbf{j} - 17\omega_{DE}\mathbf{i} &= 160\mathbf{j} - 280\mathbf{i} + 12\omega_{BD}\mathbf{j} - 3\omega_{BD}\mathbf{i} \end{aligned}$$

Equating the coefficients of the unit vectors $\mathbf{i}$ and $\mathbf{j}$, the following two scalar equations are

$$\begin{aligned} -17\omega_{DE} &= -280 - 3\omega_{BD} \\ -17\omega_{DE} &= +160 + 12\omega_{BD} \end{aligned}$$

Solving these gives you $\omega_{BD} = -(29.33 \text{ rad/s})\mathbf{k}$ $\omega_{DE} = (11.29 \text{ rad/s})\mathbf{k}$ ◀

Accelerations. At the instant considered, crank AB has a constant angular velocity, so you have

$$\begin{aligned} \alpha_{AB} &= 0 & \alpha_{BD} &= \alpha_{BD}\mathbf{k} & \alpha_{DE} &= \alpha_{DE}\mathbf{k} \\ \mathbf{a}_D &= \mathbf{a}_B + \mathbf{a}_{D/B} \end{aligned} \quad (4)$$

Evaluate each term of Eq. (4) separately:

$$\begin{aligned} \text{Bar } DE: \quad \mathbf{a}_D &= \alpha_{DE}\mathbf{k} \times \mathbf{r}_D - \omega_{DE}^2\mathbf{r}_D \\ &= \alpha_{DE}\mathbf{k} \times (-17\mathbf{i} + 17\mathbf{j}) - (11.29)^2(-17\mathbf{i} + 17\mathbf{j}) \\ &= -17\alpha_{DE}\mathbf{j} - 17\alpha_{DE}\mathbf{i} + 2170\mathbf{i} - 2170\mathbf{j} \end{aligned}$$

$$\begin{aligned} \text{Bar } AB: \quad \mathbf{a}_B &= \alpha_{AB}\mathbf{k} \times \mathbf{r}_B - \omega_{AB}^2\mathbf{r}_B = 0 - (20)^2(8\mathbf{i} + 14\mathbf{j}) \\ &= -3200\mathbf{i} - 5600\mathbf{j} \end{aligned}$$

$$\begin{aligned} \text{Bar } BD: \quad \mathbf{a}_{D/B} &= \alpha_{BD}\mathbf{k} \times \mathbf{r}_{D/B} - \omega_{BD}^2\mathbf{r}_{D/B} \\ &= \alpha_{BD}\mathbf{k} \times (12\mathbf{i} + 3\mathbf{j}) - (29.33)^2(12\mathbf{i} + 3\mathbf{j}) \\ &= 12\alpha_{BD}\mathbf{j} - 3\alpha_{BD}\mathbf{i} - 10,320\mathbf{i} - 2580\mathbf{j} \end{aligned}$$

Substituting into Eq. (4) and equating the coefficients of $\mathbf{i}$ and $\mathbf{j}$, you obtain

$$\begin{aligned} -17\alpha_{DE} + 3\alpha_{BD} &= -15,690 \\ -17\alpha_{DE} - 12\alpha_{BD} &= -6010 \end{aligned}$$

Solving these gives you $\alpha_{BD} = -(645 \text{ rad/s}^2)\mathbf{k}$ $\alpha_{DE} = (809 \text{ rad/s}^2)\mathbf{k}$ ◀

REFLECT and THINK: The vector approach is preferred when there are more than two linkages. It is a very methodic approach and is easier to program when simulating mechanism movement over time.

SOLVING PROBLEMS ON YOUR OWN

This section was devoted to determining the accelerations of the points of a rigid body in plane motion. As you did previously for velocities, you will again consider the plane motion of a rigid body as the sum of two motions, namely, a translation and a rotation.

To solve a problem involving accelerations in plane motion, use the following steps.

1. Determine the angular velocity of the body. To find ω , you can either

- a. Consider the motion of the body as the sum of a translation and a rotation, as you did in Sec. 15.2, or
- b. Use the vector approach, as you did in Sec. 15.2, or the instantaneous center of rotation of the body, as you did in Sec. 15.3. However, keep in mind that you cannot use the instantaneous center to determine accelerations.

2. A diagram may be helpful to visualize the kinematics of the rigid bodies. The diagram will include the following diagrams (Fig 15.22):

a. Plane motion diagram. Draw a sketch of the body, including all dimensions, as well as the angular velocity ω . Show the angular acceleration α with its magnitude and sense if you know them. Also show those points for which you know or seek the accelerations, indicating all that you know about these accelerations.

b. Translation diagram. Select a reference point A for which you know the direction, the magnitude, or a component of the acceleration $\mathbf{a}_A$. Draw a second diagram showing the body in translation with each point having the same acceleration as point A .

c. Rotation diagram. Considering point A as a fixed reference point, draw a third diagram showing the body in rotation about A . Indicate the normal and tangential components of the relative accelerations of other points, such as the components $(\mathbf{a}_{B/A})_n$ and $(\mathbf{a}_{B/A})_t$ of the acceleration of point B with respect to point A .

3. Write the relative-acceleration formula relating two points of interest on the body being analyzed

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A} \quad \text{or} \quad \mathbf{a}_B = \mathbf{a}_A + (\mathbf{a}_{B/A})_n + (\mathbf{a}_{B/A})_t$$

a. Graphical approach. Select a point for which you know the direction, the magnitude, or a component of the acceleration and draw a vector diagram of the equation [Sample Prob. 15.15]. Starting at the same point, draw all known acceleration

components in tip-to-tail fashion for each member of the equation. Complete the diagram by drawing the two remaining vectors in appropriate directions and in such a way that the two sums of vectors end at a common point.

b. Vector approach. For a single rigid body, it is straightforward to apply

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha}_{AB} \times \mathbf{r}_{B/A} - \omega_{AB}^2 \mathbf{r}_{B/A}$$

For linkage type problems, you will need to write multiple relative acceleration equations relating the accelerations of points along the linkage [Sample Prob. 15.16].

4. The analysis of plane motion in terms of a parameter completed this section. This method should be used *only* if it is possible to express the coordinates x and y of all significant points of the body in terms of a single parameter (Sec. 15.4B). By differentiating the coordinates x and y of a given point twice with respect to t , you can determine the rectangular components of the absolute velocity and absolute acceleration of that point.

Problems

CONCEPT QUESTIONS

15.CQ7 A rear-wheel-drive car starts from rest and accelerates to the left so that the tires do not slip on the road. What is the direction of the acceleration of the point on the tire in contact with the road, that is, point A?

- a. $\leftarrow$ b. $\nwarrow$ c. $\uparrow$ d. $\downarrow$ e. $\swarrow$

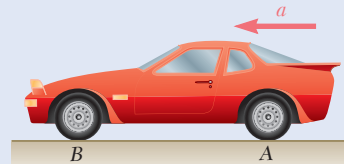


Fig. P15.CQ7

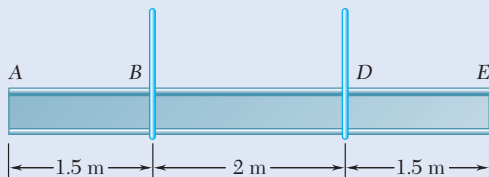


Fig. P15.105 and P15.106

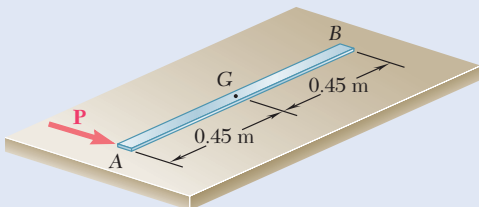


Fig. P15.107 and P15.108

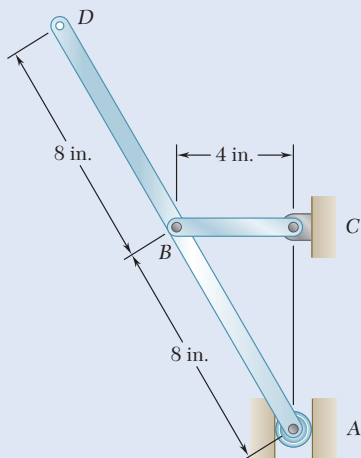


Fig. P15.109

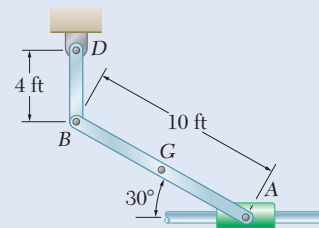


Fig. P15.110

END-OF-SECTION PROBLEMS

15.105 A 5-m steel beam is lowered by means of two cables unwinding at the same speed from overhead cranes. As the beam approaches the ground, the crane operators apply brakes to slow the unwinding motion. At the instant considered, the deceleration of the cable attached at B is 2.5 m/s^2 , while that of the cable attached at D is 1.5 m/s^2 . Determine (a) the angular acceleration of the beam, (b) the acceleration of points A and E.

15.106 For a 5-m steel beam AE, the acceleration of point A is 2 m/s^2 downward and the angular acceleration of the beam is 1.2 rad/s^2 counterclockwise. Knowing that at the instant considered the angular velocity of the beam is zero, determine the acceleration (a) of cable B, (b) of cable D.

15.107 A 900-mm rod rests on a horizontal table. A force $\mathbf{P}$ applied as shown produces the following accelerations: $\mathbf{a}_A = 3.6 \text{ m/s}^2$ to the right, $\alpha = 6 \text{ rad/s}^2$ counterclockwise as viewed from above. Determine the acceleration (a) of point G, (b) of point B.

15.108 In Prob. 15.107, determine the point of the rod that (a) has no acceleration, (b) has an acceleration of 2.4 m/s^2 to the right.

15.109 Knowing that at the instant shown crank BC has a constant angular velocity of 45 rpm clockwise, determine the acceleration (a) of point A, (b) of point D.

15.110 End A of rod AB moves to the right with a constant velocity of 6 ft/s. For the position shown, determine (a) the angular acceleration of rod AB, (b) the acceleration of the midpoint G of rod AB.

15.111 An automobile travels to the left at a constant speed of 72 km/h. Knowing that the diameter of the wheel is 560 mm, determine the acceleration (*a*) of point *B*, (*b*) of point *C*, (*c*) of point *D*.

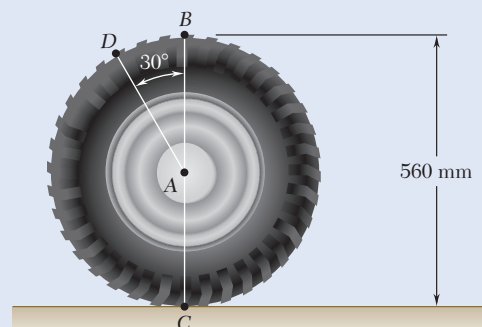


Fig. P15.111

15.112 The 18-in.-radius flywheel is rigidly attached to a 1.5-in.-radius shaft that can roll along parallel rails. Knowing that at the instant shown the center of the shaft has a velocity of 1.2 in./s and an acceleration of 0.5 in./s², both directed down to the left, determine the acceleration (*a*) of point *A*, (*b*) of point *B*.

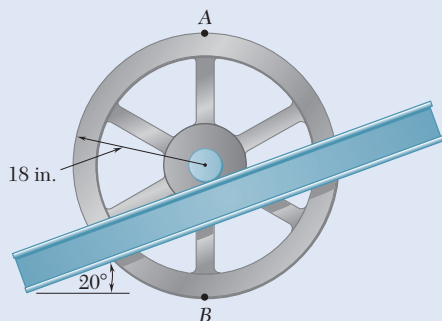


Fig. P15.112

15.113 and 15.114 A 3-in.-radius drum is rigidly attached to a 5-in.-radius drum as shown. One of the drums rolls without sliding on the surface shown, and a cord is wound around the other drum. Knowing that at the instant shown end *D* of the cord has a velocity of 8 in./s and an acceleration of 30 in./s², both directed to the left, determine the accelerations of points *A*, *B*, and *C* of the drums.

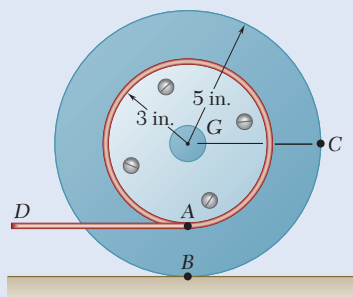


Fig. P15.113

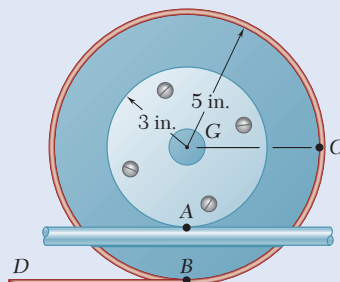


Fig. P15.114

15.115 A heavy crate is being moved a short distance using three identical cylinders as rollers. Knowing that at the instant shown the crate has a velocity of 200 mm/s and an acceleration of 400 mm/s², both directed to the right, determine (*a*) the angular acceleration of the center cylinder, (*b*) the acceleration of point *A* on the center cylinder.

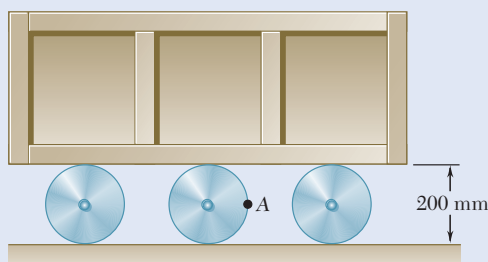


Fig. P15.115

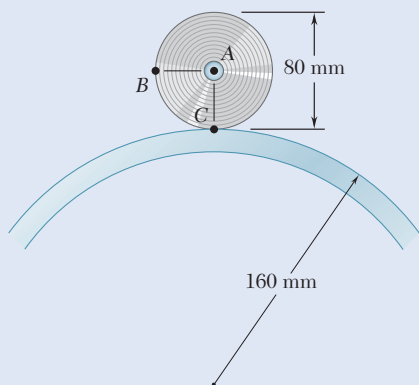


Fig. P15.116

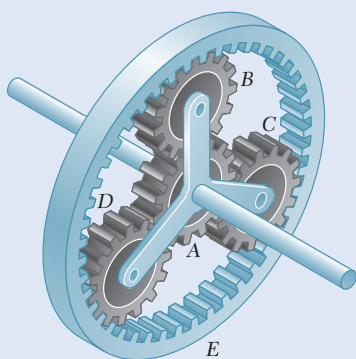


Fig. P15.118

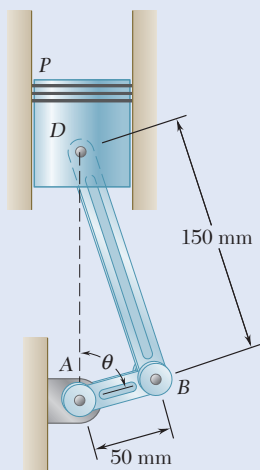


Fig. P15.120 and P15.121

15.116 A wheel rolls without slipping on a fixed cylinder. Knowing that at the instant shown the angular velocity of the wheel is 10 rad/s clockwise and its angular acceleration is 30 rad/s^2 counterclockwise, determine the acceleration of (a) point A, (b) point B, (c) point C.

15.117 The 100-mm -radius drum rolls without slipping on a portion of a belt that moves downward to the left with a constant velocity of 120 mm/s . Knowing that at a given instant the velocity and acceleration of the center A of the drum are as shown, determine the acceleration of point D.

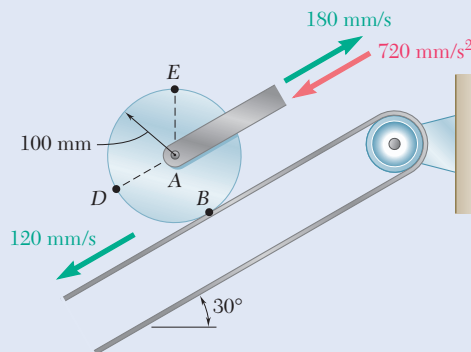


Fig. P15.117

15.118 In the planetary gear system shown, the radius of gears A, B, C, and D is 3 in. and the radius of the outer gear E is 9 in. Knowing that gear A has a constant angular velocity of 150 rpm clockwise and that the outer gear E is stationary, determine the magnitude of the acceleration of the tooth of gear D that is in contact with (a) gear A, (b) gear E.

15.119 The 200-mm -radius disk rolls without sliding on the surface shown. Knowing that the distance BG is 160 mm and that at the instant shown the disk has an angular velocity of 8 rad/s counterclockwise and an angular acceleration of 2 rad/s^2 clockwise, determine the acceleration of A.

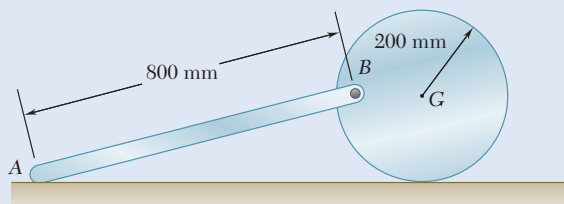


Fig. P15.119

15.120 Knowing that crank AB rotates about point A with a constant angular velocity of 900 rpm clockwise, determine the acceleration of the piston P when $\theta = 60^\circ$.

15.121 Knowing that crank AB rotates about point A with a constant angular velocity of 900 rpm clockwise, determine the acceleration of the piston P when $\theta = 120^\circ$.

15.122 In the two-cylinder air compressor shown, the connecting rods BD and BE are each 190 mm long and crank AB rotates about the fixed point A with a constant angular velocity of 1500 rpm clockwise. Determine the acceleration of each piston when $\theta = 0$.

15.123 The disk shown has a constant angular velocity of 500 rpm counterclockwise. Knowing that rod BD is 10 in. long, determine the acceleration of collar D when (a) $\theta = 90^\circ$, (b) $\theta = 180^\circ$.

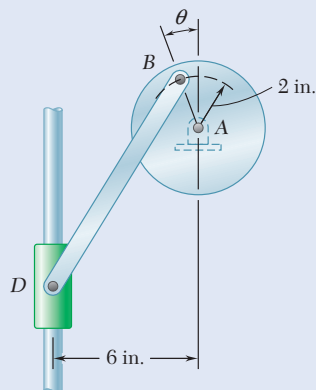


Fig. P15.123

15.124 Arm AB has a constant angular velocity of 16 rad/s counterclockwise. At the instant when $\theta = 90^\circ$, determine the acceleration (a) of collar D , (b) of the midpoint G of bar BD .

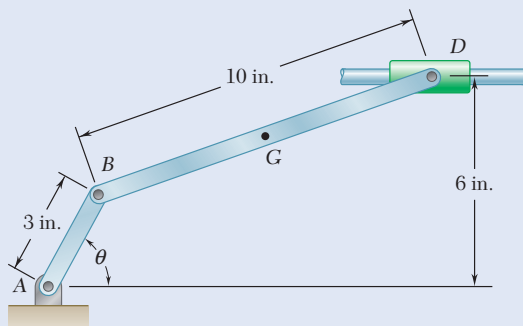


Fig. P15.124 and P15.125

15.125 Arm AB has a constant angular velocity of 16 rad/s counterclockwise. At the instant when $\theta = 60^\circ$, determine the acceleration of collar D .

15.126 A straight rack rests on a gear of radius $r = 3$ in. and is attached to a block B as shown. Knowing that at the instant shown $\theta = 20^\circ$, the angular velocity of gear D is 3 rad/s clockwise, and it is speeding up at a rate of 2 rad/s^2 , determine (a) the angular acceleration of AB , (b) the acceleration of block B .

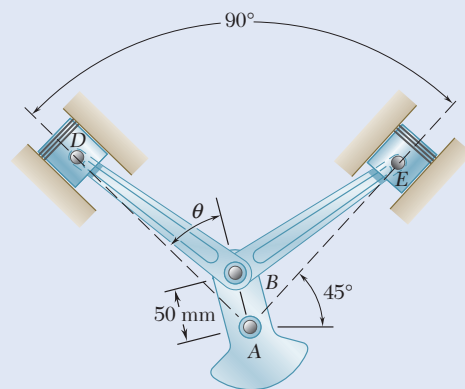


Fig. P15.122

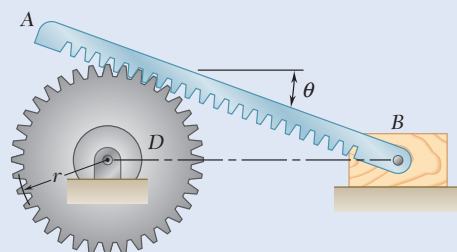


Fig. P15.126

- 15.127** The elliptical exercise machine has fixed axes of rotation at points A and E . Knowing that at the instant shown the flywheel AB has a constant angular velocity of 6 rad/s clockwise, determine the acceleration of point D .

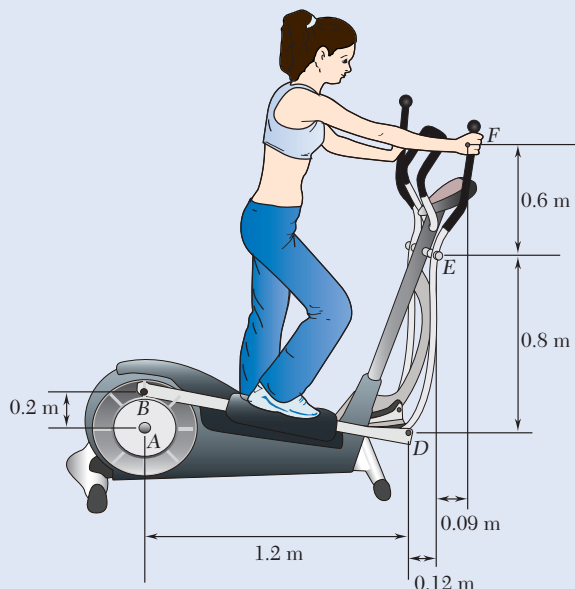


Fig. P15.127 and P15.128

- 15.128** The elliptical exercise machine has fixed axes of rotation at points A and E . Knowing that at the instant shown the flywheel AB has a constant angular velocity of 6 rad/s clockwise, determine (a) the angular acceleration of bar DEF , (b) the acceleration of point F .
- 15.129** Knowing that at the instant shown bar AB has a constant angular velocity of 19 rad/s clockwise, determine (a) the angular acceleration of bar BGD , (b) the angular acceleration of bar DE .

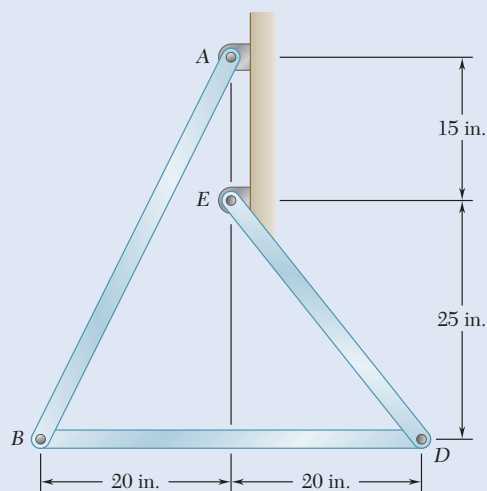


Fig. P15.131 and P15.133

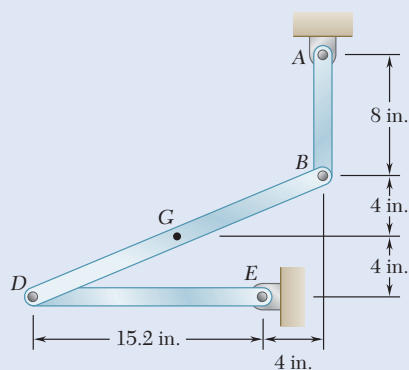


Fig. P15.129 and P15.130

- 15.130** Knowing that at the instant shown bar DE has a constant angular velocity of 18 rad/s clockwise, determine (a) the acceleration of point B , (b) the acceleration of point G .
- 15.131 and 15.132** Knowing that at the instant shown bar AB has a constant angular velocity of 4 rad/s clockwise, determine the angular acceleration (a) of bar BD , (b) of bar DE .

15.133 and 15.134 Knowing that at the instant shown bar AB has an angular velocity of 4 rad/s and an angular acceleration of 2 rad/s^2 , both clockwise, determine the angular acceleration (α) of bar BD , (b) of bar DE by using the vector approach as is done in Sample Prob. 15.16.

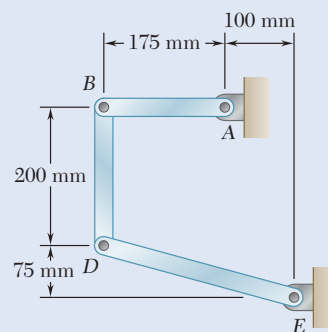


Fig. P15.132 and P15.134

15.135 Roberts linkage is named after Richard Roberts (1789–1864) and can be used to draw a close approximation to a straight line by locating a pen at point F . The distance AB is the same as BF , DF , and DE . Knowing that at the instant shown, bar AB has a constant angular velocity of 4 rad/s clockwise, determine (a) the angular acceleration of bar DE , (b) the acceleration of point F .

15.136 For the oil pump rig shown, link AB causes the beam BCE to oscillate as the crank OA revolves. Knowing that OA has a radius of 0.6 m and a constant clockwise angular velocity of 20 rpm , determine the velocity and acceleration of point D at the instant shown.

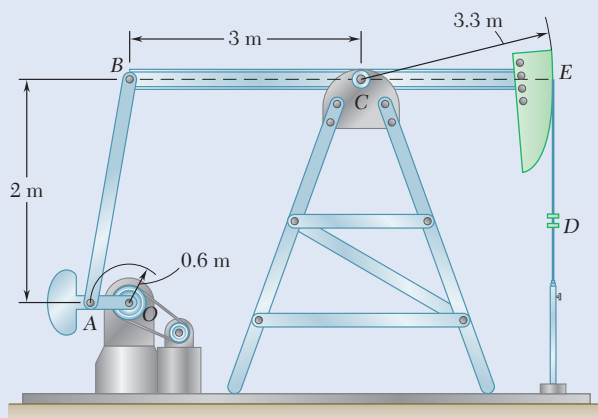


Fig. P15.136

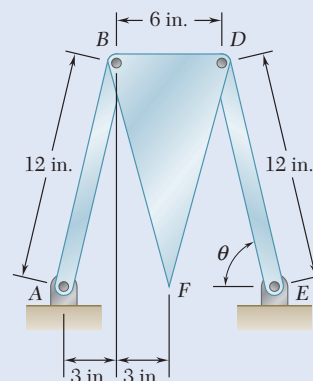


Fig. P15.135

15.137 Denoting by $\mathbf{r}_A$ the position vector of a point A of a rigid slab that is in plane motion, show that (a) the position vector $\mathbf{r}_C$ of the instantaneous center of rotation is

$$\mathbf{r}_C = \mathbf{r}_A + \frac{\boldsymbol{\omega} \times \mathbf{v}_A}{\omega^2}$$

where $\boldsymbol{\omega}$ is the angular velocity of the slab and $\mathbf{v}_A$ is the velocity of point A , (b) the acceleration of the instantaneous center of rotation is zero if, and only if,

$$\mathbf{a}_A = \frac{\alpha}{\omega} \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{v}_A$$

where $\boldsymbol{\alpha} = \alpha \mathbf{k}$ is the angular acceleration of the slab.

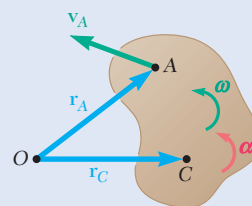


Fig. P15.137

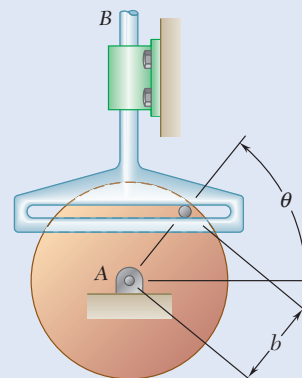


Fig. P15.138

***15.138** The drive disk of the Scotch crosshead mechanism shown has an angular velocity $\boldsymbol{\omega}$ and an angular acceleration $\boldsymbol{\alpha}$, both directed counterclockwise. Using the method of Sec. 15.4B, derive expressions for the velocity and acceleration of point B .

- *15.139** The wheels attached to the ends of rod AB roll along the surfaces shown. Using the method of Sec. 15.4B, derive an expression for the angular velocity of the rod in terms of v_B , θ , l , and β .

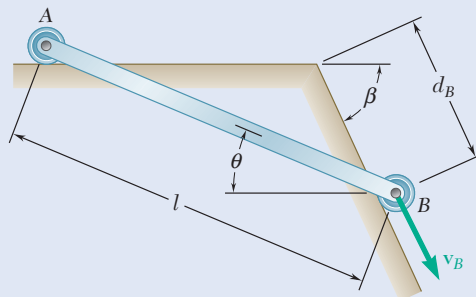


Fig. P15.139 and P15.140

- *15.140** The wheels attached to the ends of rod AB roll along the surfaces shown. Using the method of Sec. 15.4B and knowing that the acceleration of wheel B is zero, derive an expression for the angular acceleration of the rod in terms of v_B , θ , l , and β .

- *15.141** A disk of radius r rolls to the right with a constant velocity $\mathbf{v}$. Denoting by P the point of the rim in contact with the ground at $t = 0$, derive expressions for the horizontal and vertical components of the velocity of P at any time t .

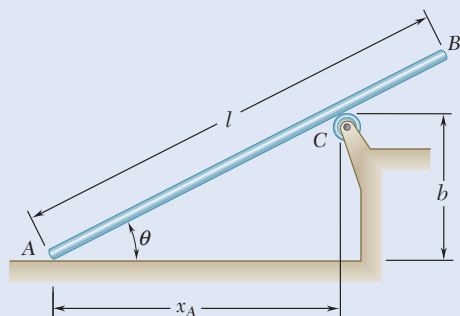


Fig. P15.142 and P15.143

- *15.142** Rod AB moves over a small wheel at C while end A moves to the right with a constant velocity $\mathbf{v}_A$. Using the method of Sec. 15.4B, derive expressions for the angular velocity and angular acceleration of the rod.

- *15.143** Rod AB moves over a small wheel at C while end A moves to the right with a constant velocity $\mathbf{v}_A$. Using the method of Sec. 15.4B, derive expressions for the horizontal and vertical components of the velocity of point B .

- 15.144** Crank AB rotates with a constant clockwise angular velocity ω . Using the method of Sec. 15.4B, derive expressions for the angular velocity of rod BD and the velocity of the point on the rod coinciding with point E in terms of θ , ω , b , and l .

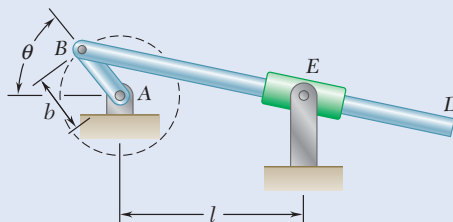


Fig. P15.144 and P15.145

- 15.145** Crank AB rotates with a constant clockwise angular velocity ω . Using the method of Sec. 15.4B, derive an expression for the angular acceleration of rod BD in terms of θ , ω , b , and l .

- 15.146** Solve the engine system from Sample Prob. 15.15 using the methods of Section 15.4B. *Hint:* Define the angle between the horizontal and the crank AB as θ and derive the motion in terms of this parameter.

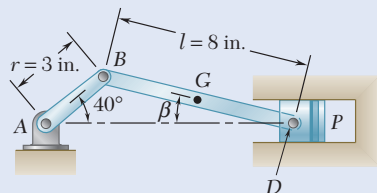


Fig. P15.146

- *15.147** The position of rod AB is controlled by a disk of radius r that is attached to yoke CD . Knowing that the yoke moves vertically upward with a constant velocity v_0 , derive expressions for the angular velocity and angular acceleration of rod AB .

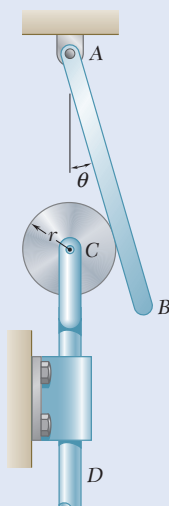


Fig. P15.147

- *15.148** A wheel of radius r rolls without slipping along the inside of a fixed cylinder of radius R with a constant angular velocity ω . Denoting by P the point of the wheel in contact with the cylinder at $t = 0$, derive expressions for the horizontal and vertical components of the velocity of P at any time t . (The curve described by point P is a *hypocycloid*.)
- *15.149** In Prob. 15.148, show that the path of P is a vertical straight line when $r = R/2$. Derive expressions for the corresponding velocity and acceleration of P at any time t .

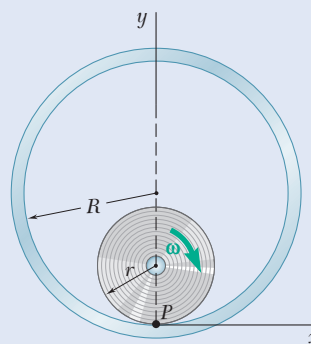


Fig. P15.148

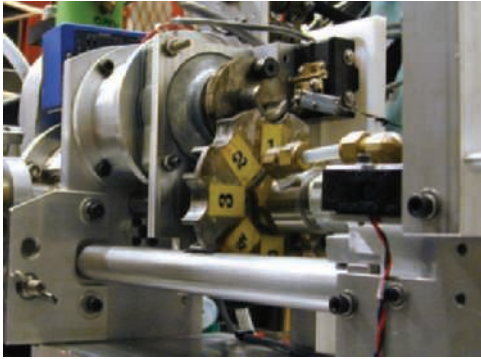


Photo 15.7 A Geneva mechanism is used to convert rotary motion into intermittent motion.

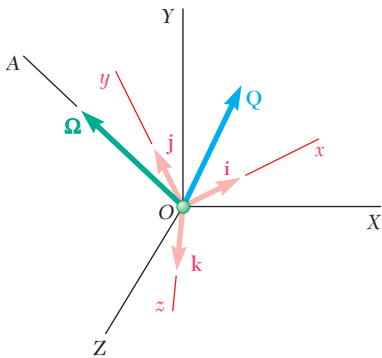


Fig. 15.26 A fixed frame of reference $OXYZ$ and a rotating frame $Oxyz$ with angular velocity Ω .

15.5 ANALYZING MOTION WITH RESPECT TO A ROTATING FRAME

We saw in Sec. 11.4B that the rate of change of a vector is the same with respect to a fixed frame and with respect to a frame in translation. In this section, we consider the rates of change of a vector $\mathbf{Q}$ with respect to a fixed frame and with respect to a rotating frame of reference.[†] You will see how to determine the rate of change of $\mathbf{Q}$ with respect to one frame of reference when $\mathbf{Q}$ is defined by its components in another frame. This kind of analysis is very useful for designing mechanisms that convert one kind of motion into another, such as continuous rotation into intermittent rotation. It is also helpful when you have, say, an extending linear actuator that is also rotating.

15.5A Rate of Change of a Vector with Respect to a Rotating Frame

Consider two frames of reference centered at O : a fixed frame $OXYZ$ and a frame $Oxyz$ that rotates about the fixed axis OA . Let Ω denote the angular velocity of the frame $Oxyz$ at a given instant (Fig. 15.26). Consider now a vector function $\mathbf{Q}(t)$ represented by the vector $\mathbf{Q}$ attached at O ; as the time t varies, both the direction and the magnitude of $\mathbf{Q}$ change. The variation of $\mathbf{Q}$ is viewed differently by an observer using $OXYZ$ as a frame of reference and by an observer using $Oxyz$, so we should expect the rate of change of $\mathbf{Q}$ to depend upon the frame of reference that has been selected. Therefore, we denote the rate of change of $\mathbf{Q}$ with respect to the fixed frame $OXYZ$ by $(\dot{\mathbf{Q}})_{OXYZ}$ and the rate of change of $\mathbf{Q}$ with respect to the rotating frame $Oxyz$ by $(\dot{\mathbf{Q}})_{Oxyz}$. We propose to determine the relation between these two rates of change.

Let us first resolve the vector $\mathbf{Q}$ into components along the x , y , and z axes of the rotating frame. Denoting the corresponding unit vectors by $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$, we have

$$\mathbf{Q} = Q_x \mathbf{i} + Q_y \mathbf{j} + Q_z \mathbf{k} \quad (15.27)$$

Differentiating Eq. (15.27) with respect to t and considering the unit vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ to be fixed, we obtain the rate of change of $\mathbf{Q}$ with respect to the rotating frame $Oxyz$, as

$$(\dot{\mathbf{Q}})_{Oxyz} = \dot{Q}_x \mathbf{i} + \dot{Q}_y \mathbf{j} + \dot{Q}_z \mathbf{k} \quad (15.28)$$

To obtain the rate of change of $\mathbf{Q}$ with respect to the fixed frame $OXYZ$, we must consider the unit vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ to be variable when differentiating Eq. (15.27). This gives

$$(\dot{\mathbf{Q}})_{OXYZ} = \dot{Q}_x \mathbf{i} + \dot{Q}_y \mathbf{j} + \dot{Q}_z \mathbf{k} + Q_x \frac{d\mathbf{i}}{dt} + Q_y \frac{d\mathbf{j}}{dt} + Q_z \frac{d\mathbf{k}}{dt} \quad (15.29)$$

From Eq. (15.28), we observe that the sum of the first three terms in the right-hand side of Eq. (15.29) represents the rate of change $(\dot{\mathbf{Q}})_{Oxyz}$.

[†]Recall that the selection of a fixed frame of reference is arbitrary. Any frame may be designated as “fixed”; all others are then considered as moving.

We note, on the other hand, that the rate of change $(\dot{\mathbf{Q}})_{OXYZ}$ would reduce to the last three terms in Eq. (15.29) if vector $\mathbf{Q}$ were fixed within the frame $Oxyz$, since $(\dot{\mathbf{Q}})_{Oxyz}$ would then be zero. But in that case, $(\dot{\mathbf{Q}})_{OXYZ}$ would represent the velocity of a particle located at the tip of $\mathbf{Q}$ and belonging to a body rigidly attached to the frame $Oxyz$. Thus, the last three terms in Eq. (15.29) represent the velocity of that particle. Since the frame $Oxyz$ has an angular velocity $\boldsymbol{\Omega}$ with respect to $OXYZ$ at the instant considered, we have, by Eq. (15.5),

$$Q_x \frac{d\mathbf{i}}{dt} + Q_y \frac{d\mathbf{j}}{dt} + Q_z \frac{d\mathbf{k}}{dt} = \boldsymbol{\Omega} \times \mathbf{Q} \quad (15.30)$$

Substituting from Eqs. (15.28) and (15.30) into Eq. (15.29), we obtain the fundamental relation

$$(\dot{\mathbf{Q}})_{OXYZ} = (\dot{\mathbf{Q}})_{Oxyz} + \boldsymbol{\Omega} \times \mathbf{Q} \quad (15.31)$$

We conclude that the rate of change of vector $\mathbf{Q}$ with respect to the fixed frame $OXYZ$ consists of two parts: The first part represents the rate of change of $\mathbf{Q}$ with respect to the rotating frame $Oxyz$; the second part, $\boldsymbol{\Omega} \times \mathbf{Q}$, is induced by the rotation of the frame $Oxyz$.

The use of the relation in Eq. (15.31) simplifies the determination of the rate of change of a vector $\mathbf{Q}$ with respect to a fixed frame of reference $OXYZ$ when vector $\mathbf{Q}$ is defined by its components along the axes of a rotating frame $Oxyz$. In particular, this relation does not require separate computations of the derivatives of the unit vectors defining the orientation of the rotating frame.

15.5B Plane Motion of a Particle Relative to a Rotating Frame

Consider two frames of reference with both centered at O and both in the plane of the figure: a fixed frame OXY and a rotating frame Oxy (Fig. 15.27). Let P be a particle moving in the plane of the figure. The position vector $\mathbf{r}$ of P is the same in both frames, but its rate of change depends upon which frame of reference you select.

The absolute velocity $\mathbf{v}_P$ of the particle is defined as the velocity observed from the fixed frame OXY and is equal to the rate of change $(\dot{\mathbf{r}})_{OXY}$ of $\mathbf{r}$ with respect to that frame. We can, however, express $\mathbf{v}_P$ in terms of the rate of change $(\dot{\mathbf{r}})_{Oxy}$ observed from the rotating frame if we make use of Eq. (15.31). Denoting the angular velocity of the frame Oxy with respect to OXY at the instant considered by $\boldsymbol{\Omega}$, we have

$$\mathbf{v}_P = (\dot{\mathbf{r}})_{OXY} = \boldsymbol{\Omega} \times \mathbf{r} + (\dot{\mathbf{r}})_{Oxy} \quad (15.32)$$

where $(\dot{\mathbf{r}})_{Oxy}$ defines the velocity of particle P relative to the rotating frame Oxy and is sometimes denoted as $\mathbf{v}_{\text{rel}}$. There also may be instances where point O is not fixed and has a velocity denoted by $\mathbf{v}_O$. Therefore, an alternative way to express Eq. (15.32) is

$$\mathbf{v}_P = \mathbf{v}_O + \boldsymbol{\Omega} \times \mathbf{r} + \mathbf{v}_{\text{rel}} \quad (15.32')$$

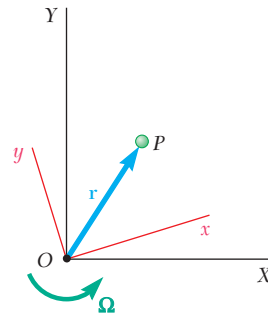


Fig. 15.27 We can express the motion of particle P in either a fixed ($OXYZ$) or a rotating ($Oxyz$) frame of reference.

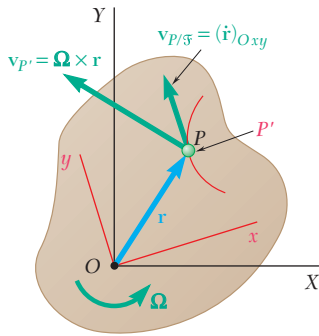


Fig. 15.28 The velocity of a point P is equal to the velocity of a point P' coincident with P but attached to the rotating frame plus the velocity of P with respect to the rotating frame.

The relative velocity, $\mathbf{v}_{\text{rel}}$ or $(\dot{\mathbf{r}})_{Oxy}$, is the velocity of point P with respect to the rotating frame. Denoting the rotating frame by $\mathcal{F}$, another way to represent the velocity $(\dot{\mathbf{r}})_{Oxy}$ of P relative to the rotating frame is $\mathbf{v}_{P/\mathcal{F}}$. Let us imagine that a rigid body has been attached to the rotating frame. Then $\mathbf{v}_{P/\mathcal{F}}$ represents the velocity of P along the path that it describes on that body (Fig. 15.28), and the term $\boldsymbol{\Omega} \times \mathbf{r}$ in Eq. (15.32) represents the velocity $\mathbf{v}_{P'}$ of the point P' of the rigid body—or rotating frame—that coincides with P at the instant considered. Thus, we have

$$\mathbf{v}_P = \mathbf{v}_{P'} + \mathbf{v}_{P/\mathcal{F}} \quad (15.33)$$

where

$\mathbf{v}_P$ = absolute velocity of particle P

$\mathbf{v}_{P'}$ = velocity of point P' of moving frame $\mathcal{F}$ coinciding with P

$\mathbf{v}_{P/\mathcal{F}}$ = velocity of P relative to moving frame $\mathcal{F}$

We define the absolute acceleration $\mathbf{a}_P$ of the particle as the rate of change of $\mathbf{v}_P$ with respect to the fixed frame OXY . Computing the rates of change with respect to OXY of the terms in Eq. (15.32), we have

$$\mathbf{a}_P = \dot{\mathbf{v}}_P = \dot{\boldsymbol{\Omega}} \times \mathbf{r} + \boldsymbol{\Omega} \times \dot{\mathbf{r}} + \frac{d}{dt}[(\dot{\mathbf{r}})_{Oxy}] \quad (15.34)$$

where all derivatives are defined with respect to OXY , except where indicated otherwise. Referring to Eq. (15.31), we note that we can express the last term in Eq. (15.34) as

$$\frac{d}{dt}[(\dot{\mathbf{r}})_{Oxy}] = (\ddot{\mathbf{r}})_{Oxy} + \boldsymbol{\Omega} \times (\dot{\mathbf{r}})_{Oxy}$$

On the other hand, $\dot{\mathbf{r}}$ represents the velocity $\mathbf{v}_P$ and can be replaced by the right-hand side of Eq. (15.32). After completing these two substitutions into Eq. (15.34), we obtain

$$\mathbf{a}_P = \dot{\boldsymbol{\Omega}} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r}) + 2\boldsymbol{\Omega} \times (\dot{\mathbf{r}})_{Oxy} + (\ddot{\mathbf{r}})_{Oxy} \quad (15.35)$$

As we had for the velocity expression, our reference point O might also be accelerating. For plane motion,

$$\mathbf{a}_P = \mathbf{a}_O + \dot{\boldsymbol{\Omega}} \times \mathbf{r} - \Omega^2 \mathbf{r} + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \mathbf{a}_{\text{rel}} \quad (15.35')$$

where

$\mathbf{a}_O$ = the linear acceleration of point O

$\dot{\boldsymbol{\Omega}}$ = angular acceleration of the rotating frame

$\boldsymbol{\Omega}$ = angular velocity of the rotating frame

$\mathbf{r}$ = position vector from the origin to point P

$\mathbf{v}_{\text{rel}}$ = relative velocity of point P with respect to the rotating frame

$\mathbf{a}_{\text{rel}}$ = relative acceleration of point P with respect to the rotating frame

From expression (15.8) obtained in Sec. 15.1B for the acceleration of a particle on a rigid body rotating about a fixed axis, we note that the sum of the first two terms in Eq. (15.35) represents the acceleration $\mathbf{a}_{P'}$ of the point P' of the rotating frame that coincides with P at the instant

considered. The last term defines the acceleration $\mathbf{a}_{P/\mathcal{F}}$ of P relative to the rotating frame. If it were not for the third term, which has not been accounted for, we could write a relation similar to Eq. (15.33) for the accelerations, and $\mathbf{a}_P$ could be expressed as the sum of $\mathbf{a}_{P'}$ and $\mathbf{a}_{P/\mathcal{F}}$. However, it is clear that *such a relation would be incorrect* and that we must include the additional term. This term, which we denote by $\mathbf{a}_C$, is called the **Coriolis acceleration**, after the French mathematician Gaspard de Coriolis (1792–1843). We have

$$\mathbf{a}_P = \mathbf{a}_{P'} + \mathbf{a}_{P/\mathcal{F}} + \mathbf{a}_C \quad (15.36)$$

where

$\mathbf{a}_P$ = absolute acceleration of particle P

$\mathbf{a}_{P'}$ = acceleration of point P' of moving frame $\mathcal{F}$ coinciding with P

$\mathbf{a}_{P/\mathcal{F}}$ = acceleration of P relative to moving frame $\mathcal{F}$

$\mathbf{a}_C = 2\boldsymbol{\Omega} \times (\dot{\mathbf{r}})_{Oxy} = 2\boldsymbol{\Omega} \times \mathbf{v}_{P/\mathcal{F}}$
= Coriolis acceleration

Note the difference between Eq. (15.36) and Eq. (15.21). When we wrote

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A} \quad (15.21)$$

in Sec. 15.4A, we were expressing the absolute acceleration of point B as the sum of the acceleration $\mathbf{a}_{B/A}$ relative to a frame in translation and the acceleration $\mathbf{a}_A$ of a point of that frame. We are now relating the absolute acceleration of point P to its acceleration $\mathbf{a}_{P/\mathcal{F}}$ relative to a rotating frame $\mathcal{F}$ and to the acceleration $\mathbf{a}_{P'}$ of point P' of that frame, which coincides with P . Equation (15.36) shows that, because the frame is rotating, it is necessary to include an additional term to represent the Coriolis acceleration $\mathbf{a}_C$.

Note that since point P' moves in a circle about the origin O , its acceleration $\mathbf{a}_{P'}$ has, in general, two components: $(\mathbf{a}_{P'})_t$ tangent to the circle and $(\mathbf{a}_{P'})_n$ directed toward O . Similarly, the acceleration $\mathbf{a}_{P/\mathcal{F}}$ generally has two components: $(\mathbf{a}_{P/\mathcal{F}})_t$ tangent to the path that P describes on the rotating rigid body and $(\mathbf{a}_{P/\mathcal{F}})_n$ directed toward the center of curvature of that path. We further note that since the vector $\boldsymbol{\Omega}$ is perpendicular to the plane of motion, and thus to $\mathbf{v}_{P/\mathcal{F}}$, the magnitude of the Coriolis acceleration $\mathbf{a}_C = 2\boldsymbol{\Omega} \times \mathbf{v}_{P/\mathcal{F}}$ is equal to $2\boldsymbol{\Omega}v_{P/\mathcal{F}}$, and its direction can be obtained by rotating the vector $\mathbf{v}_{P/\mathcal{F}}$ through 90° in the sense of rotation of the moving frame (Fig. 15.29). The Coriolis acceleration reduces to zero when either $\boldsymbol{\Omega}$ or $\mathbf{v}_{P/\mathcal{F}}$ is zero.

Consider a collar P that is made to slide at a constant relative speed u along a rod OB rotating at a constant angular velocity $\boldsymbol{\omega}$ about O (Fig. 15.30a). According to formula (15.36), we can obtain the absolute acceleration of P by adding vectorially the acceleration $\mathbf{a}_A$ of the point A of the rod coinciding with P , the relative acceleration $\mathbf{a}_{P/OB}$ of P with respect to the rod, and the Coriolis acceleration $\mathbf{a}_C$.

Since the angular velocity $\boldsymbol{\omega}$ of the rod is constant, $\mathbf{a}_A$ reduces to its normal component $(\mathbf{a}_A)_n$ with a magnitude of $r\omega^2$; and since u is constant, the relative acceleration $\mathbf{a}_{P/OB}$ is zero. According to the definition given previously, the Coriolis acceleration is a vector perpendicular to OB , has a

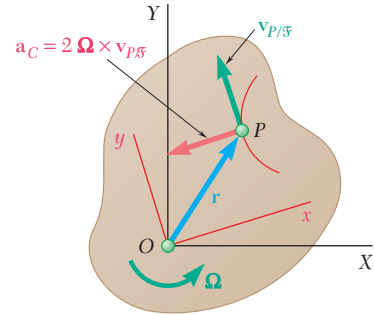


Fig. 15.29 The Coriolis acceleration is perpendicular to the relative velocity of P with respect to the rotating frame.

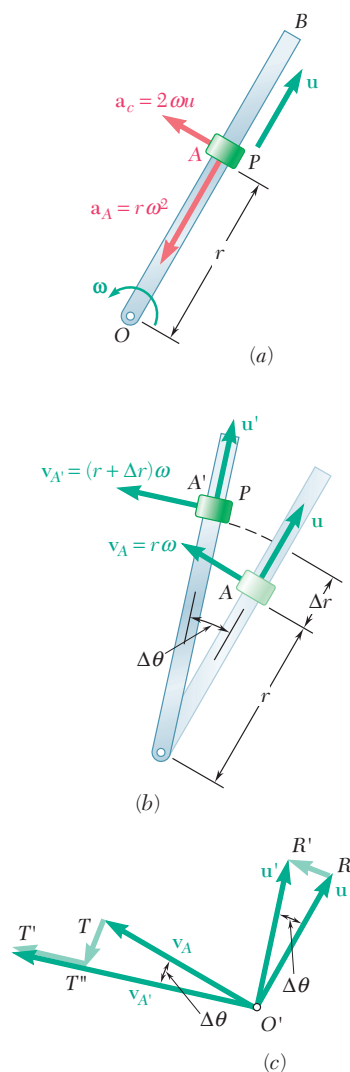


Fig. 15.30 (a) A collar sliding at constant speed along a rotating rod; (b) velocities of the collar at two points in time; (c) the acceleration components equal the changes in velocity.

magnitude of $2\omega u$, and is directed as shown in Figure 15.30. The acceleration of the collar P consists, therefore, of the two vectors shown in Fig. 15.30a. Note that you can check this result by applying the relation in Eq. (11.43).

To understand better the significance of the Coriolis acceleration, let us consider the absolute velocity of P at time t and at time $t + \Delta t$ (Fig. 15.30b). We can resolve the velocity at time t into its components $\mathbf{u}$ and $\mathbf{v}_A$; we can resolve the velocity at time $t + \Delta t$ into its components $\mathbf{u}'$ and $\mathbf{v}_{A'}$. Drawing these components from the same origin (Fig. 15.30c), we note that the change in velocity during the time Δt can be represented by the sum of three vectors: $\overrightarrow{RR'}$, $\overrightarrow{TT'}$, and $\overrightarrow{T''T'}$. The vector $\overrightarrow{TT'}$ measures the change in direction of the velocity $\mathbf{v}_A$, and the quotient $\overrightarrow{TT'}/\Delta t$ represents the acceleration $\mathbf{a}_A$ when Δt approaches zero. We check that the direction of $\overrightarrow{TT'}$ is that of $\mathbf{a}_A$ when Δt approaches zero and that

$$\lim_{\Delta t \rightarrow 0} \frac{\overrightarrow{TT'}}{\Delta t} = \lim_{\Delta t \rightarrow 0} v_A \frac{\Delta \theta}{\Delta t} = r\omega\omega = r\omega^2 = a_A$$

The vector $\overrightarrow{RR'}$ measures the change in direction of $\mathbf{u}$ due to the rotation of the rod; the vector $\overrightarrow{T''T'}$ measures the change in magnitude of $\mathbf{v}_A$ due to the motion of P on the rod. The vectors $\overrightarrow{RR'}$ and $\overrightarrow{T''T'}$ result from the *combined effect* of the relative motion of P and of the rotation of the rod; they would vanish if *either* of these two motions stopped. It is easily verified that the sum of these two vectors defines the Coriolis acceleration. Their direction is that of $\mathbf{a}_C$ when Δt approaches zero, and since $\overrightarrow{RR'} = u \Delta \theta$ and $\overrightarrow{T''T'} = v_{A'} - v_A = (r + \Delta r)\omega - r\omega = \omega \Delta r$, we check that $\mathbf{a}_C$ is equal to

$$\lim_{\Delta t \rightarrow 0} \left(\frac{\overrightarrow{RR'}}{\Delta t} + \frac{\overrightarrow{T''T'}}{\Delta t} \right) = \lim_{\Delta t \rightarrow 0} \left(u \frac{\Delta \theta}{\Delta t} + \omega \frac{\Delta r}{\Delta t} \right) = u\omega + \omega u = 2\omega u$$

We can use formulas (15.33) and (15.36) to analyze the motion of mechanisms that contain parts sliding on each other. They make it possible, for example, to relate the absolute and relative motions of sliding pins and collars (see Sample Probs. 15.18 through 15.20). The concept of Coriolis acceleration is also very useful in the study of long-range projectiles and of other objects whose motions are appreciably affected by the rotation of the earth. As we pointed out in Sec. 12.1A, a system of axes attached to the earth does not truly constitute a newtonian frame of reference; such a system of axes actually should be considered rotating. Thus, the formulas derived in this section facilitate the study of the motion of bodies with respect to axes attached to the earth.